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(54) **RNA-GUIDED NUCLEASES AND ACTIVE
FRAGMENTS AND VARIANTS THEREOF
AND METHODS OF USE**

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ABSTRACT

Compositions and methods for binding to a target sequence
of interest are provided. The compositions find use in
cleaving or modifying a target sequence of interest, visual-
ization of a target sequence of interest, and modifying the
expression of a sequence of interest. Compositions comprise
RNA-guided nuclease polypeptides, CRISPR RNAs, trans-
activating CRISPR RNAs, guide RNAs, and nucleic acid
molecules encoding the same. Vectors and host cells com-
prising the nucleic acid molecules are also provided. Further
provided are CRISPR systems for binding a target sequence
of interest, wherein the CRISPR system comprises an RNA-
guided nuclease polypeptide and one or more guide RNAs.

Specification includes a Sequence Listing.

RNA-GUIDED NUCLEASES AND ACTIVE FRAGMENTS AND VARIANTS THEREOF AND METHODS OF USE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 18/431,662, filed Feb. 2, 2024, which is a continuation of U.S. application Ser. No. 17/478,374, filed Sep. 17, 2021, now U.S. Pat. No. 11,926,843, which is a continuation of U.S. application Ser. No. 16/432,321, filed Jun. 5, 2019, now U.S. Pat. No. 11,162,114, which claims the benefit of U.S. Provisional Application Nos. 62/805,041, filed Feb. 13, 2019, 62/805,045, filed Feb. 13, 2019, 62/686,901, filed Jun. 19, 2018, 62/680,845, filed Jun. 5, 2018, 62/680,846, filed Jun. 5, 2018, 62/680,853, filed Jun. 5, 2018, 62/680,859, filed Jun. 5, 2018, 62/680,862, filed Jun. 5, 2018, and 62/680,863, filed Jun. 5, 2018, each of which is incorporated by reference herein in its entirety.

REFERENCE TO A SEQUENCE LISTING SUBMITTED ELECTRONICALLY AS AN XML FILE

[0002] The instant application contains a Sequence Listing which has been submitted in ST.26 XML format via USPTO Patent Center and is hereby incorporated by reference in its entirety. Said ST.26 XML copy, created on Aug. 6, 2024 is named L103438_1160US_C2_SL.XML, and is 1,099,916 bytes in size.

FIELD OF THE INVENTION

[0003] The present invention relates to the field of molecular biology and gene editing.

BACKGROUND OF THE INVENTION

[0004] Targeted genome editing or modification is rapidly becoming an important tool for basic and applied research. Initial methods involved engineering nucleases such as meganucleases, zinc finger fusion proteins or TALENs, requiring the generation of chimeric nucleases with engineered, programmable, sequence-specific DNA-binding domains specific for each particular target sequence. RNA-guided nucleases, such as the Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR)-associated (cas) proteins of the CRISPR-cas bacterial system, allow for the targeting of specific sequences by complexing the nucleases with guide RNA that specifically hybridizes with a particular target sequence. Producing target-specific guide RNAs is less costly and more efficient than generating chimeric nucleases for each target sequence. Such RNA-guided nucleases can be used to edit genomes through the introduction of a sequence-specific, double-stranded break that is repaired via error-prone non-homologous end-joining (NHEJ) to introduce a mutation at a specific genomic location. Alternatively, heterologous DNA may be introduced into the genomic site via homology-directed repair.

BRIEF SUMMARY OF THE INVENTION

[0005] Compositions and methods for binding a target sequence of interest are provided. The compositions find use in cleaving or modifying a target sequence of interest, detection of a target sequence of interest, and modifying the

expression of a sequence of interest. Compositions comprise RNA-guided nuclease (RGN) polypeptides, CRISPR RNAs (crRNAs), trans-activating CRISPR RNAs (tracrRNAs), guide RNAs (gRNAs), nucleic acid molecules encoding the same, and vectors and host cells comprising the nucleic acid molecules. Also provided are CRISPR systems for binding a target sequence of interest, wherein the CRISPR system comprises an RNA-guided nuclease polypeptide and one or more guide RNAs. Thus, methods disclosed herein are drawn to binding a target sequence of interest, and in some embodiments, cleaving or modifying the target sequence of interest. The target sequence of interest can be modified, for example, as a result of non-homologous end joining or homology-directed repair with an introduced donor sequence.

DETAILED DESCRIPTION

[0006] Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended embodiments. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

I. Overview

[0007] RNA-guided nucleases (RGNs) allow for the targeted manipulation of a single site within a genome and are useful in the context of gene targeting for therapeutic and research applications. In a variety of organisms, including mammals, RNA-guided nucleases have been used for genome engineering by stimulating non-homologous end joining and homologous recombination, for example. The compositions and methods described herein are useful for creating single- or double-stranded breaks in polynucleotides, modifying polynucleotides, detecting a particular site within a polynucleotide, or modifying the expression of a particular gene.

[0008] The RNA-guided nucleases disclosed herein can alter gene expression by modifying a target sequence. In specific embodiments, the RNA-guided nucleases are directed to the target sequence by a guide RNA (gRNA) as part of a Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) RNA-guided nuclease system. Guide RNAs form a complex with the RNA-guided nucleases to direct the RNA-guided nuclease to bind to a target sequence and in some embodiments, introduce a single-stranded or double-stranded break at the target sequence. After the target sequence has been cleaved, the break can be repaired such that the DNA sequence of the target sequence is modified during the repair process. Thus, provided herein are methods for using the RNA-guided nucleases to modify a target sequence in the DNA of host cells. For example, RNA-guided nucleases can be used to modify a target sequence at a genomic locus of eukaryotic cells or prokaryotic cells.

II. RNA-Guided Nucleases

[0009] Provided herein are RNA-guided nucleases. The term RNA-guided nuclease (RGN) refers to a polypeptide

that binds to a particular target nucleotide sequence in a sequence-specific manner and is directed to the target nucleotide sequence by a guide RNA molecule that is complexed with the polypeptide and hybridizes with the target sequence. Although an RNA-guided nuclease can be capable of cleaving the target sequence upon binding, the term RNA-guided nuclease also encompasses nuclease-dead RNA-guided nucleases that are capable of binding to, but not cleaving, a target sequence. Cleavage of a target sequence by an RNA-guided nuclease can result in a single- or double-stranded break. RNA-guided nucleases only capable of cleaving a single strand of a double-stranded nucleic acid molecule are referred to herein as nickases.

[0010] The RNA-guided nucleases disclosed herein include the APG05083.1, APG07433.1, APG07513.1, APG08290.1, APG05459.1, APG04583.1, and APG1688.1 RNA-guided nucleases, the amino acid sequences of which are set forth, respectively, as SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54, and active fragments or variants thereof that retain the ability to bind to a target nucleotide sequence in an RNA-guided sequence-specific manner. In some of these embodiments, the active fragment or variant of the APG05083.1, APG07433.1, APG07513.1, APG08290.1, APG05459.1, APG04583.1, and APG1688.1 RGN is capable of cleaving a single- or double-stranded target sequence. In some embodiments, an active variant of the APG05083.1, APG07433.1, APG07513.1, APG08290.1, APG05459.1, APG04583.1, or APG1688.1 RGN comprises an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to the amino acid sequence set forth as SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54. In certain embodiments, an active fragment of the APG05083.1, APG07433.1, APG07513.1, APG08290.1, APG05459.1, APG04583.1, or APG1688.1 RGN comprises at least 50, 100, 150, 200, 250, 300, 350, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, 900, 950, 1000, 1050 or more contiguous amino acid residues of the amino acid sequence set forth as SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54. RNA-guided nucleases provided herein can comprise at least one nuclease domain (e.g., DNase, RNase domain) and at least one RNA recognition and/or RNA binding domain to interact with guide RNAs. Further domains that can be found in RNA-guided nucleases provided herein include, but are not limited to: DNA binding domains, helicase domains, protein-protein interaction domains, and dimerization domains. In specific embodiments, the RNA-guided nucleases provided herein can comprise at least 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% to one or more of a DNA binding domains, helicase domains, protein-protein interaction domains, and dimerization domains.

[0011] A target nucleotide sequence is bound by an RNA-guided nuclease provided herein and hybridizes with the guide RNA associated with the RNA-guided nuclease. The target sequence can then be subsequently cleaved by the RNA-guided nuclease if the polypeptide possesses nuclease activity. The terms “cleave” or “cleavage” refer to the hydrolysis of at least one phosphodiester bond within the backbone of a target nucleotide sequence that can result in either single-stranded or double-stranded breaks within the target sequence. The presently disclosed RGNs can cleave nucleotides within a polynucleotide, functioning as an endonuclease or can be an exonuclease, removing successive

nucleotides from the end (the 5' and/or the 3' end) of a polynucleotide. In other embodiments, the disclosed RGNs can cleave nucleotides of a target sequence within any position of a polynucleotide and thus function as both an endonuclease and exonuclease. The cleavage of a target polynucleotide by the presently disclosed RGNs can result in staggered breaks or blunt ends.

[0012] The presently disclosed RNA-guided nucleases can be wild-type sequences derived from bacterial or archaeal species. Alternatively, the RNA-guided nucleases can be variants or fragments of wild-type polypeptides. The wild-type RGN can be modified to alter nuclease activity or alter PAM specificity, for example. In some embodiments, the RNA-guided nuclease is not naturally-occurring.

[0013] In certain embodiments, the RNA-guided nuclease functions as a nickase, only cleaving a single strand of the target nucleotide sequence. Such RNA-guided nucleases have a single functioning nuclease domain. In some of these embodiments, additional nuclease domains have been mutated such that the nuclease activity is reduced or eliminated.

[0014] In other embodiments, the RNA-guided nuclease lacks nuclease activity altogether or exhibits reduced nuclease activity, and is referred to herein as nuclease-dead. Any method known in the art for introducing mutations into an amino acid sequence, such as PCR-mediated mutagenesis and site-directed mutagenesis, can be used for generating nickases or nuclease-dead RGNs. See, e.g., U.S. Publ. No. 2014/0068797 and U.S. Pat. No. 9,790,490; each of which is incorporated by reference in its entirety.

[0015] RNA-guided nucleases that lack nuclease activity can be used to deliver a fused polypeptide, polynucleotide, or small molecule payload to a particular genomic location. In some of these embodiments, the RGN polypeptide or guide RNA can be fused to a detectable label to allow for detection of a particular sequence. As a non-limiting example, a nuclease-dead RGN can be fused to a detectable label (e.g., fluorescent protein) and targeted to a particular sequence associated with a disease to allow for detection of the disease-associated sequence.

[0016] Alternatively, nuclease-dead RGNs can be targeted to particular genomic locations to alter the expression of a desired sequence. In some embodiments, the binding of a nuclease-dead RNA-guided nuclease to a target sequence results in the repression of expression of the target sequence or a gene under transcriptional control by the target sequence by interfering with the binding of RNA polymerase or transcription factors within the targeted genomic region. In other embodiments, the RGN (e.g., a nuclease-dead RGN) or its complexed guide RNA further comprises an expression modulator that, upon binding to a target sequence, serves to either repress or activate the expression of the target sequence or a gene under transcriptional control by the target sequence. In some of these embodiments, the expression modulator modulates the expression of the target sequence or regulated gene through epigenetic mechanisms.

[0017] In other embodiments, the nuclease-dead RGNs or a RGN with only nickase activity can be targeted to particular genomic locations to modify the sequence of a target polynucleotide through fusion to a base-editing polypeptide, for example a deaminase polypeptide or active variant or fragment thereof that deaminates a nucleotide base, resulting in conversion from one nucleotide base to another. The base-editing polypeptide can be fused to the RGN at its

N-terminal or C-terminal end. Additionally, the base-editing polypeptide may be fused to the RGN via a peptide linker. A non-limiting example of a deaminase polypeptide that is useful for such compositions and methods include cytidine deaminase or the adenosine deaminase base editor described in Gaudelli et al. (2017) *Nature* 551:464-471, U.S. Publ. Nos. 2017/0121693 and 2018/0073012, and International Publ. No. WO/2018/027078, each of which is herein incorporated by reference in its entirety.

[0018] RNA-guided nucleases that are fused to a polypeptide or domain can be separated or joined by a linker. The term “linker,” as used herein, refers to a chemical group or a molecule linking two molecules or moieties, e.g., a binding domain and a cleavage domain of a nuclease. In some embodiments, a linker joins a gRNA binding domain of an RNA guided nuclease and a base-editing polypeptide, such as a deaminase. In some embodiments, a linker joins a nuclease-dead RGN and a deaminase. Typically, the linker is positioned between, or flanked by, two groups, molecules, or other moieties and connected to each one via a covalent bond, thus connecting the two. In some embodiments, the linker is an amino acid or a plurality of amino acids (e.g., a peptide or protein). In some embodiments, the linker is an organic molecule, group, polymer, or chemical moiety. In some embodiments, the linker is 5-100 amino acids in length, for example, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 30-35, 35-40, 40-45, 45-50, 50-60, 60-70, 70-80, 80-90, 90-100, 100-150, or 150-200 amino acids in length. Longer or shorter linkers are also contemplated.

[0019] The presently disclosed RNA-guided nucleases can comprise at least one nuclear localization signal (NLS) to enhance transport of the RGN to the nucleus of a cell. Nuclear localization signals are known in the art and generally comprise a stretch of basic amino acids (see, e.g., Lange et al., *J. Biol. Chem.* (2007) 282:5101-5105). In particular embodiments, the RGN comprises 2, 3, 4, 5, 6 or more nuclear localization signals. The nuclear localization signal(s) can be a heterologous NLS. Non-limiting examples of nuclear localization signals useful for the presently disclosed RGNs are the nuclear localization signals of SV40 Large T-antigen, nucleopasmin, and c-Myc (see, e.g., Ray et al. (2015) *Bioconjug Chem* 26 (6): 1004-7). In particular embodiments, the RGN comprises the NLS sequence set forth as SEQ ID NO: 67. The RGN can comprise one or more NLS sequences at its N-terminus, C-terminus, or both the N-terminus and C-terminus. For example, the RGN can comprise two NLS sequences at the N-terminal region and four NLS sequences at the C-terminal region.

[0020] Other localization signal sequences known in the art that localize polypeptides to particular subcellular location(s) can also be used to target the RGNs, including, but not limited to, plastid localization sequences, mitochondrial localization sequences, and dual-targeting signal sequences that target to both the plastid and mitochondria (see, e.g., Nassoury and Morse (2005) *Biochim Biophys Acta* 1743:5-19; Kunze and Berger (2015) *Front Physiol* dx.doi.org/10.3389/fphys.2015.00259; Herrmann and Neupert (2003) *IUBMB Life* 55:219-225; Soll (2002) *Curr Opin Plant Biol* 5:529-535; Carrie and Small (2013) *Biochim Biophys Acta* 1833:253-259; Carrie et al. (2009) *FEBS J* 276:1187-1195; Silva-Filho (2003) *Curr Opin Plant Biol* 6:589-595; Peeters and Small (2001) *Biochim Biophys Acta* 1541:54-63; Mur-

cha et al. (2014) *J Exp Bot* 65:6301-6335; Mackenzie (2005) *Trends Cell Biol* 15:548-554; Glaser et al. (1998) *Plant Mol Biol* 38:311-338).

[0021] In certain embodiments, the presently disclosed RNA-guided nucleases comprise at least one cell-penetrating domain that facilitates cellular uptake of the RGN. Cell-penetrating domains are known in the art and generally comprise stretches of positively charged amino acid residues (i.e., polycationic cell-penetrating domains), alternating polar amino acid residues and non-polar amino acid residues (i.e., amphipathic cell-penetrating domains), or hydrophobic amino acid residues (i.e., hydrophobic cell-penetrating domains) (see, e.g., Milletti F. (2012) *Drug Discov Today* 17:850-860). A non-limiting example of a cell-penetrating domain is the trans-activating transcriptional activator (TAT) from the human immunodeficiency virus 1.

[0022] The nuclear localization signal, plastid localization signal, mitochondrial localization signal, dual-targeting localization signal, and/or cell-penetrating domain can be located at the amino-terminus (N-terminus), the carboxyl-terminus (C-terminus), or in an internal location of the RNA-guided nuclease. The presently disclosed RGNs can be fused to an effector domain, such as a cleavage domain, a deaminase domain, or an expression modulator domain, either directly or indirectly via a linker peptide. Such a domain can be located at the N-terminus, the C-terminus, or an internal location of the RNA-guided nuclease. In some of these embodiments, the RGN component of the fusion protein is a nuclease-dead RGN.

[0023] In some embodiments, the RGN fusion protein comprises a cleavage domain, which is any domain that is capable of cleaving a polynucleotide (i.e., RNA, DNA, or RNA/DNA hybrid) and includes, but is not limited to, restriction endonucleases and homing endonucleases, such as Type IIS endonucleases (e.g., FokI) (see, e.g., Belfort et al. (1997) *Nucleic Acids Res.* 25:3379-3388; Linn et al. (eds.) *Nucleases*, Cold Spring Harbor Laboratory Press, 1993).

[0024] In other embodiments, the RGN fusion protein comprises a deaminase domain that deaminates a nucleotide base, resulting in conversion from one nucleotide base to another, and includes, but is not limited to, a cytidine deaminase or an adenosine deaminase base editor (see, e.g., Gaudelli et al. (2017) *Nature* 551:464-471, U.S. Publ. Nos. 2017/0121693 and 2018/0073012, U.S. Pat. No. 9,840,699, and International Publ. No. WO/2018/027078).

[0025] In some embodiments, the effector domain of the RGN fusion protein can be an expression modulator domain, which is a domain that either serves to upregulate or downregulate transcription. The expression modulator domain can be an epigenetic modification domain, a transcriptional repressor domain or a transcriptional activation domain.

[0026] In some of these embodiments, the expression modulator of the RGN fusion protein comprises an epigenetic modification domain that covalently modifies DNA or histone proteins to alter histone structure and/or chromosomal structure without altering the DNA sequence, leading to changes in gene expression (i.e., upregulation or downregulation). Non-limiting examples of epigenetic modifications include acetylation or methylation of lysine residues, arginine methylation, serine and threonine phosphorylation, and lysine ubiquitination and sumoylation of histone proteins, and methylation and hydroxymethylation of cytosine

residues in DNA. Non-limiting examples of epigenetic modification domains include histone acetyltransferase domains, histone deacetylase domains, histone methyltransferase domains, histone demethylase domains, DNA methyltransferase domains, and DNA demethylase domains.

[0027] In other embodiments, the expression modulator of the fusion protein comprises a transcriptional repressor domain, which interacts with transcriptional control elements and/or transcriptional regulatory proteins, such as RNA polymerases and transcription factors, to reduce or terminate transcription of at least one gene. Transcriptional repressor domains are known in the art and include, but are not limited to, Sp1-like repressors, IκB, and Krüppel associated box (KRAB) domains.

[0028] In yet other embodiments, the expression modulator of the fusion protein comprises a transcriptional activation domain, which interacts with transcriptional control elements and/or transcriptional regulatory proteins, such as RNA polymerases and transcription factors, to increase or activate transcription of at least one gene. Transcriptional activation domains are known in the art and include, but are not limited to, a herpes simplex virus VP16 activation domain and an NFAT activation domain.

[0029] The presently disclosed RGN polypeptides can comprise a detectable label or a purification tag. The detectable label or purification tag can be located at the N-terminus, the C-terminus, or an internal location of the RNA-guided nuclease, either directly or indirectly via a linker peptide. In some of these embodiments, the RGN component of the fusion protein is a nuclease-dead RGN. In other embodiments, the RGN component of the fusion protein is a RGN with nickase activity.

[0030] A detectable label is a molecule that can be visualized or otherwise observed. The detectable label may be fused to the RGN as a fusion protein (e.g., fluorescent protein) or may be a small molecule conjugated to the RGN polypeptide that can be detected visually or by other means. Detectable labels that can be fused to the presently disclosed RGNs as a fusion protein include any detectable protein domain, including but not limited to, a fluorescent protein or a protein domain that can be detected with a specific antibody. Non-limiting examples of fluorescent proteins include green fluorescent proteins (e.g., GFP, EGFP, ZsGreen1) and yellow fluorescent proteins (e.g., YFP, EYFP, ZsYellow1). Non-limiting examples of small molecule detectable labels include radioactive labels, such as 3H and 35S.

[0031] RGN polypeptides can also comprise a purification tag, which is any molecule that can be utilized to isolate a protein or fused protein from a mixture (e.g., biological sample, culture medium). Non-limiting examples of purification tags include biotin, myc, maltose binding protein (MBP), and glutathione-S-transferase (GST).

II. Guide RNA

[0032] The present disclosure provides guide RNAs and polynucleotides encoding the same. The term “guide RNA” refers to a nucleotide sequence having sufficient complementarity with a target nucleotide sequence to hybridize with the target sequence and direct sequence-specific binding of an associated RNA-guided nuclease to the target nucleotide sequence. Thus, a RGN’s respective guide RNA is one or more RNA molecules (generally, one or two), that can bind to the RGN and guide the RGN to bind to a

particular target nucleotide sequence, and in those instances wherein the RGN has nickase or nuclease activity, also cleave the target nucleotide sequence. In general, a guide RNA comprises a CRISPR RNA (crRNA) and a transactivating CRISPR RNA (tracrRNA). Native guide RNAs that comprise both a crRNA and a tracrRNA generally comprise two separate RNA molecules that hybridize to each other through the repeat sequence of the crRNA and the anti-repeat sequence of the tracrRNA.

[0033] Native direct repeat sequences within a CRISPR array generally range in length from 28 to 37 base pairs, although the length can vary between about 23 bp to about 55 bp. Spacer sequences within a CRISPR array generally range from about 32 to about 38 bp in length, although the length can be between about 21 bp to about 72 bp. Each CRISPR array generally comprises less than 50 units of the CRISPR repeat-spacer sequence. The CRISPRs are transcribed as part of a long transcript termed the primary

[0034] CRISPR transcript, which comprises much of the CRISPR array. The primary CRISPR transcript is cleaved by Cas proteins to produce crRNAs or in some cases, to produce pre-crRNAs that are further processed by additional Cas proteins into mature crRNAs. Mature crRNAs comprise a spacer sequence and a CRISPR repeat sequence. In some embodiments in which pre-crRNAs are processed into mature (or processed) crRNAs, maturation involves the removal of about one to about six or more 5', 3', or 5' and 3' nucleotides. For the purposes of genome editing or targeting a particular target nucleotide sequence of interest, these nucleotides that are removed during maturation of the pre-crRNA molecule are not necessary for generating or designing a guide RNA.

[0035] A CRISPR RNA (crRNA) comprises a spacer sequence and a CRISPR repeat sequence. The “spacer sequence” is the nucleotide sequence that directly hybridizes with the target nucleotide sequence of interest. The spacer sequence is engineered to be fully or partially complementary with the target sequence of interest. In various embodiments, the spacer sequence can comprise from about 8 nucleotides to about 30 nucleotides, or more. For example, the spacer sequence can be about 8, about 9, about 10, about 11, about 12, about 13, about 14, about 15, about 16, about 17, about 18, about 19, about 20, about 21, about 22, about 23, about 24, about 25, about 26, about 27, about 28, about 29, about 30, or more nucleotides in length. In some embodiments, the spacer sequence is about 10 to about 26 nucleotides in length, or about 12 to about 30 nucleotides in length. In particular embodiments, the spacer sequence is about 30 nucleotides in length. In some embodiments, the degree of complementarity between a spacer sequence and its corresponding target sequence, when optimally aligned using a suitable alignment algorithm, is about or more than about 50%, about 60%, about 70%, about 75%, about 80%, about 81%, about 82%, about 83%, about 84%, about 85%, about 86%, about 87%, about 88%, about 89%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or more. In particular embodiments, the spacer sequence is free of secondary structure, which can be predicted using any suitable polynucleotide folding algorithm known in the art, including but not limited to mFold (see, e.g., Zuker and Stiegler (1981) *Nucleic Acids Res.* 9:133-148) and RNAfold (see, e.g., Gruber et al. (2008) *Cell* 106 (1): 23-24).

[0036] RGN proteins can have varying sensitivity to mismatches between a spacer sequence in a gRNA and its target sequence that affects the efficiency of cleavage. As discussed in Example 5, APG05459.1 RGN has an unusual sensitivity to mismatches between the spacer sequence and target sequence, extending at least 15 nucleotides 5' of the PAM site. Thus, APG05459.1 has the potential to more finely (i.e., specifically) target particular sequences with greater precision than other RGNs with less sensitivity to mismatches between the spacer sequence and target sequence.

[0037] The CRISPR RNA repeat sequence comprises a nucleotide sequence that comprises a region with sufficient complementarity to hybridize to a tracrRNA. In various embodiments, the CRISPR RNA repeat sequence can comprise from about 8 nucleotides to about 30 nucleotides, or more. For example, the CRISPR repeat sequence can be about 8, about 9, about 10, about 11, about 12, about 13, about 14, about 15, about 16, about 17, about 18, about 19, about 20, about 21, about 22, about 23, about 24, about 25, about 26, about 27, about 28, about 29, about 30, or more nucleotides in length. In some embodiments, the CRISPR repeat sequence is about 21 nucleotides in length. In some embodiments, the degree of complementarity between a CRISPR repeat sequence and its corresponding tracrRNA sequence, when optimally aligned using a suitable alignment algorithm, is about or more than about 50%, about 60%, about 70%, about 75%, about 80%, about 81%, about 82%, about 83%, about 84%, about 85%, about 86%, about 87%, about 88%, about 89%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or more. In particular embodiments, the CRISPR repeat sequence comprises the nucleotide sequence of SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55, or an active variant or fragment thereof that when comprised within a guide RNA, is capable of directing the sequence-specific binding of an associated RNA-guided nuclease provided herein to a target sequence of interest. In certain embodiments, an active CRISPR repeat sequence variant of a wild-type sequence comprises a nucleotide sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to the nucleotide sequence set forth as SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55. In certain embodiments, an active CRISPR repeat sequence fragment of a wild-type sequence comprises at least 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 contiguous nucleotides of the nucleotide sequence set forth as SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55.

[0038] In certain embodiments, the crRNA is not naturally-occurring. In some of these embodiments, the specific CRISPR repeat sequence is not linked to the engineered spacer sequence in nature and the CRISPR repeat sequence is considered heterologous to the spacer sequence. In certain embodiments, the spacer sequence is an engineered sequence that is not naturally occurring.

[0039] A trans-activating CRISPR RNA or tracrRNA molecule comprises a nucleotide sequence comprising a region that has sufficient complementarity to hybridize to a CRISPR repeat sequence of a crRNA, which is referred to herein as the anti-repeat region. In some embodiments, the tracrRNA molecule further comprises a region with secondary structure (e.g., stem-loop) or forms secondary structure upon hybridizing with its corresponding crRNA. In particular embodiments, the region of the tracrRNA that is fully or

partially complementary to a CRISPR repeat sequence is at the 5' end of the molecule and the 3' end of the tracrRNA comprises secondary structure. This region of secondary structure generally comprises several hairpin structures, including the *nexus* hairpin, which is found adjacent to the anti-repeat sequence. The *nexus* hairpin often has a conserved nucleotide sequence in the base of the hairpin stem, with the motif UNANNG, UNANNU, or UNANNA (SEQ ID NOs: 68, 557, and 558, respectively) found in many *nexus* hairpins in tracrRNAs. There are often terminal hairpins at the 3' end of the tracrRNA that can vary in structure and number, but often comprise a GC-rich Rho-independent transcriptional terminator hairpin followed by a string of U's at the 3' end. See, for example, Briner et al. (2014) *Molecular Cell* 56:333-339, Briner and Barrangou (2016) Cold Spring Harb Protoc; doi: 10.1101/pdb.top090902, and U.S. Publication No. 2017/0275648, each of which is herein incorporated by reference in its entirety.

[0040] In various embodiments, the anti-repeat region of the tracrRNA that is fully or partially complementary to the CRISPR repeat sequence comprises from about 8 nucleotides to about 30 nucleotides, or more. For example, the region of base pairing between the tracrRNA anti-repeat sequence and the CRISPR repeat sequence can be about 8, about 9, about 10, about 11, about 12, about 13, about 14, about 15, about 16, about 17, about 18, about 19, about 20, about 21, about 22, about 23, about 24, about 25, about 26, about 27, about 28, about 29, about 30, or more nucleotides in length. In particular embodiments, the anti-repeat region of the tracrRNA that is fully or partially complementary to a CRISPR repeat sequence is about 20 nucleotides in length. In some embodiments, the degree of complementarity between a CRISPR repeat sequence and its corresponding tracrRNA anti-repeat sequence, when optimally aligned using a suitable alignment algorithm, is about or more than about 50%, about 60%, about 70%, about 75%, about 80%, about 81%, about 82%, about 83%, about 84%, about 85%, about 86%, about 87%, about 88%, about 89%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or more.

[0041] In various embodiments, the entire tracrRNA can comprise from about 60 nucleotides to more than about 140 nucleotides. For example, the tracrRNA can be about 60, about 65, about 70, about 75, about 80, about 85, about 90, about 95, about 100, about 105, about 110, about 115, about 120, about 125, about 130, about 135, about 140, or more nucleotides in length. In particular embodiments, the tracrRNA is about 80 to about 90 nucleotides in length, including about 80, about 81, about 82, about 83, about 84, about 85, about 86, about 87, about 88, about 89, and about 90 nucleotides in length. In certain embodiments, the tracrRNA is about 85 nucleotides in length.

[0042] In particular embodiments, the tracrRNA comprises the nucleotide sequence of SEQ ID NOs: 3, 13, 21, 29, 38, 47, or 56, or an active variant or fragment thereof that when comprised within a guide RNA is capable of directing the sequence-specific binding of an associated RNA-guided nuclease provided herein to a target sequence of interest. In certain embodiments, an active tracrRNA sequence variant of a wild-type sequence comprises a nucleotide sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to the nucleotide sequence set forth as SEQ ID NOs: 3, 13, 21, 29, 38, 47, or

56. In certain embodiments, an active tracrRNA sequence fragment of a wild-type sequence comprises at least 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, or more contiguous nucleotides of the nucleotide sequence set forth as SEQ ID NOs: 3, 13, 21, 29, 38, 47, or 56.

[0043] Two polynucleotide sequences can be considered to be substantially complementary when the two sequences hybridize to each other under stringent conditions. Likewise, an RGN is considered to bind to a particular target sequence within a sequence-specific manner if the guide RNA bound to the RGN binds to the target sequence under stringent conditions. By “stringent conditions” or “stringent hybridization conditions” is intended conditions under which the two polynucleotide sequences will hybridize to each other to a detectably greater degree than to other sequences (e.g., at least 2-fold over background). Stringent conditions are sequence-dependent and will be different in different circumstances. Typically, stringent conditions will be those in which the salt concentration is less than about 1.5 M Na ion, typically about 0.01 to 1.0 M Na ion concentration (or other salts) at pH 7.0 to 8.3, and the temperature is at least about 30° C. for short sequences (e.g., 10 to 50 nucleotides) and at least about 60° C. for long sequences (e.g., greater than 50 nucleotides). Stringent conditions may also be achieved with the addition of destabilizing agents such as formamide. Exemplary low stringency conditions include hybridization with a buffer solution of 30 to 35% formamide, 1 M NaCl, 1% SDS (sodium dodecyl sulfate) at 37° C., and a wash in 1× to 2×SSC (20×SSC=3.0 M NaCl/0.3 M trisodium citrate) at 50 to 55° C. Exemplary moderate stringency conditions include hybridization in 40 to 45% formamide, 1.0 M NaCl, 1% SDS at 37° C., and a wash in 0.5× to 1×SSC at 55 to 60° C. Exemplary high stringency conditions include hybridization in 50% formamide, 1 M NaCl, 1% SDS at 37° C., and a wash in 0.1×SSC at 60 to 65° C. Optionally, wash buffers may comprise about 0.1% to about 1% SDS. Duration of hybridization is generally less than about 24 hours, usually about 4 to about 12 hours. The duration of the wash time will be at least a length of time sufficient to reach equilibrium.

[0044] The T_m is the temperature (under defined ionic strength and pH) at which 50% of a complementary target sequence hybridizes to a perfectly matched sequence. For DNA-DNA hybrids, the T_m can be approximated from the equation of Meinkoth and Wahl (1984) *Anal. Biochem.* 138:267-284: $T_m = 81.5^\circ \text{C.} + 16.6 (\log M) + 0.41 (\% \text{ GC}) - 0.61 (\% \text{ form}) - 500/L$; where M is the molarity of monovalent cations, % GC is the percentage of guanosine and cytosine nucleotides in the DNA, % form is the percentage of formamide in the hybridization solution, and L is the length of the hybrid in base pairs. Generally, stringent conditions are selected to be about 5° C. lower than the thermal melting point (T_m) for the specific sequence and its complement at a defined ionic strength and pH. However, severely stringent conditions can utilize a hybridization and/or wash at 1, 2, 3, or 4° C. lower than the thermal melting point (T_m); moderately stringent conditions can utilize a hybridization and/or wash at 6, 7, 8, 9, or 10° C. lower than the thermal melting point (T_m); low stringency conditions can utilize a hybridization and/or wash at 11, 12, 13, 14, 15, or 20° C. lower than the thermal melting point (T_m). Using the equation, hybridization and wash compositions, and desired T_m , those of ordinary skill will understand that variations in the stringency of hybridization and/or wash solutions are inherently described. An extensive guide to the hybridization of

nucleic acids is found in Tijssen (1993) *Laboratory Techniques in Biochemistry and Molecular Biology-Hybridization with Nucleic Acid Probes, Part I*, Chapter 2 (Elsevier, New York); and Ausubel et al., eds. (1995) *Current Protocols in Molecular Biology, Chapter 2* (Greene Publishing and Wiley-Interscience, New York). See Sambrook et al. (1989) *Molecular Cloning: A Laboratory Manual* (2d ed., Cold Spring Harbor Laboratory Press, Plainview, New York).

[0045] The guide RNA can be a single guide RNA or a dual-guide RNA system. A single guide RNA comprises the crRNA and tracrRNA on a single molecule of RNA, whereas a dual-guide RNA system comprises a crRNA and a tracrRNA present on two distinct RNA molecules, hybridized to one another through at least a portion of the CRISPR repeat sequence of the crRNA and at least a portion of the tracrRNA, which may be fully or partially complementary to the CRISPR repeat sequence of the crRNA. In some of those embodiments wherein the guide RNA is a single guide RNA, the crRNA and tracrRNA are separated by a linker nucleotide sequence. In general, the linker nucleotide sequence is one that does not include complementary bases in order to avoid the formation of secondary structure within or comprising nucleotides of the linker nucleotide sequence. In some embodiments, the linker nucleotide sequence between the crRNA and tracrRNA is at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, or more nucleotides in length. In particular embodiments, the linker nucleotide sequence of a single guide RNA is at least 4 nucleotides in length. In certain embodiments, the linker nucleotide sequence is the nucleotide sequence set forth as SEQ ID NO: 63, 64, or 65. In other embodiments, the linker nucleotide sequence is at least 6 nucleotides in length. In certain embodiments, the linker nucleotide sequence is the nucleotide sequence set forth as SEQ ID NO: 65.

[0046] The single guide RNA or dual-guide RNA can be synthesized chemically or via in vitro transcription. Assays for determining sequence-specific binding between a RGN and a guide RNA are known in the art and include, but are not limited to, in vitro binding assays between an expressed RGN and the guide RNA, which can be tagged with a detectable label (e.g., biotin) and used in a pull-down detection assay in which the guide RNA: RGN complex is captured via the detectable label (e.g., with streptavidin beads). A control guide RNA with an unrelated sequence or structure to the guide RNA can be used as a negative control for non-specific binding of the RGN to RNA. In certain embodiments, the guide RNA is SEQ ID NO: 10, 18, 26, 35, 44, 53, or 62, wherein the spacer sequence can be any sequence and is indicated as a poly-N sequence.

[0047] As described in Example 8, certain RGNs of the invention can share certain guide RNAs. APG05083.1, APG07433.1, APG07513.1, and APG08290.1 can each function using guide RNAs comprising a crRNA comprising the nucleotide sequence of SEQ ID NOs: 2, 12, 20, or 28, with the corresponding tracrRNA comprising the nucleotide sequence of SEQ ID NOs: 3, 13, 21, or 29, respectively. Further, APG04583.1 and APG01688.1 can each function using guide RNAs comprising a crRNA comprising the nucleotide sequence of SEQ ID NOs: 46 or 55, with the corresponding tracrRNA comprising the nucleotide sequence of SEQ 47 or 56, respectively.

[0048] In certain embodiments, the guide RNA can be introduced into a target cell, organelle, or embryo as an RNA molecule. The guide RNA can be transcribed in vitro or chemically synthesized. In other embodiments, a nucleotide sequence encoding the guide RNA is introduced into the cell, organelle, or embryo. In some of these embodiments, the nucleotide sequence encoding the guide RNA is operably linked to a promoter (e.g., an RNA polymerase III promoter). The promoter can be a native promoter or heterologous to the guide RNA-encoding nucleotide sequence.

[0049] In various embodiments, the guide RNA can be introduced into a target cell, organelle, or embryo as a ribonucleoprotein complex, as described herein, wherein the guide RNA is bound to an RNA-guided nuclease polypeptide.

[0050] The guide RNA directs an associated RNA-guided nuclease to a particular target nucleotide sequence of interest through hybridization of the guide RNA to the target nucleotide sequence. A target nucleotide sequence can comprise DNA, RNA, or a combination of both and can be single-stranded or double-stranded. A target nucleotide sequence can be genomic DNA (i.e., chromosomal DNA), plasmid DNA, or an RNA molecule (e.g., messenger RNA, ribosomal RNA, transfer RNA, micro RNA, small interfering RNA). The target nucleotide sequence can be bound (and in some embodiments, cleaved) by an RNA-guided nuclease in vitro or in a cell. The chromosomal sequence targeted by the RGN can be a nuclear, plastid or mitochondrial chromosomal sequence. In some embodiments, the target nucleotide sequence is unique in the target genome.

[0051] The target nucleotide sequence is adjacent to a protospacer adjacent motif (PAM). A protospacer adjacent motif is generally within about 1 to about 10 nucleotides from the target nucleotide sequence, including about 1, about 2, about 3, about 4, about 5, about 6, about 7, about 8, about 9, or about 10 nucleotides from the target nucleotide sequence. The PAM can be 5' or 3' of the target sequence. In some embodiments, the PAM is 3' of the target sequence for the presently disclosed RGNs. Generally, the PAM is a consensus sequence of about 3-4 nucleotides, but in particular embodiments, can be 2, 3, 4, 5, 6, 7, 8, 9, or more nucleotides in length. In various embodiments, the PAM sequence recognized by the presently disclosed RGNs comprises the consensus sequence set forth as SEQ ID NOs: 6, 32, 41, 50, or 59. Non-limiting exemplary PAM sequences are the nucleotide sequences set forth as SEQ ID NO: 7, 69, 70, 71, and 72.

[0052] In particular embodiments, an RNA-guided nuclease having SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54 or an active variant or fragment thereof binds respectively a target nucleotide sequence adjacent to a PAM sequence set forth as SEQ ID NOs: 6, 32, 41, 50, 59, or 7. In some of these embodiments, the RGN binds to a guide sequence comprising a CRISPR repeat sequence set forth in SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55, respectively, or an active variant or fragment thereof, and a tracrRNA sequence set forth in SEQ ID NOs: 3, 13, 21, 29, 38, 47, or 56, respectively, or an active variant or fragment thereof. The RGN systems are described further in Example 1 and Table 1 of the present specification.

[0053] It is well-known in the art that PAM sequence specificity for a given nuclease enzyme is affected by enzyme concentration (see, e.g., Karvelis et al. (2015) *Genome Biol* 16:253), which may be modified by altering

the promoter used to express the RGN, or the amount of ribonucleoprotein complex delivered to the cell, organelle, or embryo.

[0054] Upon recognizing its corresponding PAM sequence, the RGN can cleave the target nucleotide sequence at a specific cleavage site. As used herein, a cleavage site is made up of the two particular nucleotides within a target nucleotide sequence between which the nucleotide sequence is cleaved by an RGN. The cleavage site can comprise the 1st and 2nd, 2nd and 3rd, 3rd and 4th, 4th and 5th, 5th and 6th, 7th and 8th, or 8th and 9th nucleotides from the PAM in either the 5' or 3' direction. In some embodiments, the cleavage site may be over 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 nucleotides from the PAM in either the 5' or 3' direction. In some embodiments, the cleavage site is 4 nucleotides away from the PAM. In other embodiments, the cleavage site is at least 15 nucleotides away from the PAM. As RGNs can cleave a target nucleotide sequence resulting in staggered ends, in some embodiments, the cleavage site is defined based on the distance of the two nucleotides from the PAM on the positive (+) strand of the polynucleotide and the distance of the two nucleotides from the PAM on the negative (-) strand of the polynucleotide.

III. Nucleotides Encoding RNA-Guided Nucleases, CRISPR RNA, and/or tracrRNA

[0055] The present disclosure provides polynucleotides comprising the presently disclosed CRISPR RNAs, tracrRNAs, and/or sgRNAs and polynucleotides comprising a nucleotide sequence encoding the presently disclosed RNA-guided nucleases, CRISPR RNAs, tracrRNAs, and/or sgRNAs. Presently disclosed polynucleotides include those comprising or encoding a CRISPR repeat sequence comprising the nucleotide sequence of SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55, or an active variant or fragment thereof that when comprised within a guide RNA is capable of directing the sequence-specific binding of an associated RNA-guided nuclease to a target sequence of interest. Also disclosed are polynucleotides comprising or encoding a tracrRNA comprising the nucleotide sequence of SEQ ID NOs: 3, 13, 21, 29, 38, 47, or 56, or an active variant or fragment thereof that when comprised within a guide RNA is capable of directing the sequence-specific binding of an associated RNA-guided nuclease to a target sequence of interest. Polynucleotides are also provided that encode an RNA-guided nuclease comprising the amino acid sequence set forth as SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54, and active fragments or variants thereof that retain the ability to bind to a target nucleotide sequence in an RNA-guided sequence-specific manner.

[0056] The use of the term "polynucleotide" is not intended to limit the present disclosure to polynucleotides comprising DNA. Those of ordinary skill in the art will recognize that polynucleotides can comprise ribonucleotides (RNA) and combinations of ribonucleotides and deoxyribonucleotides. Such deoxyribonucleotides and ribonucleotides include both naturally occurring molecules and synthetic analogues. These include peptide nucleic acids (PNAs), PNA-DNA chimeras, locked nucleic acids (LNAs), and phosphothiorate linked sequences. The polynucleotides disclosed herein also encompass all forms of sequences including, but not limited to, single-stranded forms, double-stranded forms, DNA-RNA hybrids, triplex structures, stem-and-loop structures, and the like.

[0057] The nucleic acid molecules encoding RGNs can be codon optimized for expression in an organism of interest. A

“codon-optimized” coding sequence is a polynucleotide coding sequence having its frequency of codon usage designed to mimic the frequency of preferred codon usage or transcription conditions of a particular host cell. Expression in the particular host cell or organism is enhanced as a result of the alteration of one or more codons at the nucleic acid level such that the translated amino acid sequence is not changed. Nucleic acid molecules can be codon optimized, either wholly or in part. Codon tables and other references providing preference information for a wide range of organisms are available in the art (see, e.g., Campbell and Gowri (1990) *Plant Physiol.* 92:1-11 for a discussion of plant-preferred codon usage). Methods are available in the art for synthesizing plant-preferred genes. See, for example, U.S. Pat. Nos. 5,380,831, and 5,436,391, and Murray et al. (1989) *Nucleic Acids Res.* 17:477-498, herein incorporated by reference.

[0058] Polynucleotides encoding the RGNs, crRNAs, tracrRNAs, and/or sgRNAs provided herein can be provided in expression cassettes for in vitro expression or expression in a cell, organelle, embryo, or organism of interest. The cassette will include 5' and 3' regulatory sequences operably linked to a polynucleotide encoding an RGN, crRNA, tracrRNAs, and/or sgRNAs provided herein that allows for expression of the polynucleotide. The cassette may additionally contain at least one additional gene or genetic element to be cotransformed into the organism. Where additional genes or elements are included, the components are operably linked. The term “operably linked” is intended to mean a functional linkage between two or more elements. For example, an operable linkage between a promoter and a coding region of interest (e.g., region coding for an RGN, crRNA, tracrRNAs, and/or sgRNAs) is a functional link that allows for expression of the coding region of interest. Operably linked elements may be contiguous or non-contiguous. When used to refer to the joining of two protein coding regions, by operably linked is intended that the coding regions are in the same reading frame. Alternatively, the additional gene(s) or element(s) can be provided on multiple expression cassettes. For example, the nucleotide sequence encoding a presently disclosed RGN can be present on one expression cassette, whereas the nucleotide sequence encoding a crRNA, tracrRNA, or complete guide RNA can be on a separate expression cassette. Such an expression cassette is provided with a plurality of restriction sites and/or recombination sites for insertion of the polynucleotides to be under the transcriptional regulation of the regulatory regions. The expression cassette may additionally contain a selectable marker gene.

[0059] The expression cassette will include in the 5'-3' direction of transcription, a transcriptional (and, in some embodiments, translational) initiation region (i.e., a promoter), an RGN-, crRNA-, tracrRNA- and/or sgRNA-encoding polynucleotide of the invention, and a transcriptional (and in some embodiments, translational) termination region (i.e., termination region) functional in the organism of interest. The promoters of the invention are capable of directing or driving expression of a coding sequence in a host cell. The regulatory regions (e.g., promoters, transcriptional regulatory regions, and translational termination regions) may be endogenous or heterologous to the host cell or to each other. As used herein, “heterologous” in reference to a sequence is a sequence that originates from a foreign species, or, if from the same species, is substantially modified

from its native form in composition and/or genomic locus by deliberate human intervention. As used herein, a chimeric gene comprises a coding sequence operably linked to a transcription initiation region that is heterologous to the coding sequence.

[0060] Convenient termination regions are available from the Ti-plasmid of *A. tumefaciens*, such as the octopine synthase and nopaline synthase termination regions. See also Guerineau et al. (1991) *Mol. Gen. Genet.* 262:141-144; Proudfoot (1991) *Cell* 64:671-674; Sanfacon et al. (1991) *Genes Dev.* 5:141-149; Mogen et al. (1990) *Plant Cell* 2:1261-1272; Munroe et al. (1990) *Gene* 91:151-158; Ballas et al. (1989) *Nucleic Acids Res.* 17:7891-7903; and Joshi et al. (1987) *Nucleic Acids Res.* 15:9627-9639.

[0061] Additional regulatory signals include, but are not limited to, transcriptional initiation start sites, operators, activators, enhancers, other regulatory elements, ribosomal binding sites, an initiation codon, termination signals, and the like. See, for example, U.S. Pat. Nos. 5,039,523 and 4,853,331; EPO 0480762A2; Sambrook et al. (1992) *Molecular Cloning: A Laboratory Manual*, ed. Maniatis et al. (Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y.), hereinafter “Sambrook 11”; Davis et al., eds. (1980) *Advanced Bacterial Genetics* (Cold Spring Harbor Laboratory Press), Cold Spring Harbor, N.Y., and the references cited therein.

[0062] In preparing the expression cassette, the various DNA fragments may be manipulated, so as to provide for the DNA sequences in the proper orientation and, as appropriate, in the proper reading frame. Toward this end, adapters or linkers may be employed to join the DNA fragments or other manipulations may be involved to provide for convenient restriction sites, removal of superfluous DNA, removal of restriction sites, or the like. For this purpose, in vitro mutagenesis, primer repair, restriction, annealing, substitutions, e.g., transitions and transversions, may be involved.

[0063] A number of promoters can be used in the practice of the invention. The promoters can be selected based on the desired outcome. The nucleic acids can be combined with constitutive, inducible, growth stage-specific, cell type-specific, tissue-preferred, tissue-specific, or other promoters for expression in the organism of interest. See, for example, promoters set forth in WO 99/43838 and in U.S. Pat. Nos. 8,575,425; 7,790,846; 8,147,856; 8,586,832; 7,772,369; 7,534,939; 6,072,050; 5,659,026; 5,608,149; 5,608,144; 5,604,121; 5,569,597; 5,466,785; 5,399,680; 5,268,463; 5,608,142; and 6,177,611; herein incorporated by reference.

[0064] For expression in plants, constitutive promoters also include CaMV 35S promoter (Odell et al. (1985) *Nature* 313:810-812); rice actin (McElroy et al. (1990) *Plant Cell* 2:163-171); ubiquitin (Christensen et al. (1989) *Plant Mol. Biol.* 12:619-632 and Christensen et al. (1992) *Plant Mol. Biol.* 18:675-689); pEMU (Last et al. (1991) *Theor. Appl. Genet.* 81:581-588); and MAS (Velten et al. (1984) *EMBO J.* 3:2723-2730).

[0065] Examples of inducible promoters are the Adh1 promoter which is inducible by hypoxia or cold stress, the Hsp70 promoter which is inducible by heat stress, the PPK promoter and the pepcarboxylase promoter which are both inducible by light. Also useful are promoters which are chemically inducible, such as the In2-2 promoter which is safener induced (U.S. Pat. No. 5,364,780), the Axig1 promoter which is auxin induced and tapetum specific but also active in callus (PCT US01/22169), the steroid-responsive

promoters (see, for example, the ERE promoter which is estrogen induced, and the glucocorticoid-inducible promoter in Schena et al. (1991) *Proc. Natl. Acad. Sci. USA* 88:10421-10425 and McNellis et al. (1998) *Plant J.* 14 (2): 247-257) and tetracycline-inducible and tetracycline-repressible promoters (see, for example, Gatz et al. (1991) *Mol. Gen. Genet.* 227:229-237, and U.S. Pat. Nos. 5,814,618 and 5,789,156), herein incorporated by reference.

[0066] Tissue-specific or tissue-preferred promoters can be utilized to target expression of an expression construct within a particular tissue. In certain embodiments, the tissue-specific or tissue-preferred promoters are active in plant tissue. Examples of promoters under developmental control in plants include promoters that initiate transcription preferentially in certain tissues, such as leaves, roots, fruit, seeds, or flowers. A “tissue specific” promoter is a promoter that initiates transcription only in certain tissues. Unlike constitutive expression of genes, tissue-specific expression is the result of several interacting levels of gene regulation. As such, promoters from homologous or closely related plant species can be preferable to use to achieve efficient and reliable expression of transgenes in particular tissues. In some embodiments, the expression comprises a tissue-preferred promoter. A “tissue preferred” promoter is a promoter that initiates transcription preferentially, but not necessarily entirely or solely in certain tissues.

[0067] In some embodiments, the nucleic acid molecules encoding a RGN, crRNA, and/or tracrRNA comprise a cell type-specific promoter. A “cell type specific” promoter is a promoter that primarily drives expression in certain cell types in one or more organs. Some examples of plant cells in which cell type specific promoters functional in plants may be primarily active include, for example, BETL cells, vascular cells in roots, leaves, stalk cells, and stem cells. The nucleic acid molecules can also include cell type preferred promoters. A “cell type preferred” promoter is a promoter that primarily drives expression mostly, but not necessarily entirely or solely in certain cell types in one or more organs. Some examples of plant cells in which cell type preferred promoters functional in plants may be preferentially active include, for example, BETL cells, vascular cells in roots, leaves, stalk cells, and stem cells.

[0068] The nucleic acid sequences encoding the RGNs, crRNAs, tracrRNAs, and/or sgRNAs can be operably linked to a promoter sequence that is recognized by a phage RNA polymerase for example, for in vitro mRNA synthesis. In such embodiments, the in vitro-transcribed RNA can be purified for use in the methods described herein. For example, the promoter sequence can be a T7, T3, or SP6 promoter sequence or a variation of a T7, T3, or SP6 promoter sequence. In such embodiments, the expressed protein and/or RNAs can be purified for use in the methods of genome modification described herein. In certain embodiments, the polynucleotide encoding the RGN, crRNA, tracrRNA, and/or sgRNA also can be linked to a polyadenylation signal (e.g., SV40 polyA signal and other signals functional in plants) and/or at least one transcriptional termination sequence. Additionally, the sequence encoding the RGN also can be linked to sequence(s) encoding at least one nuclear localization signal, at least one cell-penetrating domain, and/or at least one signal peptide capable of trafficking proteins to particular subcellular locations, as described elsewhere herein.

[0069] The polynucleotide encoding the RGN, crRNA, tracrRNA, and/or sgRNA can be present in a vector or multiple vectors. A “vector” refers to a polynucleotide composition for transferring, delivering, or introducing a nucleic acid into a host cell. Suitable vectors include plasmid vectors, phagemids, cosmids, artificial/mini-chromosomes, transposons, and viral vectors (e.g., lentiviral vectors, adeno-associated viral vectors, baculoviral vector). The vector can comprise additional expression control sequences (e.g., enhancer sequences, Kozak sequences, polyadenylation sequences, transcriptional termination sequences), selectable marker sequences (e.g., antibiotic resistance genes), origins of replication, and the like. Additional information can be found in “*Current Protocols in Molecular Biology*” Ausubel et al., John Wiley & Sons, New York, 2003 or “*Molecular Cloning: A Laboratory Manual*” Sambrook & Russell, Cold Spring Harbor Press, Cold Spring Harbor, N.Y., 3rd edition, 2001. The vector can also comprise a selectable marker gene for the selection of transformed cells. Selectable marker genes are utilized for the selection of transformed cells or tissues. Marker genes include genes encoding antibiotic resistance, such as those encoding neomycin phosphotransferase II (NEO) and hygromycin phosphotransferase (HPT), as well as genes conferring resistance to herbicidal compounds, such as glufosinate ammonium, bromoxynil, imidazolinones, and 2,4-dichlorophenoxyacetate (2,4-D). In some embodiments, the expression cassette or vector comprising the sequence encoding the RGN polypeptide can further comprise a sequence encoding a crRNA and/or a tracrRNA, or the crRNA and tracrRNA combined to create a guide RNA. The sequence(s) encoding the crRNA and/or tracrRNA can be operably linked to at least one transcriptional control sequence for expression of the crRNA and/or tracrRNA in the organism or host cell of interest. For example, the polynucleotide encoding the crRNA and/or tracrRNA can be operably linked to a promoter sequence that is recognized by RNA polymerase III (Pol III). Examples of suitable Pol III promoters include, but are not limited to, mammalian U6, U3, H1, and 7SL RNA promoters and rice U6 and U3 promoters.

[0070] As indicated, expression constructs comprising nucleotide sequences encoding the RGNs, crRNA, tracrRNA, and/or sgRNA can be used to transform organisms of interest. Methods for transformation involve introducing a nucleotide construct into an organism of interest. By “introducing” is intended to introduce the nucleotide construct to the host cell in such a manner that the construct gains access to the interior of the host cell. The methods of the invention do not require a particular method for introducing a nucleotide construct to a host organism, only that the nucleotide construct gains access to the interior of at least one cell of the host organism. The host cell can be a eukaryotic or prokaryotic cell. In particular embodiments, the eukaryotic host cell is a plant cell, a mammalian cell, or an insect cell. Methods for introducing nucleotide constructs into plants and other host cells are known in the art including, but not limited to, stable transformation methods, transient transformation methods, and virus-mediated methods.

[0071] The methods result in a transformed organism, such as a plant, including whole plants, as well as plant organs (e.g., leaves, stems, roots, etc.), seeds, plant cells, propagules, embryos and progeny of the same. Plant cells

can be differentiated or undifferentiated (e.g. callus, suspension culture cells, protoplasts, leaf cells, root cells, phloem cells, pollen).

[0072] “Transgenic organisms” or “transformed organisms” or “stably transformed” organisms or cells or tissues refers to organisms that have incorporated or integrated a polynucleotide encoding a RGN, crRNA, and/or tracrRNA of the invention. It is recognized that other exogenous or endogenous nucleic acid sequences or DNA fragments may also be incorporated into the host cell. *Agrobacterium*- and biolistic-mediated transformation remain the two predominantly employed approaches for transformation of plant cells. However, transformation of a host cell may be performed by infection, transfection, microinjection, electroporation, microprojection, biolistics or particle bombardment, electroporation, silica/carbon fibers, ultrasound mediated, PEG mediated, calcium phosphate co-precipitation, polycation DMSO technique, DEAE dextran procedure, and viral mediated, liposome mediated and the like. Viral-mediated introduction of a polynucleotide encoding an RGN, crRNA, and/or tracrRNA includes retroviral, lentiviral, adenoviral, and adeno-associated viral mediated introduction and expression, as well as the use of Caulimoviruses, Gemini-viruses, and RNA plant viruses.

[0073] Transformation protocols as well as protocols for introducing polypeptides or polynucleotide sequences into plants may vary depending on the type of host cell (e.g., monocot or dicot plant cell) targeted for transformation. Methods for transformation are known in the art and include those set forth in U.S. Pat. Nos. 8,575,425; 7,692,068; 8,802,934; 7,541,517; each of which is herein incorporated by reference. See, also, Rakoczy-Trojanowska, M. (2002) *Cell Mol Biol Lett.* 7:849-858; Jones et al. (2005) *Plant Methods* 1:5; Rivera et al. (2012) *Physics of Life Reviews* 9:308-345; Bartlett et al. (2008) *Plant Methods* 4:1-12; Bates, G.W. (1999) *Methods in Molecular Biology* 111:359-366; Binns and Thomashow (1988) *Annual Reviews in Microbiology* 42:575-606; Christou, P. (1992) *The Plant Journal* 2:275-281; Christou, P. (1995) *Euphytica* 85:13-27; Tzfira et al. (2004) *TRENDS in Genetics* 20:375-383; Yao et al. (2006) *Journal of Experimental Botany* 57:3737-3746; Zupan and Zambryski (1995) *Plant Physiology* 107:1041-1047; Jones et al. (2005) *Plant Methods* 1:5;

[0074] Transformation may result in stable or transient incorporation of the nucleic acid into the cell. “Stable transformation” is intended to mean that the nucleotide construct introduced into a host cell integrates into the genome of the host cell and is capable of being inherited by the progeny thereof. “Transient transformation” is intended to mean that a polynucleotide is introduced into the host cell and does not integrate into the genome of the host cell.

[0075] Methods for transformation of chloroplasts are known in the art. See, for example, Svab et al. (1990) *Proc. Natl. Acad. Sci. USA* 87:8526-8530; Svab and Maliga (1993) *Proc. Natl. Acad. Sci. USA* 90:913-917; Svab and Maliga (1993) *EMBO J.* 12:601-606. The method relies on particle gun delivery of DNA containing a selectable marker and targeting of the DNA to the plastid genome through homologous recombination. Additionally, plastid transformation can be accomplished by transactivation of a silent plastid-borne transgene by tissue-preferred expression of a nuclear-encoded and plastid-directed RNA polymerase. Such a system has been reported in McBride et al. (1994) *Proc. Natl. Acad. Sci. USA* 91:7301-7305.

[0076] The cells that have been transformed may be grown into a transgenic organism, such as a plant, in accordance with conventional ways. See, for example, McCormick et al. (1986) *Plant Cell Reports* 5:81-84. These plants may then be grown, and either pollinated with the same transformed strain or different strains, and the resulting hybrid having constitutive expression of the desired phenotypic characteristic identified. Two or more generations may be grown to ensure that expression of the desired phenotypic characteristic is stably maintained and inherited and then seeds harvested to ensure expression of the desired phenotypic characteristic has been achieved. In this manner, the present invention provides transformed seed (also referred to as “transgenic seed”) having a nucleotide construct of the invention, for example, an expression cassette of the invention, stably incorporated into their genome.

[0077] Alternatively, cells that have been transformed may be introduced into an organism. These cells could have originated from the organism, wherein the cells are transformed in an ex vivo approach.

[0078] The sequences provided herein may be used for transformation of any plant species, including, but not limited to, monocots and dicots. Examples of plants of interest include, but are not limited to, corn (maize), sorghum, wheat, sunflower, tomato, crucifers, peppers, potato, cotton, rice, soybean, sugarbeet, sugarcane, tobacco, barley, and oilseed rape, *Brassica* sp., alfalfa, rye, millet, safflower, peanuts, sweet potato, cassava, coffee, coconut, pineapple, citrus trees, cocoa, tea, banana, avocado, fig, guava, mango, olive, *papaya*, cashew, macadamia, almond, oats, vegetables, ornamentals, and conifers.

[0079] Vegetables include, but are not limited to, tomatoes, lettuce, green beans, lima beans, peas, and members of the genus *Curcumis* such as cucumber, cantaloupe, and musk melon. Ornamentals include, but are not limited to, azalea, *hydrangea*, hibiscus, roses, tulips, daffodils, petunias, carnation, poinsettia, and *chrysanthemum*. Preferably, plants of the present invention are crop plants (for example, maize, sorghum, wheat, sunflower, tomato, crucifers, peppers, potato, cotton, rice, soybean, sugarbeet, sugarcane, tobacco, barley, oilseed rape, etc.).

[0080] As used herein, the term plant includes plant cells, plant protoplasts, plant cell tissue cultures from which plants can be regenerated, plant calli, plant clumps, and plant cells that are intact in plants or parts of plants such as embryos, pollen, ovules, seeds, leaves, flowers, branches, fruit, kernels, ears, cobs, husks, stalks, roots, root tips, anthers, and the like. Grain is intended to mean the mature seed produced by commercial growers for purposes other than growing or reproducing the species. Progeny, variants, and mutants of the regenerated plants are also included within the scope of the invention, provided that these parts comprise the introduced polynucleotides. Further provided is a processed plant product or byproduct that retains the sequences disclosed herein, including for example, soymeal.

[0081] The polynucleotides encoding the RGNs, crRNAs, and/or tracrRNAs can also be used to transform any prokaryotic species, including but not limited to, archaea and bacteria (e.g., *Bacillus* sp., *Klebsiella* sp., *Streptomyces* sp., *Rhizobium* sp., *Escherichia* sp., *Pseudomonas* sp., *Salmonella* sp., *Shigella* sp., *Vibrio* sp., *Yersinia* sp., *Mycoplasma* sp., *Agrobacterium*, *Lactobacillus* sp.).

[0082] The polynucleotides encoding the RGNs, crRNAs, and/or tracrRNAs can be used to transform any eukaryotic

species, including but not limited to animals (e.g., mammals, insects, fish, birds, and reptiles), fungi, amoeba, algae, and yeast.

[0083] Conventional viral and non-viral based gene transfer methods can be used to introduce nucleic acids in mammalian cells or target tissues. Such methods can be used to administer nucleic acids encoding components of a CRISPR system to cells in culture, or in a host organism. Non-viral vector delivery systems include DNA plasmids, RNA (e.g. a transcript of a vector described herein), naked nucleic acid, and nucleic acid complexed with a delivery vehicle, such as a liposome. Viral vector delivery systems include DNA and RNA viruses, which have either episomal or integrated genomes after delivery to the cell. For a review of gene therapy procedures, see Anderson, *Science* 256:808-813 (1992); Nabel & Feigner, *TIBTECH* 11:211-217 (1993); Mitani & Caskey, *TIBTECH* 11:162-166 (1993); Dillon, *TIBTECH* 11:167-175 (1993); Miller, *Nature* 357:455-460 (1992); Van Brunt, *Biotechnology* 6 (10): 1149-1154 (1988); Vigne, *Restorative Neurology and Neuroscience* 8:35-36 (1995); Kremer & Perricaudet, *British Medical Bulletin* 51 (1): 31-44 (1995); Haddada et al., in *Current Topics in Microbiology and Immunology*, Doerfler and Bohm (eds) (1995); and Yu et al., *Gene Therapy* 1:13-26 (1994).

[0084] Methods of non-viral delivery of nucleic acids include lipofection, nucleofection, microinjection, biolistics, virosomes, liposomes, immunoliposomes, polycation or lipid: nucleic acid conjugates, naked DNA, artificial virions, and agent-enhanced uptake of DNA. Lipofection is described in e.g., U.S. Pat. Nos. 5,049,386, 4,946,787; and 4,897,355 and lipofection reagents are sold commercially (e.g., Transfectam™ and Lipofectin™). Cationic and neutral lipids that are suitable for efficient receptor-recognition lipofection of polynucleotides include those of Feigner, WO 91/17424; WO 91/16024. Delivery can be to cells (e.g. in vitro or ex vivo administration) or target tissues (e.g. in vivo administration). The preparation of lipid: nucleic acid complexes, including targeted liposomes such as immunolipid complexes, is well known to one of skill in the art (see, e.g., Crystal, *Science* 270:404-410 (1995); Blaese et al., *Cancer Gene Ther.* 2:291-297 (1995); Behr et al., *Bioconjugate Chem.* 5:382-389 (1994); Remy et al., *Bioconjugate Chem.* 5:647-654 (1994); Gao et al., *Gene Therapy* 2:710-722 (1995); Ahmad et al., *Cancer Res.* 52:4817-4820 (1992); U.S. Pat. Nos. 4,186,183, 4,217,344, 4,235,871, 4,261,975, 4,485,054, 4,501,728, 4,774,085, 4,837,028, and 4,946,787).

[0085] The use of RNA or DNA viral based systems for the delivery of nucleic acids takes advantage of highly evolved processes for targeting a virus to specific cells in the body and trafficking the viral payload to the nucleus. Viral vectors can be administered directly to patients (in vivo) or they can be used to treat cells in vitro, and the modified cells may optionally be administered to patients (ex vivo). Conventional viral based systems could include retroviral, lentivirus, adenoviral, adeno-associated and herpes simplex virus vectors for gene transfer. Integration in the host genome is possible with the retrovirus, lentivirus, and adeno-associated virus gene transfer methods, often resulting in long term expression of the inserted transgene. Additionally, high transduction efficiencies have been observed in many different cell types and target tissues.

[0086] The tropism of a retrovirus can be altered by incorporating foreign envelope proteins, expanding the

potential target population of target cells. Lentiviral vectors are retroviral vectors that are able to transduce or infect non-dividing cells and typically produce high viral titers. Selection of a retroviral gene transfer system would therefore depend on the target tissue. Retroviral vectors are comprised of cis-acting long terminal repeats with packaging capacity for up to 6-10 kb of foreign sequence. The minimum cis-acting LTRs are sufficient for replication and packaging of the vectors, which are then used to integrate the therapeutic gene into the target cell to provide permanent transgene expression. Widely used retroviral vectors include those based upon murine leukemia virus (MuLV), gibbon ape leukemia virus (GaLV), Simian Immuno deficiency virus (SIV), human immuno deficiency virus (HIV), and combinations thereof (see, e.g., Buchscher et al., *J. Viral.* 66:2731-2739 (1992); Johann et al., *J. Viral.* 66:1635-1640 (1992); Sommerfelt et al., *Viral.* 176:58-59 (1990); Wilson et al., *J. Viral.* 63:2374-2378 (1989); Miller et al., *J. Viral.* 65:2220-2224 (1991); PCT/US94/05700).

[0087] In applications where transient expression is preferred, adenoviral based systems may be used. Adenoviral based vectors are capable of very high transduction efficiency in many cell types and do not require cell division. With such vectors, high titer and levels of expression have been obtained. This vector can be produced in large quantities in a relatively simple system. Adeno-associated virus ("AAV") vectors may also be used to transduce cells with target nucleic acids, e.g., in the in vitro production of nucleic acids and peptides, and for in vivo and ex vivo gene therapy procedures (see, e.g., West et al., *Virology* 160:38-47 (1987); U.S. Pat. No. 4,797,368; WO 93/24641; Katin, *Human Gene Therapy* 5:793-801 (1994); Muzyczka, *Clin. Invest.* 94:1351 (1994). Construction of recombinant AAV vectors are described in a number of publications, including U.S. Pat. No. 5,173,414; Tratschin et al., *Mol. Cell. Biol.* 5:3251-3260 (1985); Tratschin, et al., *Mol. Cell. Biol.* 4:2072-2081 (1984); Hermonat & Muzyczka, *PNAS* 81:6466-6470 (1984); and Samulski et al., *J. Viral.* 63:03822-3828 (1989). Packaging cells are typically used to form virus particles that are capable of infecting a host cell. Such cells include 293 cells, which package adenovirus, and ψ 12 cells or PA317 cells, which package retrovirus.

[0088] Viral vectors used in gene therapy are usually generated by producing a cell line that packages a nucleic acid vector into a viral particle. The vectors typically contain the minimal viral sequences required for packaging and subsequent integration into a host, other viral sequences being replaced by an expression cassette for the polynucleotide(s) to be expressed. The missing viral functions are typically supplied in trans by the packaging cell line. For example, AAV vectors used in gene therapy typically only possess ITR sequences from the AAV genome which are required for packaging and integration into the host genome. Viral DNA is packaged in a cell line, which contains a helper plasmid encoding the other AAV genes, namely rep and cap, but lacking ITR sequences.

[0089] The cell line may also be infected with adenovirus as a helper. The helper virus promotes replication of the AAV vector and expression of AAV genes from the helper plasmid. The helper plasmid is not packaged in significant amounts due to a lack of ITR sequences. Contamination with adenovirus can be reduced by, e.g., heat treatment to which adenovirus is more sensitive than AAV. Additional methods

for the delivery of nucleic acids to cells are known to those skilled in the art. See, for example, US20030087817, incorporated herein by reference.

[0090] In some embodiments, a host cell is transiently or non-transiently transfected with one or more vectors described herein. In some embodiments, a cell is transfected as it naturally occurs in a subject. In some embodiments, a cell that is transfected is taken from a subject. In some embodiments, the cell is derived from cells taken from a subject, such as a cell line. A wide variety of cell lines for tissue culture are known in the art. Examples of cell lines include, but are not limited to, C8161, CCRF-CEM, MOLT, mIMCD-3, NHDF, HeLaS3, Huh1, Huh4, Huh7, HUVEC, HASMC, HEK293, HEK293T, MiaPaCell, Panel, PC-3, TFI, CTLL-2, CIR, Rat6, CV1, RPTE, AIO, T24, 182, A375, ARH-77, Calu, SW480, SW620, SKOV3, SK-UT, CaCo2, P388DI, SEM-K2, WEHI-231, HB56, TIB55, Jurkat, 145.01, LRMB, Bcl-1, BC-3, IC21, DLD2, Raw264.7, NRK, NRK-52E, MRC5, MEF, Hep G2, HeLa B, HeLa T4, COS, COS-1, COS-6, COS-M6A, BS-C-1 monkey kidney epithelial, BALB/3T3 mouse embryo fibroblast, 3T3 Swiss, 3T3-L1, 132-d5 human fetal fibroblasts; 10.1 mouse fibroblasts, 293-T, 3T3, 721, 9L, A2780, A2780ADR, A2780cis, A172, A20, A253, A431, A-549, ALC, B16, B35, BCP-1 cells, BEAS-2B, bEnd.3, BHK-21, BR 293, BxPC3, C3H-10T1/2, C6/36, Cal-27, CHO, CHO-7, CHO-IR, CHO-K1, CHO-K2, CHO-T, CHO Dhfr^{-/-}, COR-L23, COR-L23/CPR, COR-L235010, CORL23/R23, COS-7, COV-434, CML TI, CMT, CT26, D17, DH82, DU145, DuCaP, EL4, EM2, EM3, EMT6/AR1, EMT6/AR10.0, FM3, H1299, H69, HB54, HB55, HCA2, HEK-293, HeLa, Hepalclc7, HL-60, HMEC, HT-29, Jurkat, IY cells, K562 cells, Ku812, KCL22, KGI, KYOI, LNCap, Ma-Mel 1-48, MC-38, MCF-7, MCF-10A, MDA-MB-231, MDA-MB-468, MDA-MB-435, MDCKII, MDCKII, MOR/0.2R, MONO-MAC 6, MTD-1A, MyEnd, NCI-H69/CPR, NCI-H69/LX10, NCI-H69/LX20, NCI-H69/LX4, NIH-3T3, NALM-1, NW-145, OPCN/OPCT cell lines, Peer, PNT-1A/PNT 2, RenCa, RIN-5F, RMA/RMAS, Saos-2 cells, Sf-9, SkBr3, T2, T-47D, T84, THPI cell line, U373, U87, U937, VCaP, Vero cells, WM39, WT-49, X63, YAC-1, YAR, and transgenic varieties thereof. Cell lines are available from a variety of sources known to those with skill in the art (see, e.g., the American Type Culture Collection (ATCC) (Manassas, Va.)).

[0091] In some embodiments, a cell transfected with one or more vectors described herein is used to establish a new cell line comprising one or more vector-derived sequences. In some embodiments, a cell transiently transfected with the components of a CRISPR system as described herein (such as by transient transfection of one or more vectors, or transfection with RNA), and modified through the activity of a CRISPR complex, is used to establish a new cell line comprising cells containing the modification but lacking any other exogenous sequence. In some embodiments, cells transiently or non-transiently transfected with one or more vectors described herein, or cell lines derived from such cells are used in assessing one or more test compounds.

[0092] In some embodiments, one or more vectors described herein are used to produce a non-human transgenic animal or transgenic plant. In some embodiments, the transgenic animal is a mammal, such as a mouse, rat, or rabbit.

IV. Variants and Fragments of Polypeptides and Polynucleotides

[0093] The present disclosure provides active variants and fragments of a naturally-occurring (i.e., wild-type) RNA-guided nuclease, the amino acid sequence of which is set forth as SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54, as well as active variants and fragments of naturally-occurring CRISPR repeats, such as the sequence set forth as SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55, and active variant and fragments of naturally-occurring tracrRNAs, such as the sequence set forth as SEQ ID NOs: 3, 13, 21, 29, 38, 47, or 56, and polynucleotides encoding the same.

[0094] While the activity of a variant or fragment may be altered compared to the polynucleotide or polypeptide of interest, the variant and fragment should retain the functionality of the polynucleotide or polypeptide of interest. For example, a variant or fragment may have increased activity, decreased activity, different spectrum of activity or any other alteration in activity when compared to the polynucleotide or polypeptide of interest.

[0095] Fragments and variants of naturally-occurring RGN polypeptides, such as those disclosed herein, will retain sequence-specific, RNA-guided DNA-binding activity. In particular embodiments, fragments and variants of naturally-occurring RGN polypeptides, such as those disclosed herein, will retain nuclease activity (single-stranded or double-stranded).

[0096] Fragments and variants of naturally-occurring CRISPR repeats, such as those disclosed herein, will retain the ability, when part of a guide RNA (comprising a tracrRNA), to bind to and guide an RNA-guided nuclease (complexed with the guide RNA) to a target nucleotide sequence in a sequence-specific manner.

[0097] Fragments and variants of naturally-occurring tracrRNAs, such as those disclosed herein, will retain the ability, when part of a guide RNA (comprising a CRISPR RNA), to guide an RNA-guided nuclease (complexed with the guide RNA) to a target nucleotide sequence in a sequence-specific manner.

[0098] The term “fragment” refers to a portion of a polynucleotide or polypeptide sequence of the invention. “Fragments” or “biologically active portions” include polynucleotides comprising a sufficient number of contiguous nucleotides to retain the biological activity (i.e., binding to and directing an RGN in a sequence-specific manner to a target nucleotide sequence when comprised within a guideRNA). “Fragments” or “biologically active portions” include polypeptides comprising a sufficient number of contiguous amino acid residues to retain the biological activity (i.e., binding to a target nucleotide sequence in a sequence-specific manner when complexed with a guide RNA). Fragments of the RGN proteins include those that are shorter than the full-length sequences due to the use of an alternate downstream start site. A biologically active portion of an RGN protein can be a polypeptide that comprises, for example, 10, 25, 50, 100, 150, 200, 250, 300, 350, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, 900, 950, 1000, 1050 or more contiguous amino acid residues of SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54. Such biologically active portions can be prepared by recombinant techniques and evaluated for sequence-specific, RNA-guided DNA-binding activity. A biologically active fragment of a CRISPR repeat sequence can comprise at least 8 contiguous amino acids of SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55. A biologically active

portion of a CRISPR repeat sequence can be a polynucleotide that comprises, for example, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 contiguous nucleotides of SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55. A biologically active portion of a tracrRNA can be a polynucleotide that comprises, for example, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80 or more contiguous nucleotides of SEQ ID NOs: 3, 13, 21, 29, 38, 47, or 56.

[0099] In general, “variants” is intended to mean substantially similar sequences. For polynucleotides, a variant comprises a deletion and/or addition of one or more nucleotides at one or more internal sites within the native polynucleotide and/or a substitution of one or more nucleotides at one or more sites in the native polynucleotide. As used herein, a “native” or “wild type” polynucleotide or polypeptide comprises a naturally occurring nucleotide sequence or amino acid sequence, respectively. For polynucleotides, conservative variants include those sequences that, because of the degeneracy of the genetic code, encode the native amino acid sequence of the gene of interest. Naturally occurring allelic variants such as these can be identified with the use of well-known molecular biology techniques, as, for example, with polymerase chain reaction (PCR) and hybridization techniques as outlined below. Variant polynucleotides also include synthetically derived polynucleotides, such as those generated, for example, by using site-directed mutagenesis but which still encode the polypeptide or the polynucleotide of interest. Generally, variants of a particular polynucleotide disclosed herein will have at least about 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to that particular polynucleotide as determined by sequence alignment programs and parameters described elsewhere herein.

[0100] Variants of a particular polynucleotide disclosed herein (i.e., the reference polynucleotide) can also be evaluated by comparison of the percent sequence identity between the polypeptide encoded by a variant polynucleotide and the polypeptide encoded by the reference polynucleotide. Percent sequence identity between any two polypeptides can be calculated using sequence alignment programs and parameters described elsewhere herein. Where any given pair of polynucleotides disclosed herein is evaluated by comparison of the percent sequence identity shared by the two polypeptides they encode, the percent sequence identity between the two encoded polypeptides is at least about 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity.

[0101] In particular embodiments, the presently disclosed polynucleotides encode an RNA-guided nuclease polypeptide comprising an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or greater identity to an amino acid sequence of SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54.

[0102] A biologically active variant of an RGN polypeptide of the invention may differ by as few as about 1-15 amino acid residues, as few as about 1-10, such as about 6-10, as few as 5, as few as 4, as few as 3, as few as 2, or as few as 1 amino acid residue. In specific embodiments, the polypeptides can comprise an N-terminal or a C-terminal

truncation, which can comprise at least a deletion of 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 150, 200, 250, 300, 350, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, 900, 950, 1000, 1050 amino acids or more from either the N or C terminus of the polypeptide.

[0103] In certain embodiments, the presently disclosed polynucleotides comprise or encode a CRISPR repeat comprising a nucleotide sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or greater identity to the nucleotide sequence set forth as SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55.

[0104] The presently disclosed polynucleotides can comprise or encode a tracrRNA comprising a nucleotide sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or greater identity to the nucleotide sequence set forth as SEQ ID NOs: 3, 13, 21, 29, 38, 47, or 56.

[0105] Biologically active variants of a CRISPR repeat or tracrRNA of the invention may differ by as few as about 1-15 nucleotides, as few as about 1-10, such as about 6-10, as few as 5, as few as 4, as few as 3, as few as 2, or as few as 1 nucleotides. In specific embodiments, the polynucleotides can comprise a 5' or 3' truncation, which can comprise at least a deletion of 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80 nucleotides or more from either the 5' or 3' end of the polynucleotide.

[0106] It is recognized that modifications may be made to the RGN polypeptides, CRISPR repeats, and tracrRNAs provided herein creating variant proteins and polynucleotides. Changes designed by man may be introduced through the application of site-directed mutagenesis techniques. Alternatively, native, as yet-unknown or as yet unidentified polynucleotides and/or polypeptides structurally and/or functionally-related to the sequences disclosed herein may also be identified that fall within the scope of the present invention. Conservative amino acid substitutions may be made in nonconserved regions that do not alter the function of the RGN proteins. Alternatively, modifications may be made that improve the activity of the RGN.

[0107] Variant polynucleotides and proteins also encompass sequences and proteins derived from a mutagenic and recombinogenic procedure such as DNA shuffling. With such a procedure, one or more different RGN proteins disclosed herein (e.g., SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54) is manipulated to create a new RGN protein possessing the desired properties. In this manner, libraries of recombinant polynucleotides are generated from a population of related sequence polynucleotides comprising sequence regions that have substantial sequence identity and can be homologously recombined in vitro or in vivo. For example, using this approach, sequence motifs encoding a domain of interest may be shuffled between the RGN sequences provided herein and other known RGN genes to obtain a new gene coding for a protein with an improved property of interest, such as an increased K_m in the case of an enzyme. Strategies for such DNA shuffling are known in the art. See, for example, Stemmer (1994) *Proc. Natl. Acad. Sci. USA* 91:10747-10751; Stemmer (1994) *Nature* 370:389-391; Cramer et al. (1997) *Nature Biotech.* 15:436-438; Moore et al. (1997) *J. Mol. Biol.* 272:336-347; Zhang et al. (1997) *Proc. Natl. Acad. Sci. USA* 94:4504-4509; Cramer et al.

(1998) *Nature* 391:288-291; and U.S. Pat. Nos. 5,605,793 and 5,837,458. A “shuffled” nucleic acid is a nucleic acid produced by a shuffling procedure such as any shuffling procedure set forth herein. Shuffled nucleic acids are produced by recombining (physically or virtually) two or more nucleic acids (or character strings), for example in an artificial, and optionally recursive, fashion. Generally, one or more screening steps are used in shuffling processes to identify nucleic acids of interest; this screening step can be performed before or after any recombination step. In some (but not all) shuffling embodiments, it is desirable to perform multiple rounds of recombination prior to selection to increase the diversity of the pool to be screened. The overall process of recombination and selection are optionally repeated recursively. Depending on context, shuffling can refer to an overall process of recombination and selection, or, alternately, can simply refer to the recombinational portions of the overall process.

[0108] As used herein, “sequence identity” or “identity” in the context of two polynucleotides or polypeptide sequences makes reference to the residues in the two sequences that are the same when aligned for maximum correspondence over a specified comparison window. When percentage of sequence identity is used in reference to proteins it is recognized that residue positions which are not identical often differ by conservative amino acid substitutions, where amino acid residues are substituted for other amino acid residues with similar chemical properties (e.g., charge or hydrophobicity) and therefore do not change the functional properties of the molecule. When sequences differ in conservative substitutions, the percent sequence identity may be adjusted upwards to correct for the conservative nature of the substitution. Sequences that differ by such conservative substitutions are said to have “sequence similarity” or “similarity”. Means for making this adjustment are well known to those of skill in the art. Typically this involves scoring a conservative substitution as a partial rather than a full mismatch, thereby increasing the percentage sequence identity. Thus, for example, where an identical amino acid is given a score of 1 and a non-conservative substitution is given a score of zero, a conservative substitution is given a score between zero and 1. The scoring of conservative substitutions is calculated, e.g., as implemented in the program PC/GENE (Intelligenetics, Mountain View, California).

[0109] As used herein, “percentage of sequence identity” means the value determined by comparing two optimally aligned sequences over a comparison window, wherein the portion of the polynucleotide sequence in the comparison window may comprise additions or deletions (i.e., gaps) as compared to the reference sequence (which does not comprise additions or deletions) for optimal alignment of the two sequences. The percentage is calculated by determining the number of positions at which the identical nucleic acid base or amino acid residue occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison, and multiplying the result by 100 to yield the percentage of sequence identity.

[0110] Unless otherwise stated, sequence identity/similarity values provided herein refer to the value obtained using GAP Version 10 using the following parameters: % identity and % similarity for a nucleotide sequence using GAP Weight of 50 and Length Weight of 3, and the nwsgapdna.

cmp scoring matrix; % identity and % similarity for an amino acid sequence using GAP Weight of 8 and Length Weight of 2, and the BLOSUM62 scoring matrix; or any equivalent program thereof. By “equivalent program” is intended any sequence comparison program that, for any two sequences in question, generates an alignment having identical nucleotide or amino acid residue matches and an identical percent sequence identity when compared to the corresponding alignment generated by GAP Version 10.

[0111] Two sequences are “optimally aligned” when they are aligned for similarity scoring using a defined amino acid substitution matrix (e.g., BLOSUM62), gap existence penalty and gap extension penalty so as to arrive at the highest score possible for that pair of sequences. Amino acid substitution matrices and their use in quantifying the similarity between two sequences are well-known in the art and described, e.g., in Dayhoff et al. (1978) “A model of evolutionary change in proteins.” In *Atlas of Protein Sequence and Structure*, Vol. 5, Suppl. 3 (ed. M. O. Dayhoff), pp. 345-352. Natl. Biomed. Res. Found., Washington, D.C. and Henikoff et al. (1992) *Proc. Natl. Acad. Sci. USA* 89:10915-10919. The BLOSUM62 matrix is often used as a default scoring substitution matrix in sequence alignment protocols. The gap existence penalty is imposed for the introduction of a single amino acid gap in one of the aligned sequences, and the gap extension penalty is imposed for each additional empty amino acid position inserted into an already opened gap. The alignment is defined by the amino acids positions of each sequence at which the alignment begins and ends, and optionally by the insertion of a gap or multiple gaps in one or both sequences, so as to arrive at the highest possible score. While optimal alignment and scoring can be accomplished manually, the process is facilitated by the use of a computer-implemented alignment algorithm, e.g., gapped BLAST 2.0, described in Altschul et al. (1997) *Nucleic Acids Res.* 25:3389-3402, and made available to the public at the National Center for Biotechnology Information Website (www.ncbi.nlm.nih.gov). Optimal alignments, including multiple alignments, can be prepared using, e.g., PSI-BLAST, available through www.ncbi.nlm.nih.gov and described by Altschul et al. (1997) *Nucleic Acids Res.* 25:3389-3402.

[0112] With respect to an amino acid sequence that is optimally aligned with a reference sequence, an amino acid residue “corresponds to” the position in the reference sequence with which the residue is paired in the alignment. The “position” is denoted by a number that sequentially identifies each amino acid in the reference sequence based on its position relative to the N-terminus. Owing to deletions, insertion, truncations, fusions, etc., that must be taken into account when determining an optimal alignment, in general the amino acid residue number in a test sequence as determined by simply counting from the N-terminal will not necessarily be the same as the number of its corresponding position in the reference sequence. For example, in a case where there is a deletion in an aligned test sequence, there will be no amino acid that corresponds to a position in the reference sequence at the site of deletion. Where there is an insertion in an aligned reference sequence, that insertion will not correspond to any amino acid position in the reference sequence. In the case of truncations or fusions there can be stretches of amino acids in either the reference or aligned sequence that do not correspond to any amino acid in the corresponding sequence.

V. Antibodies

[0113] Antibodies to the RGN polypeptides or ribonucleoproteins comprising the RGN polypeptides of the present invention, including those having the amino acid sequence set forth as SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54 or active variants or fragments thereof, are also encompassed. Methods for producing antibodies are well known in the art (see, for example, Harlow and Lane (1988) *Antibodies: A Laboratory Manual*, Cold Spring Harbor Laboratory, Cold Spring Harbor, N.Y.; and U.S. Pat. No. 4,196,265). These antibodies can be used in kits for the detection and isolation of RGN polypeptides or ribonucleoproteins. Thus, this disclosure provides kits comprising antibodies that specifically bind to the polypeptides or ribonucleoproteins described herein, including, for example, polypeptides having the sequence of SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54.

VI. Systems and Ribonucleoprotein Complexes for Binding a Target Sequence of Interest and Methods of Making the Same

[0114] The present disclosure provides a system for binding a target sequence of interest, wherein the system comprises at least one guide RNA or a nucleotide sequence encoding the same, and at least one RNA-guided nuclease or a nucleotide sequence encoding the same. The guide RNA hybridizes to the target sequence of interest and also forms a complex with the RGN polypeptide, thereby directing the RGN polypeptide to bind to the target sequence. In some of these embodiments, the RGN comprises an amino acid sequence of SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54, or an active variant or fragment thereof. In various embodiments, the guide RNA comprises a CRISPR repeat sequence comprising the nucleotide sequence of SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55, or an active variant or fragment thereof. In particular embodiments, the guide RNA comprises a tracrRNA comprising a nucleotide sequence of SEQ ID NOs: 3, 13, 21, 29, 38, 47, or 56, or an active variant or fragment thereof. The guide RNA of the system can be a single guide RNA or a dual-guide RNA. In particular embodiments, the system comprises a RNA-guided nuclease that is heterologous to the guideRNA, wherein the RGN and guideRNA are not naturally complexed in nature.

[0115] The system for binding a target sequence of interest provided herein can be a ribonucleoprotein complex, which is at least one molecule of an RNA bound to at least one protein. The ribonucleoprotein complexes provided herein comprise at least one guide RNA as the RNA component and an RNA-guided nuclease as the protein component. Such ribonucleoprotein complexes can be purified from a cell or organism that naturally expresses an RGN polypeptide and has been engineered to express a particular guide RNA that is specific for a target sequence of interest. Alternatively, the ribonucleoprotein complex can be purified from a cell or organism that has been transformed with polynucleotides that encode an RGN polypeptide and a guide RNA and cultured under conditions to allow for the expression of the RGN polypeptide and guide RNA. Thus, methods are provided for making an RGN polypeptide or an RGN ribonucleoprotein complex. Such methods comprise culturing a cell comprising a nucleotide sequence encoding an RGN polypeptide, and in some embodiments a nucleotide sequence encoding a guide RNA, under conditions in which the RGN polypeptide (and in some embodiments, the guide

RNA) is expressed. The RGN polypeptide or RGN ribonucleoprotein can then be purified from a lysate of the cultured cells.

[0116] Methods for purifying an RGN polypeptide or RGN ribonucleoprotein complex from a lysate of a biological sample are known in the art (e.g., size exclusion and/or affinity chromatography, 2D-PAGE, HPLC, reversed-phase chromatography, immunoprecipitation). In particular methods, the RGN polypeptide is recombinantly produced and comprises a purification tag to aid in its purification, including but not limited to, glutathione-S-transferase (GST), chitin binding protein (CBP), maltose binding protein, thioredoxin (TRX), poly (NANP), tandem affinity purification (TAP) tag, myc, AcV5, AU1, AU5, E, ECS, E2, FLAG, HA, nus, Softag 1, Softag 3, Strep, SBP, Glu-Glu, HSV, KT3, S, S1, T7, V5, VSV-G, 6xHis, 10xHis, biotin carboxyl carrier protein (BCCP), and calmodulin. Generally, the tagged RGN polypeptide or RGN ribonucleoprotein complex is purified using immobilized metal affinity chromatography. It will be appreciated that other similar methods known in the art may be used, including other forms of chromatography or for example immunoprecipitation, either alone or in combination.

[0117] An “isolated” or “purified” polypeptide, or biologically active portion thereof, is substantially or essentially free from components that normally accompany or interact with the polypeptide as found in its naturally occurring environment. Thus, an isolated or purified polypeptide is substantially free of other cellular material, or culture medium when produced by recombinant techniques, or substantially free of chemical precursors or other chemicals when chemically synthesized. A protein that is substantially free of cellular material includes preparations of protein having less than about 30%, 20%, 10%, 5%, or 1% (by dry weight) of contaminating protein. When the protein of the invention or biologically active portion thereof is recombinantly produced, optimally culture medium represents less than about 30%, 20%, 10%, 5%, or 1% (by dry weight) of chemical precursors or non-protein-of-interest chemicals.

[0118] Particular methods provided herein for binding and/or cleaving a target sequence of interest involve the use of an in vitro assembled RGN ribonucleoprotein complex. In vitro assembly of an RGN ribonucleoprotein complex can be performed using any method known in the art in which an RGN polypeptide is contacted with a guide RNA under conditions to allow for binding of the RGN polypeptide to the guide RNA. As used herein, “contact”, “contacting”, “contacted”, “contacting”, refer to placing the components of a desired reaction together under conditions suitable for carrying out the desired reaction. The RGN polypeptide can be purified from a biological sample, cell lysate, or culture medium, produced via in vitro translation, or chemically synthesized. The guide RNA can be purified from a biological sample, cell lysate, or culture medium, transcribed in vitro, or chemically synthesized. The RGN polypeptide and guide RNA can be brought into contact in solution (e.g., buffered saline solution) to allow for in vitro assembly of the RGN ribonucleoprotein complex.

VII. Methods of Binding, Cleaving, or Modifying a Target Sequence

[0119] The present disclosure provides methods for binding, cleaving, and/or modifying a target nucleotide sequence of interest. The methods include delivering a system comprising at least one guide

[0120] RNA or a polynucleotide encoding the same, and at least one RGN polypeptide or a polynucleotide encoding the same to the target sequence or a cell, organelle, or embryo comprising the target sequence. In some of these embodiments, the RGN comprises the amino acid sequence of SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54, or an active variant or fragment thereof. In various embodiments, the guide RNA comprises a CRISPR repeat sequence comprising the nucleotide sequence of SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55, or an active variant or fragment thereof. In particular embodiments, the guide RNA comprises a tracrRNA comprising the nucleotide sequence of SEQ ID NOs: 3, 13, 21, 29, 38, 47, or 56, or an active variant or fragment thereof. The guide RNA of the system can be a single guide RNA or a dual-guide RNA. The RGN of the system may be nuclease dead RGN, have nickase activity, or may be a fusion polypeptide. In some embodiments, the fusion polypeptide comprises a base-editing polypeptide, for example a cytidine deaminase or an adenosine deaminase. In particular embodiments, the RGN and/or guide RNA is heterologous to the cell, organelle, or embryo to which the RGN and/or guide RNA (or polynucleotide(s) encoding at least one of the RGN and guide RNA) are introduced.

[0121] In those embodiments wherein the method comprises delivering a polynucleotide encoding a guide RNA and/or an RGN polypeptide, the cell or embryo can then be cultured under conditions in which the guide RNA and/or RGN polypeptide are expressed. In various embodiments, the method comprises contacting a target sequence with an RGN ribonucleoprotein complex. The RGN ribonucleoprotein complex may comprise an RGN that is nuclease dead or has nickase activity. In some embodiments, the RGN of the ribonucleoprotein complex is a fusion polypeptide comprising a base-editing polypeptide. In certain embodiments, the method comprises introducing into a cell, organelle, or embryo comprising a target sequence an RGN ribonucleoprotein complex. The RGN ribonucleoprotein complex can be one that has been purified from a biological sample, recombinantly produced and subsequently purified, or in vitro-assembled as described herein. In those embodiments wherein the RGN ribonucleoprotein complex that is contacted with the target sequence or a cell organelle, or embryo has been assembled in vitro, the method can further comprise the in vitro assembly of the complex prior to contact with the target sequence, cell, organelle, or embryo.

[0122] A purified or in vitro assembled RGN ribonucleoprotein complex can be introduced into a cell, organelle, or embryo using any method known in the art, including, but not limited to electroporation. Alternatively, an RGN polypeptide and/or polynucleotide encoding or comprising the guide RNA can be introduced into a cell, organelle, or embryo using any method known in the art (e.g., electroporation).

[0123] Upon delivery to or contact with the target sequence or cell, organelle, or embryo comprising the target sequence, the guide RNA directs the RGN to bind to the target sequence in a sequence-specific manner. In those embodiments wherein the RGN has nuclease activity, the RGN polypeptide cleaves the target sequence of interest upon binding. The target sequence can subsequently be modified via endogenous repair mechanisms, such as non-homologous end joining, or homology-directed repair with a provided donor polynucleotide.

[0124] Methods to measure binding of an RGN polypeptide to a target sequence are known in the art and include chromatin immunoprecipitation assays, gel mobility shift assays, DNA pull-down assays, reporter assays, microplate capture and detection assays. Likewise, methods to measure cleavage or modification of a target sequence are known in the art and include in vitro or in vivo cleavage assays wherein cleavage is confirmed using PCR, sequencing, or gel electrophoresis, with or without the attachment of an appropriate label (e.g., radioisotope, fluorescent substance) to the target sequence to facilitate detection of degradation products. Alternatively, the nicking triggered exponential amplification reaction (NTEXPAR) assay can be used (see, e.g., Zhang et al. (2016) *Chem. Sci.* 7:4951-4957). In vivo cleavage can be evaluated using the Surveyor assay (Guschin et al. (2010) *Methods Mol Biol* 649:247-256).

[0125] In some embodiments, the methods involve the use of a single type of RGN complexed with more than one guide RNA. The more than one guide RNA can target different regions of a single gene or can target multiple genes.

[0126] In those embodiments wherein a donor polynucleotide is not provided, a double-stranded break introduced by an RGN polypeptide can be repaired by a non-homologous end-joining (NHEJ) repair process. Due to the error-prone nature of NHEJ, repair of the double-stranded break can result in a modification to the target sequence. As used herein, a “modification” in reference to a nucleic acid molecule refers to a change in the nucleotide sequence of the nucleic acid molecule, which can be a deletion, insertion, or substitution of one or more nucleotides, or a combination thereof. Modification of the target sequence can result in the expression of an altered protein product or inactivation of a coding sequence.

[0127] In those embodiments wherein a donor polynucleotide is present, the donor sequence in the donor polynucleotide can be integrated into or exchanged with the target nucleotide sequence during the course of repair of the introduced double-stranded break, resulting in the introduction of the exogenous donor sequence. A donor polynucleotide thus comprises a donor sequence that is desired to be introduced into a target sequence of interest. In some embodiments, the donor sequence alters the original target nucleotide sequence such that the newly integrated donor sequence will not be recognized and cleaved by the RGN. Integration of the donor sequence can be enhanced by the inclusion within the donor polynucleotide of flanking sequences that have substantial sequence identity with the sequences flanking the target nucleotide sequence, allowing for a homology-directed repair process. In those embodiments wherein the RGN polypeptide introduces double-stranded staggered breaks, the donor polynucleotide can comprise a donor sequence flanked by compatible overhangs, allowing for direct ligation of the donor sequence to the cleaved target nucleotide sequence comprising overhangs by a non-homologous repair process during repair of the double-stranded break.

[0128] In those embodiments wherein the method involves the use of an RGN that is a nickase (i.e., is only able to cleave a single strand of a double-stranded polynucleotide), the method can comprise introducing two RGN nickases that target identical or overlapping target sequences and cleave different strands of the polynucleotide. For example, an RGN nickase that only cleaves the positive (+)

strand of a double-stranded polynucleotide can be introduced along with a second RGN nickase that only cleaves the negative (−) strand of a double-stranded polynucleotide.

[0129] In various embodiments, a method is provided for binding a target nucleotide sequence and detecting the target sequence, wherein the method comprises introducing into a cell, organelle, or embryo at least one guide RNA or a polynucleotide encoding the same, and at least one RGN polypeptide or a polynucleotide encoding the same, expressing the guide RNA and/or RGN polypeptide (if coding sequences are introduced), wherein the RGN polypeptide is a nuclease-dead RGN and further comprises a detectable label, and the method further comprises detecting the detectable label. The detectable label may be fused to the RGN as a fusion protein (e.g., fluorescent protein) or may be a small molecule conjugated to or incorporated within the RGN polypeptide that can be detected visually or by other means.

[0130] Also provided herein are methods for modulating the expression of a target sequence or a gene of interest under the regulation of a target sequence. The methods comprise introducing into a cell, organelle, or embryo at least one guide RNA or a polynucleotide encoding the same, and at least one

[0131] RGN polypeptide or a polynucleotide encoding the same, expressing the guide RNA and/or RGN polypeptide (if coding sequences are introduced), wherein the RGN polypeptide is a nuclease-dead RGN. In some of these embodiments, the nuclease-dead RGN is a fusion protein comprising an expression modulator domain (i.e., epigenetic modification domain, transcriptional activation domain or a transcriptional repressor domain) as described herein.

[0132] The present disclosure also provides methods for binding and/or modifying a target nucleotide sequence of interest. The methods include delivering a system comprising at least one guide RNA or a polynucleotide encoding the same, and at least one fusion polypeptide comprises an RGN of the invention and a base-editing polypeptide, for example a cytidine deaminase or an adenosine deaminase, or a polynucleotide encoding the fusion polypeptide, to the target sequence or a cell, organelle, or embryo comprising the target sequence.

[0133] One of ordinary skill in the art will appreciate that any of the presently disclosed methods can be used to target a single target sequence or multiple target sequences. Thus, methods comprise the use of a single RGN polypeptide in combination with multiple, distinct guide RNAs, which can target multiple, distinct sequences within a single gene and/or multiple genes. Also encompassed herein are methods wherein multiple, distinct guide RNAs are introduced in combination with multiple, distinct RGN polypeptides. These guide RNAs and guide RNA/RGN polypeptide systems can target multiple, distinct sequences within a single gene and/or multiple genes.

[0134] In one aspect, the invention provides kits containing any one or more of the elements disclosed in the above methods and compositions. In some embodiments, the kit comprises a vector system and instructions for using the kit. In some embodiments, the vector system comprises (a) a first regulatory element operably linked to a tracr mate sequence and one or more insertion sites for inserting a guide sequence upstream of the tracr mate sequence, wherein when expressed, the guide sequence directs sequence-specific binding of a CRISPR complex to a target sequence in a eukaryotic cell, wherein the CRISPR complex

comprises a CRISPR enzyme complexed with (1) the guide sequence that is hybridized to the target sequence, and (2) the tracr mate sequence that is hybridized to the tracr sequence; and/or (b) a second regulatory element operably linked to an enzyme coding sequence encoding said CRISPR enzyme comprising a nuclear localization sequence. Elements may be provided individually or in combinations, and may be provided in any suitable container, such as a vial, a bottle, or a tube.

[0135] In some embodiments, the kit includes instructions in one or more languages. In some embodiments, a kit comprises one or more reagents for use in a process utilizing one or more of the elements described herein. Reagents may be provided in any suitable container. For example, a kit may provide one or more reaction or storage buffers. Reagents may be provided in a form that is usable in a particular assay, or in a form that requires addition of one or more other components before use (e.g. in concentrate or lyophilized form). A buffer can be any buffer, including but not limited to a sodium carbonate buffer, a sodium bicarbonate buffer, a borate buffer, a Tris buffer, a MOPS buffer, a HEPES buffer, and combinations thereof. In some embodiments, the buffer is alkaline. In some embodiments, the buffer has a pH from about 7 to about 10.

[0136] In some embodiments, the kit comprises one or more oligonucleotides corresponding to a guide sequence for insertion into a vector so as to operably link the guide sequence and a regulatory element. In some embodiments, the kit comprises a homologous recombination template polynucleotide. In one aspect, the invention provides methods for using one or more elements of a CRISPR system. The CRISPR complex of the invention provides an effective means for modifying a target polynucleotide. The CRISPR complex of the invention has a wide variety of utility including modifying (e.g., deleting, inserting, translocating, inactivating, activating) a target polynucleotide in a multiplicity of cell types. As such the CRISPR complex of the invention has a broad spectrum of applications in, e.g., gene therapy, drug screening, disease diagnosis, and prognosis. An exemplary CRISPR complex comprises a CRISPR enzyme complexed with a guide sequence hybridized to a target sequence within the target polynucleotide.

VIII. Target Polynucleotides

[0137] In one aspect, the invention provides for methods of modifying a target polynucleotide in a eukaryotic cell, which may be in vivo, ex vivo or in vitro. In some embodiments, the method comprises sampling a cell or population of cells from a human or non-human animal or plant (including microalgae) and modifying the cell or cells. Culturing may occur at any stage ex vivo. The cell or cells may even be re-introduced into the non-human animal or plant (including micro-algae).

[0138] Using natural variability, plant breeders combine most useful genes for desirable qualities, such as yield, quality, uniformity, hardiness, and resistance against pests. These desirable qualities also include growth, day length preferences, temperature requirements, initiation date of floral or reproductive development, fatty acid content, insect resistance, disease resistance, nematode resistance, fungal resistance, herbicide resistance, tolerance to various environmental factors including drought, heat, wet, cold, wind, and adverse soil conditions including high salinity. The sources of these useful genes include native or foreign

varieties, heirloom varieties, wild plant relatives, and induced mutations, e.g., treating plant material with mutagenic agents. Using the present invention, plant breeders are provided with a new tool to induce mutations. Accordingly, one skilled in the art can analyze the genome for sources of useful genes, and in varieties having desired characteristics or traits employ the present invention to induce the rise of useful genes, with more precision than previous mutagenic agents and hence accelerate and improve plant breeding programs.

[0139] The target polynucleotide of an RGN system can be any polynucleotide endogenous or exogenous to the eukaryotic cell. For example, the target polynucleotide can be a polynucleotide residing in the nucleus of the eukaryotic cell. The target polynucleotide can be a sequence coding a gene product (e.g., a protein) or a non-coding sequence (e.g., a regulatory polynucleotide or a junk DNA). Without wishing to be bound by theory, it is believed that the target sequence should be associated with a PAM (protospacer adjacent motif); that is, a short sequence recognized by the CRISPR complex. The precise sequence and length requirements for the PAM differ depending on the CRISPR enzyme used, but PAMs are typically 2-5 base pair sequences adjacent the protospacer (that is, the target sequence).

[0140] The target polynucleotide of a CRISPR complex may include a number of disease-associated genes and polynucleotides as well as signaling biochemical pathway-associated genes and polynucleotides. Examples of target polynucleotides include a sequence associated with a signaling biochemical pathway, e.g., a signaling biochemical pathway-associated gene or polynucleotide. Examples of target polynucleotides include a disease associated gene or polynucleotide. A “disease-associated” gene or polynucleotide refers to any gene or polynucleotide which is yielding transcription or translation products at an abnormal level or in an abnormal form in cells derived from a disease-affected tissues compared with tissues or cells of a non-disease control. It may be a gene that becomes expressed at an abnormally high level; it may be a gene that becomes expressed at an abnormally low level, where the altered expression correlates with the occurrence and/or progression of the disease. A disease-associated gene also refers to a gene possessing mutation(s) or genetic variation that is directly responsible or is in linkage disequilibrium with a gene(s) that is responsible for the etiology of a disease (e.g., a causal mutation). The transcribed or translated products may be known or unknown, and further may be at a normal or abnormal level. Examples of disease-associated genes and polynucleotides are available from McKusick-Nathans Institute of Genetic Medicine, Johns Hopkins University (Baltimore, Md.) and National Center for Biotechnology Information, National Library of Medicine (Bethesda, Md.), available on the World Wide Web.

[0141] Although CRISPR systems are particularly useful for their relative ease in targeting to genomic sequences of interest, there still remains an issue of what the RGN can do to address a causal mutation. One approach is to produce a fusion protein between an RGN (preferably an inactive or nickase variant of the RGN) and a base-editing enzyme or the active domain of a base editing enzyme, such as an cytidine deaminase or an adenosine deaminase base editor (U.S. Pat. No. 9,840,699, herein incorporated by reference). In some embodiments, the methods comprise contacting a DNA molecule with (a) a fusion protein comprising an RGN

of the invention and a base-editing polypeptide such as a deaminase; and (b) a gRNA targeting the fusion protein of (a) to a target nucleotide sequence of the DNA strand; wherein the DNA molecule is contacted with the fusion protein and the gRNA in an amount effective and under conditions suitable for the deamination of a nucleotide base. In some embodiments, the target DNA sequence comprises a sequence associated with a disease or disorder, and wherein the deamination of the nucleotide base results in a sequence that is not associated with a disease or disorder. In some embodiments, the target DNA sequence resides in an allele of a crop plant, wherein the particular allele of the trait of interest results in a plant of lesser agronomic value. The deamination of the nucleotide base results in an allele that improves the trait and increases the agronomic value of the plant.

[0142] In some embodiments, the DNA sequence comprises a T→C or A→G point mutation associated with a disease or disorder, and wherein the deamination of the mutant C or G base results in a sequence that is not associated with a disease or disorder. In some embodiments, the deamination corrects a point mutation in the sequence associated with the disease or disorder.

[0143] In some embodiments, the sequence associated with the disease or disorder encodes a protein, and wherein the deamination introduces a stop codon into the sequence associated with the disease or disorder, resulting in a truncation of the encoded protein. In some embodiments, the contacting is performed in vivo in a subject susceptible to having, having, or diagnosed with the disease or disorder. In some embodiments, the disease or disorder is a disease associated with a point mutation, or a single-base mutation, in the genome. In some embodiments, the disease is a genetic disease, a cancer, a metabolic disease, or a lysosomal storage disease.

[0144] Further examples of loci which are causal for certain genetic diseases, particularly loci which can be readily targeted by RGNs or RGN-base editor fusion proteins of the invention, can be found in Example 9 and corresponding Table 12.

Hurler Syndrome

[0145] An example of a genetically inherited disease which could be corrected using an approach that relies on an RGN-base editor fusion protein of the invention is Hurler Syndrome. Hurler Syndrome, also known as MPS-1, is the result of a deficiency of α -L-iduronidase (IDUA) resulting in a lysosomal storage disease characterized at the molecular level by the accumulation of dermatan sulfate and heparan sulfate in lysosomes. This disease is generally an inherited genetic disorder caused by mutations in the IDUA gene encoding α -L-iduronidase. Common IDUA mutations are W402X and Q70X, both nonsense mutations resulting in premature termination of translation. Such mutations are well addressed by precise genome editing (PGE) approaches, since reversion of a single nucleotide, for example by a base-editing approach, would restore the wild-type coding sequence and result in protein expression controlled by the endogenous regulatory mechanisms of the genetic locus. Additionally, since heterozygotes are known to be asymptomatic, a PGE therapy that targets one of these mutations would be useful to a large proportion of patients with this disease, as only one of the mutated alleles needs to

be corrected (Bunge et al. (1994) *Hum. Mol. Genet.* 3 (6): 861-866, herein incorporated by reference).

[0146] Current treatments for Hurler Syndrome include enzyme replacement therapy and bone marrow transplants (Vellodi et al. (1997) *Arch. Dis. Child.* 76 (2): 92-99; Peters et al. (1998) *Blood* 91 (7): 2601-2608, herein incorporated by reference). While enzyme replacement therapy has had a dramatic effect on the survival and quality of life of Hurler Syndrome patients, this approach requires costly and time-consuming weekly infusions. Additional approaches include the delivery of the IDUA gene on an expression vector or the insertion of the gene into a highly expressed locus such as that of serum albumin (U.S. Pat. No. 9,956,247, herein incorporated by reference). However, these approaches do not restore the original IDUA locus to the correct coding sequence. A genome-editing strategy would have a number of advantages, most notably that regulation of gene expression would be controlled by the natural mechanisms present in healthy individuals. Additionally, using base editing does not necessitate causing a double stranded DNA breaks, which could lead to large chromosomal rearrangements, cell death, or oncogenicity by the disruption of tumor suppression mechanisms. An enabling description of a method to correct the causal mutation of this disease is provided in Example 10. The described methods are an example of a general strategy directed toward using RGN-base editor fusion proteins of the invention to target and correct certain disease-causing mutations in the human genome. It will be appreciated that similar approaches to target diseases such as those described in Table 12 may also be pursued. It will be further appreciated that similar approaches to target disease-causing mutations in other species, particularly common household pets or livestock, can also be deployed using the RGNs of the invention. Common household pets and livestock include dogs, cats, horses, pigs, cows, sheep, chickens, donkeys, snakes, ferrets, fish including salmon, and shrimp.

Friedreich's Ataxia

[0147] RGNs of the invention could also be useful in human therapeutic approaches where the causal mutation is more complicated. For example, some diseases such as Friedreich's Ataxia and Huntington's Disease are the result of a significant increase in repeats of a three nucleotide motif at a particular region of a gene, which affects the ability of the expressed protein to function or to be expressed. Friedreich's Ataxia (FRDA) is an autosomal recessive disease resulting in progressive degeneration of nervous tissue in the spinal cord. Reduced levels of the frataxin (FXN) protein in the mitochondria cause oxidative damages and iron deficiencies at the cellular level. The reduced FXN expression has been linked to a GAA triplet expansion within the intron 1 of the somatic and germline FXN gene. In FRDA patients, the GAA repeat frequently consists of more than 70, sometimes even more than 1000 (most commonly 600-900) triplets, whereas unaffected individuals have about 40 repeats or less (Pandolfo et al. (2012) *Handbook of Clinical Neurology* 103:275-294; Campuzano et al. (1996) *Science* 271:1423-1427; Pandolfo (2002) *Adv. Exp. Med. Biol.* 516: 99-118; all herein incorporated by reference).

[0148] The expansion of the trinucleotide repeat sequence causing Friedreich's Ataxia (FRDA) occurs in a defined genetic locus within the FXN gene, referred to as the FRDA instability region. RNA guided nucleases (RGNs) may be used for excising the instability region in FRDA patient

cells. This approach requires 1) an RGN and guide RNA sequence that can be programmed to target the allele in the human genome; and 2) a delivery approach for the RGN and guide sequence. Many nucleases used for genome editing, such as the commonly used Cas9 nuclease from *S. pyogenes* (SpCas9), are too large to be packaged into adeno-associated viral (AAV) vectors, especially when considering the length of the SpCas9 gene and the guide RNA in addition to other genetic elements required for functional expression cassettes. This makes an approach using SpCas9 more difficult.

[0149] The compact RNA guided nucleases of the invention, particularly APG07433.1 and APG08290.1, are uniquely well suited for the excision of the FRDA instability region. Each RGN has a PAM requirement that is in the vicinity of the FRDA instability region. Additionally, each of these RGNs can be packaged into an AAV vector along with a guide RNA. Packing two guide RNAs would likely require a second vector, but this approach still compares favorably to what would be required of a larger nuclease such as SpCas9, which may require splitting the protein sequence between two vectors. An enabling description of a method to correct the causal mutation of this disease is provided in Example 11. The described methods encompass a strategy using RGNs of the invention in which a region of genomic instability is removed. Such a strategy is applicable to other diseases and disorders which have a similar genetic basis, such as Huntington's Disease. Similar strategies using RGNs of the invention may also be applicable to similar diseases and disorders in non-human animals of agronomic or economic importance, including dogs, cats, horses, pigs, cows, sheep, chickens, donkeys, snakes, ferrets, fish including salmon, and shrimp

Hemoglobinopathies

[0150] RGNs of the invention could also be to introduce disruptive mutations that may result in a beneficial effect. Genetic defects in the genes encoding hemoglobin, particularly the beta globin chain (the HBB gene), can be responsible for a number of diseases known as hemoglobinopathies, including sickle cell anemia and thalassemias.

[0151] In adult humans, hemoglobin is a heterotetramer comprising two alpha (α)-like globin chains and two beta (β)-like globin chains and 4 heme groups. In adults the $\alpha\beta$ 2 tetramer is referred to as Hemoglobin A (HbA) or adult hemoglobin. Typically, the alpha and beta globin chains are synthesized in an approximate 1:1 ratio and this ratio seems to be critical in terms of hemoglobin and red blood cell (RBC) stabilization. In a developing fetus, a different form of hemoglobin, fetal hemoglobin (HbF), is produced which has a higher binding affinity for oxygen than Hemoglobin A such that oxygen can be delivered to the baby's system via the mother's blood stream. Fetal hemoglobin also contains two α globin chains, but in place of the adult β -globin chains, it has two fetal gamma (γ)-globin chains (i.e., fetal hemoglobin is $\alpha\gamma$ 2). The regulation of the switch from production of gamma-to beta-globin is quite complex, and primarily involves a down-regulation of gamma globin transcription with a simultaneous up-regulation of beta globin transcription. At approximately 30 weeks of gestation, the synthesis of gamma globin in the fetus starts to drop while the production of beta globin increases. By approximately 10 months of age, the newborn's hemoglobin is nearly all $\alpha\beta$ 2 although some HbF persists into adulthood (approximately 1-3% of total hemoglobin). In the majority

of patients with hemoglobinopathies, the genes encoding gamma globin remain present, but expression is relatively low due to normal gene repression occurring around parturition as described above.

[0152] Sickle cell disease is caused by a V6E mutation in the β globin gene (HBB) (a GAG to GTG at the DNA level), where the resultant hemoglobin is referred to as “hemoglobinS” or “HbS.” Under lower oxygen conditions, HbS molecules aggregate and form fibrous precipitates. These aggregates cause the abnormality or ‘sickling’ of the RBCs, resulting in a loss of flexibility of the cells. The sickling RBCs are no longer able to squeeze into the capillary beds and can result in vaso-occlusive crisis in sickle cell patients. In addition, sickled RBCs are more fragile than normal RBCs, and tend towards hemolysis, eventually leading to anemia in the patient.

[0153] Treatment and management of sickle cell patients is a life-long proposition involving antibiotic treatment, pain management and transfusions during acute episodes. One approach is the use of hydroxyurea, which exerts its effects in part by increasing the production of gamma globin. Long term side effects of chronic hydroxyurea therapy are still unknown, however, and treatment gives unwanted side effects and can have variable efficacy from patient to patient. Despite an increase in the efficacy of sickle cell treatments, the life expectancy of patients is still only in the mid to late 50's and the associated morbidities of the disease have a profound impact on a patient's quality of life.

[0154] Thalassemias (alpha thalassemias and beta thalassemia) are also diseases relating to hemoglobin and typically involve a reduced expression of globin chains. This can occur through mutations in the regulatory regions of the genes or from a mutation in a globin coding sequence that results in reduced expression or reduced levels of functional globin protein. Treatment of thalassemias usually involves blood transfusions and iron chelation therapy. Bone marrow transplants are also being used for treatment of people with severe thalassemias if an appropriate donor can be identified, but this procedure can have significant risks.

[0155] One approach that has been proposed for the treatment of both SCD and beta thalassemias is to increase the expression of gamma globin so that HbF functionally replaces the aberrant adult hemoglobin As mentioned above, treatment of SCD patients with hydroxyurea is thought to be successful in part due to its effect on increasing gamma globin expression (DeSimone (1982) *Proc Nat'l Acad Sci USA* 79 (14): 4428-31; Ley, et al., (1982) *N. Engl. J. Medicine*, 307:1469-1475; Ley, et al., (1983) *Blood* 62:370-380; Constantoulakis et al., (1988) *Blood* 72 (6): 1961-1967, all herein incorporated by reference). Increasing the expression of HbF involves identification of genes whose products play a role in the regulation of gamma globin expression. One such gene is BCL11A. BCL11A encodes a zinc finger protein that expressed in adult erythroid precursor cells, and down-regulation of its expression leads to an increase in gamma globin expression (Sankaran et al (2008) *Science* 322:1839, herein incorporated by reference). Use of an inhibitory RNA targeted to the BCL11A gene has been proposed (e.g., U.S. Patent Publication 2011/0182867, herein incorporated by reference) but this technology has several potential drawbacks, including that complete knock down may not be achieved, delivery of such RNAs may be problematic, and the RNAs must be present continuously, requiring multiple treatments for life.

[0156] RGNs of the invention may be used to target the BCL11A enhancer region to disrupt expression of BCL11A, thereby increasing gamma globin expression. This targeted disruption can be achieved by non-homologous end joining (NHEJ), whereby an RGN of the invention targets to a particular sequence within the BCL11A enhancer region, makes a double-stranded break, and the cell's machinery repairs the break, typically simultaneously introducing deleterious mutations. Similar to what is described for other disease targets, the RGNs of the invention have advantages over other known RGNs due to their relatively small size, which enables packaging expression cassettes for the RGN and its guide RNA into a single AAV vector for in vivo delivery. An enabling description of this method is provided in Example 12. Similar strategies using RGNs of the invention may also be applicable to similar diseases and disorders in both humans and in non-human animals of agronomic or economic importance.

IX. Cells Comprising a Polynucleotide Genetic Modification

[0157] Provided herein are cells and organisms comprising a target sequence of interest that has been modified using a process mediated by an RGN, crRNA, and/or tracrRNA as described herein. In some of these embodiments, the RGN comprises the amino acid sequence of SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54, or an active variant or fragment thereof. In various embodiments, the guide RNA comprises a CRISPR repeat sequence comprising the nucleotide sequence of SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55, or an active variant or fragment thereof. In particular embodiments, the guide RNA comprises a tracrRNA comprising the nucleotide sequence of SEQ ID NOs: 3, 13, 21, 29, 38, 47, or 56, or an active variant or fragment thereof. The guide RNA of the system can be a single guide RNA or a dual-guide RNA.

[0158] The modified cells can be eukaryotic (e.g., mammalian, plant, insect cell) or prokaryotic. Also provided are organelles and embryos comprising at least one nucleotide sequence that has been modified by a process utilizing an RGN, crRNA, and/or tracrRNA as described herein. The genetically modified cells, organisms, organelles, and embryos can be heterozygous or homozygous for the modified nucleotide sequence.

[0159] The chromosomal modification of the cell, organism, organelle, or embryo can result in altered expression (up-regulation or down-regulation), inactivation, or the expression of an altered protein product or an integrated sequence. In those instances wherein the chromosomal modification results in either the inactivation of a gene or the expression of a non-functional protein product, the genetically modified cell, organism, organelle, or embryo is referred to as a “knock out”. The knock out phenotype can be the result of a deletion mutation (i.e., deletion of at least one nucleotide), an insertion mutation (i.e., insertion of at least one nucleotide), or a nonsense mutation (i.e., substitution of at least one nucleotide such that a stop codon is introduced).

[0160] Alternatively, the chromosomal modification of a cell, organism, organelle, or embryo can produce a “knock in”, which results from the chromosomal integration of a nucleotide sequence that encodes a protein. In some of these embodiments, the coding sequence is integrated into the chromosome such that the chromosomal sequence encoding

the wild-type protein is inactivated, but the exogenously introduced protein is expressed.

[0161] In other embodiments, the chromosomal modification results in the production of a variant protein product. The expressed variant protein product can have at least one amino acid substitution and/or the addition or deletion of at least one amino acid. The variant protein product encoded by the altered chromosomal sequence can exhibit modified characteristics or activities when compared to the wild-type protein, including but not limited to altered enzymatic activity or substrate specificity.

[0162] In yet other embodiments, the chromosomal modification can result in an altered expression pattern of a protein. As a non-limiting example, chromosomal alterations in the regulatory regions controlling the expression of a protein product can result in the overexpression or down-regulation of the protein product or an altered tissue or temporal expression pattern.

[0163] The article “a” and “an” are used herein to refer to one or more than one (i.e., to at least one) of the grammatical object of the article. By way of example, “a polypeptide” means one or more polypeptides.

[0164] All publications and patent applications mentioned in the specification are indicative of the level of those skilled in the art to which this disclosure pertains. All publications and patent applications are herein incorporated by reference to the same extent as if each individual publication or patent application was specifically and individually indicated to be incorporated by reference.

[0165] Although the foregoing invention has been described in some detail by way of illustration and example for purposes of clarity of understanding, it will be obvious that certain changes and modifications may be practiced within the scope of the appended embodiments.

[0166] Non-limiting embodiments include:

[0167] 1. A nucleic acid molecule comprising a polynucleotide encoding an RNA-guided nuclease (RGN) polypeptide, wherein said polynucleotide comprises a nucleotide sequence encoding an RGN polypeptide comprising an amino acid sequence having at least 95% sequence identity to SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54;

[0168] wherein said RGN polypeptide binds a target DNA sequence in an RNA-guided sequence specific manner when bound to a guide RNA (gRNA) capable of hybridizing to said target DNA sequence, and wherein said polynucleotide encoding an RGN polypeptide is operably linked to a promoter heterologous to said polynucleotide.

[0169] 2. The nucleic acid molecule of embodiment 1, wherein said RGN polypeptide is capable of cleaving said target DNA sequence upon binding.

[0170] 3. The nucleic acid molecule of embodiment 2, wherein cleavage by said RGN polypeptide generates a double-stranded break.

[0171] 4. The nucleic acid molecule of embodiment 2, wherein cleavage by said RGN polypeptide generates a single-stranded break.

[0172] 5. The nucleic acid molecule of any one of embodiments 1-4, wherein the RGN polypeptide is operably fused to a base-editing polypeptide.

[0173] 6. The nucleic acid molecule of any one of embodiments 1-5, wherein the RGN polypeptide comprises one or more nuclear localization signals.

[0174] 7. The nucleic acid molecule of any one of embodiments 1-6, wherein the RGN polypeptide is codon optimized for expression in a eukaryotic cell.

[0175] 8. The nucleic acid molecule of any one of embodiments 1-7, wherein said target DNA sequence is located adjacent to a protospacer adjacent motif (PAM).

[0176] 9. A vector comprising the nucleic acid molecule of any one of embodiments 1-8.

[0177] 10. The vector of embodiment 9, further comprising at least one nucleotide sequence encoding said gRNA capable of hybridizing to said target DNA sequence.

[0178] 11. The vector of embodiment 10, where said gRNA is a single guide RNA.

[0179] 12. The vector of embodiment 10, wherein said gRNA is a dual-guide RNA.

[0180] 13. The vector of any one of embodiments 10-12, wherein the guide RNA comprises a CRISPR RNA comprising a CRISPR repeat sequence having at least 95% sequence identity to SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55.

[0181] 14. The vector of any one of embodiments 10-13, wherein the guide RNA comprises a tracrRNA having at least 95% sequence identity to SEQ ID NOs: 3, 13, 21, 29, 38, 47, or 56.

[0182] 15. A cell comprising the nucleic acid molecule of any one of embodiments 1-8 or the vector of any one of embodiments 9-14.

[0183] 16. A method for making an RGN polypeptide comprising culturing the cell of embodiment 15 under conditions in which the RGN polypeptide is expressed.

[0184] 17. A method for making an RGN polypeptide comprising introducing into a cell a heterologous nucleic acid molecule comprising a nucleotide sequence encoding an RNA-guided nuclease (RGN) polypeptide comprising an amino acid sequence having at least 95% sequence identity to SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54;

[0185] wherein said RGN polypeptide binds a target DNA sequence in an RNA-guided sequence specific manner when bound to a guide RNA (gRNA) capable of hybridizing to said target DNA sequence;

[0186] and culturing said cell under conditions in which the RGN polypeptide is expressed.

[0187] 18. The method of embodiment 16 or 17, further comprising purifying said RGN polypeptide.

[0188] 19. The method of embodiment 16 or 17, wherein said cell further expresses one or more guide RNAs that binds to said RGN polypeptide to form an RGN ribonucleoprotein complex.

[0189] 20. The method of embodiment 19, further comprising purifying said RGN ribonucleoprotein complex.

[0190] 21. A nucleic acid molecule comprising a polynucleotide encoding a CRISPR RNA (crRNA), wherein said crRNA comprises a spacer sequence and a CRISPR repeat sequence, wherein said CRISPR repeat sequence comprises a nucleotide sequence having at least 95% sequence identity to SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55;

- [0191] wherein a guide RNA comprising:
- [0192] a) said crRNA; and
- [0193] b) a trans-activating CRISPR RNA (tracrRNA) hybridized to said CRISPR repeat sequence of said crRNA;
- [0194] is capable of hybridizing to a target DNA sequence in a sequence specific manner through the spacer sequence of said crRNA when said guide RNA is bound to an RNA-guided nuclease (RGN) polypeptide, and
- [0195] wherein said polynucleotide encoding a crRNA is operably linked to a promoter heterologous to said polynucleotide.
- [0196] 22. A vector comprising the nucleic acid molecule of embodiment 21.
- [0197] 23. The vector of embodiment 22, wherein said vector further comprises a polynucleotide encoding said tracrRNA.
- [0198] 24. The vector of embodiment 23, wherein said tracrRNA comprises a nucleotide sequence having at least 95% sequence identity to SEQ ID NOs: 3, 13, 21, 29, 38, 47, or 56.
- [0199] 25. The vector of embodiment 23 or 24, wherein said polynucleotide encoding said crRNA and said polynucleotide encoding said tracrRNA are operably linked to the same promoter and are encoded as a single guide RNA.
- [0200] 26. The vector of embodiment 23 or 24, wherein said polynucleotide encoding said crRNA and said polynucleotide encoding said tracrRNA are operably linked to separate promoters.
- [0201] 27. The vector of any one of embodiments 22-26, wherein said vector further comprises a polynucleotide encoding said RGN polypeptide, wherein said RGN polypeptide comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54.
- [0202] 28. A nucleic acid molecule comprising a polynucleotide encoding a trans-activating CRISPR RNA (tracrRNA) comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NOs: 3, 13, 21, 29, 38, 47, or 56;
- [0203] wherein a guide RNA comprising:
- [0204] a) said tracrRNA; and
- [0205] b) a crRNA comprising a spacer sequence and a CRISPR repeat sequence, wherein said tracrRNA hybridizes with said CRISPR repeat sequence of said crRNA;
- [0206] is capable of hybridizing to a target DNA sequence in a sequence specific manner through the spacer sequence of said crRNA when said guide RNA is bound to an RNA-guided nuclease (RGN) polypeptide, and
- [0207] wherein said polynucleotide encoding a tracrRNA is operably linked to a promoter heterologous to said polynucleotide.
- [0208] 29. A vector comprising the nucleic acid molecule of embodiment 28.
- [0209] 30. The vector of embodiment 29, wherein said vector further comprises a polynucleotide encoding said crRNA.
- [0210] 31. The vector of embodiment 30, wherein the CRISPR repeat sequence of said crRNA comprises a nucleotide sequence having at least 95% sequence identity to SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55.
- [0211] 32. The vector of embodiment 30 or 31, wherein said polynucleotide encoding said crRNA and said polynucleotide encoding said tracrRNA are operably linked to the same promoter and are encoded as a single guide RNA.
- [0212] 33. The vector of embodiment 30 or 31, wherein said polynucleotide encoding said crRNA and said polynucleotide encoding said tracrRNA are operably linked to separate promoters.
- [0213] 34. The vector of any one of embodiments 29-33, wherein said vector further comprises a polynucleotide encoding said RGN polypeptide, wherein said RGN polypeptide comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54.
- [0214] 35. A system for binding a target DNA sequence, said system comprising:
- [0215] a) one or more guide RNAs capable of hybridizing to said target DNA sequence or one or more nucleotide sequences encoding the one or more guide RNAs (gRNAs); and
- [0216] b) an RNA-guided nuclease (RGN) polypeptide comprising an amino acid sequence having at least 95% sequence identity to SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54 or a nucleotide sequence encoding the RGN polypeptide;
- [0217] wherein said nucleotide sequences encoding the one or more guide RNAs and encoding the RGN polypeptide are each operably linked to a promoter heterologous to said nucleotide sequence;
- [0218] wherein the one or more guide RNAs hybridize to the target DNA sequence, and
- [0219] wherein the one or more guide RNAs form a complex with the RGN polypeptide, thereby directing said RGN polypeptide to bind to said target DNA sequence.
- [0220] 36. The system of embodiment 35, wherein said gRNA is a single guide RNA (sgRNA).
- [0221] 37. The system of embodiment 35, wherein said gRNA is a dual-guide RNA.
- [0222] 38. The system of any one of embodiments 35-37, wherein said gRNA comprises a CRISPR repeat sequence comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55.
- [0223] 39. The system of any one of embodiments 35-38, wherein said gRNA comprises a tracrRNA comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NOs: 3, 13, 21, 29, 38, 47, or 56.
- [0224] 40. The system of any one of embodiments 35-39, wherein said target DNA sequence is located adjacent to a protospacer adjacent motif (PAM).
- [0225] 41. The system of any one of embodiments 35-40, wherein the target DNA sequence is within a cell.
- [0226] 42. The system of embodiment 41, wherein the cell is a eukaryotic cell.
- [0227] 43. The system of embodiment 42, wherein the eukaryotic cell is a plant cell.
- [0228] 44. The system of embodiment 42, wherein the eukaryotic cell is a mammalian cell.

- [0229] 45. The system of embodiment 42, wherein the eukaryotic cell is an insect cell.
- [0230] 46. The system of embodiment 41, wherein the cell is a prokaryotic cell.
- [0231] 47. The system of any one of embodiments 35-46, wherein when transcribed the one or more guide RNAs hybridize to the target DNA sequence and the guide RNA forms a complex with the RGN polypeptide which causes cleavage of the target DNA sequence.
- [0232] 48. The system of embodiment 47, wherein the cleavage generates a double-stranded break.
- [0233] 49. The system of embodiment 47, wherein cleavage by said RGN polypeptide generates a single-stranded break.
- [0234] 50. The system of any one of embodiments 35-49, wherein the RGN polypeptide is operably linked to a base-editing polypeptide.
- [0235] 51. The system of any one of embodiments 35-50, wherein the RGN polypeptide comprises one or more nuclear localization signals.
- [0236] 52. The system of any one of embodiments 35-51, wherein the RGN polypeptide is codon optimized for expression in a eukaryotic cell.
- [0237] 53. The system of any one of embodiments 35-52, wherein nucleotide sequences encoding the one or more guide RNAs and the nucleotide sequence encoding an RGN polypeptide are located on one vector.
- [0238] 54. The system of any one of embodiments 35-53, wherein said system further comprises one or more donor polynucleotides or one or more nucleotide sequences encoding the one or more donor polynucleotides.
- [0239] 55. A method for binding a target DNA sequence comprising delivering a system according to any one of embodiments 35-54, to said target DNA sequence or a cell comprising the target DNA sequence.
- [0240] 56. The method of embodiment 55, wherein said RGN polypeptide or said guide RNA further comprises a detectable label, thereby allowing for detection of said target DNA sequence.
- [0241] 57. The method of embodiment 55, wherein said guide RNA or said RGN polypeptide further comprises an expression modulator, thereby modulating expression of said target DNA sequence or a gene under transcriptional control by said target DNA sequence.
- [0242] 58. A method for cleaving or modifying a target DNA sequence comprising delivering a system according to any one of embodiments 35-54, to said target DNA sequence or a cell comprising the target DNA sequence.
- [0243] 59. The method of embodiment 58, wherein said modified target DNA sequence comprises insertion of heterologous DNA into the target DNA sequence.
- [0244] 60. The method of embodiment 58, wherein said modified target DNA sequence comprises deletion of at least one nucleotide from the target DNA sequence.
- [0245] 61. The method of embodiment 58, wherein said modified target DNA sequence comprises mutation of at least one nucleotide in the target DNA sequence.
- [0246] 62. A method for binding a target DNA sequence comprising:
- [0247] a) assembling a RNA-guided nuclease (RGN) ribonucleotide complex in vitro by combining:
- [0248] i) one or more guide RNAs capable of hybridizing to the target DNA sequence; and
- [0249] ii) an RGN polypeptide comprising an amino acid sequence having at least 95% sequence identity to SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54;
- [0250] under conditions suitable for formation of the RGN ribonucleotide complex; and
- [0251] b) contacting said target DNA sequence or a cell comprising said target DNA sequence with the in vitro-assembled RGN ribonucleotide complex;
- [0252] wherein the one or more guide RNAs hybridize to the target DNA sequence, thereby directing said RGN polypeptide to bind to said target DNA sequence.
- [0253] 63. The method of embodiment 62, wherein said RGN polypeptide or said guide RNA further comprises a detectable label, thereby allowing for detection of said target DNA sequence.
- [0254] 64. The method of embodiment 62, wherein said guide RNA or said RGN polypeptide further comprises an expression modulator, thereby allowing for the modulation of expression of said target DNA sequence.
- [0255] 65. A method for cleaving and/or modifying a target DNA sequence, comprising contacting the DNA molecule with:
- [0256] a) an RNA-guided nuclease (RGN) polypeptide, wherein said RGN comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NOs: 1, 11, 19, 27, 36, 45 or 54; and
- [0257] b) one or more guide RNAs capable of targeting the RGN of (a) to the target DNA sequence;
- [0258] wherein the one or more guide RNAs hybridize to the target DNA sequence, thereby directing said RGN polypeptide to bind to said target DNA sequence and cleavage and/or modification of said target DNA sequence occurs.
- [0259] 66. The method of embodiment 65, wherein said modified target DNA sequence comprises insertion of heterologous DNA into the target DNA sequence.
- [0260] 67. The method of embodiment 65, wherein said modified target DNA sequence comprises deletion of at least one nucleotide from the target DNA sequence.
- [0261] 68. The method of embodiment 65, wherein said modified target DNA sequence comprises mutation of at least one nucleotide in the target DNA sequence.
- [0262] 69. The method of any one of embodiments 62-68, wherein said gRNA is a single guide RNA (sgRNA).
- [0263] 70. The method of any one of embodiments 62-68, wherein said gRNA is a dual-guide RNA.
- [0264] 71. The method of any one of embodiments 62-70, wherein said gRNA comprises a CRISPR repeat sequence comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55.
- [0265] 72. The method of any one of embodiments 62-71, wherein said gRNA comprises a tracrRNA

comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NOs: 3, 13, 21, 29, 38, 47, or 56.

[0266] 73. The method of any one of embodiments 62-72, wherein said target DNA sequence is located adjacent to a protospacer adjacent motif (PAM).

[0267] 74. The method of any one of embodiments 55-73, wherein the target DNA sequence is within a cell.

[0268] 75. The method of embodiment 74, wherein the cell is a eukaryotic cell.

[0269] 76. The method of embodiment 75, wherein the eukaryotic cell is a plant cell.

[0270] 77. The method of embodiment 75, wherein the eukaryotic cell is a mammalian cell.

[0271] 78. The method of embodiment 75, wherein the eukaryotic cell is an insect cell.

[0272] 79. The method of embodiment 74, wherein the cell is a prokaryotic cell.

[0273] 80. The method of any one of embodiments 74-79, further comprising culturing the cell under conditions in which the RGN polypeptide is expressed and cleaves the target DNA sequence to produce a modified DNA sequence; and selecting a cell comprising said modified DNA sequence.

[0274] 81. A cell comprising a modified target DNA sequence according to the method of embodiment 80.

[0275] 82. The cell of embodiment 81, wherein the cell is a eukaryotic cell.

[0276] 83. The cell of embodiment 82, wherein the eukaryotic cell is a plant cell.

[0277] 84. A plant comprising the cell of embodiment 83.

[0278] 85. A seed comprising the cell of embodiment 83.

[0279] 86. The cell of embodiment 82, wherein the eukaryotic cell is a mammalian cell.

[0280] 87. The cell of embodiment 82, wherein the eukaryotic cell is an insect cell.

[0281] 88. The cell of embodiment 81, wherein the cell is a prokaryotic cell.

[0282] 89. A method for producing a genetically modified cell with a correction in a causal mutation for a genetically inherited disease, the method comprising introducing into the cell:

[0283] a) an RNA-guided nuclease (RGN) polypeptide, wherein the RGN polypeptide comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54, or a polynucleotide encoding said RGN polypeptide, wherein said polynucleotide encoding the RGN polypeptide is operably linked to a promoter to enable expression of the RGN polypeptide in the cell; and

[0284] b) a guide RNA (gRNA), wherein the gRNA comprises a CRISPR repeat sequence comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55, or a polynucleotide encoding said gRNA, wherein said polynucleotide encoding the gRNA is operably linked to a promoter to enable expression of the gRNA in the cell

[0285] whereby the RGN and gRNA target to the genomic location of the causal mutation and modify the genomic sequence to remove the causal mutation.

[0286] 90. The method of embodiment 89, wherein the RGN is fused to a polypeptide which has base-editing activity.

[0287] 91. The method of embodiment 90, wherein the polypeptide with base-editing activity is a cytidine deaminase or an adenosine deaminase.

[0288] 92. The method of embodiment 89, wherein the cell is an animal cell.

[0289] 93. The method of embodiment 89, wherein the cell is a mammalian cell.

[0290] 94. The method of embodiment 92, wherein the cell is derived from a dog, cat, mouse, rat, rabbit, horse, cow, pig, or human.

[0291] 95. The method of embodiment 92, wherein the genetically inherited disease is a disease listed in Table 12.

[0292] 96. The method of embodiment 92, wherein the genetically inherited disease is Hurler Syndrome.

[0293] 97. The method of embodiment 96, wherein the gRNA further comprises a spacer sequence that targets SEQ ID NO: 453, 454, or 455.

[0294] 98. A method for producing a genetically modified cell with a deletion in a disease-causing genomic region of instability, the method comprising introducing into the cell:

[0295] a) an RNA-guided nuclease (RGN) polypeptide, wherein the RGN polypeptide comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54, or a polynucleotide encoding said RGN polypeptide, wherein said polynucleotide encoding the RGN polypeptide is operably linked to a promoter to enable expression of the RGN polypeptide in the cell; and

[0296] b) a guide RNA (gRNA), wherein the gRNA comprises a CRISPR repeat sequence comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55, or a polynucleotide encoding said gRNA, wherein said polynucleotide encoding the gRNA is operably linked to a promoter to enable expression of the gRNA in the cell, and further wherein the gRNA comprises a spacer sequence that targets the 5' flank of the genomic region of instability; and

[0297] c) a second guide RNA (gRNA), wherein the gRNA comprises a CRISPR repeat sequence comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55, or a polynucleotide encoding said gRNA, wherein said polynucleotide encoding the gRNA is operably linked to a promoter to enable expression of the gRNA in the cell, and further wherein said second gRNA comprises a spacer sequence that targets the 3' flank of the genomic region of instability; whereby the RGN and the two gRNAs target to the genomic region of instability and at least a portion of the genomic region of instability is removed.

[0298] 99. The method of embodiment 98, wherein the cell is an animal cell.

- [0299] 100. The method of embodiment 98, wherein the cell is a mammalian cell.
- [0300] 101. The method of embodiment 100, wherein the cell is derived from a dog, cat, mouse, rat, rabbit, horse, cow, pig, or human.
- [0301] 102. The method of embodiment 99, wherein the genetically inherited disease is Friedrich's Ataxia or Huntington's Disease.
- [0302] 103. The method of embodiment 102, wherein the first gRNA further comprises a spacer sequence that targets SEQ ID NO: 468, 469, or 470.
- [0303] 104. The method of embodiment 103, wherein the second gRNA further comprises a spacer sequence that targets SEQ ID NO: 471.
- [0304] 105. A method for producing a genetically modified mammalian hematopoietic progenitor cell having decreased BCL11A mRNA and protein expression, the method comprising introducing into an isolated human hematopoietic progenitor cell:
- [0305] a) an RNA-guided nuclease (RGN) polypeptide, wherein the RGN polypeptide comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54, or a polynucleotide encoding said RGN polypeptide, wherein said polynucleotide encoding the RGN polypeptide is operably linked to a promoter to enable expression of the RGN polypeptide in the cell; and
- [0306] b) a guide RNA (gRNA), wherein the gRNA comprises a CRISPR repeat sequence comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NOs: 2, 12, 20, 28, 37, 46, or 55, or a polynucleotide encoding said gRNA, wherein said polynucleotide encoding the gRNA is operably linked to a promoter to enable expression of the gRNA in the cell,
- [0307] whereby the RGN and gRNA are expressed in the cell and cleave at the BCL11A enhancer region, resulting in genetic modification of the human hematopoietic progenitor cell and reducing the mRNA and/or protein expression of BCL11A.
- [0308] 106. The method of embodiment 105, wherein the gRNA further comprises a spacer sequence that targets SEQ ID NO: 473, 474, 475, 476, 477, or 478.
- [0309] 107. A system for binding a target DNA sequence, said system comprising:
- [0310] a) one or more guide RNAs capable of hybridizing to said target DNA sequence or one or more nucleotide sequences encoding the one or more guide RNAs (gRNAs); and
- [0311] b) an RNA-guided nuclease (RGN) polypeptide comprising an amino acid sequence having at least 95% sequence identity to SEQ ID NOs: 1, 11, 19, 27, 36, 45, or 54;
- [0312] wherein the one or more guide RNAs hybridize to the target DNA sequence, and
- [0313] wherein the one or more guide RNAs forms a complex with the RGN polypeptide, thereby directing said RGN polypeptide to bind to said target DNA sequence.
- [0314] 108. The system of embodiment 107, wherein said RGN polypeptide is nuclease dead or functions as a nickase.

- [0315] 109. The system of embodiment 107 or 108, wherein said RGN polypeptide is operably fused to a base-editing polypeptide.

- [0316] 110. The system of embodiment 109, wherein the base-editing polypeptide is a deaminase.

- [0317] 111. The system of embodiment 110, wherein the deaminase is a cytidine deaminase or an adenosine deaminase.

[0318] The following examples are offered by way of illustration and not by way of limitation.

EXPERIMENTAL

Example 1. Identification of RNA-Guided Nuclease

[0319] Seven distinct CRISPR-associated RNA-guided nucleases (RGN's) were identified and are described in Table 1 below. Table 1 provides the name of each RGN, its amino acid sequence, the source from which it was derived, and processed crRNA and tracrRNA sequences. Table 1 further provides a generic single guide RNA (sgRNA) sequence, where the poly-N indicates the location of the spacer sequence which determines the nucleic acid target sequence of the sgRNA. RGN systems APG systems APG05083.1, APG07433.1, APG08290.1, and APG08290.1 had a conserved sequence in the base of the hairpin stem of the tracrRNA, UNANNG (SEQ ID NO: 68). For the AP05459.1 system, the sequence in the same location is UNANNU (SEQ ID NO: 557). For APG04583.1 and APG01688.1 systems, the sequence is UNANNA (SEQ ID NO: 558).

TABLE 1

Summary of SEQ IDs and CRISPR associated systems					
RGN ID	SEQ ID NO.	Source	crRNA repeat seq (SEQ ID NO.)	tracrRNA (SEQ ID NO.)	sgRNA (SEQ ID NO.)
APG05083.1	1	<i>Bacillus</i> sp.	2	3	10
APG07433.1	10	<i>Bacillus</i> sp.	12	13	18
APG07513.1	18	<i>Bacillus</i> sp.	20	21	26
APG08290.1	26	<i>Bacillus</i> sp.	28	29	35
APG05459.1	34	<i>Enterococcus</i> sp.	37	38	44
APG04583.1	42	<i>Enterococcus</i> sp.	46	47	53
APG01688.1	50	<i>Empedobacter</i> sp.	55	56	62

Example 2: Guide RNA Identification and sgRNA Construction

[0320] Cultures of bacteria that natively express the RNA-guided nuclease system under investigation were grown to mid-log phase (OD600 of ~0.600), pelleted, and flash frozen. RNA was isolated from the pellets using a mirVANA miRNA Isolation Kit (Life Technologies, Carlsbad, CA), and sequencing libraries were prepared from the isolated RNA using an NEBNext Small RNA Library Prep kit (NEB, Beverly, MA). The library prep was fractionated on a 6% polyacrylamide gel into 2 size fractions corresponding to 18-65nt and 90-200nt RNA species to detect crRNAs and tracrRNAs, respectively. Deep sequencing (40 bp paired-end for the smaller fraction and 80 bp paired-end for the larger fraction) was performed on a Next Seq 500 (High Output kit) by a service provider (MoGene, St. Louis, MO). Reads were quality trimmed using Cutadapt and mapped to

reference genomes using Bowtie2. A custom RNAseq pipeline was written in python to detect the crRNA and tracrRNA transcripts. Processed crRNA boundaries were determined by sequence coverage of the native repeat spacer array. The anti-repeat portion of the tracrRNA was identified using permissive BLASTn parameters. RNA sequencing depth confirmed the boundaries of the processed tracrRNA by identifying the transcript containing the anti-repeat. Manual curation of RNAs was performed using secondary structure prediction by NUPACK, an RNA folding software. sgRNA cassettes were prepared by DNA synthesis and were generally designed as follows (5'→3'): 20-30 bp spacer sequence--processed repeat portion of the crRNA--4 bp noncomplementary linker (AAAG; SEQ ID NO: 63)--processed tracrRNA. Other 4 bp noncomplementary linkers may also be used, for example GAAA (SEQ ID NO: 64) or ACUU (SEQ ID NO: 65). In some instances, a 6 bp nucleotide linker may be used, for example CAAAGG (SEQ ID NO: 66). For in vitro assays, sgRNAs were synthesized by in vitro transcription of the sgRNA cassettes with a GeneArt™ Precision gRNA Synthesis Kit (ThermoFisher). Processed crRNA and tracrRNA sequences for each of the RGN polypeptides are identified and are set forth in Table 1. See below for the sgRNAs constructed for PAM libraries 1 and 2.

Example 3: Determination of PAM Requirements for Each RGN

[0321] PAM requirements for each RGN were determined using a PAM depletion assay essentially adapted from Kleinstiver et al. (2015) *Nature* 523:481-485 and Zetsche et al. (2015) *Cell* 163:759-771. Briefly, two plasmid libraries (L1 and L2) were generated in a pUC18 backbone (ampR), with each containing a distinct 30 bp protospacer (target) sequence flanked by 8 random nucleotides (i.e., the PAM region). The target sequence and flanking PAM region of library 1 and library 2 for each RGN are set forth in Table 2.

[0322] The libraries were separately electroporated into *E. coli* BL21 (DE3) cells harboring pRSF-1b expression vectors containing an RGN of the invention (codon optimized for *E. coli*) along with a cognate sgRNA containing a spacer sequence corresponding to the protospacer in L1 or L2. Sufficient library plasmid was used in the transformation reaction to obtain $>10^6$ cfu. Both the RGN and sgRNA in the pRSF-1b backbone were under the control of T7 promoters. The transformation reaction was allowed to recover for 1 hr after which it was diluted into LB media containing carbenicillin and kanamycin and grown overnight. The following day the mixture was diluted into self-inducing Overnight Express™ Instant TB Medium (Millipore Sigma) to allow expression of the RGN and sgRNA, and grown for an additional 4h or 20h after which the cells were spun down and plasmid DNA was isolated with a Mini-prep kit (Qiagen, Germantown, MD). In the presence of the appropriate sgRNA, plasmids containing a PAM that is recognizable by the RGN will be cleaved resulting in their removal from the population. Plasmids containing PAMs that are not recognizable by the RGN, or that are transformed into bacteria not containing an appropriate sgRNA, will survive and replicate. The PAM and protospacer regions of uncleaved plasmids were PCR-amplified and prepared for sequencing following published protocols (16S-metagenomic library prep guide 15044223B, Illumina, San Diego, CA). Deep sequencing

(80 bp single end reads) was performed on a MiSeq (Illumina) by a service provider (MoGene, St. Louis, MO). Typically, 1-4M reads were obtained per amplicon. PAM regions were extracted, counted, and normalized to total reads for each sample. PAMs that lead to plasmid cleavage were identified by being underrepresented when compared to controls (i.e., when the library is transformed into *E. coli* containing the RGN but lacking an appropriate sgRNA). To represent PAM requirements for a novel RGN, the depletion ratios (frequency in sample/frequency in control) for all sequences in the region in question were converted to enrichment values with a $-\log$ base 2 transformation. Sufficient PAMs were defined as those with enrichment values >2.3 (which corresponds to depletion ratios <0.2). PAMs above this threshold in both libraries were collected and used to generate web logos, which for example can be generated using a web-based service on the internet known as “weblogo”. PAM sequences were identified and reported when there was a consistent pattern in the top enriched PAMs. A PAM (having an enrichment factor (EF) >2.3) for each RGN is provided in Table 2. For some RGNs, non-limiting exemplary PAMs (having an EF >3.3) were also identified. For APG005083.1, the exemplary PAM is NNRNCC (SEQ ID NO: 69). For APG007433.1, the exemplary PAM is NNNNCCR (SEQ ID NO: 70). For APG007513.1, the exemplary PAM is NNRNCC (SEQ ID NO: 71). For APG001688.1, the exemplary PAM is NNRANC (SEQ ID NO: 72).

TABLE 2

PAM determination					
RGN ID	sgRNA L1 (SEQ ID NO.)	sgRNA L2 (SEQ ID NO.)	PAM (SEQ ID NO.)	Target seq and PAM region of plasmid library 1 (SEQ ID NO.)	Target seq and PAM region of plasmid library 2 (SEQ ID NO.)
APG05083.1	4	5	6	8	9
APG07433.1	13	14	6	16	17
APG07513.1	21	22	6	24	25
APG08290.1	29	30	32	33	34
APG05459.1	37	38	41	42	43
APG04583.1	45	46	50	51	52
APG01688.1	53	54	59	60	61

Example 4: Cleavage Determination

[0323] Cleavage sites were determined from in vitro cleavage reactions using RNPs (ribonucleoproteins). Expression plasmids containing an RGN fused to a His6 or a His10 tag were constructed and transformed into BL21 (DE3) strains of *E. coli*. Expression was performed using self-inducing media or with IPTG induction. After lysis and clarification, the proteins were purified by immobilized metal affinity chromatography.

[0324] Ribonucleoprotein complexes (comprising nuclease and an sgRNA or a crRNA and tracrRNA duplex) were formed by incubation of nuclease and the RNA in a buffered solution for 20 min at room temperature. The complex was transferred to a tube containing digestion buffer and a PCR amplified target, referred to as “Sequence 1”. Sequence 1 comprised a nucleotide sequence (SEQ ID NO: 73) directly linked at its 3' end to the corresponding PAM sequence for each RGN. Each RGN as a ribonucleoprotein complex was

incubated with its respective target polynucleotide at 25° C. (APG04583.1) or 37° C. (all others) for 30 min or 60 min (APG05459.1 and APG01688.1 only). The digestion reaction was heat inactivated and run on an agarose gel. The cleavage product bands were excised from the gel and sequenced using Sanger sequencing. Cleavage sites were identified by aligning the sequencing results with the expected sequence of the PCR product. Results are shown in Table 3. As shown in Table 3, RGN APG007433.1 may also produce a blunt cut with a different target sequence.

[0325] The cleavage site for Sequence 2 (SEQ ID NO: 559, operably fused at its 3' end to a PAM sequence for RGN APG0733.1) was determined by the following approach for the nuclease APG07433.1. After digestion, the gel purified DNA products were treated with a DNA end repair kit (Thermo Scientific K0771), ligated into linearized blunt vector, and the resulting circular DNA was transformed into *E. coli* competent cells. A staggered cut with a 5' overhang would result in detection of overlapping sequences in the clones from both cleavage products. A 3' overhang would result in missing sequence, and a blunt cut would result in all of the original sequence being detected with no overlap. This experiment also verified the finding from the above described method for sequence 1—most of the clones were detected as having originated from a cut with a 5' overlap, so it is not expected that the finding of a blunt cut is an artifact of utilizing this method.

TABLE 3

RGN cleavage sites						
Sequence 1			Sequence 2			
Distance from PAM			Distance from PAM			
Nuclease	NTS cut site	TS cut site	Overhang	NTS cut site	TS cut site	Overhang
APG07433.1	4	3	1 nt, 5'	3	3	None
APG08290.1	4	3	1 nt, 5'			Not determined
APG05459.1	3	3	None			Not determined
APG04583.1	3	3	None			Not determined
APG01688.1	3	3	None			Not determined

NTS = non-target strand; TS = target strand

Example 5: Mismatch Sensitivity Assay

[0326] Plasmids were designed and obtained with a target sequence (SEQ ID NO: 73) immediately 5' to a suitable PAM motif for the nuclease being evaluated. Single mismatch sequences were also generated with an altered sequence at the position indicated (Table 4). RNP complexes of purified nuclease (APG08290.1 or APG05459.1) and guide RNA were formed and incubated with PCR amplified linear DNA from the designed plasmids. After incubation for a designated period of time and nuclease inactivation, the samples were analyzed by agarose gel electrophoresis to determine the fraction of the linear PCR product remaining. The percentage of the intact band cleaved is shown in Table 5 for mismatches in each position.

TABLE 4

Sequences tested for the mismatch sensitivity assay for APG08290.1 and APG05459.1		
Protospacer sequence	SEQ ID NO.	Mismatch position
GAGCGGACAGCAGCTTCCTATATCTCGTAC	73	None
GAGCGGACAGCAGCTTCCTATATCTCGTAG	74	1
GAGCGGACAGCAGCTTCCTATATCTCGTTC	75	2
GAGCGGACAGCAGCTTCCTATATCTCGAAC	76	3
GAGCGGACAGCAGCTTCCTATATCTCCTAC	77	4
GAGCGGACAGCAGCTTCCTATATCTGGTAC	78	5
GAGCGGACAGCAGCTTCCTATATCACGTAC	79	6
GAGCGGACAGCAGCTTCCTATATGTCGTAC	80	7
GAGCGGACAGCAGCTTCCTATAACTCGTAC	81	8
GAGCGGACAGCAGCTTCCTATTTCTCGTAC	82	9
GAGCGGACAGCAGCTTCCTAAATCTCGTAC	83	10
GAGCGGACAGCAGCTTCCTTTATCTCGTAC	84	11
GAGCGGACAGCAGCTTCCTAATATCTCGTAC	85	12

TABLE 4-continued

Sequences tested for the mismatch sensitivity assay for APG08290.1 and APG05459.1		
Protospacer sequence	SEQ ID NO.	Mismatch position
GAGCGGACAGCAGCTTCGTATATCTCGTAC	86	13
GAGCGGACAGCAGCTTGCTATATCTCGTAC	87	14
GAGCGGACAGCAGCTACCTATATCTCGTAC	88	15
GAGCGGACAGCAGCATCCTATATCTCGTAC	89	16

TABLE 4-continued

Sequences tested for the mismatch sensitivity assay for APG08290.1 and APG05459.1		
Protospacer sequence	SEQ ID NO.	Mismatch position
GAGCGGACAGCAGGTTCTATATCTCGTAC	90	17
GAGCGGACAGCACCTTCCTATATCTCGTAC	91	18
GAGCGGACAGCTGCTTCCTATATCTCGTAC	92	19
GAGCGGACAGGAGCTTCCTATATCTCGTAC	93	20
GAGCGGACACAGCTTCCTATATCTCGTAC	94	21
GAGCGGACTGCAGCTTCCTATATCTCGTAC	95	22
GAGCGGAGAGCAGCTTCCTATATCTCGTAC	96	23
GAGCGGTTCAGCAGCTTCCTATATCTCGTAC	97	24
GAGCGCACAGCAGCTTCCTATATCTCGTAC	98	25
GAGCGGACAGCAGCTTCCTATATCTCGTAC	99	26
GAGGGGACAGCAGCTTCCTATATCTCGTAC	100	27
GACCGGACAGCAGCTTCCTATATCTCGTAC	101	28
GTGCGGACAGCAGCTTCCTATATCTCGTAC	102	29
CAGCGGACAGCAGCTTCCTATATCTCGTAC	103	30

TABLE 5

Mismatch sensitivity for RGN APG08290.1 and RGN APG05459.1		
Mismatch position	% cleaved	
	APG08290.1	APG05459.1
Incompatible PAM, no mismatch	0	0
No mismatch	95	67
1	0	0
2	0	74
3	73	3
4	0	0
5	0	0
6	31	30
7	0	12
8	0	51
9	0	0
10	75	52
11	77	5
12	79	62
13	28	18
14	8	5
15	90	6
16	85	5
17	81	4
18	100	0
19	100	0
20	100	2
21	100	30
22	100	48
23	100	40
24	100	45
25	100	29
26	100	33

TABLE 5-continued

Mismatch sensitivity for RGN APG08290.1 and RGN APG05459.1		
Mismatch position	% cleaved	
	APG08290.1	APG05459.1
27	100	73
28	100	46
29	100	59
30	100	57

[0327] A similar mismatch sensitivity experiment was performed for RGN APG07433.1. This experiment was similar to the one described above, except that the alternative base was introduced into the RNA guide rather than the DNA target. DNA sequences for mismatched sgRNA synthesis are shown in Table 6. Results of the mismatch sensitivity assay are shown in Table 7.

TABLE 6

Sequences tested for the mismatch sensitivity assay for RGN APG07433.1		
DNA template for sgRNA synthesis	SEQ ID NO.	Mismatch position
GAGCGGACAGCAGCTTCCTATATCTCGTAC	73	None
GAGCGGACAGCAGCTTCCTATATCTCGTAT	104	1
GAGCGGACAGCAGCTTCCTATATCTCGTG	105	2
GAGCGGACAGCAGCTTCCTATATCTCGCAC	106	3
GAGCGGACAGCAGCTTCCTATATCTCATAC	107	4
GAGCGGACAGCAGCTTCCTATATCTTGTAC	108	5
GAGCGGACAGCAGCTTCCTATATCCCCGTAC	109	6
GAGCGGACAGCAGCTTCCTATATTTCGTAC	110	7
GAGCGGACAGCAGCTTCCTATACCTCGTAC	111	8
GAGCGGACAGCAGCTTCCTATGTCTCGTAC	112	9
GAGCGGACAGCAGCTTCCTACATCTCGTAC	113	10
GAGCGGACAGCAGCTTCCTGTATCTCGTAC	114	11
GAGCGGACAGCAGCTTCCCATATCTCGTAC	115	12
GAGCGGACAGCAGCTTCTTATATCTCGTAC	116	13
GAGCGGACAGCAGCTTCTATATCTCGTAC	117	14
GAGCGGACAGCAGCTCCCTATATCTCGTAC	118	15
GAGCGGACAGCAGCTTCCTATATCTCGTAC	119	16
GAGCGGACAGCAGTTTCCTATATCTCGTAC	120	17
GAGCGGACAGCACTTCCTATATCTCGTAC	121	18
GAGCGGACAGCGGCTTCCTATATCTCGTAC	122	19
GAGCGGACAGTAGCTTCCTATATCTCGTAC	123	20
GAGCGGACAACTTCCTATATCTCGTAC	124	21

TABLE 6-continued

Sequences tested for the mismatch sensitivity assay for RGN APG07433.1		
DNA template for sgRNA synthesis	SEQ ID NO.	Mismatch position
GAGCGGACGGCAGCTTCCTATATCTCGTAC	125	22
GAGCGGATAGCAGCTTCCTATATCTCGTAC	126	23

TABLE 7

Mismatch sensitivity for RGN APG07433.1	
Mismatch position	% cleaved APG07433.1
No mismatch	86
1	6
2	21
3	-2
4	1
5	-1
6	0
7	7
8	24
9	14
10	-1
11	72
12	44
13	54
14	60
16	65
17	76
18	84
19	86
20	83
21	83
22	93
23	80

[0328] RGNs APG07433.1 and APG08290.1 show significant sensitivity to mismatches in positions 1-10 5' from the PAM with a few exceptions (Table 5 and Table 7). RGN APG05459.1 is sensitive as well to mismatches in this region, but its ability to cleave dsDNA is also heavily abrogated by mismatches distant from the PAM site (Table 5). The total number of sites with a significant influence on whether or not cleavage occurs is at least 15 positions in the spacer sequence. This compares favorably to other genome editing tools, such as the well-studied Cas9 nuclease from *S. pyogenes*, which is generally sensitive to between 10-13 base pairs (Hsu et al., *Nat Biotechnol* (2013) 31 (9): 827-832). Additionally, many of the critical sites abrogating RGN APG05459.1 mediated cleavage are very far from the PAM sequence, notably in the range of 13-20 bp, where many other nucleases show little if any sensitivity to mismatches. This property could be extraordinarily useful in targeting genetic loci that have close sequence similarity to other sites in the organism of interest.

Example 6: Demonstration of Gene Editing Activity in Mammalian Cells

[0329] RGN expression cassettes were produced and introduced into vectors for mammalian expression. Each RGN was codon-optimized for human expression (SEQ ID

NOs 127-133), and operably fused at the 5'end to an SV40 nuclear localization sequence (NLS; SEQ ID NO 134) and to 3xFLAG tags (SEQ ID NO: 135), and operably fused at the 3'end to nucleoplasmin NLS sequences (SEQ ID NO: 136). Each expression cassette was under control of a cytomegalovirus (CMV) promoter (SEQ ID NO: 137). It is known in the art that the CMV transcription enhancer (SEQ ID NO: 138) may also be included in constructs comprising the CMV promoter. Guide RNA expression constructs encoding a single gRNA each under the control of a human RNA polymerase III U6 promoter (SEQ ID NO. 139) were produced and introduced into the pTwist High Copy Amp vector. Sequences for the target sequences for each guide are in Table 9.

[0330] The constructs described above were introduced into mammalian cells. One day prior to transfection, 1×10^5 HEK293T cells/well (Sigma) were plated in 24-well dishes in Dulbecco's modified Eagle medium (DMEM) plus 10% (vol/vol) fetal bovine serum (Gibco) and 1% Penicillin-Streptomycin (Gibco). The next day when the cells were at 50-60% confluency, 500 ng of a RGN expression plasmid plus 500 ng of a single gRNA expression plasmid were co-transfected using 1.5 L of Lipofectamine 3000 (Thermo Scientific) per well, following the manufacturer's instructions. After 48 hours of growth, total genomic DNA was harvested using a genomic DNA isolation kit (Machery-Nagel) according to the manufacturer's instructions.

[0331] The total genomic DNA was then analyzed to determine the rate of editing for each RGN for each genomic target. First, oligonucleotides were produced to be used for PCR amplification and subsequent analysis of the amplified genomic target site. Oligonucleotide sequences used are listed in Tables 8.1 to 8.5.

[0332] All PCR reactions were performed using 10 μ L of 2x Master Mix Phusion High-Fidelity DNA polymerase (Thermo Scientific) in a 20 μ L reaction including 0.5 μ M of each primer. Large genomic regions encompassing each target gene were first amplified using PCR #1 primers, using a program of: 98° C., 1 min; 30 cycles of [98° C., 10 sec; 62° C., 15 sec; 72° C., 5 min]; 72° C., 5 min; 12° C., forever. One microliter of this PCR reaction was then further amplified using primers specific for each guide (PCR #2 primers), using a program of: 98° C., 1 min; 35 cycles of [98° C., 10 sec; 67° C., 15 sec; 72° C., 30 sec]; 72° C., 5 min; 12° C., forever. Primers for PCR #2 include Nextera Read 1 and Read 2 Transposase Adapter overhang sequences for Illumina sequencing.

TABLE 8.1

Oligonucleotides for detection of gene editing activity in mammalian cells, PCR #1		
Description	Sequence	SEQ ID NO
RelA FWD	5'-CTT AGT TTC ACC GCA GGT TCT A-3'	479
RelA REV	5'-CTG TGC ACT CAA CAC TGA TCT A-3'	480
AurkB FWD	5'-CCC AGC CCT AGG TTG TTT ATT-3'	481
AurkB REV	5'-CTG GCT ACA TCT TCC TTG ACT AC-3'	482

TABLE 8.1-continued

Oligonucleotides for detection of gene editing activity in mammalian cells, PCR #1			
		SEQ ID NO.	
Description	Sequence		
HPRT1 FWD	5'-GTG GCA GAA GCA GTG AGT AA-3'	483	
HPRT1 REV	5'-TCC CAT CTA GGC ACT AGG TAA A-3'	484	

TABLE 8.2

Oligonucleotides for detection of gene editing activity in mammalian cells, PCR#2 for APG05083.1, APG07433.1, APG07513.1, and APG08290.1			
Description		SEQ ID NO.	
FWD_Guide 134, Guide 135, Guide 136, Guide 137		485	
REV_Guide 134, Guide 135, Guide 136, Guide 137		486	
FWD_Guide 138, Guide 139, Guide 140, Guide 141		487	
REV_Guide 138, Guide 139, Guide 140, Guide 141		488	
REV_Guide 142, Guide 143, Guide 144, Guide 145		489	
FWD_Guide 142, Guide 143, Guide 144, Guide 145		490	
REV_Guide 164, Guide 165, Guide 166, Guide 167		491	
FWD_Guide 164, Guide 165, Guide 166, Guide 167		492	
REV_Guide 168, Guide 169, Guide 170, Guide 171		493	
FWD_Guide 168, Guide 169, Guide 170, Guide 171		494	
REV_Guide 172, Guide 173, Guide 174, Guide 175		495	
FWD_Guide 172, Guide 173, Guide 174, Guide 175		496	
REV_Guide 185, Guide 186, Guide 187, Guide 188		497	
FWD_Guide 185, Guide 186, Guide 187, Guide 188		498	
REV_Guide 189, Guide 190, Guide 191, Guide 192		499	
FWD_Guide 189, Guide 190, Guide 191, Guide 192		500	
REV_Guide 193, Guide 194, Guide 195, Guide 196		501	
FWD_Guide 193, Guide 194, Guide 195, Guide 196		502	

TABLE 8.3

Oligonucleotides for detection of gene editing activity in mammalian cells, PCR#2 for APG005459.1			
Description		SEQ ID NO.	
FWD_Guide 146		503	
REV_Guide 146		504	
FWD_Guide 147		505	
REV_Guide 147		506	
REV_Guide 148		507	
FWD_Guide 148		508	
REV_Guide 176		509	
FWD_Guide 176		510	
REV_Guide 177		511	
FWD_Guide 177		512	
REV_Guide 209		513	
FWD_Guide 209		514	
REV_Guide 197		515	
FWD_Guide 197		516	
REV_Guide 198		517	
FWD_Guide 198		518	
REV_Guide 199		519	
FWD_Guide 199		520	

TABLE 8.4

Oligonucleotides for detection of gene editing activity in mammalian cells, PCR#2 for APG004583.1			
Description		SEQ ID NO.	
FWD_Guide 149		521	
REV_Guide 149		522	
FWD_Guide 150		523	
REV_Guide 150		524	
REV_Guide 151		525	
FWD_Guide 151		526	
REV_Guide 179		527	
FWD_Guide 179		528	
REV_Guide 180		529	
FWD_Guide 180		530	
REV_Guide 181		531	
FWD_Guide 181		532	
REV_Guide 200		533	
FWD_Guide 200		534	
REV_Guide 201		535	
FWD_Guide 201		536	
REV_Guide 202		537	
FWD_Guide 202		538	

TABLE 8.5

Oligonucleotides for detection of gene editing activity in mammalian cells, PCR#2 for APG01988.1			
Description		SEQ ID NO.	
FWD_Guide 152		539	
REV_Guide 152		540	
FWD_Guide 153		541	
REV_Guide 153		542	
FWD_Guide 154		543	
REV_Guide 154		544	
FWD_Guide 182		545	
REV_Guide 182		546	
FWD_Guide 183		547	
REV_Guide 183		548	
FWD_Guide 184		549	
REV_Guide 184		550	
FWD_Guide 203		551	
REV_Guide 203		552	
FWD_Guide 204		553	
REV_Guide 204		554	
FWD_Guide 205		555	
REV_Guide 205		556	

[0333] Purified genomic DNA was subjected to PCR #1 and PCR #2 as above. Following the second PCR amplification DNA was cleaned using a PCR cleanup kit (Zymo) according to the manufacturer's instructions and eluted in water. 200-500 ng of purified PCR #2 product was combined with 2 μ L of 10 $\times$ NEB Buffer 2 and water in a 20 μ L reaction and annealed to form heteroduplex DNA using a program of: 95 $^{\circ}$ C., 5 min; 95-85 $^{\circ}$ C., cooled at a rate of 2 $^{\circ}$ C./sec; 85-25 $^{\circ}$ C., cooled at a rate of 0.1 $^{\circ}$ C./sec.; 12 $^{\circ}$ C., forever. Following annealing 5 μ L of DNA was removed as a no enzyme control, and 1 μ L of T7 Endonuclease 1 (NEB) was added and the reaction incubated at 37 $^{\circ}$ C. for 1 hr. After incubation 5 $\times$ FlashGel loading dye (Lonza) was added and 5 μ L of each reaction and controls were analyzed by a 2.2% agarose FlashGel (Lonza) using gel electrophoresis. Following visualization of the gel, the percentage of non-homologous end joining (NHEJ) was determined using the following equa-

tion: % NHEJ events = $100 \times [1 - (1 - \text{fraction cleaved})^{1/2}]$, where (fraction cleaved) is defined as: (density of digested products)/(density of digested products + undigested parental band).

[0334] For some samples, SURVEYOR® was used to analyze the results following expression in mammalian cells. Cells were incubated at 37° C. for 72 h post-transfection before genomic DNA extraction. Genomic DNA was extracted using the QuickExtract DNA Extraction Solution (Epicentre) following the manufacturer's protocol. The genomic region flanking the RGN target site was PCR amplified, and products were purified using QiaQuick Spin Column (Qiagen) following the manufacturer's protocol. 200-500 ng total of the purified PCR products were mixed with 1 µl 10× Taq DNA Polymerase PCR buffer (Enzymatics) and ultrapure water to a final volume of 10 µl, and subjected to a re-annealing process to enable heteroduplex formation: 95° C. for 10 min, 95° C. to 85° C. ramping at -2° C./s, 85° C. to 25° C. at -0.25° C./s, and 25° C. hold for 1 min.

[0335] After reannealing, products were treated with SURVEYOR® nuclease and SURVEYOR® enhancer S

where a is the integrated intensity of the undigested PCR product, and b and c are the integrated intensities of each cleavage product.

[0336] Additionally, products from PCR #2 containing Illumina overhang sequences underwent library preparation following the Illumina 16S Metagenomic Sequencing Library protocol. Deep sequencing was performed on an Illumina Mi-Seq platform by a service provider (MOGene). Typically 200,000 of 250 bp paired-end reads (2×100,000 reads) are generated per amplicon. The reads were analyzed using CRISPResso (Pinello, et al. 2016 Nature Biotech, 34:695-697) to calculate the rates of editing. Output alignments were hand-curated to confirm insertion and deletion sites as well as identify microhomology sites at the recombination sites. The rates of editing are shown in Table 9. All experiments were performed in human cells. The "target sequence" is the targeted sequence within the gene target. For each target sequence, the guide RNA comprised the complementary RNA target sequence and the appropriate sgRNA depending on the RGN used. A selected breakdown of experiments by guide RNA is shown in Tables 10.1-10.9.

TABLE 9

Overall rates of editing						
RGN	Guide RNA ID	Target Sequence (SEQ ID NO.)	Gene Target	Overall Editing Rate in Sample	Deletion Rate in Sample	Insertion Rate in Sample
APG05083.1	189	140	RelA	6.9%		100%
APG05083.1	185	141	RelA	8.2%	79.9%	20.1%
APG05083.1	168	142	HPRT1	11.3%	36.3%	72.4%
APG07433.1	135	143	AurkB	1.7%	88.3%	11.7%
APG07433.1	139	144	AurkB	3.32%	84.3%	15.6%
APG07433.1	143	145	AurkB	2.2%	35.1%	64.9%
APG07433.1	190	146	RelA	60.5%	94.8%	5.2%
APG07433.1	194	147	RelA	6.2%		100%
APG07433.1	165	148	HPRT1	3.5%	68.0%	32.0%
APG07433.1	169	149	HPRT1	18.1%	30.3%	69.7%
APG07433.1	173	150	HPRT1	26.6%	91.9%	10.0%
APG07513.1	144	151	AurkB	2.4%	59.1%	40.9%
APG07513.1	136	152	AurkB	0.9%	80.5%	19.5%
APG08290.1	145	153	AurkB	14.18%	75.85%	24.15%
APG08290.1	188	154	RelA	21.40%	99.05%	50.05%
APG08290.1	192	155	RelA	28.98%	42.05%	57.95%
APG08290.1	196	156	RelA	13.27%	91.80%	8.20%
APG08290.1	167	157	HPRT1	14.14%	65.98%	34.02%
APG08290.1	171	158	HPRT1	48.23%	58.26%	41.74%
APG08290.1	175	159	HPRT1	13.60%	74.18%	25.82%
APG05459.1	197	160	RelA	12.95%	92.16%	7.84%
APG05459.1	199	161	RelA	5.19%	100%	
APG05459.1	146	162	AurkB	1.12%	61.50%	38.50%
APG05459.1	148	163	AurkB	0.78%	49.47%	50.53%
APG05459.1	176	164	HPRT1	6.20%	48.91%	51.09%
APG05459.1	177	165	HPRT1	9.00%	9.33%	90.67%
APG05459.1	209	166	HPRT1	2.50%		100%
APG04583.1	151	167	AurkB	0.0%		
APG01688.1	152	168	AurkB	0.0%		

(Integrated DNA Technologies) following the manufacturer's recommended protocol and analyzed on 4-20% Novex TBE polyacrylamide gels (Life Technologies). Gels were stained with SYBR Gold DNA stain (Life Technologies) for 10 min and imaged with a Gel Doc gel imaging system (Bio-rad). Quantification was based on relative band intensities. Indel percentage was determined by the formula,

$$100 \times (1 - ((b+c)/(a+b+c))^{1/2}),$$

[0337] Specific insertions and deletions for respective guides are shown in Tables 10.1-10.7. In these tables, the target sequence is identified by bold upper case letters. The 8mer PAM regions are double underlined, with the main recognized nucleotides in bold. Insertions are identified by lowercase letters. Deletions are indicated with dashes (- -). The INDEL location is calculated from the PAM proximal edge of the target sequence, with the edge being location 0. The location is positive (+) if the location is on the target

side of the edge; the location is negative (–) if the location is on the PAM side of the edge.

TABLE 10.1

Specific insertions and deletions for Guide 139 using RGN APG07433.1						
Guide 139 (SEQ ID NO: 144)	# Reads	% Reads	% of INDELs	INDEL Type	INDEL Location	Size
CCTGGGTGTGAGGCTGGGCCATTAAACCTC TCC	82540	95.562				
CCTG----- AAAACCTCTCC	170	0.199	23.16	Deletion	–6	19
CCTGGGTGTGA- GCTGGGCCATTAAACCTCTCC	132	0.155	17.98	Deletion	+1	1
C----- CTGGGCCATTAAACCTCTCC	107	0.125	14.57	Deletion	–9	12
CCTGG----- CTCTCC	101	0.118	13.76	Deletion	–5	23
C----- GGGCCATTAAACCTCTCC	61	0.071	8.31	Deletion	–9	14
CCTGGGTGTGAGGccagacCTGGGCCATTAA AACCTCTCC	49	0.057	6.67	Insertion	+3	6
CCTGGGTGTGAGgggaagctgacgtcctttc catggctgctgcctgtgttgccaccGCTGG GCCATTAAACCTCTCC	44	0.051	5.99	Insertion	+2	45
CCTGGGTGTGA- cCTGGGCCATTAAACCTCTCC	39	0.045	5.31	Deletion & Mutation	+1	1
CCTGGGTGTGAGGaCTGGGCCATTAAACCT CTCC	31	0.036	4.22	Insertion	+3	1

TABLE 10.2

Specific insertions and deletions for Guide 143 using RGN APG07433.1						
Guide 143 (SEQ ID NO: 145)	# Reads	% Reads	% of INDELs	INDEL Type	INDEL Location	Size
AGTTGGCAGATGCTCTAATGTACTGCCATGGGAA	84043	99.646				
AGTTGGCAGATGTC---AATGTACTGCCATGGGAA	126	0.149	42.281	Deletion	+3	3
AGTTGGCAGATGTC---ATGTACTGCCATGGGAA	81	0.096	27.181	Deletion	+3	3
AGTTGGCAGATGCT---ATGTACTGCCATGGGAA	42	0.049	14.093	Deletion	+4	3
AGTTGGCAGATGTC--TAATGTACTGCCATGGGAA	34	0.040	11.409	Deletion	+3	2
AGTTGGCAGATGCT---ATGTAAaTGCCATGGGAA	8	0.009	2.684	Deletion & Mutation	+4	3
AGTTGGCAGATGCTCT-ATGTACTGCCATGGGAA	7	0.008	2.348	Deletion	+6	1

TABLE 10.3

Specific insertions and deletions for Guide 190 using RGN APG07433.1						
Guide 190 (SEQ ID NO: 146)	# Reads	% Reads	% of INDELs	Type	INDEL Location	Size
CGACCTGAATGCTGTGTC----- -----GGCGCTCTGGCTTCATTCAATC	64040	55.46	91.70	Deletion	-164	170
CGACCTGAATGCTGTGCGGCTCTGCTTCCAGGTGA	45619	39.51		WT		
CGACCTGAATGCTGTGCGGCaTCTGCTTCCAGGTG A	3620	3.13	5.18	Insertion	+3	1
CGACCTGAATG-----CTGCTTCCAGGTGA	1110	0.96	1.58	Deletion	+2	10
CGACCTGAATGCT-----TCTGCTTCCAGGTGA	858	0.74	1.22	Deletion	+3	7
CGACCTGAA-----TGCTTCCAGGTGA	206	0.17	0.29	Deletion	+1	13

TABLE 10.4

Specific insertions and deletions for Guide 194 using RGN APG07433.1						
Guide 194 (SEQ ID NO: 147)	# Reads	% Reads	% of INDELs	Type	INDEL Location	Size
GCGTGGGGACTACGACCTGAATGCTGTGCGGCTCT	96635	97.318				
GCG-----ACCTGAATGCTGTGCGGCTCT	1194	1.202	44.836	Deletion	-9	11
GCGTGGGGACTACGA-----GCTGTGCGGCTCT	547	0.550	20.540	Deletion	+3	7
GCGTGGGGA-----CTGAATGCTGTGCGGCTCT	473	0.476	17.761	Deletion	-3	7
GCGTGGGGACT-----CCTGAATGCTGTGCGGCTCT	270	0.271	10.138	Deletion	-1	4
GCGTGGGGACTACGAaCCTGAaTGCTGTGCGGCTCT	88	0.088	3.304	Insertion	+3	1
GCGTGGGGACTACGA-----ATGCTGTGCGGCTCT	41	0.041	1.539	Deletion	+3	5
GCGTGGGGACTAC---CTGAATGCTGTGCGGCTCT	31	0.031	1.164	Deletion	+2	3
GCG-----TGAAATGCTGTGCGGCTCT	9	0.009	0.337	Deletion	-9	14
GCG-----ACCTGAaTGCTGTGCGGCTCT	5	0.005	0.187	Deletion & Mutation	-9	11
GCGTGGGGACTACG-CCTGAATGCTGTGCGGCTCT	5	0.005	0.187	Deletion	+2	1

TABLE 10.5

Specific insertions and deletions for Guide 145 using RGN APG08290.1						
Guide 145 (SEQ ID NO: 153)	# Reads	% Reads	% of INDELs	Type	INDEL Location	Size
ATGGAGGAGTTGGCAGATGCTCTAATGTACTGC CATGGGAAG	62618	95.889				
ATGGAGGAGTTGGCAGATGC-TAATGTACTGCC ATGGGAAG	976	1.494	36.363	Deletion	+3	2
ATGGAGGAGTTGGCAGATG-----TACTGC CATGGGAAG	319	0.488	11.885	Deletion	+2	8
ATG-----TACTGC CATGGGAAG	168	0.257	6.259	Deletion	-14	24

TABLE 10.5-continued

Specific insertions and deletions for Guide 145 using RGN APG08290.1						
Guide 145 (SEQ ID NO: 153)	# Reads	% Reads	% of INDELs	Type	INDEL Location	Size
ATGGAGGAGT <u>TGG</u> -----TGTACTGC CATGGGAAG	157	0.240	5.849	Deletion	-4	12
ATGGAGGAGT <u>TGGCAGATGCTCT</u> aAATGTACTG CCATGGGAAG	147	0.225	5.476	Insertion	+6	1
ATGGAGGAGT <u>TGGCAGATG</u> CtctTCTAATGTAC TGCCATGGGAAG	123	0.188	4.582	Insertion	+2	3
ATGGAGGAGT <u>TGGCAGATG</u> ccCTAATGTACT GCCATGGGAAG	110	0.168	4.098	Insertion	+2	2
ATGGAGGAGT <u>TGGCAGAT</u> -----AATGTACTGC CATGGGAAGAAG	103	0.157	3.837	Deletion	+1	5
ATGG-----cGTACTGC CATGGGAAGAAG	96	0.147	3.57	Deletion & Mutation	-7	21
ATGGAGGAGT <u>TGGCAGATG</u> CtTCTAATGTACTG CCATGGGAAGAAG	85	0.130	3.166	Insertion	+3	1
ATGGAGGAGT <u>TGGCA</u> -----TCTGC CATGGGAAGAAG	84	0.128	3.129	Deletion	-2	13
ATGGAGGAGT <u>TGGCAGATG</u> C---AATGTACTGC CATGGGAAGAAG	79	0.120	2.943	Deletion	+3	3
ATGGAGGAGT <u>TGGCAGATG</u> Caaaactgaaaaac aaatcaaagcactcttattgagtgcctggcgatc cccgacgccacgggcccgaacccttatcataga aaCTCTAATGTACTGCCATGGGAAG	58	0.0884	2.160	Insertion	+3	81
ATGGAGGAGT <u>TGGCAGATG</u> Ctgccttatatagac ctcccaccgtacacgcctaccgcccatttTCTA ATGTACTGCCATGGGAAG	53	0.081	1.974	Insertion	+3	42
ATGGAGGAGT <u>TG</u> -----TCTAATGTACTGC CATGGGAAG	47	0.071	1.751	Deletion	-5	8
-----CTGC CATGGGAAGAAG	26	0.039	0.968	Deletion		
ATGGAGGAGT <u>TGGCAGATG</u> Cgcggctgttcctg tacagaaccgtggcgagatgtggatcaaggat gcTCTAATGTACTGCCATGGGAAG	21	0.032	0.782	Insertion	+3	48
ATGGAGGAGT <u>TGGCAGATG</u> C-CTAATGTACTGC CATGGGAAG	14	0.021	0.521	Deletion	+3	1
ATGGAGGAGT <u>TGGCAGATG</u> Ctgtcatgatcttt ttccgctcgtcgtgggacttgctcagttctctg gccagctcgTCTAATGTACTGCCATGGGAAG	10	0.015	0.372	Insertion	+3	55
ATGGAGGAGT <u>TGGCAGATG</u> CTCT-ATGTACTGC CATGGGAAG	8	0.012	0.29	Deletion	+6	1

TABLE 10.6

Specific insertions and deletions for Guide 188 using RGN APG08290.1						
Guide 188 (SEQ ID NO: 154)	# Reads	% Reads	% of INDELs	Type	INDEL Location	Size
CAGGGACAGTGCGCATCTCCCTG <u>GTCACCAAG</u>	59686	97.000				
CAGGGACA-----GTCACCAAG	1286	2.089	69.664	Deletion	0	15
CAGGGACAGTGCGCATCTC-CTG <u>GTCACCAAG</u>	473	0.768	25.622	Deletion	+3	1

TABLE 10.6-continued

Specific insertions and deletions for Guide 188 using RGN APG08290.1						
Guide 188 (SEQ ID NO: 154)	# Reads	% Reads	% of INDELs	Type	INDEL Location	Size
CAGGGACAGTGC GCATCT--CTGGTCACCAAG	57	0.092	3.087	Deletion	+3	2
CAGGGACAGTGC GCATCTCCTCTGGTCACCAAG	11	0.017	0.595	Insertion	+3	1
CAGGGACAGTGC GCATC---CTGGTCACCAAG	7	0.011	0.379	Deletion	+3	3
CAGGGAC-----GGTCACCAAG	7	0.011	0.379	Deletion	+2	15
CGGGGACAGgGCGCATCTC-CTGGTCACCAAG	5	0.008	0.270	Deletion & Mutation	+3	1

TABLE 10.7

Specific insertions and deletions for Guide 192 using RGN APG08290.1						
Guide 192 (SEQ ID NO: 155)	# Reads	% Reads	% of INDELs	Type	INDEL Location	Size
CGACCTGAATGCTGTGCGGCTCTGCTTCCAGG	62352	95.658				
CGACCTGAATGCTGTGCGGCaTCTGCTTCCAGG	1262	1.936	44.593	Insertion	+3	1
CGACCTGAATGCTGTGCGGCTCTGCTTCCAGG	842	1.291	29.752	Insertion	+3	1
CGACCTGAATGCTGTG---TCTGCTTCCAGG	686	1.052	24.240	Deletion	+3	4
CGACCTGcATGCTGTGCGGCaTCTGCTTCCAGG	18	0.027	0.636	Insertion & Mutation	+3	1
CGACCTGcATGCTGTGCGGCTCTGCTTCCAGG	11	0.016	0.388	Insertion & Mutation	+3	1
CGACCTGcATGCTGTG---TCTGCTTCCAGG	6	0.009	0.212	Deletion & Mutation	+3	4
CGACCTGAATGCTGTGCGCaTCTGCTTCCAGG	5	0.007	0.176	Insertion & Mutation	+3	2

TABLE 10.8

Specific insertions and deletions for Guide 196 using RGN APG08290.1						
Guide 196 (SEQ ID NO: 156)	# Reads	% Reads	% of INDELs	Type	INDEL Location	Size
TGGGGACTACGACCTGAATGCTGTGCGGCTCT	37206	93.073				
TGGGGACTACGA-----ATGCTGTGCGGCTCT	1288	3.222	46.514	Deletion	+3	5
TGGGGACTACGAgcaggcagaagtatgc aaagc atgcatctcaattCCTGAATGCTGTGCGGCTCT	881	2.203	31.816	Insertion	+3	34
TGGGGACTACGAagaaggcgatagaaggccatg cgctgcgaatcgggagcggCCTGAATGCTGTGCGGCTCT	302	0.755	10.906	Insertion	+3	40
TGGGGACTACGAatgactcgctgcgctcggtcggtcgctgcggcgagcggtatcagctcaactcaaa ggcggtaatacggCCTGAATGCTGTGCGGCTCT	272	0.680	9.823	Insertion	+3	67
TGGG-----GTGCGGCTCT	13	0.032	0.4694	Deletion	-5	18
TGGGGACTACGAC-----TGCTGTGCGGCTCT	13	0.032	0.469	Deletion	+4	5

TABLE 10.9

Specific insertions and deletions for Guide 190 using RGN APG07433.1							
Guide 190 (SEQ ID NO: 146)	#Reads	% Reads	% of INDELS	Type	INDEL Location	Size	
CGACCTGAATGCTGTGC----- ----- ----- ----- -----GGCGCTCTGGCTTCATTCAATC	64040	55.46	91.70	Deletion	-164	170	
CGACCTGAATGCTGTGCGGCTCTGCTTCCAGGTGA	45619	39.51		WT			
CGACCTGAATGCTGTGCGGCaTCTGCTTCCAGGTG A	3620	3.13	5.18	Insertion	+3	1	
CGACCTGAATG-----CTGCTTCCAGGTGA	1110	0.96	1.58	Deletion	+2	10	
CGACCTGAATGCT-----TCTGCTTCCAGGTGA	858	0.74	1.22	Deletion	+3	7	
CGACCTGAA-----TGCTTCCAGGTGA	206	0.17	0.29	Deletion	+1	13	

Example 7: Demonstration of Gene Editing Activity in Plant Cells

[0338] RNA-guided nuclease activity of the RGNs of the invention is demonstrated in plant cells using protocols adapted from Li, et al. (2013) *Nat. Biotech.* 31:688-691. Briefly, plant codon optimized versions of each RGN (SEQ ID NOs: 169-182) containing an N-terminal SV40 nuclear localization signal are cloned behind the strong constitutive 35S promoter in a transient transformation vector. sgRNAs targeting one or more sites in the plant PDS gene that flank an appropriate PAM sequence are cloned behind a plant U6 promoter in a second transient expression vector. The expression vectors are introduced into *Nicotiana benthamiana* mesophyll protoplasts using PEG-mediated transformation. The transformed protoplasts are incubated in the dark for up to 36 hr. Genomic DNA is isolated from the protoplasts using a DNeasy Plant Mini Kit (Qiagen). The genomic region flanking the RGN target site is PCR amplified, and products are purified using QiaQuick Spin Column (Qiagen) following the manufacturer's protocol. 200-500 ng total of the purified PCR products are mixed with 1 µl 10× Taq DNA Polymerase PCR buffer (Enzymatics) and ultrapure water to a final volume of 10 µl, and subjected to a re-annealing process to enable heteroduplex formation: 95° C. for 10 min, 95° C. to 85° C. ramping at -2° C./s, 85° C. to 25° C. at -0.25° C./s, and 25° C. hold for 1 min. After reannealing, products are treated with SURVEYOR nuclease and SURVEYOR enhancer S (Integrated DNA Technologies) following the manufacturer's recommended protocol and analyzed on 4-20% Novex TBE polyacrylamide gels (Life Technologies). Gels are stained with SYBR Gold DNA stain (Life Technologies) for 10 min and imaged with a Gel Doc gel imaging system (Bio-rad). Quantification is based on relative band intensities. Indel percentage is determined by the formula, $100 \times (1 - (1 - (b+c)/(a+b+c))^{1/2})$, where a is the integrated intensity of the undigested PCR product, and b and c are the integrated intensities of each cleavage product.

[0339] Alternatively, PCR products derived from the targeted genomic sequence can be subjected to PCR similar to that described in Example 6, so that PCR products contain Illumina overhang sequences and can undergo library preparation and deep sequencing. This method allows determination of the rates of editing as shown in Table 9.

Example 8: Guide Cross-Compatibility

[0340] To determine the cross-compatibility of guide RNAs between RGNs, a two-plasmid interference experiment was performed (Esvelt et al (2013), *Nat. Methods* 10 (11): 1116-21). The first plasmid contained the RGN with several targets containing defined PAMs on a kanamycin resistant backbone. These plasmids were transformed into *E. coli* BL21, and the transformed strains were made to be chemically competent. A second plasmid containing a guide RNA on an ampicillin resistance backbone was then introduced. Cells were plated on media containing both antibiotics. If an RGN is able to use the guide on the second plasmid, the kanamycin-resistance plasmid is cleaved and linearized, resulting in little or no colony formation. If an RGN is not able to use the guide on the second plasmid, the kanamycin-resistance plasmid is not cleaved, resulting in high levels of colony formation. Guide RNAs for *Streptococcus pyogenes* Cas9 (SpyCas9) and *Staphylococcus aureus* Cas9 (SauCas9) were also included to determine cross-compatibility with those guide RNAs.

[0341] To calculate the depletion percentage, the number of colonies for each guide transformation is compared to the transformation efficiency using a positive control. Based on this comparison, if an RGN can use a guide, the depletion percentage should be 0, as no colonies are able to survive. If an RGN cannot use a guide, the depletion percentage should be 1 as all plasmids remain intact. Results are shown in Table 11 below. "sg" indicates the guide RNA for the recited RGN.

TABLE 11

Cross-compatibility assay						
	APG05083.1	APG07513.1	APG08290.1	APG05459.1	APG04583.1	APG01688.1
sgAPG05083.1	0	0	0	0.21	1	0.74
sgAPG07433.1	0	0	0	0.16	0.78	0.33
sgAPG07513.1	0	0.01	0	0.32	0.97	0.64
sgAPG05459.1	0.24	0.53	0.26	0.09	1	0.49
sgAPG04583.1	0.74	0.8	0.36	0.21	0	0
sgAPG01688.1	0.12	0.26	0.18	0.43	0	0
sgSauCas9	1	0.23	0.27	0.53	0.51	0.92
sg Spy	0.16	0.27	0.32	0.06	1	1

[0342] As Table 11 indicates, there are four groups of orthogonal systems. RGNs can recognize guides from other systems in their groups, but cannot use guides from other groups. The first group contains APG05083.1, APG07433.1, APG07513.1, and APG08290.1. The second group contains SpyCas9 and APG05459.1. The third group contains APG04583.1 and APG01688.1. The fourth group contains SauCas9.

Example 9: Identification of Disease Targets

[0343] A database of clinical variants was obtained from NCBI ClinVar database, which is available through the world wide web at the NCBI Clin Var website. Pathogenic Single Nucleotide Polymorphisms (SNPs) were identified from this list. Using the genomic locus information,

CRISPR targets in the region overlapping and surrounding each SNP were identified. A selection of SNPs that can be corrected using base editing in combination with the RGNs of the invention to target the causal mutation is listed in Table 12. In Table 12, only one alias of each disease is listed. The “RS #” corresponds to the RS accession number through the SNP database at the NCBI website. The AlleleID corresponds to a causal allele accession number, and the Chromosome Accession number also provides accession reference information found through the NCBI website. Table 12 also provides genomic target sequence information suitable for the RGN listed for each disease. The target sequence information also provides protospacer sequence for the production of the necessary sgRNA for the corresponding RGN of the invention.

TABLE 12

Disease Targets for RGNs of the invention							
Disease	RS# (dbSNP)	RGN	Causal Mutation	AlleleID	ChromosomeAccession	Gene Symbol	Target Seq (SEQ ID NO)
Ataxia-telangiectasia syndrome	1137887	APG04583.1	G > A	18083	NC_000011.10, NC_000011.9	ATM	197
Very long chain acyl-CoA dehydrogenase deficiency	2309689	APG05459.1	G > A	33868	NC_000017.10, NC_000017.11	ACADVL	198
Abnormality of T cell physiology	3218716	APG01688.1	G > A	52071	NC_000014.8, NC_000014.9	MYH7	199
Cardiovascular phenotype	5742905	APG05083, APG07433.1, APG07513.1, APG08290.1	T > C	15159	NC_000021.8, NC_000021.9	CBS	200
3-Oxo-5 alpha-steroid delta 4- dehydrogenase deficiency	9332964	APG04583.1	G > A	18390	NC_000002.11, NC_000002.12	SRD5A2	201
Acute myeloid leukemia	11540652	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	27395	NC_000017.10, NC_000017.11	TP53	202
Acute myeloid leukemia	11540652	APG05459.1	G > A	27395	NC_000017.10, NC_000017.11	TP53	203
Cutaneous malignant melanoma 3	11547328	APG05459.1	C > T	31967	NC_000012.11, NC_000012.12	CDK4	204
Alpha-1-antitrypsin deficiency	28929474	APG05459.1	G > A	33006	NC_000014.8, NC_000014.9	SERPINA1	205
Charcot-Marie-Tooth disease, type 2	28933093	APG05459.1	G > A	29543	NC_000001.10, NC_000001.11	LMNA	206
Hereditary cancer- predisposing syndrome	28934578	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	27413	NC_000017.10, NC_000017.11	TP53	207
Hereditary cancer- predisposing syndrome	28934578	APG01688.1	G > A	27413	NC_000017.10, NC_000017.11	TP53	208
Hereditary cancer- predisposing syndrome	28934872	APG05459.1	G > A	27436	NC_000016.9, NC_000016.10	TSC2	209

TABLE 12-continued

Disease Targets for RGNs of the invention							
Disease	RS# (dbSNP)	RGN	Causal Mutation	AlleleID	ChromosomeAccession	Gene Symbol	Target Seq (SEQ ID NO)
Brugada syndrome	28937316	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	24408	NC_000003.11, NC_000003.12	SCN5A	210
Brugada syndrome	28937318	APG05459.1	G > A	24429	NC_000003.11, NC_000003.12	SCN5A	211
GRACILE syndrome	28937590	APG05459.1	A > G	21206	NC_000002.11, NC_000002.12	BCS1L	212
Enhanced s-cone syndrome	28937873	APG05459.1	G > A	20571	NC_000015.9, NC_000015.10	NR2E3	213
Charcot-Marie-Tooth disease, type 2	28940293	APG05083, APG07433.1, APG07513.1, APG08290.1	T > C	17309	NC_000001.10, NC_000001.11	MFN2	214
Charcot-Marie-Tooth disease, type 2	28940293	APG05459.1	T > C	17309	NC_000001.10, NC_000001.11	MFN2	215
Arylsulfatase a, allele a	28940893	APG05459.1	C > T	18091	NC_000022.10, NC_000022.11	ARSA	216
Familial hypercholesterolemia	28942078	APG05459.1	G > A	18733	NC_000019.9, NC_000019.10	LDLR	217
Familial hypercholesterolemia	28942079	APG05459.1	G > A	18734	NC_000019.9, NC_000019.10	LDLR	218
HEMOGLOBIN ARLINGTON PARK	33930165	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	30165	NC_000011.9, NC_000011.10	HBB	219
Familial hypertrophic cardiomyopathy 1	36211715	APG05459.1	G > A	29159	NC_000014.8, NC_000014.9	MYH7	220
Cardiovascular phenotype	36211723	APG05459.1	G > A	45266	NC_000011.9, NC_000011.10	MYBPC3	221
Cardiovascular phenotype	36211723	APG01688.1	G > A	45266	NC_000011.9, NC_000011.10	MYBPC3	222
Brugada syndrome	45546039	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	48043	NC_000003.11, NC_000003.12	SCN5A	223
Brugada syndrome	45546039	APG01688.1	G > A	48043	NC_000003.11, NC_000003.12	SCN5A	224
Hereditary cancer- predisposing syndrome	55863639	APG05459.1	G > A	176641	NC_000017.10, NC_000017.11	TP53	225
Deficiency of butyryl-CoA dehydrogenase	57443665	APG05459.1	T > C	18867	NC_000012.11, NC_000012.12	ACADS	226
Deficiency of butyryl-CoA dehydrogenase	57443665	APG01688.1	T > C	18867	NC_000012.11, NC_000012.12	ACADS	227
Benign scapuloperoneal muscular dystrophy with cardiomyopathy	59332535	APG05459.1	G > A	77828	NC_000001.10, NC_000001.11	LMNA	228
Benign scapuloperoneal muscular dystrophy with cardiomyopathy	60458016	APG05459.1	G > A	29564	NC_000001.10, NC_000001.11	LMNA	229
Cone-rod dystrophy 6	61750173	APG05459.1	G > A	24396	NC_000017.10, NC_000017.11	GUCY2D	230
Cone-rod dystrophy 6	61750173	APG01688.1	G > A	24396	NC_000017.10, NC_000017.11	GUCY2D	231
Stargardt disease 1	61750641	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	105317	NC_000001.10, NC_000001.11	ABCA4	232
Leber congenital amaurosis 2	61751276	APG05459.1	G > A	104715	NC_000001.10, NC_000001.11	RPE65	233
Cone-rod dystrophy 3	61751407	APG05459.1	G > A	105292	NC_000001.10, NC_000001.11	ABCA4	234
Nonsyndromic Oculocutaneous Albinism	61754375	APG05459.1	G > A	18835	NC_000011.9, NC_000011.10	TYR	235
Phenylketonuria	62508646	APG05459.1	T > C	15654	NC_000012.11, NC_000012.12	PAH	236
Phenylketonuria	62516101	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	15658	NC_000012.11, NC_000012.12	PAH	237
Breast-ovarian cancer, familial 1	62625303	APG05459.1	C > T	68931	NC_000017.10, NC_000017.11	BRCA1	238
Hyperphenylalaninemia, non- pku	62644499	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	15656	NC_000012.11, NC_000012.12	PAH	239
Hyperphenylalaninemia, non- pku	62644499	APG05459.1	G > A	15656	NC_000012.11, NC_000012.12	PAH	240
Hereditary cancer- predisposing syndrome	63750217	APG05459.1	G > A	32138	NC_000003.11, NC_000003.12	MLH1	241
Hereditary cancer- predisposing syndrome	63750741	APG05083, APG07433.1, APG07513.1, APG08290.1	T > C	94663	NC_000002.11, NC_000002.12	MSH6	242

TABLE 12-continued

Disease Targets for RGNs of the invention							
Disease	RS# (dbSNP)	RGN	Causal Mutation	AlleleID	ChromosomeAccession	Gene Symbol	Target Seq (SEQ ID NO)
Hereditary cancer- predisposing syndrome	63750809	APG05459.1	T > C	95480	NC_000003.11, NC_000003.12	MLH1	243
Hereditary cancer- predisposing syndrome	63751657	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	95331	NC_000003.11, NC_000003.12	MLH1	244
Hereditary cancer- predisposing syndrome	63751711	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	95792	NC_000003.11, NC_000003.12	MLH1	245
Hereditary cancer- predisposing syndrome	63751711	APG01688.1	G > A	95792	NC_000003.11, NC_000003.12	MLH1	246
Anterior segment dysgenesis 6	72549387	APG05459.1	G > A	22776	NC_000002.11, NC_000002.12	CYP1B1	247
Brugada syndrome	72549410	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	78547	NC_000003.11, NC_000003.12	SCN5A	248
Brugada syndrome	72549410	APG05459.1	G > A	78547	NC_000003.11, NC_000003.12	SCN5A	249
Ornithine carbamoyltransferase deficiency	72554308	APG01688.1	G > A	26053	NC_000023.10, NC_000023.11	OTC	250
Osteogenesis imperfecta type I	72645321	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	414022	NC_000017.10, NC_000017.11	COL1A1	251
Osteogenesis imperfecta type I	72645321	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	414022	NC_000017.10, NC_000017.11	COL1A1	252
Constipation	74799832	APG05083, APG07433.1, APG07513.1, APG08290.1	T > C	28958	NC_000010.10, NC_000010.11	RET	253
Dopamine beta hydroxylase deficiency	74853476	APG05083, APG07433.1, APG07513.1, APG08290.1	T > C	16789	NC_000009.11, NC_000009.12	DBH	254
Cystic fibrosis	75096551	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	33858	NC_000007.13, NC_000007.14	CFTR	255
Phenylketonuria	75193786	APG01688.1	T > C	15675	NC_000012.11, NC_000012.12	PAH	256
Deficiency of UDPglucose- hexose-1-phosphate uridylyltransferase	75391579	APG05459.1	A > G	18653	NC_000009.11, NC_000009.12	GALT	257
Amyloid Cardiomyopathy, Transthyretin-related	76992529	APG05459.1	G > A	28465	NC_000018.9, NC_000018.10	TTR	258
Carbohydrate-deficient glycoprotein syndrome type I	80338707	APG01688.1	G > A	22758	NC_000016.9, NC_000016.10	PMM2	259
Metachromatic leukodystrophy	80338815	APG01688.1	G > A	18090	NC_000022.10, NC_000022.11	ARSA	260
Smith-Lemli-Opitz syndrome	80338857	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	34128	NC_000011.9, NC_000011.10	DHCR7	261
Deafness, autosomal recessive 1A	80338940	APG05459.1	G > A	32068	NC_000013.10, NC_000013.11	GJB2	262
Congenital omphalocele	80338945	APG05083, APG07433.1, APG07513.1, APG08290.1	T > C	32055	NC_000013.10, NC_000013.11	GJB2	263
Congenital omphalocele	80338945	APG05083, APG07433.1, APG07513.1, APG08290.1	T > C	32055	NC_000013.10, NC_000013.11	GJB2	264
Congenital omphalocele	80338945	APG05459.1	T > C	32055	NC_000013.10, NC_000013.11	GJB2	265
Congenital omphalocele	80338945	APG05459.1	T > C	32055	NC_000013.10, NC_000013.11	GJB2	266

TABLE 12-continued

Disease Targets for RGNs of the invention							
Disease	RS# (dbSNP)	RGN	Causal Mutation	AlleleID	ChromosomeAccession	Gene Symbol	Target Seq (SEQ ID NO)
Congenital myotonia, autosomal dominant form	80356701	APG05083, APG07433.1, APG07513.1, APG08290.1	T > C	33902	NC_000007.13, NC_000007.14	CLCN1	267
Breast-ovarian cancer, familial 1	80356914	APG05459.1	G > A	70276	NC_000017.10, NC_000017.11	BRCA1	268
Breast and/or ovarian cancer	80356962	APG05459.1	G > A	70247	NC_000017.10, NC_000017.11	BRCA1	269
Breast-ovarian cancer, familial 1	80357212	APG05459.1	G > A	70255	NC_000017.10, NC_000017.11	BRCA1	270
Breast-ovarian cancer, familial 1	80357281	APG05459.1	T > C	70177	NC_000017.10, NC_000017.11	BRCA1	271
Breast-ovarian cancer, familial 1	80357307	APG05459.1	G > A	70275	NC_000017.10, NC_000017.11	BRCA1	272
Breast-ovarian cancer, familial 1	80357352	APG05459.1	C > T	69958	NC_000017.10, NC_000017.11	BRCA1	273
Breast-ovarian cancer, familial 1	80358145	APG05459.1	G > A	46229	NC_000017.10, NC_000017.11	BRCA1	274
Inborn genetic diseases	80358259	APG05083, APG07433.1, APG07513.1, APG08290.1	T > C	18006	NC_000018.9, NC_000018.10	NPC1	275
Breast-ovarian cancer, familial 2	80358543	APG05459.1	G > A	131539	NC_000013.10, NC_000013.11	BRCA2	276
Breast-ovarian cancer, familial 2	80358544	APG05459.1	G > A	46368	NC_000013.10, NC_000013.11	BRCA2	277
Breast-ovarian cancer, familial 2	80358997	APG05459.1	G > A	67062	NC_000013.10, NC_000013.11	BRCA2	278
Breast and/or ovarian cancer	80359003	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	67069	NC_000013.10, NC_000013.11	BRCA2	279
Breast-ovarian cancer, familial 2	80359004	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	46672	NC_000013.10, NC_000013.11	BRCA2	280
Breast-ovarian cancer, familial 2	80359071	APG05459.1	G > A	67203	NC_000013.10, NC_000013.11	BRCA2	281
Breast-ovarian cancer, familial 2	80359112	APG05459.1	C > T	67292	NC_000013.10, NC_000013.11	BRCA2	282
Breast-ovarian cancer, familial 2	80359115	APG05459.1	C > T	67294	NC_000013.10, NC_000013.11	BRCA2	283
Smith-Lemli-Opitz syndrome	104886033	APG05459.1	A > G	21833	NC_000011.9, NC_000011.10	DHCR7	284
Alport syndrome 1, X-linked recessive	104886142	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	35796	NC_000023.10, NC_000023.11	COL4A5	285
Acute neuronopathic Gaucher's disease	104886460	APG05459.1	G > A	99352	NC_000001.10, NC_000001.11	GBA	286
Gonadotropin deficiency	104893836	APG05459.1	A > G	31062	NC_000004.11, NC_000004.12	GNRHR	287
Distal arthrogryposis type 1A	104894129	APG05459.1	G > A	27501	NC_000009.11, NC_000009.12	TPM2	288
Distal arthrogryposis type 1A	104894129	APG05459.1	G > A	27501	NC_000009.11, NC_000009.12	TPM2	289
Hereditary cancer- predisposing syndrome	104894261	APG05459.1	C > T	31727	NC_000011.9, NC_000011.10	MEN1	290
Inborn genetic diseases	104894313	APG05459.1	C > T	18816	NC_000011.9, NC_000011.10	TYR	291
Death in early adulthood	104894368	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	29104	NC_000012.11, NC_000012.12	MYL2	292
Death in early adulthood	104894368	APG05459.1	G > A	29104	NC_000012.11, NC_000012.12	MYL2	293
Severe autosomal recessive muscular dystrophy of childhood-North African type	104894423	APG05459.1	G > A	17048	NC_000013.10, NC_000013.11, NC_000013.9	SGCG	294
Cardiovascular phenotype	104894503	APG05459.1	G > A	27495	NC_000015.9, NC_000015.10	TPM1	295
Carbohydrate-deficient glycoprotein syndrome type I	104894525	APG01688.1	G > A	22747	NC_000016.9, NC_000016.10	PMM2	296
Charcot-Marie-Tooth disease, type I	104894621	APG05459.1	C > T	23472	NC_000017.10, NC_000017.11	PMP22	297

TABLE 12-continued

Disease Targets for RGNs of the invention							
Disease	RS# (dbSNP)	RGN	Causal Mutation	AlleleID	ChromosomeAccession	Gene Symbol	Target Seq (SEQ ID NO)
Inborn genetic diseases	104894635	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	20146	NC_000017.10, NC_000017.11	SGSH	298
Inborn genetic diseases	104894635	APG05459.1	G > A	20146	NC_000017.10, NC_000017.11	SGSH	299
Familial Mediterranean fever	104895097	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	17588	NC_000016.9, NC_000016.10	MEFV	300
Deafness, autosomal recessive 2	111033178	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	52388	NC_000011.9, NC_000011.10	MYO7A	301
Deafness, autosomal recessive 2	111033178	APG01688.1	G > A	52388	NC_000011.9, NC_000011.10	MYO7A	302
Deafness, X-linked 2	111033299	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	53902	NC_000013.10, NC_000013.11	GJB2	303
Enlarged vestibular aqueduct	111033305	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	52666	NC_000007.13, NC_000007.14	SLC26A4	304
Congenital sensorineural hearing impairment	111033364	APG05459.1, APG01688.1	G > A	17396	NC_000001.10, NC_000001.11	USH2A	305
Deficiency of UDPglucose-hexose-1-phosphate uridylyltransferase	111033728	APG05459.1	T > C	36556	NC_000009.11, NC_000009.12	GALT	306
Very long chain acyl-CoA dehydrogenase deficiency	112406105	APG05459.1	G > A	200333	NC_000017.10, NC_000017.11	ACADVL	307
Cardiovascular phenotype	112645512	APG05459.1	C > T	178700	NC_000015.10, NC_000015.9	FBN1	308
Pyruvate kinase deficiency of red cells	113403872	APG05459.1	G > A	16550	NC_000001.10, NC_000001.11	PKLR	309
Autosomal dominant progressive external ophthalmoplegia with mitochondrial DNA deletions 1	113994095	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	28535	NC_000015.9, NC_000015.10	POLG	310
Very long chain acyl-CoA dehydrogenase deficiency	113994167	APG05459.1	T > C	33877	NC_000017.10, NC_000017.11	ACADVL	311
Cystinosis, ocular nonnephropathic	113994205	APG05459.1	G > A	19482	NC_000017.10, NC_000017.11	CTNS	312
Pyruvate kinase deficiency of red cells	116100695	APG05459.1	C > T	16552	NC_000001.10, NC_000001.11	PKLR	313
Distal myopathy, Tateyama type	116840778	APG01688.1	G > A	23322	NC_000003.11, NC_000003.12	CAV3; SSUH2	314
Malignant hyperthermia, susceptibility to, 1	118192122	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	76888	NC_000019.9, NC_000019.10	RYR1	315
Malignant hyperthermia, susceptibility to, 1	118192122	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	76888	NC_000019.9, NC_000019.10	RYR1	316
Myopathy, Central Core	118192158	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	76835	NC_000019.9, NC_000019.10	RYR1	317
Myopathy, Central Core	118192158	APG05459.1	G > A	76835	NC_000019.9, NC_000019.10	RYR1	318
Myopathy, Central Core	118192158	APG01688.1	G > A	76835	NC_000019.9, NC_000019.10	RYR1	319
Malignant hyperthermia, susceptibility to, 1	118192170	APG05083, APG07433.1, APG07513.1, APG08290.1	T > C	28014	NC_000019.9, NC_000019.10	RYR1	320
Ceroid lipofuscinosis neuronal 2	119455954	APG05459.1	G > A	17681	NC_000011.9, NC_000011.10	TPP1	321
Ceroid lipofuscinosis neuronal 2	119455954	APG01688.1	G > A	17681	NC_000011.9, NC_000011.10	TPP1	322

TABLE 12-continued

Disease Targets for RGNs of the invention							
Disease	RS# (dbSNP)	RGN	Causal Mutation	AlleleID	ChromosomeAccession	Gene Symbol	Target Seq (SEQ ID NO)
Niemann-Pick disease type C1	120074135	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	18010	NC_000018.9, NC_000018.10	NPC1	323
Glutaric aciduria, type 1	121434372	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	17127	NC_000019.9, NC_000019.10	GCDH	324
CAPN3-Related Disorders	121434548	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	32661	NC_000015.9, NC_000015.10	CAPN3; POMT1	325
CAPN3-Related Disorders	121434548	APG05459.1	G > A	32661	NC_000015.9, NC_000015.10	CAPN3; POMT1	326
Glycogen storage disease, type II	121907943	APG05459.1	C > T	19073	NC_000017.10, NC_000017.11	GAA	327
Nonsyndromic Oculocutaneous Albinism	121908011	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	18814	NC_000011.9, NC_000011.10	TYR	328
Familial hypercholesterolemia	121908033	APG05459.1	G > A	18765	NC_000019.9, NC_000019.10	LDLR	329
Familial hypercholesterolemia	121908039	APG05459.1	G > A	18778	NC_000019.9, NC_000019.10	LDLR	330
Deafness, autosomal recessive 7	121908073	APG05459.1	C > T	19142	NC_000009.11, NC_000009.12	TMC1	331
Chronic infantile neurological, cutaneous and articular syndrome	121908153	APG05459.1	G > A	19416	NC_000001.10, NC_000001.11	NLRP3	332
Eichsfeld type congenital muscular dystrophy	121908185	APG05459.1	G > A	19531	NC_000001.10, NC_000001.11	SELENON	333
Inborn genetic diseases	121908192	APG05459.1	G > A	23730	NC_000016.9, NC_000016.10	GFER	334
Hyperkalemic Periodic Paralysis Type 1	121908557	APG05459.1	G > A	20958	NC_000017.10, NC_000017.11	SCN4A	335
Inclusion body myopathy 2 Severe	121908627	APG05459.1	G > A	21067	NC_000009.11, NC_000009.12	GNE	336
APG05083.1, APG07433.1, APG07513.1, APG08290.1	121908716	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	16996	NC_000020.10, NC_000020.11	ADA	337
immunodeficiency due to ADA deficiency Severe	121908739	APG05083, APG07433.1, APG07513.1, APG08290.1	T > C	17004	NC_000020.10, NC_000020.11	ADA	338
immunodeficiency due to ADA deficiency							
Cardiovascular phenotype	121908987	APG05459.1	G > A	21885	NC_000007.13, NC_000007.14	PRKAG2	339
Cystic fibrosis	121909019	APG05459.1	G > A	22197	NC_000007.13, NC_000007.14	CFTR	340
Cystic fibrosis	121909036	APG05459.1	T > C	22247	NC_000007.13, NC_000007.14	CFTR	341
Adrenocortical carcinoma, pediatric	121912664	APG01688.1	G > A	27418	NC_000017.10, NC_000017.11	TP53	342
Fumarase deficiency	121913123	APG05459.1	G > A	31275	NC_000001.10, NC_000001.11	FH	343
Adenocarcinoma of prostate	121913272	APG05459.1	T > C	40610	NC_000003.11, NC_000003.12	PIK3CA	344
Familial hypertrophic cardiomyopathy 1	121913638	APG05459.1	G > A	29144	NC_000014.8, NC_000014.9	MYH7	345
Adult hypophosphatasia	121918007	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	28709	NC_000001.10, NC_000001.11	ALPL	346
Adult hypophosphatasia	121918007	APG01688.1	G > A	28709	NC_000001.10, NC_000001.11	ALPL	347
Adult hypophosphatasia	121918007	APG01688.1	G > A	28709	NC_000001.10, NC_000001.11	ALPL	348
Inborn genetic diseases	121918166	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	15994	NC_000015.9, NC_000015.10	OCA2	349
Inborn genetic diseases	121918243	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	16464	NC_000001.10, NC_000001.11	MMACHC	350
Crouzon syndrome	121918505	APG05459.1	T > C	28329	NC_000010.10, NC_000010.11	FGFR2	351
Propionyl-CoA carboxylase deficiency	121964961	APG05459.1	A > G	27057	NC_000003.11, NC_000003.12	PCCB	352

TABLE 12-continued

Disease Targets for RGNs of the invention							
Disease	RS# (dbSNP)	RGN	Causal Mutation	AlleleID	ChromosomeAccession	Gene Symbol	Target Seq (SEQ ID NO)
Cardiovascular phenotype	121964962	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	15156	NC_000021.8, NC_000021.9	CBS	353
Dysostosis multiplex	121965019	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	26947	NC_000004.11, NC_000004.12	IDUA	354
Multiple sulfatase deficiency	137852850	APG05459.1	T > C	17711	NC_000003.11, NC_000003.12	SUMF1	355
Bifunctional peroxisomal enzyme deficiency	137853096	APG05459.1	G > A	22694	NC_000005.9, NC_000005.10	HSD17B4	356
Bifunctional peroxisomal enzyme deficiency	137853096	APG01688.1	G > A	22694	NC_000005.9, NC_000005.10	HSD17B4	357
Hereditary cancer- predisposing syndrome	137853293	APG05459.1	C > T	28112	NC_000013.10, NC_000013.11	RB1	358
Cardiovascular phenotype	137854478	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	31496	NC_000015.9, NC_000015.10	FBN1	359
Cardiovascular phenotype	137854478	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	31496	NC_000015.9, NC_000015.10	FBN1	360
Limb-girdle muscular dystrophy, type 2L	137854529	APG05459.1	C > T	17205	NC_000011.9, NC_000011.10	ANO5	361
Familial hypercholesterolemia	137929307	APG01688.1	G > A	171217	NC_000019.9, NC_000019.10	LDLR	362
Spastic Paraplegia, Recessive	141659620	APG05459.1	G > A	21858	NC_000016.9, NC_000016.10	SPG7	363
Isovaleryl-CoA dehydrogenase deficiency	142761835	APG05459.1	G > A	177782	NC_000015.9, NC_000015.10	IVD	364
Familial hypercholesterolemia	145787161	APG05459.1	G > A	18783	NC_000019.10, NC_000019.9	LDLR	365
Biotinidase deficiency	146015592	APG05459.1	G > A	46845	NC_000003.11, NC_000003.12	BTD	366
Biotinidase deficiency	146015592	APG05459.1	G > A	46845	NC_000003.11, NC_000003.12	BTD	367
Leber congenital amaurosis	150726175	APG01688.1	G > A	45795	NC_000001.10, NC_000001.11	NMNAT1	368
Familial hyperinsulinism	151344623	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	24127	NC_000011.9, NC_000011.10	ABCC8	369
Familial cancer of breast	180177122	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	132185	NC_000016.10, NC_000016.9	PALB2	370
Cohen syndrome	180177366	APG05459.1	G > A	71322	NC_000008.10, NC_000008.11	VPS13B	371
Cardiovascular phenotype	187830361	APG05083, APG07433.1, APG07513.1, APG08290.1	T > C	45267	NC_000011.9, NC_000011.10	MYBPC3	372
Wilson disease	193922103	APG05459.1	A > G	44370	NC_000013.10, NC_000013.11	ATP7B	373
Wilson disease	193922110	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	44393	NC_000013.10, NC_000013.11	ATP7B	374
Wilson disease	193922110	APG05459.1	G > A	44393	NC_000013.10, NC_000013.11	ATP7B	375
Familial hypercholesterolemia	193922566	APG05459.1	G > A	45113	NC_000019.9, NC_000019.10	LDLR	376
Familial hypercholesterolemia	193922566	APG05459.1	G > A	45113	NC_000019.9, NC_000019.10	LDLR	377
Floating-Harbor syndrome	199469464	APG05459.1	C > T	39865	NC_000016.9, NC_000016.10	SRCAP	378
Congenital long QT syndrome	199472712	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	67758	NC_000011.9, NC_000011.10	KCNQ1	379
Congenital long QT syndrome	199472712	APG05459.1	G > A	67758	NC_000011.9, NC_000011.10	KCNQ1	380
Andersen Tawil syndrome	199473384	APG01688.1	G > A	78481	NC_000017.10, NC_000017.11	KCNJ2	381
Cardiovascular phenotype	199473460	APG05083, APG07433.1, APG07513.1, APG08290.1	T > C	67776	NC_000011.9, NC_000011.10	KCNQ1	382
Familial hypercholesterolemia	200238879	APG05459.1	T > C	18777	NC_000019.9, NC_000019.10	LDLR	383
Cardiovascular phenotype	200411226	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	174776	NC_000011.9, NC_000011.10	MYBPC3	384

TABLE 12-continued

Disease Targets for RGNs of the invention							
Disease	RS# (dbSNP)	RGN	Causal Mutation	AlleleID	ChromosomeAccession	Gene Symbol	Target Seq (SEQ ID NO)
Gastrointestinal stroma tumor	201286421	APG05459.1	C > T	50215	NC_000001.10, NC_000001.11	SDHC	385
Dyskeratosis congenita	201540674	APG05459.1	G > A	51186	NC_000020.10, NC_000020.11	RTEL1	386
Dyskeratosis congenita	201540674	APG01688.1	G > A	51186	NC_000020.10, NC_000020.11	RTEL1	387
Glycogen storage disease IIIa	267606640	APG04583.1	G > A	16147	NC_000001.10, NC_000001.11	AGL	388
Dilated cardiomyopathy 1DD	267607004	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	15310	NC_000010.10, NC_000010.11	RBM20	389
Renal carnitine transport defect	267607054	APG05459.1	C > T	21466	NC_000005.9, NC_000005.10	SLC22A5	390
Baraitser-Winter syndrome 1	281875334	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	38553	NC_000007.13, NC_000007.14	ACTB	391
Very long chain acyl-CoA dehydrogenase deficiency	369560930	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	98197	NC_000017.10, NC_000017.11	ACADVL	392
Familial hypercholesterolemia	373822756	APG05459.1	A > G	181233	NC_000019.9, NC_000019.10	LDLR	393
Limb-girdle muscular dystrophy, type 2A	376107921	APG05459.1	G > A	213634	NC_000015.9, NC_000015.10	CAPN3	394
Familial hypercholesterolemia	376459828	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	198012	NC_000019.10, NC_000019.9	LDLR	395
Aortic aneurysm, familial thoracic 6	387906592	APG05459.1	G > A	38552	NC_000010.10, NC_000010.11	ACTA2	396
Acromicric dysplasia	387906623	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	38652	NC_000015.9, NC_000015.10	FBN1	397
Charcot-Marie-Tooth disease type 2C	387906905	APG01688.1	G > A	39430	NC_000012.11, NC_000012.12	TRPV4	398
Breast-ovarian cancer, familial 2	397507389	APG01688.1	G > A	46666	NC_000013.10, NC_000013.11	BRCA2	399
Breast-ovarian cancer, familial 1	397509284	APG05459.1	G > A	70248	NC_000017.10, NC_000017.11	BRCA1	400
Charcot-Marie-Tooth disease type 2C	397514494	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	48018	NC_000012.11, NC_000012.12	TRPV4	401
Charcot-Marie-Tooth disease type 2C	397514494	APG01688.1	G > A	48018	NC_000012.11, NC_000012.12	TRPV4	402
Hereditary cancer- predisposing syndrome	397514495	APG05459.1	G > A	152034	NC_000017.10, NC_000017.11	TP53	403
Early infantile epileptic encephalopathy	397514581	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	48359	NC_000020.10, NC_000020.11	KCNQ2	404
Early infantile epileptic encephalopathy	397514581	APG05459.1	G > A	48359	NC_000020.10, NC_000020.11	KCNQ2	405
Early infantile epileptic encephalopathy	397514581	APG01688.1	G > A	48359	NC_000020.10, NC_000020.11	KCNQ2	406
Acromicric dysplasia	397515757	APG05459.1	G > A	51454	NC_000015.9, NC_000015.10	FBN1	407
Hypertrophic cardiomyopathy	397515982	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	51820	NC_000011.9, NC_000011.10	MYBPC3	408
Cardiovascular phenotype	397516031	APG04583.1	G > A	51898	NC_000011.9, NC_000011.10	MYBPC3	409
Cardiovascular phenotype	397516074	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	51962	NC_000011.9, NC_000011.10	MYBPC3	410
Cardiovascular phenotype	397516083	APG01688.1	G > A	51977	NC_000011.9, NC_000011.10	MYBPC3	411
Familial hypertrophic cardiomyopathy 1	397516269	APG05083, APG07433.1, APG07513.1, APG08290.1	T > C	52276	NC_000014.8, NC_000014.9	MYH7	412

TABLE 12-continued

Disease Targets for RGNs of the invention							
Disease	RS# (dbSNP)	RGN	Causal Mutation	AlleleID	ChromosomeAccession	Gene Symbol	Target Seq (SEQ ID NO)
Benign scapuloperoneal muscular dystrophy with cardiomyopathy	397517889	APG05459.1	C > T	57195	NC_000001.10, NC_000001.11	LMNA	413
Glycogen storage disease, type II	398123172	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	415590	NC_000017.10, NC_000017.11	GAA	414
Diffuse mesangial sclerosis	587776576	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	18532	NC_000011.10, NC_000011.9	WT1	415
Colobomatous microphthalmia	587776954	APG05459.1	A > G	51108	NC_000012.11, NC_000012.12	C12orf57	416
Ataxia-telangiectasia syndrome	587779826	APG05459.1	T > C	132814	NC_000011.10, NC_000011.9	ATM	417
Familial cancer of breast	587780226	APG05459.1	C > T	133611	NC_000017.10, NC_000017.11	BRIP1	418
Limb-girdle muscular dystrophy, type 2A	587780290	APG01688.1	G > A	134019	NC_000015.9, NC_000015.10	CAPN3	419
Hereditary cancer- predisposing syndrome	587781462	APG05459.1	C > T	150772	NC_000002.11, NC_000002.12	MSH6	420
Asymmetric septal hypertrophy	587782958	APG01688.1	G > A	165560	NC_000011.10, NC_000011.9	MYBPC3	421
Hereditary cancer- predisposing syndrome	587783050	APG05459.1	G > A	166264	NC_000016.10, NC_000016.9	CDH1	422
Familial hypertrophic cardiomyopathy 2	727504247	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	172354	NC_000001.10, NC_000001.11	TNNT2	423
Familial hypertrophic cardiomyopathy 2	727504247	APG01688.1	G > A	172354	NC_000001.10, NC_000001.11	TNNT2	424
Familial hypertrophic cardiomyopathy 2	727504247	APG01688.1	G > A	172354	NC_000001.10, NC_000001.11	TNNT2	425
Erythrocytosis, familial, 2	730882035	APG01688.1	G > A	180121	NC_000003.12, NC_000003.11	VHL	426
Death in infancy	730882246	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	181441	NC_000014.9, NC_000014.8	ISCA2	427
Muscular Diseases	751995154	APG05459.1	G > A	200340	NC_000017.10, NC_000017.11	ACADVL	428
Familial hypercholesterolemia	756039188	APG04583.1	G > A	243266	NC_000019.9, NC_000019.10	LDLR	429
Familial cancer of breast	761494650	APG05459.1	C > T	185659	NC_000022.10, NC_000022.11	CHEK2	430
Hereditary cancer- predisposing syndrome	762307622	APG01688.1	G > A	232266	NC_000001.10, NC_000001.11	MUTYH	431
Familial hypercholesterolemia	769370816	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	228176	NC_000019.10, NC_000019.9	LDLR	432
Familial hypercholesterolemia	775092314	APG05083, APG07433.1, APG07513.1, APG08290.1	T > C	228197	NC_000019.9, NC_000019.10	LDLR	433
Familial hypercholesterolemia	775924858	APG05083, APG07433.1, APG07513.1, APG08290.1	G > A	246116	NC_000019.9, NC_000019.10	LDLR	434
Inclusion body myopathy 2	779694939	APG04583.1	T > C	214934	NC_000009.12, NC_000009.11	GNE	435
Ataxia-telangiectasia syndrome	780619951	APG05459.1	C > T	212851	NC_000011.10, NC_000011.9	ATM	436
Benign familial neonatal- infantile seizures	794727152	APG04583.1	G > A	191718	NC_000002.11, NC_000002.12	SCN2A	437
Marfan Syndrome/Loeys-Dietz Syndrome/Familial Thoracic Aortic Aneurysms and Dissections	794728228	APG05459.1	C > T	197690	NC_000015.10, NC_000015.9	FBN1	438
Dilated cardiomyopathy 1G	869320740	APG01688.1	T > C	136355	NC_000002.11, NC_000002.12	TTN	439
Familial hypercholesterolemia	875989911	APG05459.1	G > A	228151	NC_000019.9, NC_000019.10	LDLR	440
Breast-ovarian cancer, familial 2	876657678	APG05459.1	C > T	230443	NC_000013.10, NC_000013.11	BRCA2	441
Familial hypercholesterolemia	879254600	APG05459.1	G > A	245669	NC_000019.10, NC_000019.9	LDLR	442

TABLE 12-continued

Disease Targets for RGNs of the invention							
Disease	RS# (dbSNP)	RGN	Causal Mutation	AlleleID	Chromosome	Accession	Target Seq (SEQ ID NO)
Familial hypercholesterolemia	879254803	APG05083, APG07433.1, APG07513.1, APG08290.1	T > C	246008	NC_000019.10, NC_000019.9	LDLR	443
Familial hypercholesterolemia	879254803	APG01688.1	T > C	246008	NC_000019.10, NC_000019.9	LDLR	444
Familial hypercholesterolemia	879254849	APG01688.1	T > C	246074	NC_000019.10, NC_000019.9	LDLR	445
Familial cancer of breast	1057517585	APG01688.1	G > A	358911	NC_000016.10, NC_000016.9	PALB2	446
Hereditary hemorrhagic telangiectasia type 2	1057517944	APG05459.1	C > T	360048	NC_000012.11, NC_000012.12	ACVRL1	447

Example 10: Targeting Mutations Responsible for Hurler Syndrome

[0344] The following describes a potential treatment for Hurler Syndrome, also referred to as MPS-1, is described, using an RNA directed base editing system that corrects a mutation responsible for Hurler syndrome in a large proportion of patients with the disease. This approach utilizes a base editing fusion protein that is RNA guided and that can be packaged into a single AAV vector for delivery to a wide range of tissue types. Depending on the exact regulatory elements and base editor domain used, it may also be possible to engineer a single vector that encodes for both the base editing fusion protein and a single guide RNA to target the diseased locus.

Example 10.1: Identifying RGN with Ideal PAM

[0345] The genetic disease MPS-1 is a lysosomal storage disease characterized at the molecular level by the accumulation of dermatan sulfate and heparan sulfate in lysosomes. This disease is generally an inherited genetic disorder caused by mutations in the IDUA gene (NCBI Reference sequence NG_008103.1), which encodes α -L-iduronidase. The disease is a result of a deficiency of α -L-iduronidase. The most common IDUA mutations found in studies of individuals of Northern European background are W402 $\times$ and Q70X, both nonsense mutations resulting in premature termination of translation (Bunge et al. (1994), *Hum. Mol. Genet.*, 3 (6): 861-866, herein incorporated by reference). Reversion of a single nucleotide would restore the wild-type coding sequence and result in protein expression controlled by the endogenous regulatory mechanisms of the genetic locus.

[0346] The W402 $\times$ mutation of the human Idua gene accounts for a high proportion of MPS-1H cases. Base editors can target a narrow sequence window relative to the binding site of the protospacer component of the guide RNA and thus the presence of a PAM sequence a specific distance from the target locus is essential for the success of the strategy. Given the constraints that the target mutation must be on the exposed non-target strand (NTS) during the interaction of the base editing protein and that the footprint of the RGN domain will block access to the region near the PAM, an accessible locus is thought to be 10-30 bp from the PAM. To avoid editing and mutagenesis of other nearby adenosine bases in this window, different linkers are screened. The ideal window is 12-16 bp from the PAM.

[0347] A PAM sequence compatible with APG07433.1 and APG08290.1 is readily apparent at the genetic locus and within the ideal base editing window as defined above. These nucleases have a PAM sequence of NNNNCC (SEQ ID NO: 6) and NNRNCC (SEQ ID NO: 32), respectively, and are compact in size-potentially allowing delivery via a single AAV vector. This delivery approach bestows multiple advantages relative to others, such as access to a wide range of tissues (liver, muscle, CNS) and well-established safety profile and manufacturing techniques.

[0348] Cas9 from *S. pyogenes* (SpyCas9) requires a PAM sequence of NGG (SEQ ID NO: 448), which is present near the W402 $\times$ locus, but the size of SpyCas9 prevents packaging of a gene encoding a fusion protein of a base editing domain and the SpyCas9 nuclease into a single AAV vector, and thus forgoes the aforementioned advantages of this approach. Including a guide RNA encoding sequence on this vector would be even less feasible, even if there are to be significant technological improvements that reduce the size of gene regulatory elements or increase the packaging limits of AAV vectors. While a dual delivery strategy may be employed (for example, Ryu et al. (2018), *Nat. Biotechnol.*, 36 (6): 536-539, herein incorporated by reference), it would add significant manufacturing complexity and cost. Additionally, dual viral vector delivery significantly decreases the efficiency of gene correction, since a successful edit in a given cell requires infection with both vectors and assembly of the fusion protein in the cell. A commonly used Cas9 ortholog from *S. aureus* (SauCas9) is considerably smaller in size relative to SpyCas9, but has a more complex PAM requirement-NGRRT (SEQ ID NO: 449). This sequence, however, is not within a range expected to be useful for base editing of the causative locus.

Example 10.2: RGN Fusion Constructs and sgRNA Sequences

[0349] A DNA sequence encoding a fusion protein with the following domains is produced using standard molecular biology techniques: 1) an RGN domain with mutations that inactivate the DNA cleavage activity ("dead" or "nickase"); 2) an adenosine deaminase useful for base editing. All constructs described in the table below comprise a fusion protein with the base editing active domain, in this example ADAT (SEQ ID NO: 450) operably fused to the N-terminal end of the RGN APG08290.1. It is known in the art that a fusion protein could also be made with the base-editing enzyme at the C-terminal end of the RGN. Additionally, the

RGN and the base editor of the fusion protein are typically separated by a linker amino sequence. It is known in the art that lengths of standard linkers range from 15-30 amino acids. Further, it is known in the art that certain fusion proteins between an RGN and a base-editing enzyme, for example a cytidine deaminase, may also comprise at least one uracil glycosylase inhibitor (UGI) domain, which may increase base editing efficiency (U.S. Pat. No. 10,167,457, herein incorporated by reference). Therefore, a fusion protein may comprise APG08290.1, a base-modifying enzyme, and at least one UGI.

TABLE 13

Constructs for RNA-targeted base editing					
Seq ID No.	Construct	RGN	Dead (D) or Nickase (N)	Base editor	Linker
451	Nuc-ADAT-Linker-dAPG08290.1-Linker-SV40	APG08290.1	D	ADAT	XTEN1
452	Nuc-ADAT-XTEN1-nAPG08290.1-Linker-SV40	APG08290.1	N	ADAT	XTEN1

[0350] The accessible editing sites of an RGN are determined by the PAM sequence. When combining an RGN with a base editing domain, the target residue for editing must reside on the non-target strand (NTS), since the NTS is single stranded while the RGN is associated with the locus. Evaluating a number of nucleases and corresponding guide RNAs enables the selection of the most appropriate gene editing tool for this particular locus. Several potential PAM sequences that can be targeted by the constructs described above in the human *Idua* gene are in the proximity of the mutant nucleotide responsible for the W402X mutation. A sequence encoding a guide RNA transcript containing 1) a “spacer” that is complementary to the non-coding DNA strand at the disease locus; and 2) RNA sequence required for association of the guide RNA with the RGN is also produced. Useful guide RNA sequences (sgRNA) are shown in Table 14 below. These guide RNA sequences can be evaluated for their efficiency in directing the base editors above to the locus of interest.

TABLE 14

Sequence of guide RNAs		
Sequence of target genomic sequence	SEQ ID NO.	Coding sequence of sgRNA (SEQ ID NO.)
5' - GGAGCAGCTCTAGGCCGAAGTGTGCG - 3'	453	456
5' - TAGGCCGAAGTGTGCGAGGCCGGA - 3'	454	457
5' - GCTCTAGGCCGAAGTGTGCGAGGCC - 3'	455	458

Example 10.3: Assay for Activity in Cells from Hurler Disease Patients

[0351] To verify the genotype strategy and evaluate the constructs described above, fibroblasts from Hurler disease

patients are used. A vector is designed containing appropriate promoters upstream of the fusion protein coding sequence and the sgRNA encoding sequence for expression of these in human cells, similar to those vectors described in Example 5. It is recognized that promoters and other DNA elements (for example enhancers, or terminators) which either are known for high levels of expression in human cells or may specifically express well in fibroblast cells may also be used. The vector is transfected into the fibroblasts using standard techniques, for example transfection similar to what is described in Example 6. Alternatively, electroporation may be used. The cells are cultured for 1-3 days. Genomic DNA (gDNA) is isolated using standard techniques. The editing efficiency is determined by performing a qPCR genotyping assay and/or next generation sequencing on the purified gDNA, as described further below.

[0352] Taqman™ qPCR analysis utilizes probes specific for the wild-type and mutant allele. These probes bear fluorophores which are resolved by their spectral excitation and/or emission properties using a qPCR instrument. A genotyping kit containing PCR primers and probes can be obtained commercially (i.e. Thermo Fisher Taqman™ SNP genotyping assayID C_27862753_10 for SNP ID rs121965019) or designed. An example of a designed primer and probe set is shown in Table 15.

TABLE 15

RT-PCR primers and probes		
Description	Sequence	SEQ ID NO.
Forward Amplification Primer	5'-GACTCCTTACCAAG-3'	459
Reverse Amplification Primer	5'-GTAGATCAGCACCG-3'	460
Wild Type Probe	5'-CTCTGGGCCGAAGT-3'	461
W402X Probe	5'-CTCTAGGCCGAAGT-3'	462

[0353] Following the editing experiment, the gDNA is subjected to qPCR analysis using standard methods and the primers and probes described above. Expected results are shown in Table 16. This in vitro system can be used to expediently evaluate constructs and choose one with high editing efficiency for further studies. The systems will be evaluated in comparison with cells with and without the W402X mutation, and preferably with some that are heterozygous for this mutation. The Ct values will be compared to either a reference gene or the total amplification of the locus using a dye such as Sybr green.

TABLE 16

Expected qPCR results		
Genotype	Transfected with base editor	Expected PCR result
<i>Idua</i> ^{WT/WT}	No	Homozygous WT
<i>Idua</i> ^{WT/W402X}	No	Heterozygous: 50% WT, 50% W402X
<i>Idua</i> ^{W402X/W402X}	No	Homozygous W402X
<i>Idua</i> ^{W402X/W402X}	Yes	Variable

[0354] The tissues can also be analyzed by next generation sequencing. Primer binding sites such as the ones shown below (Table 17), or other suitable primer binding sites that can be identified by a person of skill in the art, can be used. Following PCR amplification, products containing Illumina Nextera XT overhang sequences undergo library preparation following the Illumina 16S Metagenomic Sequencing Library protocol. Deep sequencing is performed on an Illumina Mi-Seq platform. Typically, 200,000 of 250 bp paired-end reads (2×100,000 reads) are generated per amplicon. The reads are analyzed using CRISPResso (Pinello et al., 2016) to calculate the rates of editing. Output alignments are hand-curated to confirm insertion and deletion sites as well as identify microhomology sites at the recombination sites.

TABLE 17		
NGS primer binding sites		
Direction	Sequence	SEQ ID NO.
Forward	5'-ACTTCCTCCAGCC-3'	463
Reverse	5'-GAACCCCGGCTTA-3'	464

[0355] Western blotting of cell lysate of transfected cells and control cells using an anti-IDUA antibody is performed to verify expression of the full-length protein and an enzyme activity assay on the cell lysate using substrate 4-methylumbelliferyl α -L-iduronide verifies that the enzyme is catalytically active (Hopwood et al., *Clin.Chim. Acta* (1979), 92 (2): 257-265, incorporated by reference herein). These experiments are performed in comparison with the original Idua^{W402X/W402X} cell line (without transfection), the Idua^{W402X/W402X} cell line transfected with the base editing construct and a random guide sequence, and a cell line expressing wild-type IDUA.

Example 10.4: Disease Treatment Validation in a Murine Model

[0356] To verify the efficacy of this therapeutic approach, a mouse model with a nonsense mutation in the analogous amino acid is used. The mouse strain bears a W392× mutation in its Idua gene (Gene ID: 15932) which corresponds to the homologous mutation in Hurler syndrome patients (Bunge et al., (1994), *Hum. Mol. Genet.* 3 (6): 861-866, incorporated by reference herein). This locus comprises a distinct nucleotide sequence relative to that in humans, which lacks the PAM sequence necessary for correction with the base editors described in the previous examples, and thus necessitates design of a distinct fusion protein to perform the nucleotide correction. Amelioration of the disease in this animal can validate the therapeutic approach of correcting the mutation in tissues accessible by a gene delivery vector.

[0357] Mice homozygous for this mutation display a number of phenotypic characteristics similar to Hurler syndrome patients. A base editing-RGN fusion protein as described above (Table 13) along with an RNA guide sequence are incorporated into an expression vector that allows protein expression and RNA transcription in mice. A study design is shown below in Table 18. The study includes groups that are treated with a high dose of the expression vector comprising

the base-editing fusion protein and RNA guide sequence, a low dose of same expression vector, control which is the model mouse treated with an expression vector that does not comprise the base editing fusion protein or the guide RNA, and a second control which is a wild type mouse treated with the same empty vector.

TABLE 18			
Genome editing experiment in murine model			
Group	Mouse strain	N	Treatment
1	Idua-W392X ¹	≥5	Low dose of vector
2	Idua-W392X	≥5	High dose of vector
3	Idua-W392X	≥5	Vehicle
4	129/Sv (WT)	5	Vehicle

[0358] Endpoints to evaluate include body weight, urine GAG excretion, serum IDUA enzymatic activity, IDUA activity in tissues of interest, tissue pathology, genotyping of tissues of interest to verify correction of the SNP, and behavioral and neurological evaluation. Since some endpoints are terminal, additional groups may be added for evaluation of, for example, tissue pathology and tissue IDUA activities before the end of the study. Additional examples of endpoints can be found in published papers establishing Hurler syndrome animal models (Shull et al. (1994), *Proc. Natl. Acad. Sci. U.S.A.*, 91 (26): 12937-12941; Wang et al. (2010), *Mol. Genet. Metab.*, 99 (1): 62-71; Hartung et al. (2004), *Mol. Ther.*, 9 (6): 866-875; Liu et al. (2005), *Mol. Ther.*, 11 (1): 35-47; Clarke et al. (1997), *Hum. Mol. Genet.* 6 (4): 503-511; all herein incorporated by reference).

[0359] One possible delivery vector utilizes the adeno associated virus (AAV). A vector is produced to include a base editor-dRGN fusion protein coding sequence (for example, SEQ ID NO: 452) preceded by a CMV enhancer (SEQ ID NO: 138) and promoter (SEQ ID NO: 137), or other suitable enhancer and promoter combination), optionally a Kozak sequence, and operably fused at the 3' end to a terminator sequence and a poly adenylation sequence such as the minimal sequence described in Levitt, N.; Briggs, D.; Gil, A.; Proudfoot, N. J. Definition of an Efficient Synthetic Poly (A) Site. *Genes Dev.* 1989, 3 (7), 1019-1025. The vector may further comprise an expression cassette encoding for a single guide RNA operably linked at its 5' end to a human U6 promoter (SEQ ID NO: 139), or another promoter suitable for production of small non-coding RNAs, and further comprising inverted terminal repeat (ITR) sequences necessary and well-known in the art for packaging into the AAV capsid. Production and viral packaging is performed by standard methods, such as those described in U.S. Pat. No. 9,587,250, herein incorporated by reference.

[0360] Other possible viral vectors include adenovirus and lentivirus vectors, which are commonly used and would contain similar elements, with different packaging capabilities and requirements. Non-viral delivery methods also be used, such as mRNA and sgRNA encapsulated by lipid nanoparticles (Cullis, P. R. and Allen, T. M. (2013), *Adv. Drug Deliv. Rev.* 65 (1): 36-48; Finn et al. (2018), *Cell Rep.* 22 (9): 2227-2235, both incorporated by reference) hydrodynamic injection of plasmid DNA (Suda T and Liu D.) 2007) *Mol. Ther.* 15 (12): 2063-2069, herein incorporated by reference), or ribonucleoprotein complexes of sgRNA and associated with gold nanoparticles (Lee, K.; Conboy, M.;

Park, H. M.; Jiang, F.; Kim, H. J.; Dewitt, M. A.; Mackley, V. A.; Chang, K.; Rao, A.; Skinner, C.; et al. Nanoparticle Delivery of Cas9 Ribonucleoprotein and Donor DNA in Vivo Induces Homology-Directed DNA Repair. *Nat. Biomed. Eng.* 2017, 1 (11), 889-90).

Example 10.5: Disease Correction in a Murine Model with a Humanized Locus

[0361] To evaluate the efficacy of an identical base editor construct as would be used for human therapy, a mouse model in which the nucleotides near W392 are altered to match the sequence in humans around W402 is needed. This can be accomplished by a variety of techniques, including use of an RGN and an HDR template to cut and replace the locus in mouse embryos.

[0362] Due to the high degree of amino acid conservation, most nucleotides in the mouse locus can be altered to those of the human sequence with silent mutations as shown in Table 19. The only base changes resulting in altered coding sequence in the resulting engineered mouse genome occur after the introduced stop codon.

[0363] Upon engineering of this mouse strain, similar experiments will be performed as described in Example 10.4.

Example 11: Targeting Mutations Responsible for Friedreich Ataxia

[0364] The expansion of the trinucleotide repeat sequence causing Friedreich’s Ataxia (FRDA) occurs in a defined genetic locus within the FXN gene, referred to as the FRDA instability region. RNA guided nucleases (RGNs) may be used for excising the instability region in FRDA patient cells. This approach requires 1) an RGN and guide RNA sequence that can be programmed to target the allele in the human genome; and 2) a delivery approach for the RGN and guide sequence. Many nucleases used for genome editing, such as the commonly used Cas9 nuclease from *S. pyogenes* (SpCas9), are too large to be packaged into adeno-associated viral (AAV) vectors, especially when considering the length of the SpCas9 gene and the guide RNA in addition to other genetic elements required for functional expression cassettes. This makes a viable approach using SpCas9 unlikely.

TABLE 19

Nucleotide mutations to generate a humanized mouse locus						
Feature	Human (W402X)		Mouse (W392X)		Humanized Mouse	
	Nucleotide (SEQ ID NO: 465)	Encoded AA	Nucleotide (SEQ ID NO: 466)	Encoded AA	Nucleotide (SEQ ID NO: 467)	Encoded AA
Protospacer	G	E	A	G	G	G
	G	E	G	E	G	E
	A		A		A	
	G		A		G	
	C	Q	C	Q	C	Q
	A		A		A	
	G		A		G	
	C	L	C	L	C	L
	T		T		T	
	C		C		C	
	T	STOP	T	STOP	T	STOP
	A		A		A	
	G		G		G	
	G	A	G	A	G	A
	C		C		C	
	C		A		C	
	G	E	G	E	G	E
	A		A		A	
	A		G		A	
	G	V	G	V	G	V
	T		T		T	
	G		C		G	
	T	S	T	S	T	S
PAM, non-critical	C		C		C	
	G		A		G	
	C	Q	A	K	C	Q
	A		A		A	
PAM, critical	G		G		G	
	G	A	G	A	G	A
	C		C		C	
	C		T		C	

[0365] The compact RNA guided nucleases of the invention, particularly APG07433.1 and APG08290.1, are uniquely well suited for the excision of the FRDA instability region. Each RGN has a PAM requirement that is in the vicinity of the FRDA instability region. Additionally, each of these RGNs can be packaged into an AAV vector along with a guide RNA. Packing two guide RNAs would likely require a second vector, but this approach still compares favorably to what would be required of a larger nuclease such as SpCas9, which would require splitting the protein sequence between two vectors.

[0366] Table 20 shows the location of genomic target sequences suitable for targeting APG07433.1 or APG08290.1 to the 5' and 3' flanks of the FRDA instability region. Once at the locus, the RGN would excise the FA instability region. Excision of the region can be verified with Illumina sequencing of the locus.

TABLE 20				
Genomic target sequences for RGN systems				
Guide No.	Location relative to FRDA instability region	Genome target sequence	SEQ ID NO.	
1	5'	ATCACCTGAGGTCGGAGTTCAAGA	468	
2	5'	GTCTTGAACCTCCGACCTCAGGTGA	469	
3	5'	TGAACCTCCGACCTCAGGTGATCCA	470	
4	3'	GAAAGTTAGCCGGCGTGGTGTCTG	471	

Example 12: Targeting Mutations Responsible for Sickle Cell Diseases

[0367] Targeting sequences within the BCL11A enhancer region (SEQ ID NO: 472) may provide a mechanism for increasing fetal hemoglobin (HbF) to either cure or alleviate the symptoms of sickle cell diseases. For example, genome wide association studies have identified a set of genetic variations at BCL11A that are associated with increased HbF levels. These variations are a collection of SNPs found in non-coding regions of BCL11A that function as a stage-specific, lineage-restricted enhancer region. Further investigation revealed that this BCL11A enhancer is required in erythroid cells for BCL11A expression (Bauer et al, (2013) *Science* 343:253-257, incorporated by reference herein). The enhancer region was found within intron 2 of the BCL11A gene, and three areas of DNaseI hypersensitivity (often indicative of a chromatin state that is associated with regulatory potential) in intron 2 were identified. These three areas were identified as “+62”, “+58” and “+55” in accordance with the distance in kilobases from the transcription start site of BCL11A. These enhancer regions are roughly 350 (+55); 550 (+58); and 350 (+62) nucleotides in length (Bauer et al., 2013).

Example 12.1: Identifying Preferred RGN Systems

[0368] Here we describe a potential treatment for beta-hemoglobinopathies using an RGN system that disrupts BCL11A binding to its binding site within the HBB locus, which is the gene responsible for making beta-globin in

adult hemoglobin. This approach uses NHEJ which is more efficient in mammalian cells. In addition, this approach uses a nuclease of sufficiently small size that can be packaged into a single AAV vector for in vivo delivery.

[0369] The GATA1 enhancer motif in the human BCL11A enhancer region (SEQ ID NO: 472) is an ideal target for disruption using RNA guided nucleases (RGNs) to reduce BCL11A expression with concurrent re-expression of HbF in adult human erythrocytes (Wu et al. (2019) *Nat Med* 387:2554). Several PAM sequences compatible with APG07433.1 and APG08290.1 are readily apparent at the genetic locus surrounding this GATA1 site. These nucleases have a PAM sequence of 5'-NNNNCC-3'(SEQ ID NO: 6) and are compact in size, potentially allowing their delivery along with an appropriate guide RNA in a single AAV or adenoviral vector. This delivery approach bestows multiple advantages relative to others, such as access to hematopoietic stem cells and a well-established safety profile and manufacturing techniques.

[0370] The commonly used Cas9 nuclease from *S. pyogenes* (SpyCas9) requires a PAM sequence of 5'-NGG-3', (SEQ ID NO: 448) several of which are present near the GATA1 motif. However, the size of SpyCas9 prevents packaging into a single AAV or adenoviral vector and thus forgoes the aforementioned advantages of this approach. While a dual delivery strategy may be employed, it would add significant manufacturing complexity and cost. Additionally, dual viral vector delivery significantly decreases the efficiency of gene correction, since a successful edit in a given cell requires infection with both vectors.

[0371] An expression cassette encoding a human codon optimized APG07433.1 (SEQ ID NO: 128) or APG08290.1 (SEQ ID NO: 130) is produced, similar to those described in Example 6. Expression cassettes which express guide RNAs for RGNs APG07433.1 and APG08290.1 are also produced. These guide RNAs comprise 1) a protospacer sequence that is complementary to either the non-coding or coding DNA strand within the BCL11A enhancer locus (the target sequence) and 2) an RNA sequence required for association of the guide RNA with the RGN (SEQ ID NO. 18 for APG07433.1 and SEQ ID NO: 35 for APG08290.1). Because several potential PAM sequences for targeting by APG07433.1 or APG08290.1 surround the BCL11A GATA1 enhancer motif, several potential guide RNA constructs are produced to determine the best protospacer sequence that produces robust cleavage and NHEJ mediated disruption of the BCL11A GATA1 enhancer sequence. The target genomic sequences in the table below (Table 21) are evaluated to direct the RGN to this locus.

TABLE 21			
Target Sequences for BCL11A GATA1 enhancer locus			
Guide	Nuclease	Target genomic sequence	Target SEQ ID NO.
1	APG07433.1	GCACTAGACTAGCTTCAAAGTTGTAG	473
2	APG07433.1	CCTAATCAGAGGCCAAACCTTCCTG	474
3	APG07433.1	CAAGCTAACAGTTGCTTTTATCACAG	475
4	APG08290.1	GCACTAGACTAGCTTCAAAGTTGTAG	476

TABLE 21-continued

Target Sequences for BCL11A GATA1 enhancer locus		
Guide Nuclease	Target genomic sequence	Target SEQ ID NO.
5	APG08290.1CCTAATCAGAGGCCAAACCCCTTCCTG	477
6	APG08290.1CAAGCTAACAGTTGCTTTTATCACAG	478

[0372] To evaluate the efficiency with which APG07433.1 or APG08290.1 generates insertions or deletions that disrupt the BCL11A enhancer region, human cell lines such as human embryonic kidney cells (HEK cells) are used. A DNA vector comprising an RGN expression cassette (for example, as described in Example 6) is produced. A separate vector comprising an expression cassette comprising a coding sequence for a guide RNA sequence of Table 21 is also produced. Such an expression cassette may further comprise a human RNA polymerase III U6 promoter (SEQ ID NO: 139), as described in Example 6. Alternatively, a single vector comprising expression cassettes of both the RGN and guide RNA may be used. The vector is introduced into HEK cells using standard techniques such as those described in Example 6, and the cells are cultured for 1-3 days. Following this culture period, genomic DNA is isolated and the frequency of insertions or deletions is determined by using T7 Endonuclease I digestion and/or direct DNA sequencing, as described in Example 6.

[0373] A region of DNA encompassing the target BCL11A region is amplified by PCR with primers containing Illumina Nextera XT overhang sequences. These PCR amplicons are either examined for NHEJ formation using T7 Endonuclease I digestion, or undergo library preparation following the Illumina 16S Metagenomic Sequencing Library protocol or a similar Next Generation Sequencing (NGS) library preparation. Following deep sequencing, the reads generated are analyzed by CRISPResso to calculate rates of editing. Output alignments are hand-curated to confirm insertion and deletion sites. This analysis identifies the preferred RGN and the corresponding preferred guide RNA (sgRNA). The analysis may result in both APG07433.1 and APG08290.1 being equally preferred. Additionally, the analysis may determine there is more than one preferred guide RNA, or that all target genomic sequences in Table 21 are equally preferred.

Example 12.2: Assay for Expression of Fetal Hemoglobin

[0374] In this example, APG07433.1 or APG08290.1 generated insertions or deletions disrupting the BCL11A enhancer region are assayed for expression of fetal hemoglobin. Healthy human donor CD34+ hematopoietic stem cells (HSCs) are used. These HSCs are cultured and vector (s) comprising expression cassettes comprising the coding regions of the preferred RGN and the preferred sgRNA are introduced using methods similar to those described in Example 11.1. Following electroporation, these cells are differentiated in vitro into erythrocytes using established protocols (for example, Giarratana et al. (2004) *Nat Biotechnology* 23:69-74, herein incorporated by reference). The expression of HbF is then measured using western blotting with an anti-human HbF antibody, or quantified via High

Performance Liquid Chromatography (HPLC). It is expected that successful disruption of the BCL11A enhancer locus will lead to an increase in HbF production when compared to HSCs electroporated with only the RGN but no guide.

Example 12.3: Assay for Decreased Sick Cell Formation

[0375] In this example, APG07433.1 or APG08290.1 generated insertions or deletions disrupting the BCL11A enhancer region are assayed for decreased sickle-cell formation. Donor CD34+ hematopoietic stem cells (HSCs) from patients afflicted with sickle cell disease are used. These HSCs are cultured and vector(s) comprising expression cassettes comprising the coding regions of preferred RGN and the preferred sgRNA are introduced using methods similar to those described in Example 11.1. Following electroporation, these cells are differentiated in vitro into erythrocytes using established protocols (Giarratana et al. (2004) *Nat Biotechnology* 23:69-74). The expression of HbF is then measured using western blotting with an anti-human HbF antibody, or quantified via High Performance Liquid Chromatography (HPLC). It is expected that successful disruption of the BCL11A enhancer locus will lead to an increase in HbF production when compared to HSCs electroporated with only the RGN but no guide.

[0376] Sickle cell formation is induced in these differentiated erythrocytes by the addition of metabisulfite. The numbers of sickled vs normal erythrocytes are counted using a microscope. It is expected that the numbers of sickled cells are less in cells treated with APG07433.1 or APG08290.1 plus sgRNAs than with cells untreated, or treated with RGNs alone.

Example 12.4: Disease treatment validation in a murine model

[0377] To evaluate the efficacy of using APG07433.1 or APG08290.1 disruption of the BCL11A locus, suitable humanized mouse models of sickle cell anemia are used. Expression cassettes encoding for the preferred RGN and for the preferred sgRNA are packaged into AAV vectors or adenovirus vectors. In particular, adenovirus type Ad5/35 is effective at targeting HSCs. A suitable mouse model containing a humanized HBB locus with sickle cell alleles is chosen such as B6.FVB-Tg (LCR-HBA2,LCR-HBB*E26K) 53Hhb/J or B6.Cg-Hbatm1Paz Hbbtm1Tow Tg (HBA-HBBs) 41Paz/HhbJ. These mice are treated with granulocyte colony-stimulating factor alone or in combination with plerixafor to mobilize HSCs into circulation. AAVs or adenoviruses carrying the RGN and guide plasmid are then injected intravenously, and the mice are allowed to recover for a week. Blood obtained from these mice is tested in an in vitro sickling assay using metabisulfite, and the mice are followed longitudinally to monitor mortality rates and hematopoietic function. It is expected that treatment with AAVs or adenoviruses carrying an RGN and guide RNA will reduce sickling, mortality, and improve hematopoietic function when compared to mice treated with viruses lacking both expression cassettes, or with viruses carrying the RGN expression cassette alone.

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SEQ ID NO: 14      moltype = RNA length = 140
FEATURE           Location/Qualifiers
misc_feature       1..140
                   note = source = /note="Description of Artificial Sequence:
                   Syntheticpolynucleotide"
misc_feature       1..140
                   note = source = /note="sgRNA L1"
source            1..140
                   mol_type = other RNA
                   organism = synthetic construct

```

```

SEQUENCE: 14
gagcggacag cagcttcta tatctcgtagt gtcatatgtc cattaaagcc agaaatggct 60
ttgatgtttc tatgataagg gtttcgaccc gtggcgctcg ggatcgccctg cccattgaaa 120
tgggctttct cccatttatt 140

```

```

SEQ ID NO: 15      moltype = RNA length = 140
FEATURE           Location/Qualifiers
misc_feature       1..140
                   note = source = /note="Description of Artificial Sequence:
                   Syntheticpolynucleotide"
misc_feature       1..140
                   note = source = /note="sgRNA L2"
source            1..140
                   mol_type = other RNA
                   organism = synthetic construct

```

```

SEQUENCE: 15
ccatgatata gacgttgtag ctgttgtagt gtcatatgtc cattaaagcc agaaatggct 60
ttgatgtttc tatgataagg gtttcgaccc gtggcgctcg ggatcgccctg cccattgaaa 120
tgggctttct cccatttatt 140

```

```

SEQ ID NO: 16      moltype = DNA length = 38
FEATURE           Location/Qualifiers
misc_difference    31..38
                   note = a,c,t,g, unknown, or other
misc_feature       1..38
                   note = source = /note="Description of Artificial Sequence:
                   Syntheticoligonucleotide"
misc_feature       1..38
                   note = source = /note="Target seq and PAM region of plasmid
                   library 1"
source            1..38
                   mol_type = other DNA
                   organism = synthetic construct

```

-continued

SEQUENCE: 16
gagcggacag cagcttccta tatctcgtag nnnnnnnn 38

SEQ ID NO: 17 moltype = DNA length = 38
FEATURE Location/Qualifiers
misc_difference 31..38
 note = a,c,t,g, unknown, or other
misc_feature 1..38
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
misc_feature 1..38
 note = source = /note="Target seq and PAM region of plasmid
 library 2"
source 1..38
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 17
ccatgatata gacgttgtag ctgttgtagt nnnnnnnn 38

SEQ ID NO: 18 moltype = RNA length = 135
FEATURE Location/Qualifiers
misc_difference 1..25
 note = a,c,t,g, unknown, or other
misc_feature 1..135
 note = source = /note="Description of Artificial Sequence:
 Syntheticpolynucleotide"
misc_feature 1..135
 note = source = /note="sgRNA with N's"
source 1..135
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 18
nnnnnnnnnn nnnnnnnnnn nnnnngtcat agttccatta aagccaaaag tggctttgat 60
gtttctatga taagggtttc gaccctgggc gtcggggatc gcctgcccac tgaatgggc 120
ttctcccat ttatt 135

SEQ ID NO: 19 moltype = AA length = 1071
FEATURE Location/Qualifiers
REGION 1..1071
 note = source = /note="APG07513.1"
source 1..1071
 mol_type = protein
 organism = Bacillus sp.

SEQUENCE: 19
MRELDYRIGL DIGTNSIGWG VIELSWNKDR EQYEKTRIVD KGVRMFDKAE IPKTGASLAE 60
PRRIARSSRR RLNRKSQRKK DIRNLLVQHE IISQKELTSL YPLSKSSMDI WDIRLDGLDR 120
LLDRFEWARL LIHLAQRRGF KSNRKSELKD VETGKVLSSI QVNEKRLSLY RTVGEMMMKN 180
ADCSKYGKRR NSPNEYVFSV SRADLEKEIV TLFEAQRFH SSYASVDLQK TYIQIWAHQL 240
PFASGNAIVN KVGVCSSLKG KEKRVPKATY TFQYFNTLDQ INRTRLGPNF QPFTKEQORDI 300
ILDKMFQRTD YYKKKTIPEV TTYDIRKWLA LDETIQFKGL TYDPNEELKK IEMKPFINLK 360
PFYEIKKVVT NYAKKTNEVF SALDYDTVAY ALTVYKTDKD IRSYLKRSNN LSKRCYDDQL 420
IEELTLSTY KFGHLSFKAI NHVLPIMQEG RTYQEAHQL GYDTNLKKE NRSMLPIIP 480
DEITNPIVKR ALTQARKVNV AIIRRYGSPN SVHIELAREL SKSHDERKKI MTAHDENYKK 540
NKGAVSILID NGILNPTGYD IVRYLWKEQ GERCAYSKK IPANTFFNEL KKERSGPPVL 600
EVDHILPYSQ SFIDSYHNKV LVGDENQKK GNRIPYTFFS EEDKEWESFE SYVRSNSFFS 660
KKKRGYLLKK AYLPRESNLI KERHLNDRY ASSYLKNFIE KNLKFKEAVG ITRKKYVQTV 720
NGVITAHLRK RWGLEKDRQE TYLHHAMDAI IVACTDHHMV TKVTEYYQIK EGNKSIKKPY 780
FPLPWWGPRE EILSHLESQP IARKISEELK IGYQSPDYIL VSRMPKRSVT GSAHDQTVMK 840
KGDIDKKGKT IIKRVHLKD IKPDENGDFK MVGKEQDLAT YEAIKQRYLE YRKESKKAPE 900
TPLYPKSKKG KGNLIKKIKV EVQTKSFVRE INGGVAQNGD LVRVDLFEKD NRYMVPPIYV 960
VDTVRSELPN KAVTSSKGYE QWLSIDNSFT FKFSLYPYDL VRLVKGDEDR FLYFSTLDIN 1020
SDRLNFKDVN KPSQAEYRY SLKTIENTLEK YEIGVLGLDR LVRQETRKIF K 1071

SEQ ID NO: 20 moltype = RNA length = 27
FEATURE Location/Qualifiers
misc_feature 1..27
 note = source = /note="crRNA"
source 1..27
 mol_type = other RNA
 organism = Bacillus sp.

SEQUENCE: 20
gtcatagttc cattaaagcc attgctg 27

SEQ ID NO: 21 moltype = RNA length = 93
FEATURE Location/Qualifiers
misc_feature 1..93

-continued

```

source          note = source = /note="tracrRNA"
                1..93
                mol_type = other RNA
                organism = Bacillus sp.

SEQUENCE: 21
acagcaatgg ctttgatgtt tctatgataa gggcttcggc ccggtggcgtt ggggatcgcc 60
tgcccatttt aatgggcttc tccccatcta ttt                                     93

SEQ ID NO: 22    moltype = RNA length = 154
FEATURE         Location/Qualifiers
misc_feature     1..154
                note = source = /note="Description of Artificial Sequence:
                Syntheticpolynucleotide"
misc_feature     1..154
                note = source = /note="sgRNA L1"
source          1..154
                mol_type = other RNA
                organism = synthetic construct

SEQUENCE: 22
gagcggacag cagcttccta tatctcgtag gtcatagttc cattaaagcc attgctgaaa 60
gacagcaatg gctttgatgt ttctatgata agggcttcgg cccgtggcgt tggggatcgc 120
ctgccatttt taatgggctt ctccccatct attt                               154

SEQ ID NO: 23    moltype = RNA length = 154
FEATURE         Location/Qualifiers
misc_feature     1..154
                note = source = /note="Description of Artificial Sequence:
                Syntheticpolynucleotide"
misc_feature     1..154
                note = source = /note="sgRNA L2"
source          1..154
                mol_type = other RNA
                organism = synthetic construct

SEQUENCE: 23
ccatgatata gacgttgttg ctgttgtagt gtcatagttc cattaaagcc attgctgaaa 60
gacagcaatg gctttgatgt ttctatgata agggcttcgg cccgtggcgt tggggatcgc 120
ctgccatttt taatgggctt ctccccatct attt                               154

SEQ ID NO: 24    moltype = DNA length = 38
FEATURE         Location/Qualifiers
misc_difference  31..38
                note = a,c,t,g, unknown, or other
misc_feature     1..38
                note = source = /note="Description of Artificial Sequence:
                Syntheticoligonucleotide"
misc_feature     1..38
                note = source = /note="Target seq and PAM region of plasmid
                library 1"
source          1..38
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 24
gagcggacag cagcttccta tatctcgtag nnnnnnnn                               38

SEQ ID NO: 25    moltype = DNA length = 38
FEATURE         Location/Qualifiers
misc_difference  31..38
                note = a,c,t,g, unknown, or other
misc_feature     1..38
                note = source = /note="Description of Artificial Sequence:
                Syntheticoligonucleotide"
misc_feature     1..38
                note = source = /note="Target seq and PAM region of plasmid
                library 2"
source          1..38
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 25
ccatgatata gacgttgttg ctgttgtagt nnnnnnnn                               38

SEQ ID NO: 26    moltype = RNA length = 149
FEATURE         Location/Qualifiers
misc_difference  1..25
                note = a,c,t,g, unknown, or other
misc_feature     1..149
                note = source = /note="Description of Artificial Sequence:

```

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```

misc_feature      Syntheticpolynucleotide"
                  1..149
                  note = source = /note="sgRNA with N's"
source            1..149
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 26
nnnnnnnnnn nnnnnnnnnn nnnnngtcat agttccatta aagccattgc tgaagacag 60
caatggcttt gatgtttcta tgataagggc ttcggcccggt ggcgttgggg atcgccctgcc 120
cattttaatg ggcttctccc catctatatt 149

SEQ ID NO: 27      moltype = AA length = 1072
FEATURE           Location/Qualifiers
REGION            1..1072
                  note = source = /note="APG08290.1"
source            1..1072
                  mol_type = protein
                  organism = Bacillus sp.

SEQUENCE: 27
MSELDYRIGL DIGTNSIGWG VIELFWNKDR ERYEKVRIVD KGVRMFDKAE IPNKGASLAE 60
PRRIARSSRR RLNRKSQRKK EIRNLLVQHG MITQEEELDLL YPLSKKSIDI WDIRLDGLDR 120
LNLHLEWARL LIHLAQRGRF KSNRKSELKD AETGKVLSSI QVNEKRLFLY RTVGEMMIKD 180
AEFSKYDRRR NSPNEYVFSV SRADLEKEIV TLFEAQRKFQ SSYASKNLQE TYLQIWAHQL 240
PFASGNAILN KVGYCSLLKG KERRIPKATY TFQYFSALDQ VNRTRLGPDF QPFTQEQKEI 300
IDDKMFQRTD YKKKKTIEPV SYDIRKWLE LDETIQFKGL NYDPNEELKK IEKKPFINLK 360
AFYEIKKVVA NYAERTNEAF STLDYDAIAY ALTVYKTDKD IRSYLKKSNN LSKRCYDDQL 420
IEELFTLSYT KFGHLSFKAI NHVLPIMQEG RTYQEAHQH GYDTTNLKE NRSMPFLPLIP 480
DEITNPIVKR AITQARKVVN AIIRRYGSPN SVHIELAREL SKSHDERKKI MTAHDENYKK 540
NKGAISILIE NGILNPTGYD IVRYKLWKEQ GERCAYSLEK IPPDTFFNEL KKERNGPSIL 600
EVDHILPYSQ SFIDSYHNKV LVYSDENRKN GNRIPYTYFL ETNKDWEAFE RYVRSNKLFS 660
KKKREYLLKK TYLPRESLEI KERHLNDRY ASTFLKNFIE QNLQFKEVEV NLRKKRVQTV 720
NGVITAHLRK RWGLEKNRQE TYLHHAMDAI IVACTDHHMV TRITEYYQIK ESNKSVKKPY 780
FPMPWEGFRD ELLSHLASQP IAKKISEELK AGYQSSDYIF VSRMPKRSVT GAAHDQTIRR 840
KGGIDKKGKT IIKRVRLKD IKPDENGDFK MVGKEQDLAT YEAIKQRYLE HRKNSKKAPE 900
TPLYKPSKKG TGNLIKRVKI EGQTKAFVRE VNGGVAQNSD LVRVDLFEKD DKYYMVPIYV 960
PDTVCSELKP KVVKSGGYE QWLTLDNSFT FKSSLYPYDL VRLVKGNEDR FLYFGTLDID 1020
SDRLNFKDVN KPSKQNEYRY SLKTIENTLEK YEVGVLGDLR LVKQETRRIF NR 1072

SEQ ID NO: 28      moltype = RNA length = 21
FEATURE           Location/Qualifiers
misc_feature      1..21
                  note = source = /note="crRNA"
source            1..21
                  mol_type = other RNA
                  organism = Bacillus sp.

SEQUENCE: 28
gtcatagttc catgaaagcc a 21

SEQ ID NO: 29      moltype = RNA length = 85
FEATURE           Location/Qualifiers
misc_feature      1..85
                  note = source = /note="tracrRNA"
source            1..85
                  mol_type = other RNA
                  organism = Bacillus sp.

SEQUENCE: 29
tggctttgat gttctatga taagggtttc ggcccggtggc gtcggggatc gctgccccat 60
tccgatgggc ttctcccat ttatt 85

SEQ ID NO: 30      moltype = RNA length = 140
FEATURE           Location/Qualifiers
misc_feature      1..140
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticpolynucleotide"
misc_feature      1..140
                  note = source = /note="sgRNA L1"
source            1..140
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 30
gagcggacag cagcttccta tatctcgtag gtcatagttc catgaaagcc aaaagtggct 60
ttgatgtttc tatgataagg gtttcggccc gtggcgctcg ggatcgctg cccattccga 120
tgggcttctc cccatttatt 140

SEQ ID NO: 31      moltype = RNA length = 140
FEATURE           Location/Qualifiers

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misc_feature      1..140
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticpolynucleotide"
misc_feature      1..140
                  note = source = /note="sgRNA L2"
source            1..140
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 31
ccatgatata gacgttgtagg ctgttgtagt gtcatagttc catgaaagcc aaaagtggtc 60
ttgatgtttc tatgataagg gtttcggccc gtggcgctcg ggatcgctg cccattccga 120
tgggcttctc cccatttatt                                     140

SEQ ID NO: 32      moltype =      length =
SEQUENCE: 32
000

SEQ ID NO: 33      moltype = DNA length = 38
FEATURE            Location/Qualifiers
misc_difference    31..38
                  note = a,c,t,g, unknown, or other
misc_feature      1..38
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..38
                  note = source = /note="Target seq and PAM region of plasmid
                  library 1"
source            1..38
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 33
gagcggacag cagcttccta tatctcgtag nnnnnnnn 38

SEQ ID NO: 34      moltype = DNA length = 38
FEATURE            Location/Qualifiers
misc_difference    31..38
                  note = a,c,t,g, unknown, or other
misc_feature      1..38
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..38
                  note = source = /note="Target seq and PAM region of plasmid
                  library 2"
source            1..38
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 34
ccatgatata gacgttgtagg ctgttgtagt nnnnnnnn 38

SEQ ID NO: 35      moltype = RNA length = 135
FEATURE            Location/Qualifiers
misc_difference    1..25
                  note = a,c,t,g, unknown, or other
misc_feature      1..135
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticpolynucleotide"
misc_feature      1..135
                  note = source = /note="sgRNA with N's"
source            1..135
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 35
nnnnnnnnnn nnnnnnnnnn nnnnngtcat agttccatga aagccaaaag tggctttgat 60
gtttctatga taagggtttc ggcccggtgc gtcggggatc gcctgcccac tccgatgggc 120
ttctcccat ttatt                                     135

SEQ ID NO: 36      moltype = AA length = 1337
FEATURE            Location/Qualifiers
REGION            1..1337
                  note = source = /note="APG05459.1"
source            1..1337
                  mol_type = protein
                  organism = Enterococcus sp.

SEQUENCE: 36
MKKDYVIGLD IGTNSVGWAV MTEDYQLVKK KMPIYGNTK KIKKNFWGV RLFEEGHTAE 60
DRRLKRTARR RISRRNRRLR YLQAFEEAM TALDENFFAR LQESFLVPED KKWHRHPIFA 120

```

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KLEDEVAYHE TYPTIYHLRK KLADSSEQAD LRLIYLALAH IVKYRGHFLI EGKLSTENIS 180
VKEQFPQPMI IYNQTFVNGE SRLVSAPLPE SVLIEEELTE KASRTKKSEK VLQQFPQEK 240
NGLFGQFLKL MVGNKADFKK VFGLEEEAKI TYASESYEED LEGILAKVGD EYSDVFLAAK 300
NVYDAVELST ILADSDKKSH AKLSSSMIVR FTEHQEDLKK FKRPIRENC DEYDNLFKNE 360
QKDG YAGYIA HAGKVSQ LKF QYVKKIIQD IAGAEYFLEK IAQENFLRKQ RTFDNGVIPH 420
QIHLAELQAI IHRQAAYYPF LKENQEKIEQ LVTFRIPYV GPLSKGDAST FAWLKRQSEE 480
PIRPWNLQET VDLDQSATAF IERMTNFDY LPSEKVLPHK SLLYEKPMVF NELTKISYTD 540
DRGIKANFSG KEKEKIFDYL FKTRRKVKKK DIIQFYRNEY NTEIVTSLGL EEDQFNASFS 600
TYQDLLKCGL TRAE LDHPDN AEKLEDIIKI LTIFEDRQRI RTQLSTFKGQ FSAEVLKKLE 660
RKHYTGWGR L SKKLINGIYD KESGKTILDY LIKDDGVSKH YNRNFMQLIN DSQLSFKNAI 720
QKAQSSEHEE TLSETVNELA GSPA KKKGIY QSLKIVDELV AIMGYAPKRI VVEMARENQT 780
TSTGKRRSIQ RLKIVEKAMA EIGSNLLKEQ PTNEQLRDT RLFLYYMQNG KDMYTGDELS 840
LHRLSHYDID HIIPQSF MKD DSLDNLVLVG STENRGKSD VPSKEVVKDM KAYWEKLYAA 900
GLISQRKQ R LTKGEQGGLT LEDKAHFIQR QLVETRQITK NVAGILDQRY NANSKEKKVQ 960
IITLKASLTS QRSIFGLYK VREVNDYHHG QDAYLNCVVA TTLLKVYPNL APEFVYGEYP 1020
KQAPKENKA TAKTIIYTNL MRFTEDEPR FMKDGEILWS NSYLNKIKKE LNVHQMNIVK 1080
KVEVQKGGFS KESIKPKGPS NKLIPVKNGL DPQKYGGFDS PVVAYTVLFT HEKGKKPLIK 1140
QBILGITIME KTRFEQNPI L FLEEKGF LRP RVLMLKLPKYT LYEFPEGRRR LLASAKEAQK 1200
GNQMLVPEHL LTLLYHAKQC LLPNQSESLA YVEQHQP EFQ EILERVVDFA EVHTLAKSKV 1260
QQIVKLFEAN QTADVKEIAA SFIQLMQFNA MGAPSTFKPF QKDIERARYT SIKEIFDATI 1320
IYQSTTGLYE TRRKVVD 1337

```

```

SEQ ID NO: 37      moltype = RNA length = 16
FEATURE           Location/Qualifiers
misc_feature       1..16
                   note = source = /note="crRNA"
source            1..16
                   mol_type = other RNA
                   organism = Enterococcus sp.

```

```

SEQUENCE: 37
gttttagagt catgtt 16

```

```

SEQ ID NO: 38      moltype = RNA length = 74
FEATURE           Location/Qualifiers
misc_feature       1..74
                   note = source = /note="tracrRNA"
source            1..74
                   mol_type = other RNA
                   organism = Enterococcus sp.

```

```

SEQUENCE: 38
aacatagcaa gttaaaataa ggttttaacc gtaatcaact gtaaagtggc gctgtttcgg 60
cgcttttttt gttt 74

```

```

SEQ ID NO: 39      moltype = RNA length = 124
FEATURE           Location/Qualifiers
misc_feature       1..124
                   note = source = /note="Description of Artificial Sequence:
                   Syntheticpolynucleotide"
misc_feature       1..124
                   note = source = /note="sgRNA L1"
source            1..124
                   mol_type = other RNA
                   organism = synthetic construct

```

```

SEQUENCE: 39
gagcggacag cagcttccta tatctcgtagt gtttttagagt catgttaaag aacatagcaa 60
gttaaaataa ggttttaacc gtaatcaact gtaaagtggc gctgtttcgg cgcttttttt 120
gttt 124

```

```

SEQ ID NO: 40      moltype = RNA length = 124
FEATURE           Location/Qualifiers
misc_feature       1..124
                   note = source = /note="Description of Artificial Sequence:
                   Syntheticpolynucleotide"
misc_feature       1..124
                   note = source = /note="sgRNA L2"
source            1..124
                   mol_type = other RNA
                   organism = synthetic construct

```

```

SEQUENCE: 40
ccatgatata gacgttgtagt ctgttgtagt gtttttagagt catgttaaag aacatagcaa 60
gttaaaataa ggttttaacc gtaatcaact gtaaagtggc gctgtttcgg cgcttttttt 120
gttt 124

```

```

SEQ ID NO: 41      moltype = length =
SEQUENCE: 41
000

```


[illegible]

-continued

```

VISFKQNLRV INKATNYYEK IENGKKVKVE QKGINWAVRK ALHKETVSGQ VHLDRIVKAK 1200
GKILTATRKLT LDASFNEKTI ESITDTGIQK ILLNYLKSVD NNPEVAFSPE GIEELNKNIR 1260
LYNDGKAHQP ILKVRVFEQG SKFTLGETGN KTKKFVEAAK GTNLFFGIYE DKSCKRSYET 1320
IPLNIVIERQ KQGLQAVPET NEKGQLLFT LSPNDLVVVP EGVFDENNI KVDRIYKVVS 1380
FSTYQCFVR NDVSTSVVNK VEYSALNKME KSIDNIMIKE NCVKLNVDRL GKISKA 1436

```

```

SEQ ID NO: 46          moltype = RNA length = 16
FEATURE               Location/Qualifiers
misc_feature           1..16
                      note = source = /note="crRNA"
source                1..16
                      mol_type = other RNA
                      organism = Enterococcus sp.

```

```

SEQUENCE: 46
gttgtgagtt cctttc                                     16

```

```

SEQ ID NO: 47          moltype = RNA length = 78
FEATURE               Location/Qualifiers
misc_feature           1..78
                      note = source = /note="tracrRNA"
source                1..78
                      mol_type = other RNA
                      organism = Enterococcus sp.

```

```

SEQUENCE: 47
gaaaggaact cacaataagg attattccgt tgtgaaaaca tttagcgct cgactatctt 60
cggggctttt ttattttt                                     78

```

```

SEQ ID NO: 48          moltype = RNA length = 128
FEATURE               Location/Qualifiers
misc_feature           1..128
                      note = source = /note="Description of Artificial Sequence:
                      Syntheticpolynucleotide"
misc_feature           1..128
                      note = source = /note="sgRNA L1"
source                1..128
                      mol_type = other RNA
                      organism = synthetic construct

```

```

SEQUENCE: 48
gagcggacag cagcttccta tatctcgtag gttgtgagtt cctttcactt gaaaggaact 60
cacaataagg attattccgt tgtgaaaaca tttagcgct cgactatctt cggggctttt 120
ttattttt                                     128

```

```

SEQ ID NO: 49          moltype = RNA length = 128
FEATURE               Location/Qualifiers
misc_feature           1..128
                      note = source = /note="Description of Artificial Sequence:
                      Syntheticpolynucleotide"
misc_feature           1..128
                      note = source = /note="sgRNA L2"
source                1..128
                      mol_type = other RNA
                      organism = synthetic construct

```

```

SEQUENCE: 49
ccatgatata gacgttgttg ctgttgtagt gttgtgagtt cctttcactt gaaaggaact 60
cacaataagg attattccgt tgtgaaaaca tttagcgct cgactatctt cggggctttt 120
ttattttt                                     128

```

```

SEQ ID NO: 50          moltype = length =
SEQUENCE: 50
000

```

```

SEQ ID NO: 51          moltype = DNA length = 38
FEATURE               Location/Qualifiers
misc_difference        31..38
                      note = a,c,t,g, unknown, or other
misc_feature           1..38
                      note = source = /note="Description of Artificial Sequence:
                      Syntheticoligonucleotide"
misc_feature           1..38
                      note = source = /note="Target seq and PAM region of plasmid
                      library 1"
source                1..38
                      mol_type = other DNA
                      organism = synthetic construct

```

```

SEQUENCE: 51
gagcggacag cagcttccta tatctcgtag nnnnnnnn                                     38

```

SEQUENCE: 54									
MMIKNLGLLD	LGTNISIGWAL	IKQDFENKHG	EILMGMSRI	PMSQDILGDF	GKGNSSISQTA	60			
DRTKYRSVRR	LRERFLRLRL	LRHRVLHLNL	FLPHQYASQI	DFEKKFGFKF	SETEPKLAW	120			
NWGGKFSFLF	QNSFNEMLED	FKAAGQGLKI	PYDWTIYYLR	KKALSQKIEK	EELAWILLNF	180			
NQKRGGYYQLR	GEEBEEENPNK	LVBEFYSLKIV	DVVADEPQKG	KSDIHWYSLIL	ENGWVYIRAS	240			
KIPLFDWKDK	TRDFIVTTDL	NDRRSVKTDK	BGNEKRSFRA	PSENDWTLVK	KKTEQEIDRQ	300			
KHTLVGTYYIE	TLLLNPQPKI	KGKLVRTIER	EKFQKDELQI	LEKQKEFHQE	LKNDDLYNDC	360			
IRELYRNEEA	HQLTLSKKDF	VHLLMDMLFI	YQRPRLRSQS	SISNCTLEFR	KYKDENGIEH	420			
TQYLKAIPKS	NPYYEQFRLW	QWMYNLNIYR	KDDEANVTKE	FLNTNKFDFS	LFEFLNNRKE	480			
IEQKPLIKFL	LEQKDINKKL	LNAAEAKPYRW	NIYBDDKYPK	NETKTMISSR	LKDVENISDD	540			
FLTRDIBQKI	WHI IYSVNDK	IEYEKALKFS	ATRNDDLEGS	FIEAFKFKSP	FKSEYGSFSE	600			
KAIKKLLPLM	RLGKYWYEDE	IVKHSDIYFK	NIENLLDGSF	NRNKKISIED	EKKWNKSINL	660			
KQLEELKDFQ	TAIEDLFDGL	RLHIAQILVY	GRHSEASMIG	KWNSADEED	FLKDFKQHSL	720			
RNPIVEQVIT	ETLRVVVDKI	LKYGNGAKDF	FNEIHEILGR	EMKLPAADDR	KLTNQISENE	780			
NTNFRIKALL	AEAMMNDSSVE	NVRPFPSPMQQ	EILKIYEDDD	LKSDIEIEDD	ILIKSKTAQP	840			
SPSCLKRYKL	WLEQKYKSPY	TGQIIPLNKL	PTPEYEIEHI	IPQSRDIEDS	FSKNVICESA	900			
VNKLKDNYYIG	LEFIKQFGGT	IIELGPGKSI	KVPETKEYED	FVKKHYANNQ	GKRNNKLLMED	960			
IPEKMIERQM	NDRTRYISKYI	SGVLSNINVR	EDGSDSEGVN	KNYIVGNKGI	TTQLQKQDWGL	1020			
NVDVNDLILP	REFERNMNLK	SKVFTAWNEN	YQKFLPTVPI	EKISGFKSKR	IDHRHHALDA	1080			
LVIACATKDH	VNLLMNNQSAK	SDTKRYDLKK	KSMKFEKVYV	NDAKTGEKIE	REVPKQFLPK	1140			
WENFTLDVHK	NLETTIVSFK	QNLRVINKAT	NIYBEKYVEKD	GTGNKERVEQ	TGTNWAIRPK	1200			
MHKDTVSGVK	DLPWVSPKVG	KILTRATKSL	DSSPEFLSIG	SITDTGIQKI	LKNYLBAFKD	1260			
NPELAFSPGK	IDDLNKNIEK	YNDGKHQHPQI	NKVRVPELGS	KFQVQSGSNK	KDKYVBAAGK	1320			
TNLFVAVYED	EKGKRYNETI	PLNEVIERQK	QGLSVVDLKG	TNDFYLCPNP	FVYIPSGDEL	1380			
ENINNVDVFK	IKKINERIER	KVVSFTGNRL	SCIPIVMATT	IVNKLEFTQL	NKIEPTKEKE	1440			
ICKLNVDRIL	GNISKA					1455			

SEQUENCE: 55

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gttgtgaatt gctttcaa                                     18

SEQ ID NO: 56      moltype = RNA  length = 75
FEATURE           Location/Qualifiers
misc_feature       1..75
                   note = source = /note="tracrRNA"
source            1..75
                   mol_type = other RNA
                   organism = Empedobacter sp.

SEQUENCE: 56
ttgaaaagca attcacaata aggattattc cgttgtgaaa acattcaagg cggggcaact 60
cgtctttttt ctttt                                     75

SEQ ID NO: 57      moltype = RNA  length = 127
FEATURE           Location/Qualifiers
misc_feature       1..127
                   note = source = /note="Description of Artificial Sequence:
                   Syntheticpolynucleotide"
misc_feature       1..127
                   note = source = /note="sgRNA L1"
source            1..127
                   mol_type = other RNA
                   organism = synthetic construct

SEQUENCE: 57
gagcggacag cagcttccta tatctcgtag gtgtgtgaatt gctttcaaaa agttgaaaag 60
caattcacaa taaggattat tccgttgtga aaacattcaa ggcgggggcaa ctgctctttt 120
ttctttt                                     127

SEQ ID NO: 58      moltype = RNA  length = 127
FEATURE           Location/Qualifiers
misc_feature       1..127
                   note = source = /note="Description of Artificial Sequence:
                   Syntheticpolynucleotide"
misc_feature       1..127
                   note = source = /note="sgRNA L2"
source            1..127
                   mol_type = other RNA
                   organism = synthetic construct

SEQUENCE: 58
ccatgatata gacgttgtgg ctgttgtagt gtgtgtgaatt gctttcaaaa agttgaaaag 60
caattcacaa taaggattat tccgttgtga aaacattcaa ggcgggggcaa ctgctctttt 120
ttctttt                                     127

SEQ ID NO: 59      moltype =   length =
SEQUENCE: 59
000

SEQ ID NO: 60      moltype = DNA  length = 38
FEATURE           Location/Qualifiers
misc_difference    31..38
                   note = a,c,t,g, unknown, or other
misc_feature       1..38
                   note = source = /note="Description of Artificial Sequence:
                   Syntheticoligonucleotide"
misc_feature       1..38
                   note = source = /note="Target seq and PAM region of plasmid
                   library 1"
source            1..38
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 60
gagcggacag cagcttccta tatctcgtag nnnnnnnn                                     38

SEQ ID NO: 61      moltype = DNA  length = 38
FEATURE           Location/Qualifiers
misc_difference    31..38
                   note = a,c,t,g, unknown,or other
misc_feature       1..38
                   note = source = /note="Description of Artificial Sequence:
                   Syntheticoligonucleotide"
misc_feature       1..38
                   note = source = /note="Target seq and PAM region of plasmid
                   library 2"
source            1..38
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 61
ccatgatata gacgttgtagg ctgttgtagt nnnnnnnn 38

SEQ ID NO: 62 moltype = RNA length = 122
FEATURE Location/Qualifiers
misc_difference 1..25
note = a,c,t,g, unknown, or other
misc_feature 1..122
note = source = /note="Description of Artificial Sequence:
Syntheticpolynucleotide"
misc_feature 1..122
note = source = /note="sgRNA with N's"
source 1..122
mol_type = other RNA
organism = synthetic construct

SEQUENCE: 62
nnnnnnnnnn nnnnnnnnnn nnnnngttgt gaattgcttt caaaaagttg aaaagcaatt 60
cacaataagg attattccgt tgtgaaaaca ttcaaggcgg ggcaactcgt cttttttcctt 120
tt 122

SEQ ID NO: 63 moltype = length =
SEQUENCE: 63
000

SEQ ID NO: 64 moltype = length =
SEQUENCE: 64
000

SEQ ID NO: 65 moltype = length =
SEQUENCE: 65
000

SEQ ID NO: 66 moltype = length =
SEQUENCE: 66
000

SEQ ID NO: 67 moltype = AA length = 7
FEATURE Location/Qualifiers
REGION 1..7
note = source = /note="Description of Artificial Sequence:
Syntheticpeptide"
REGION 1..7
note = source = /note="NLS"
source 1..7
mol_type = protein
organism = synthetic construct

SEQUENCE: 67
PKKKRKV 7

SEQ ID NO: 68 moltype = length =
SEQUENCE: 68
000

SEQ ID NO: 69 moltype = length =
SEQUENCE: 69
000

SEQ ID NO: 70 moltype = length =
SEQUENCE: 70
000

SEQ ID NO: 71 moltype = length =
SEQUENCE: 71
000

SEQ ID NO: 72 moltype = length =
SEQUENCE: 72
000

SEQ ID NO: 73 moltype = DNA length = 30
FEATURE Location/Qualifiers
misc_feature 1..30
note = source = /note="Description of Artificial Sequence:
Syntheticoligonucleotide"
misc_feature 1..30
note = source = /note="mismatch repair Table 4"

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source                1..30
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 73
gagcggacag cagcttccta tatctcgtag                30

SEQ ID NO: 74          moltype = DNA length = 30
FEATURE               Location/Qualifiers
misc_feature          1..30
                      note = source = /note="Description of Artificial Sequence:
                      Syntheticoligonucleotide"
misc_feature          1..30
                      note = source = /note="mismatch repair Table 4"
source                1..30
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 74
gagcggacag cagcttccta tatctcgtag                30

SEQ ID NO: 75          moltype = DNA length = 30
FEATURE               Location/Qualifiers
misc_feature          1..30
                      note = source = /note="Description of Artificial Sequence:
                      Syntheticoligonucleotide"
misc_feature          1..30
                      note = source = /note="mismatch repair Table 4"
source                1..30
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 75
gagcggacag cagcttccta tatctcggtc                30

SEQ ID NO: 76          moltype = DNA length = 30
FEATURE               Location/Qualifiers
misc_feature          1..30
                      note = source = /note="Description of Artificial Sequence:
                      Syntheticoligonucleotide"
misc_feature          1..30
                      note = source = /note="mismatch repair Table 4"
source                1..30
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 76
gagcggacag cagcttccta tatctcgaac                30

SEQ ID NO: 77          moltype = DNA length = 30
FEATURE               Location/Qualifiers
misc_feature          1..30
                      note = source = /note="Description of Artificial Sequence:
                      Syntheticoligonucleotide"
misc_feature          1..30
                      note = source = /note="mismatch repair Table 4"
source                1..30
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 77
gagcggacag cagcttccta tatctcctac                30

SEQ ID NO: 78          moltype = DNA length = 30
FEATURE               Location/Qualifiers
misc_feature          1..30
                      note = source = /note="Description of Artificial Sequence:
                      Syntheticoligonucleotide"
misc_feature          1..30
                      note = source = /note="mismatch repair Table 4"
source                1..30
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 78
gagcggacag cagcttccta tatctggtag                30

SEQ ID NO: 79          moltype = DNA length = 30
FEATURE               Location/Qualifiers
misc_feature          1..30
                      note = source = /note="Description of Artificial Sequence:
                      Syntheticoligonucleotide"

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misc_feature      1..30
                  note = source = /note="mismatch repair Table 4"
source            1..30
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 79
gagcggacag cagcttccta tatcacgtac                      30

SEQ ID NO: 80      moltype = DNA length = 30
FEATURE           Location/Qualifiers
misc_feature      1..30
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..30
                  note = source = /note="mismatch repair Table 4"
source            1..30
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 80
gagcggacag cagcttccta tatgtcgtac                      30

SEQ ID NO: 81      moltype = DNA length = 30
FEATURE           Location/Qualifiers
misc_feature      1..30
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..30
                  note = source = /note="mismatch repair Table 4"
source            1..30
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 81
gagcggacag cagcttccta taactcgtac                      30

SEQ ID NO: 82      moltype = DNA length = 30
FEATURE           Location/Qualifiers
misc_feature      1..30
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..30
                  note = source = /note="mismatch repair Table 4"
source            1..30
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 82
gagcggacag cagcttccta tttctcgtac                      30

SEQ ID NO: 83      moltype = DNA length = 30
FEATURE           Location/Qualifiers
misc_feature      1..30
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..30
                  note = source = /note="mismatch repair Table 4"
source            1..30
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 83
gagcggacag cagcttccta aatctcgtac                      30

SEQ ID NO: 84      moltype = DNA length = 30
FEATURE           Location/Qualifiers
misc_feature      1..30
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..30
                  note = source = /note="mismatch repair Table 4"
source            1..30
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 84
gagcggacag cagcttcctt tatctcgtac                      30

SEQ ID NO: 85      moltype = DNA length = 30
FEATURE           Location/Qualifiers
misc_feature      1..30

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                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
misc_feature                    1..30
                                note = source = /note="mismatch repair Table 4"
source                          1..30
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 85
gagcggacag cagcttccaa tatctcgta                      30

SEQ ID NO: 86                moltype = DNA length = 30
FEATURE                      Location/Qualifiers
misc_feature                  1..30
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
misc_feature                  1..30
                                note = source = /note="mismatch repair Table 4"
source                        1..30
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 86
gagcggacag cagcttcgta tatctcgta                      30

SEQ ID NO: 87                moltype = DNA length = 30
FEATURE                      Location/Qualifiers
misc_feature                  1..30
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
misc_feature                  1..30
                                note = source = /note="mismatch repair Table 4"
source                        1..30
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 87
gagcggacag cagcttgcta tatctcgta                      30

SEQ ID NO: 88                moltype = DNA length = 30
FEATURE                      Location/Qualifiers
misc_feature                  1..30
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
misc_feature                  1..30
                                note = source = /note="mismatch repair Table 4"
source                        1..30
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 88
gagcggacag cagctaccta tatctcgta                      30

SEQ ID NO: 89                moltype = DNA length = 30
FEATURE                      Location/Qualifiers
misc_feature                  1..30
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
misc_feature                  1..30
                                note = source = /note="mismatch repair Table 4"
source                        1..30
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 89
gagcggacag cagcatccta tatctcgta                      30

SEQ ID NO: 90                moltype = DNA length = 30
FEATURE                      Location/Qualifiers
misc_feature                  1..30
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
misc_feature                  1..30
                                note = source = /note="mismatch repair Table 4"
source                        1..30
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 90
gagcggacag caggttccta tatctcgta                      30

SEQ ID NO: 91                moltype = DNA length = 30

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FEATURE	Location/Qualifiers	
misc_feature	1..30	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..30	
	note = source = /note="mismatch repair Table 4"	
source	1..30	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 91		
gagcggacag caccttccta tatctcgtag		30
SEQ ID NO: 92	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..30	
	note = source = /note="mismatch repair Table 4"	
source	1..30	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 92		
gagcggacag ctgcttccta tatctcgtag		30
SEQ ID NO: 93	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..30	
	note = source = /note="mismatch repair Table 4"	
source	1..30	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 93		
gagcggacag gagcttccta tatctcgtag		30
SEQ ID NO: 94	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..30	
	note = source = /note="mismatch repair Table 4"	
source	1..30	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 94		
gagcggacac cagcttccta tatctcgtag		30
SEQ ID NO: 95	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..30	
	note = source = /note="mismatch repair Table 4"	
source	1..30	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 95		
gagcggactg cagcttccta tatctcgtag		30
SEQ ID NO: 96	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..30	
	note = source = /note="mismatch repair Table 4"	
source	1..30	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 96		
gagcggagag cagcttccta tatctcgtag		30

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SEQ ID NO: 97      moltype = DNA  length = 30
FEATURE           Location/Qualifiers
misc_feature       1..30
                   note = source = /note="Description of Artificial Sequence:
                   Syntheticoligonucleotide"
misc_feature       1..30
                   note = source = /note="mismatch repair Table 4"
source            1..30
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 97
gagcggtcag cagcttccta tatctcgtag                      30

SEQ ID NO: 98      moltype = DNA  length = 30
FEATURE           Location/Qualifiers
misc_feature       1..30
                   note = source = /note="Description of Artificial Sequence:
                   Syntheticoligonucleotide"
misc_feature       1..30
                   note = source = /note="mismatch repair Table 4"
source            1..30
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 98
gagcgcacag cagcttccta tatctcgtag                      30

SEQ ID NO: 99      moltype = DNA  length = 30
FEATURE           Location/Qualifiers
misc_feature       1..30
                   note = source = /note="Description of Artificial Sequence:
                   Syntheticoligonucleotide"
misc_feature       1..30
                   note = source = /note="mismatch repair Table 4"
source            1..30
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 99
gagccgacag cagcttccta tatctcgtag                      30

SEQ ID NO: 100     moltype = DNA  length = 30
FEATURE           Location/Qualifiers
misc_feature       1..30
                   note = source = /note="Description of Artificial Sequence:
                   Syntheticoligonucleotide"
misc_feature       1..30
                   note = source = /note="mismatch repair Table 4"
source            1..30
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 100
gaggggacag cagcttccta tatctcgtag                      30

SEQ ID NO: 101     moltype = DNA  length = 30
FEATURE           Location/Qualifiers
misc_feature       1..30
                   note = source = /note="Description of Artificial Sequence:
                   Syntheticoligonucleotide"
misc_feature       1..30
                   note = source = /note="mismatch repair Table 4"
source            1..30
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 101
gaccggacag cagcttccta tatctcgtag                      30

SEQ ID NO: 102     moltype = DNA  length = 30
FEATURE           Location/Qualifiers
misc_feature       1..30
                   note = source = /note="Description of Artificial Sequence:
                   Syntheticoligonucleotide"
misc_feature       1..30
                   note = source = /note="mismatch repair Table 4"
source            1..30
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 102			
gtcgggacag cagcttccta tatctcgtag			30
SEQ ID NO: 103	moltype = DNA	length = 30	
FEATURE	Location/Qualifiers		
misc_feature	1..30		
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"		
misc_feature	1..30		
	note = source = /note="mismatch repair Table 4"		
source	1..30		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 103			
cagcggacag cagcttccta tatctcgtag			30
SEQ ID NO: 104	moltype = DNA	length = 30	
FEATURE	Location/Qualifiers		
misc_feature	1..30		
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"		
misc_feature	1..30		
	note = source = /note="mismatch repair Table 6"		
source	1..30		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 104			
gagcggacag cagcttccta tatctcgtag			30
SEQ ID NO: 105	moltype = DNA	length = 30	
FEATURE	Location/Qualifiers		
misc_feature	1..30		
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"		
misc_feature	1..30		
	note = source = /note="mismatch repair Table 6"		
source	1..30		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 105			
gagcggacag cagcttccta tatctcgtgc			30
SEQ ID NO: 106	moltype = DNA	length = 30	
FEATURE	Location/Qualifiers		
misc_feature	1..30		
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"		
misc_feature	1..30		
	note = source = /note="mismatch repair Table 6"		
source	1..30		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 106			
gagcggacag cagcttccta tatctcgtag			30
SEQ ID NO: 107	moltype = DNA	length = 30	
FEATURE	Location/Qualifiers		
misc_feature	1..30		
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"		
misc_feature	1..30		
	note = source = /note="mismatch repair Table 6"		
source	1..30		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 107			
gagcggacag cagcttccta tatctcatag			30
SEQ ID NO: 108	moltype = DNA	length = 30	
FEATURE	Location/Qualifiers		
misc_feature	1..30		
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"		
misc_feature	1..30		
	note = source = /note="mismatch repair Table 6"		
source	1..30		

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	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 108		
gagcggacag cagcttccta tatcttgtag		30
SEQ ID NO: 109	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..30	
	note = source = /note="mismatch repair Table 6"	
source	1..30	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 109		
gagcggacag cagcttccta tatcccgtag		30
SEQ ID NO: 110	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..30	
	note = source = /note="mismatch repair Table 6"	
source	1..30	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 110		
gagcggacag cagcttccta tatttcgtag		30
SEQ ID NO: 111	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..30	
	note = source = /note="mismatch repair Table 6"	
source	1..30	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 111		
gagcggacag cagcttccta tacctcgtag		30
SEQ ID NO: 112	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..30	
	note = source = /note="mismatch repair Table 6"	
source	1..30	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 112		
gagcggacag cagcttccta tgtctcgtag		30
SEQ ID NO: 113	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..30	
	note = source = /note="mismatch repair Table 6"	
source	1..30	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 113		
gagcggacag cagcttccta catctcgtag		30
SEQ ID NO: 114	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..30	

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source          note = source = /note="mismatch repair Table 6"
                1..30
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 114
gagcggacag cagcttcctg tatctcgtag          30

SEQ ID NO: 115      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..30
                    note = source = /note="mismatch repair Table 6"
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 115
gagcggacag cagcttccca tatctcgtag          30

SEQ ID NO: 116      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..30
                    note = source = /note="mismatch repair Table 6"
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 116
gagcggacag cagctttcta tatctcgtag          30

SEQ ID NO: 117      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..30
                    note = source = /note="mismatch repair Table 6"
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 117
gagcggacag cagctttcta tatctcgtag          30

SEQ ID NO: 118      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..30
                    note = source = /note="mismatch repair Table 6"
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 118
gagcggacag cagctcccta tatctcgtag          30

SEQ ID NO: 119      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..30
                    note = source = /note="mismatch repair Table 6"
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 119
gagcggacag cagctcccta tatctcgtag          30

SEQ ID NO: 120      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = source = /note="Description of Artificial Sequence:

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misc_feature      Syntheticoligonucleotide"
                  1..30
note = source = /note="mismatch repair Table 6"
source            1..30
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 120
gagcggacag cagtctccta tatctcgtag                      30

SEQ ID NO: 121      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..30
note = source = /note="mismatch repair Table 6"
source              1..30
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 121
gagcggacag caacttccta tatctcgtag                      30

SEQ ID NO: 122      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..30
note = source = /note="mismatch repair Table 6"
source              1..30
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 122
gagcggacag cggcttccta tatctcgtag                      30

SEQ ID NO: 123      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..30
note = source = /note="mismatch repair Table 6"
source              1..30
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 123
gagcggacag tagcttccta tatctcgtag                      30

SEQ ID NO: 124      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..30
note = source = /note="mismatch repair Table 6"
source              1..30
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 124
gagcggacaa cagcttccta tatctcgtag                      30

SEQ ID NO: 125      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..30
note = source = /note="mismatch repair Table 6"
source              1..30
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 125
gagcggacgg cagcttccta tatctcgtag                      30

SEQ ID NO: 126      moltype = DNA length = 30
FEATURE            Location/Qualifiers

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misc_feature      1..30
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misc_feature      1..30
                  note = source = /note="mismatch repair Table 6"
source            1..30
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 126
gagcggatag cagcttccta tatctcgtac                               30

SEQ ID NO: 127      moltype = DNA length = 3210
FEATURE            Location/Qualifiers
misc_feature        1..3210
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticpolynucleotide"
misc_feature        1..3210
                  note = source = /note="APG05083.1 mammalian codon optimized
                  sequence"
source              1..3210
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 127
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atcatcgagc tgtcttggaa caaagacaga gaacagtacg agaaggcaag aatcgtggac 120
aagggcggtg gaattgtcga caaggccgaa atacccaaga cggagccag cctggccgag 180
cctagaagaa ttgccagatc aagcagaaga cggtccaaca gaaagtctca gagaaaaaaa 240
gatattcggg acctgtggtg cagcacgag atcatcagcc aaaaggaact cgtagcctg 300
tatctctcga ccaagagcag catggacatt tgggacatca gactggacgg cctggacaga 360
ctgctgggata gattcgagtg gaccagattg ctgatccacc tggctcagcg gagaggcttc 420
aaaagcaacc ggaagagcga gctgaaggac gtggaaaaccg gcaagggtgt ctccagcatc 480
caggctaatt agaagcggct gtccctgtac agaactgtgg gcgagatgtg gatgaagaat 540
gaagatttta gcaagtacga caaaagaaga aatagtagca acgaatacgt gttctctgtg 600
tcccgggccg acctggaaaa ggaatcgtg acactgtttg aagctcagcg gaagttccag 660
agcagctatg ccagcgccga tcttcaaaaa acctacctcc agatctgggc ccatcaactg 720
cctttcgcct ctggcaacgc catcgtaaac aaggtgggct actgcagcct actgaaaggc 780
aaggagaaga gagtctctaa ggcacacctac acctccaat actcagcac cctggatcag 840
atcaacagaa ccagactggg cccaaacttc cagcccttca ccaaggaaac gagagatgtg 900
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acgtactacg atatcagaaa atggctggcc ctggacgaga caatccagtt caagggcctg 1020
acctacgacc ataataagaa actgaagaaa attgagctaa agtcttttat caatctgaag 1080
ctttctacg agataaagaa ggtggtgaca aactacgcca agaagacaaa tgaggccttt 1140
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gaaatcgatc acatcctgcc ctacagccag agcttcatcg acagctacca caacaaagt 1860
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aagaatctga agttcaagga ggtggaaggc tctacccgga agaagcacgt gcaaacccgtg 2160
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gaagtgcata ccaagagctt tgtgagagaa gtcaacggcg gagtggccca gaacggcgat 2820
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caatggctga ccacgaccaa cagcttcacc ttcaagttca gcctataccc ctacgacctg 3000
gtgcggtcgg tcaagggtta cgaggacaga ttcctgtact tttccacctt ggatattaac 3060
agtgtatgac tcaacttcaa agacgtcaac aagcccagca agcaggccga gaacagatat 3120
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tttgtgagac aggagatcag aaagaacttc                               3210

SEQ ID NO: 128      moltype = DNA  length = 3213
FEATURE
misc_feature        1..3213
                     note = source = /note="Description of Artificial Sequence:
                     Syntheticpolynucleotide"
misc_feature        1..3213
                     note = source = /note="APG07433.1 mammalian codon optimized
                     sequence"
source              1..3213
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 128
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gtgatcgagc tgtcctggaa caaagaccgg gagagatacg agaaggtcag aatcgtggat 120
caaggcgtga gaatgttcga cagagccgag atgcccaga caggccgacg cttagctgaa 180
cccagaagaa tcgccagatc cagcagacgg agactgaatc gcaagtccca gagaaagaaa 240
aacatccgga acctgtggtg gcaacacggc gtgatcacac aggaggaact ggatagcctg 300
taccctctga gcaaaaagag catggacatc tggggcattc ggctcgacgg cctggacaga 360
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aagtccaaca gaaagagtga actgaaagat acagagacag gcaagggtgt gagcagcatc 480
caactgaacg agaaacgggt gagcttgtat agaaccgtgg gcgagatgtg gatgaaggac 540
ccgactttct ctaataacga taggaagaga aatagcccca acgaatacgt gttcagcgtg 600
tctagagccg agctggaaaa ggaatcgtg accctgttcg ccgccagcg gagattccag 660
agcccttacg ccagcaaaaga tctgcaggag acatatctgc agatctggac ccaccaactg 720
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agcgacagac ttaacttcaa ggacgtgaac aagccaagca agaagaacga gtaccggtac 3120
agcttgaaaa ccacgagga cttggagaag tacgaggtgg gcgtgctggg cgatctaaga 3180
ctggtccgga aggaacttcg aagaaacttc cac                               3213

SEQ ID NO: 129      moltype = DNA  length = 3213
FEATURE
misc_feature        1..3213
                     note = source = /note="Description of Artificial Sequence:
                     Syntheticpolynucleotide"
misc_feature        1..3213
                     note = source = /note="APG07513.1 mammalian codon optimized

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sequence"
source      1..3213
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 129
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gtgatcgagc tgagctggaa caaggacaga gagcagtagc agaagaccag aatcggtggac 120
aaggcggtta ggatgttcga caaggccgag atccccaaga ccggagccag cctggccgag 180
cctagacgta tcgccagatc ctccagacgt agactgaaca gaaagtccca gaggaagaag 240
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gaagtggacc acatcctgcc ctacagccag agcttccatc acagctacca caacaagggt 1860
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agcgacagac tgaacttcaa ggacgtgaac aagcctagca agcaggctga ataccggtag 3120
agcctgaaga cgtcgagaa cctggaaaaa tacgagatcg gtgttctggg agatctgaga 3180
ctcgttagac agggagacaag aaaaattttc aag 3213

SEQ ID NO: 130      moltype = DNA length = 3216
FEATURE             Location/Qualifiers
misc_feature         1..3216
                     note = source = /note="Description of Artificial Sequence:
                     Syntheticpolynucleotide"
misc_feature         1..3216
                     note = source = /note="APG08290.1 mammalian codon optimized
                     sequence"
source               1..3216
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 130
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aaaggggtca gaatgtttga caaggccgag atccccaaca agggcgccag cctggccgag 180
cctcgagaaa tcgcccggag cagcagaaga gcctgaaca ggaagtccca aaggaagaag 240

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ctgctgaacc acttagagtg ggctagactg ctgattcacc tggcccacgg gcggggcttc 420
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agactccata	tgcgccagta	cctgggtgtac	ggccggcaca	gcgaagccag	catgatccga	2100
aagtggaaaca	cgcccgagga	tctggaagaa	ttcctgaagg	acttcaagca	gcacagccctg	2160
agaaacccca	cgtctgagac	atggatcacc	gagaccctga	gggtgggttaa	ggatatatgg	2220
ctgaaatacg	gcaacggcgc	caaggacttc	ttcaacgaga	ttcacatcga	gctgggcaga	2280
gaaatgaagc	tgccctgccga	cgaccggaaa	aaagtgcacca	accagatcag	cgagaacgaa	2340
aacacaaaat	tcagaattca	ggccctgctg	gctgaaatga	tgaacgacag	cagtgtggaa	2400
aatgtgcggc	ctttcagccc	tatgcagcag	gagatcctga	agatctatga	agatgacgtc	2460
ctgaagtctg	acattgaaat	cgaggatgac	atccttaaga	tctctaaaac	cgcccagcct	2520
tctcttagcg	atctcaagcg	atacaagctc	tggtcggagc	agaagtacaa	gtctccatac	2580
accggccaga	tcatacctct	gaacaagctg	tttaccctgt	aatacgagat	cgagcacatt	2640
atccctcagt	ctagatatct	cgacgacagc	ttcagcaaca	aagtgatctg	cgagagcgcc	2700
gtgaacaagc	tgaagaccaa	ctacattggc	ctggaattca	tcaagcaatt	tggaagccac	2760
atcatcgaa	tggtgctttg	caagtcctac	aaagtgttcg	agaccaaaga	gtacgaggac	2820
ttcgtgaaga	aactactacg	caacaatcag	ggcaagagaa	acaagctgct	catggaagac	2880
atccccgaga	aaatgatcga	gagacagatg	aacgacacca	gatcatctc	taagtacatc	2940
agcggagtgc	tgaaccaatc	ggtcagagtg	gaagatggat	ctgatgaggg	cgtgaacagc	3000
aagaacatcg	tgcccggcaa	cggaagatt	acaacacaa	tgaagcagga	ctggggcctg	3060
aatgacgtgt	ggaacgacct	tatcctgcgc	cggttcgaga	gaatgaatca	actgaccaac	3120
agcaaggtgt	tcacagcctg	gaacgaaaa	taccagaagt	tcctgcctac	agtgccaatc	3180
gaatacacga	aagcttccag	caagaagcgt	atcgatcaca	gacaccacgc	cctggatgct	3240
ctgggttatc	cctgcgccac	caaggaccac	gtgaacctgc	tgaaccaaca	gtctgcacaag	3300
agcgacacca	agagatacga	ctttaaagaa	aaaagtatga	aattcgagaa	agtggtgtac	3360
aatgatgcca	agaccggcga	aaagatcgag	cgggaagtgc	ccaagcagtt	cctgaagcct	3420
tggaagaact	tcaccctgga	tgtgaagcac	aaacctgaaa	cgatcatcgt	ctctttttaa	3480
caaaatctgc	gggtgatcaa	taaagccacg	aactactacg	agaagtacgt	cgagaaggac	3540
ggcaccacaaa	acaagagcgc	cgtggagcag	acagggacaa	actgggccat	tagaaaagcc	3600
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aagatcctga	ccgccacacg	gaagagcttg	gatagcagct	tcgacctcaa	gagcatcggc	3720
tctataacag	atacaggcat	tcagaagatc	ctaaagaatt	acctggcatt	caaggacggc	3780
aacctgagc	tggttttcag	ccctgagggc	atcgacgac	tgaaccaaga	catagagaag	3840
tacaacgatg	gcaaacctca	ccagcctatc	aacaaggtgc	gagtggttga	gctgggcagc	3900
aagttccagg	tggaacaatc	tggttaacaag	aaggacaagt	atgtggaagc	tgctaagggc	3960
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acaaacgact	tctacctgtg	ccctaaccgac	tttgtgtaca	ttccaaagcgg	cgacgagctg	4140
gaaaacatca	acaacgtgga	cttcaaggac	atcaagaagg	agattaaacga	gcgcattctac	4200
aaagtgggtg	cttttacaag	caatagactt	tcttgttatcc	cttacatggg	ggccacaacg	4260
atcgtcaaca	aactogaatt	caactcaactg	aacaaaatcg	agttttacaa	ggaaaaagaa	4320
atgtgtatca	agctcaacgt	ggcacagactg	ggcaatatct	ccaaggcc		4368

SEQ ID NO: 134

moltype = DNA length = 21

FEATURE

Location/Qualifiers

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misc_feature      1..21
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..21
                  note = source = /note="SV40 large T-antigen nuclear
                  localization sequence"
source            1..21
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 134
cctaagaaga aaagaaaggt g                               21

SEQ ID NO: 135      moltype = DNA length = 66
FEATURE            Location/Qualifiers
misc_feature        1..66
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..66
                  note = source = /note="Triple FLAG epitope tag"
source              1..66
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 135
gactacaagg accacgacgg cgactacaaa gatcacgata tcgactacaa ggacgacgat 60
gataag                                           66

SEQ ID NO: 136      moltype = DNA length = 48
FEATURE            Location/Qualifiers
misc_feature        1..48
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..48
                  note = source = /note="Nucleoplasmin nuclear localization
                  sequence"
source              1..48
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 136
aaaagacctg ccgctacaaa gaaggccggc caggccaaga aaaagaag           48

SEQ ID NO: 137      moltype = DNA length = 203
FEATURE            Location/Qualifiers
misc_feature        1..203
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticpolynucleotide"
misc_feature        1..203
                  note = source = /note="Cytomegalovirus mammalian
                  transcription promoter"
source              1..203
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 137
tgatgcggtt ttggcagttac atcaatgggc gtggatagcg gtttgactca cggggatttc 60
caagtctcca cccattgac gtcaatggga gtttgttttg gcacaaaaat caacgggact 120
ttccaaaatg tcgtaacaac tccgccccat tgacgcaaat gggcggtagg cgtgtacggg 180
gggaggtcta tataagcaga gct                                           203

SEQ ID NO: 138      moltype = DNA length = 304
FEATURE            Location/Qualifiers
misc_feature        1..304
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticpolynucleotide"
misc_feature        1..304
                  note = source = /note="Cytomegalovirus transcription
                  enhancer"
source              1..304
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 138
cgttacataa cttacggtta atggcccgcc ttggtgacgg cccaacgacc cccgccatt 60
gacgtcaata atgacgtatg ttcccatagt aacgccaata gggactttcc attgacgtca 120
atgggtggag tatttacggt aaactgccca cttggcagta catcaagtgt atcatatgcc 180
aagtagcccc cctattgacg tcaatgacgg taaatggccc gcctggcatt atgcccagta 240
catgacctta tgggactttc ctacttgcca gtacatctac gtattagtca tcgctattac 300
catg                                           304

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SEQ ID NO: 139	moltype = DNA length = 318	
FEATURE	Location/Qualifiers	
misc_feature	1..318	note = source = /note="Description of Artificial Sequence: Syntheticpolynucleotide"
misc_feature	1..318	note = source = /note="Human U6 RNA promoter"
source	1..318	mol_type = other DNA
	organism = synthetic construct	
SEQUENCE: 139		
tgtacaaaaa agcaggcttt	aaaggaacca attcagtcga ctggatccgg taccaaggtc	60
gggcaggaag agggcctatt	tcccatgatt ccttcatatt tgcataatcg atacaaggct	120
gttagagaga taattagaat	taatttgact gtaaacacaa agatattagt acaaaatacg	180
tgacgtagaa agtaataatt	tcttgggtag tttagcagtt taaaattatg ttttaaatg	240
gactatcata tgcttaccgt	aacttgaaag tatttcgatt tcttggcttt atatatcttg	300
tggaaaggac gaaacacc		318
SEQ ID NO: 140	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"
misc_feature	1..20	note = source = /note="target for Guide 189"
source	1..20	mol_type = other DNA
	organism = synthetic construct	
SEQUENCE: 140		
cctgaatgct gtgcggctct		20
SEQ ID NO: 141	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"
misc_feature	1..20	note = source = /note="target for Guide 185"
source	1..20	mol_type = other DNA
	organism = synthetic construct	
SEQUENCE: 141		
ggacagtgcg catctccctg		20
SEQ ID NO: 142	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"
misc_feature	1..20	note = source = /note="target for Guide 168"
source	1..20	mol_type = other DNA
	organism = synthetic construct	
SEQUENCE: 142		
cacatctcga gcaagacgtt		20
SEQ ID NO: 143	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"
misc_feature	1..20	note = source = /note="target for Guide 135"
source	1..20	mol_type = other DNA
	organism = synthetic construct	
SEQUENCE: 143		
gggccattaa aacctctcca		20
SEQ ID NO: 144	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"
misc feature	1..20	

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source	note = source = /note="target for Guide 139"	
	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 144		
aggttttaat ggcccagcct		20
SEQ ID NO: 145	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..20	
	note = source = /note="target for Guide 143"	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 145		
catggcagta cattagagca		20
SEQ ID NO: 146	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..20	
	note = source = /note="target for Guide 190"	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 146		
cctgaatgct gtgcggtctt		20
SEQ ID NO: 147	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..20	
	note = source = /note="target for Guide 194"	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 147		
gccgcacagc attcaggtcg		20
SEQ ID NO: 148	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..20	
	note = source = /note="target for Guide 165"	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 148		
atggggaatg tagcaagacc		20
SEQ ID NO: 149	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..20	
	note = source = /note="target for Guide 169"	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 149		
cacatctcga gcaagacgtt		20
SEQ ID NO: 150	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="Description of Artificial Sequence:	

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misc_feature      Syntheticoligonucleotide"
                  1..20
source            note = source = /note="target for Guide 173"
                  1..20
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 150
cttctatagc ctccttcccc                               20

SEQ ID NO: 151    moltype = DNA length = 20
FEATURE          Location/Qualifiers
misc_feature      1..20
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..20
                  note = source = /note="target for Guide 144"
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 151
catggcagta cattagagca                               20

SEQ ID NO: 152    moltype = DNA length = 20
FEATURE          Location/Qualifiers
misc_feature      1..20
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..20
                  note = source = /note="target for Guide 136"
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 152
gggccattaa aacctctcca                               20

SEQ ID NO: 153    moltype = DNA length = 20
FEATURE          Location/Qualifiers
misc_feature      1..20
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..20
                  note = source = /note="target for Guide 145"
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 153
catggcagta cattagagca                               20

SEQ ID NO: 154    moltype = DNA length = 20
FEATURE          Location/Qualifiers
misc_feature      1..20
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..20
                  note = source = /note="target for Guide 188"
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 154
ggacagtgcg catctccctg                               20

SEQ ID NO: 155    moltype = DNA length = 20
FEATURE          Location/Qualifiers
misc_feature      1..20
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..20
                  note = source = /note="target for Guide 192"
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 155
cctgaatgct gtgcggctct                               20

SEQ ID NO: 156    moltype = DNA length = 20
FEATURE          Location/Qualifiers

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misc_feature      1..20
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..20
                  note = source = /note="target for Guide 196"
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 156
gccgcacagc attcaggtcg                                20

SEQ ID NO: 157      moltype = DNA length = 20
FEATURE            Location/Qualifiers
misc_feature      1..20
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..20
                  note = source = /note="target for Guide 167"
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 157
atggggaatg tagcaagacc                                20

SEQ ID NO: 158      moltype = DNA length = 20
FEATURE            Location/Qualifiers
misc_feature      1..20
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..20
                  note = source = /note="target for Guide 171"
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 158
cacatctcga gcaagacggt                                20

SEQ ID NO: 159      moltype = DNA length = 20
FEATURE            Location/Qualifiers
misc_feature      1..20
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..20
                  note = source = /note="target for Guide 175"
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 159
cttctatagc ctccttcccc                                20

SEQ ID NO: 160      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature      1..25
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..25
                  note = source = /note="target for Guide 197"
source            1..25
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 160
cctcaccccc acgagcttgt aggaa                            25

SEQ ID NO: 161      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature      1..25
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..25
                  note = source = /note="target for Guide 199"
source            1..25
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 161
acagtgcgca tctccctggt cacca                            25

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SEQ ID NO: 162      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
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                    Syntheticoligonucleotide"
misc_feature        1..25
                    note = source = /note="target for Guide 146"
source              1..25
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 162
ggctgggcca ttaaacctc tccag                               25

SEQ ID NO: 163      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..25
                    note = source = /note="target for Guide 148"
source              1..25
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 163
gaggctgggc cattaaaacc tctcc                               25

SEQ ID NO: 164      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..25
                    note = source = /note="target for Guide 176"
source              1..25
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 164
gaggctgaga caggagagtt gcttg                               25

SEQ ID NO: 165      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..25
                    note = source = /note="target for Guide 177"
source              1..25
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 165
cccacaaaacc gatgtagctc aagag                               25

SEQ ID NO: 166      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..25
                    note = source = /note="target for Guide 209"
source              1..25
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 166
tagggtctta ctctgttgtc cacgc                               25

SEQ ID NO: 167      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..25
                    note = source = /note="target for Guide 151"
source              1..25
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 167

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ttctcccgag ccaagtacac gtttc                                     25

SEQ ID NO: 168          moltype = DNA length = 25
FEATURE
misc_feature            1..25
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                        Syntheticoligonucleotide"
misc_feature            1..25
                        note = source = /note="target for Guide 152"
source                  1..25
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 168
gaaaatctgc tcttagggct caagg                                     25

SEQ ID NO: 169          moltype = DNA length = 3210
FEATURE
misc_feature            1..3210
                        note = source = /note="Description of Artificial Sequence:
                        Syntheticpolynucleotide"
misc_feature            1..3210
                        note = source = /note="APG05083.1 Soy optimized"
source                  1..3210
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 169
atgagagagc tagattacag gatccgactc gatatcgga ctaattcaat cgggtgggga 60
ataattgagc tgtcctggaa taaagatcgc gagcaatatg aaaaggctag gattgtggat 120
aaggagtgga ggatgttoga taaggctgaa attcctaaga ctggagcttc tcttgctgaa 180
ccaagaagaa ttgctagatc tagcaggagg agattaaaca ggaagtctca gcgcaagaag 240
gatataagga accttctcgt tcagcatgaa atcatttctc agaaggagct tgcttctctc 300
tacctcttaa ccaaatcaag catggacatc tgggatataa ggcttgatgg actcgatagg 360
cttcttgata gatttgagtg gaccaggctt ctaatacacc ttgctcaaa aaggggcttc 420
aaatcaaac ccaagtccga attaaaggat gttgagactg gaaagggtgt ttcaagcatc 480
caagctaattg aaaagaggct ttctttgtac aggactgttg gggagatgtg gatgaaaaac 540
gaggacttct caaaatagca taaaaggagg aactcctcca acgagtacgt gttctctgtt 600
tctagggctg atcttgagaa ggagattgtg actctttttg aggcctcaac caagtccaa 660
tcaagctacg cttctgctga tcttcaaaa acctaccttc aaatttgggc tcatcagctt 720
ccttttgctt ctggaaacgc catttgtaat aaagtgggat actgctccct attaaaggga 780
aaggagaaga gagtgctcta ggctacctat acattccagt acttctctac cctcgatcaa 840
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atcctcgatg aaatgttcaa caggaccgac tactataaaa agaagaccat tcccagggtg 960
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organism = synthetic construct

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FEATURE	Location/Qualifiers	
misc_feature	1..20	note = source = /note="target seq for Acute myeloid leukemia"
source	1..20	mol_type = genomic DNA organism = Homo sapiens
SEQUENCE: 203		
gcatgggcgg catgaaccgg		20
SEQ ID NO: 204	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	note = source = /note="target seq for Cutaneous malignant melanoma 3"
source	1..20	mol_type = genomic DNA organism = Homo sapiens
SEQUENCE: 204		

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gtggccactg tggggatcac	20
SEQ ID NO: 205	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
	note = source = /note="target seq for Alpha-1-antitrypsin deficiency"
source	1..20
	mol_type = genomic DNA
	organism = Homo sapiens
SEQUENCE: 205	
gtgctgacca tcgacgagaa	20
SEQ ID NO: 206	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
	note = source = /note="target seq for Charcot-Marie-Tooth disease, type 2"
source	1..20
	mol_type = genomic DNA
	organism = Homo sapiens
SEQUENCE: 206	
ggcgagctgc atgatctgcg	20
SEQ ID NO: 207	moltype = DNA length = 25
FEATURE	Location/Qualifiers
misc_feature	1..25
	note = source = /note="target seq for Hereditary cancer-predisposing syndrome"
source	1..25
	mol_type = genomic DNA
	organism = Homo sapiens
SEQUENCE: 207	
gcacatgacg gaggttgtga ggcgc	25
SEQ ID NO: 208	moltype = DNA length = 25
FEATURE	Location/Qualifiers
misc_feature	1..25
	note = source = /note="target seq for Hereditary cancer-predisposing syndrome"
source	1..25
	mol_type = genomic DNA
	organism = Homo sapiens
SEQUENCE: 208	
gaggttgtga ggcaactgccc ccacc	25
SEQ ID NO: 209	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
	note = source = /note="target seq for Hereditary cancer-predisposing syndrome"
source	1..20
	mol_type = genomic DNA
	organism = Homo sapiens
SEQUENCE: 209	
gcagcatccg gctgcaggta	20
SEQ ID NO: 210	moltype = DNA length = 25
FEATURE	Location/Qualifiers
misc_feature	1..25
	note = source = /note="target seq for Brugada syndrome"
source	1..25
	mol_type = genomic DNA
	organism = Homo sapiens
SEQUENCE: 210	
ggccaagggg atccgcacgc tgctc	25
SEQ ID NO: 211	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
	note = source = /note="target seq for Brugada syndrome"
source	1..20
	mol_type = genomic DNA
	organism = Homo sapiens
SEQUENCE: 211	
gcctgatgac gcaggactgc	20

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SEQ ID NO: 212      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                    note = source = /note="target seq for GRACILE syndrome"
source              1..20
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 212
ggtagcgaagt ctcgacactg                                20

SEQ ID NO: 213      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                    note = source = /note="target seq for Enhanced s-cone
                    syndrome"
source              1..20
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 213
ggcattggcg gtggacccca                                20

SEQ ID NO: 214      moltype = DNA  length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="target seq for Charcot-Marie-Tooth
                    disease, type 2"
source              1..25
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 214
gaagaacagg ttctggacgt caaag                            25

SEQ ID NO: 215      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                    note = source = /note="target seq for Charcot-Marie-Tooth
                    disease, type 2"
source              1..20
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 215
gaacagggttc tggacgtcaa                                20

SEQ ID NO: 216      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                    note = source = /note="target seq for Arylsulfatase a,
                    allele a"
source              1..20
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 216
gacaggtcac agagcagcgg                                20

SEQ ID NO: 217      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                    note = source = /note="target seq for Familial
                    hypercholesterolemia"
source              1..20
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 217
gaacgtggtc gctctggaca                                20

SEQ ID NO: 218      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                    note = source = /note="target seq for Familial
                    hypercholesterolemia"
source              1..20
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 218
gaacgtggtc gctctggaca                                20

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SEQ ID NO: 219      moltype = DNA  length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="target seq for HEMOGLOBIN ARLINGTON
                    PARK"
source              1..25
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 219
ggcgcatctg actcctgagg agaag                               25

SEQ ID NO: 220      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                    note = source = /note="target seq for Familial hypertrophic
                    cardiomyopathy 1"
source              1..20
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 220
gaagtcgag gctcgccgca                                       20

SEQ ID NO: 221      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                    note = source = /note="target seq for Cardiovascular
                    phenotype"
source              1..20
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 221
gtcaagggtca tcggtgaggc                                       20

SEQ ID NO: 222      moltype = DNA  length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="target seq for Cardiovascular
                    phenotype"
source              1..25
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 222
ggcgcatcagt gaggccggcc ggggt                               25

SEQ ID NO: 223      moltype = DNA  length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="target seq for Brugada syndrome"
source              1..25
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 223
gtctcagcct tacgcacctt ccgag                               25

SEQ ID NO: 224      moltype = DNA  length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="target seq for Brugada syndrome"
source              1..25
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 224
gcaccttcca agtcctccgg gccct                               25

SEQ ID NO: 225      moltype = DNA  length = 26
FEATURE            Location/Qualifiers
misc_feature        1..26
                    note = source = /note="target seq for Hereditary
                    cancer-predisposing syndrome"
source              1..26
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 225
gaagccagcc cctcagggca actgac                               26

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SEQ ID NO: 226	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Deficiency of butyryl-CoA dehydrogenase"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 226		
ggcgactcat gggttctgaa		20
SEQ ID NO: 227	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Deficiency of butyryl-CoA dehydrogenase"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 227		
gcgactcacg ggttctgaat ggaac		25
SEQ ID NO: 228	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Benign scapuloperoneal muscular dystrophy with cardiomyopathy"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 228		
ggaactgcgg gcccgcatg		20
SEQ ID NO: 229	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Benign scapuloperoneal muscular dystrophy with cardiomyopathy"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 229		
ggacgagtac caggagcttc		20
SEQ ID NO: 230	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Cone-rod dystrophy 6"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 230		
ggatctgac cgaggagcgca		20
SEQ ID NO: 231	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Cone-rod dystrophy 6"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 231		
gaatgcacta tggtctattc catcc		25
SEQ ID NO: 232	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Stargardt disease 1"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 232		
gagctgctca caggacgaga acatc		25
SEQ ID NO: 233	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	

-continued

misc_feature	1..20	
	note = source = /note="target seq for Leber congenital amaurosis 2"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 233		
gtctatccag taagtatctc		20
SEQ ID NO: 234	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Cone-rod dystrophy 3"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 234		
ggtacgtcca tgccacacc		20
SEQ ID NO: 235	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Nonsyndromic Oculocutaneous Albinism"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 235		
ggacctttac ggcgtaatcc		20
SEQ ID NO: 236	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Phenylketonuria"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 236		
gtcatccttt ggtgaattac		20
SEQ ID NO: 237	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Phenylketonuria"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 237		
gtattacgtg gcagagagtt ttaat		25
SEQ ID NO: 238	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Breast-ovarian cancer, familial 1"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 238		
ggacgctctt gtattatctg		20
SEQ ID NO: 239	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Hyperphenylalaninemia, non-pku"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 239		
gcccttctca gttcgctacg accca		25
SEQ ID NO: 240	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for	

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source Hyperphenylalaninemia, non-pku"
 1..20
 mol_type = genomic DNA
 organism = Homo sapiens

SEQUENCE: 240
 gctacgaccc atacaccaa 20

SEQ ID NO: 241 moltype = DNA length = 20
 FEATURE Location/Qualifiers
 misc_feature 1..20
 note = source = /note="target seq for Hereditary
 cancer-predisposing syndrome"
 source 1..20
 mol_type = genomic DNA
 organism = Homo sapiens

SEQUENCE: 241
 gcgctatggt ctattccatc 20

SEQ ID NO: 242 moltype = DNA length = 25
 FEATURE Location/Qualifiers
 misc_feature 1..25
 note = source = /note="target seq for Hereditary
 cancer-predisposing syndrome"
 source 1..25
 mol_type = genomic DNA
 organism = Homo sapiens

SEQUENCE: 242
 ggggctggta ttcataaaag gcaac 25

SEQ ID NO: 243 moltype = DNA length = 20
 FEATURE Location/Qualifiers
 misc_feature 1..20
 note = source = /note="target seq for Hereditary
 cancer-predisposing syndrome"
 source 1..20
 mol_type = genomic DNA
 organism = Homo sapiens

SEQUENCE: 243
 gcgctatggt ctattccatc 20

SEQ ID NO: 244 moltype = DNA length = 25
 FEATURE Location/Qualifiers
 misc_feature 1..25
 note = source = /note="target seq for Hereditary
 cancer-predisposing syndrome"
 source 1..25
 mol_type = genomic DNA
 organism = Homo sapiens

SEQUENCE: 244
 gtgttctcag gttatcgga agttt 25

SEQ ID NO: 245 moltype = DNA length = 25
 FEATURE Location/Qualifiers
 misc_feature 1..25
 note = source = /note="target seq for Hereditary
 cancer-predisposing syndrome"
 source 1..25
 mol_type = genomic DNA
 organism = Homo sapiens

SEQUENCE: 245
 ggaaatgctg ttagtcgga tgtcg 25

SEQ ID NO: 246 moltype = DNA length = 25
 FEATURE Location/Qualifiers
 misc_feature 1..25
 note = source = /note="target seq for Hereditary
 cancer-predisposing syndrome"
 source 1..25
 mol_type = genomic DNA
 organism = Homo sapiens

SEQUENCE: 246
 ggaaatgctg ttagtcagga tgtcg 25

SEQ ID NO: 247 moltype = DNA length = 20
 FEATURE Location/Qualifiers
 misc_feature 1..20

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	note = source = /note="target seq for Anterior segment dysgenesis 6"	
source	1..20 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 247		
gtggccactg atcggaacg		20
SEQ ID NO: 248	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Brugada syndrome"	
source	1..25 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 248		
gaacctgata ctggccgtgg tcgca		25
SEQ ID NO: 249	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Brugada syndrome"	
source	1..20 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 249		
gatacctggcc gtggtcgcaa		20
SEQ ID NO: 250	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Ornithine carbamoyltransferase deficiency"	
source	1..25 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 250		
gaagggccat gacattctca ctcta		25
SEQ ID NO: 251	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Osteogenesis imperfecta type I"	
source	1..25 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 251		
ggtgctcgag gattgcccgg aacag		25
SEQ ID NO: 252	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Osteogenesis imperfecta type I"	
source	1..25 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 252		
gctcgaggat tgcccgaac agctg		25
SEQ ID NO: 253	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Constipation"	
source	1..25 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 253		
gattccagtt aaatggatgg caatt		25
SEQ ID NO: 254	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Dopamine beta hydroxylase deficiency"	

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source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 254		
ggacactgcc tatattgegg ttagt		25
SEQ ID NO: 255	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Cystic fibrosis"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 255		
gacttcaccc aggtatgtaa aaata		25
SEQ ID NO: 256	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Phenylketonuria"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 256		
gatgtaaacc tgacccacac tgaat		25
SEQ ID NO: 257	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Deficiency of UDPglucose-hexose-1-phosphate uridylyltransferase"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 257		
gagcccctga cacccttacc		20
SEQ ID NO: 258	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Amyloid Cardiomyopathy, Transthyretin-related"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 258		
gtcaccaatc ccaaggaatg		20
SEQ ID NO: 259	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Carbohydrate-deficient glycoprotein syndrome type I"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 259		
ggctactcca tgacagcgcc ttagg		25
SEQ ID NO: 260	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Metachromatic leukodystrophy"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 260		
gcattccgta ctcccacgac cagat		25
SEQ ID NO: 261	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Smith-Lemli-Opitz syndrome"	
source	1..25	

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	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 261		
gttcttcaat gggcgcccg ggatc		25
SEQ ID NO: 262	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Deafness, autosomal recessive 1A"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 262		
gcaggtgagc ccgccggccc		20
SEQ ID NO: 263	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Congenital omphalocele"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 263		
gccagcgctc ctagtggcca tgcac		25
SEQ ID NO: 264	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Congenital omphalocele"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 264		
gcgctcctag tggccatgca cgtgg		25
SEQ ID NO: 265	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Congenital omphalocele"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 265		
gtccacgccca gcgctcctag		20
SEQ ID NO: 266	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Congenital omphalocele"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 266		
gctcctagtgc gccatgcacg		20
SEQ ID NO: 267	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Congenital myotonia, autosomaldominant form"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 267		
gagaggattc tttgcagcca cgttc		25
SEQ ID NO: 268	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Breast-ovarian cancer, familial 1"	

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source	1..20 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 268		
ggtgacccga gagtgggtgt		20
SEQ ID NO: 269	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20 note = source = /note="target seq for Breast and/or ovarian cancer"	
source	1..20 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 269		
gatgcctgga cagaggacaa		20
SEQ ID NO: 270	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20 note = source = /note="target seq for Breast-ovarian cancer, familial 1"	
source	1..20 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 270		
ggacaatggc ttccatggta		20
SEQ ID NO: 271	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20 note = source = /note="target seq for Breast-ovarian cancer, familial 1"	
source	1..20 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 271		
gggctagaaa tctgttgcta		20
SEQ ID NO: 272	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20 note = source = /note="target seq for Breast-ovarian cancer, familial 1"	
source	1..20 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 272		
ggtgacccga gagtgggtgt		20
SEQ ID NO: 273	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20 note = source = /note="target seq for Breast-ovarian cancer, familial 1"	
source	1..20 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 273		
gattctgcaa ctttcaattg		20
SEQ ID NO: 274	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20 note = source = /note="target seq for Breast-ovarian cancer, familial 1"	
source	1..20 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 274		
ggacaatggc ttccatggta		20
SEQ ID NO: 275	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25 note = source = /note="target seq for Inborn genetic	

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source	diseases" 1..25 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 275		
gaaagcccgga cttatagcca gtaat		25
SEQ ID NO: 276	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20 note = source = /note="target seq for Breast-ovarian cancer, familial 2"	
source	1..20 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 276		
gattacatga acaaatgggc		20
SEQ ID NO: 277	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20 note = source = /note="target seq for Breast-ovarian cancer, familial 2"	
source	1..20 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 277		
gattacatga acaaatgggc		20
SEQ ID NO: 278	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20 note = source = /note="target seq for Breast-ovarian cancer, familial 2"	
source	1..20 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 278		
gactggaaaa ggaatacagt		20
SEQ ID NO: 279	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25 note = source = /note="target seq for Breast and/or ovarian cancer"	
source	1..25 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 279		
gttggtgat ggtggatggc tcata		25
SEQ ID NO: 280	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25 note = source = /note="target seq for Breast-ovarian cancer, familial 2"	
source	1..25 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 280		
gttggtgat ggtggatggc tcata		25
SEQ ID NO: 281	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20 note = source = /note="target seq for Breast-ovarian cancer, familial 2"	
source	1..20 mol_type = genomic DNA organism = Homo sapiens	
SEQUENCE: 281		
ggtcagaaga ttattcttca		20
SEQ ID NO: 282	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	

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	note = source = /note="target seq for Breast-ovarian cancer, familial 2"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 282		
ggcttctagt ctcttttggt		20
SEQ ID NO: 283	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Breast-ovarian cancer, familial 2"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 283		
ggcttctagt ctcttttggt		20
SEQ ID NO: 284	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Smith-Lemli-Opitz syndrome"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 284		
gttgcgattt tgcagccatt		20
SEQ ID NO: 285	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Alport syndrome 1, X-linked recessive"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 285		
ggcctatggg tccccctggt ttcgg		25
SEQ ID NO: 286	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Acute neuronopathic Gaucher's disease"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 286		
ggcatcaggt gagtgagtca		20
SEQ ID NO: 287	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Gonadotropin deficiency"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 287		
gtaactctcc agcataccat		20
SEQ ID NO: 288	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Distal arthrogryposis type 1A"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 288		
gaagctggag gaggccgaga		20
SEQ ID NO: 289	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	

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misc_feature	1..20	
	note = source = /note="target seq for Distal arthrogryposis type 1A"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 289		
gctggaggag gccgagaagg		20
SEQ ID NO: 290	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Hereditary cancer-predisposing syndrome"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 290		
gctgccacct tcagggcctc		20
SEQ ID NO: 291	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Inborn genetic diseases"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 291		
gataaacttc ttgaagagga		20
SEQ ID NO: 292	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Death in early adulthood"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 292		
gtgttctcca tggtcgaaca gaccc		25
SEQ ID NO: 293	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Death in early adulthood"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 293		
gttcgaacag acccaaatcc		20
SEQ ID NO: 294	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Severe autosomal recessive muscular dystrophy of childhood - North African type"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 294		
gaaatctgtg tgtgtccaga		20
SEQ ID NO: 295	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Cardiovascular phenotype"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 295		
ggtcacatt gagagcgacc		20

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SEQ ID NO: 296	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for	
	Carbohydrate-deficient glycoproteinsyndrome type I"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 296		
gaatgctgaa gatgatcgac		20
SEQ ID NO: 297	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Charcot-Marie-Tooth	
	disease, type I"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 297		
gaatgctgaa gatgatcgac		20
SEQ ID NO: 298	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Inborn genetic	
	diseases"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 298		
gctcagtaca ccaccgtcgg ccgca		25
SEQ ID NO: 299	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Inborn genetic	
	diseases"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 299		
gtacaccacc gtcggccgca		20
SEQ ID NO: 300	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Familial	
	Mediterranean fever"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 300		
ggacacgtga tggagggaag aacac		25
SEQ ID NO: 301	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Deafness, autosomal	
	recessive 2"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 301		
ggacctttgt caatgggaca cggac		25
SEQ ID NO: 302	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Deafness, autosomal	
	recessive 2"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 302		
ggacctttgt caatgggaca cagac		25

SEQ ID NO: 303	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	note = source = /note="target seq for Deafness, X-linked 2"
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 303		
gcgctcctag tggccatgca cgtgg		25
SEQ ID NO: 304	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	note = source = /note="target seq for Enlarged vestibular aqueduct"
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 304		
gccaccactg ctctttcccg cacgg		25
SEQ ID NO: 305	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	note = source = /note="target seq for Congenital sensorineuralhearing impairment"
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 305		
gtgggtcatta acacaaactc		20
SEQ ID NO: 306	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	note = source = /note="target seq for Deficiency ofUDPglucose-hexose-1-phosphate uridylyltransferase"
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 306		
gccagatatt gccccagctg		20
SEQ ID NO: 307	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	note = source = /note="target seq for Very long chain acyl-CoAdehydrogenase deficiency"
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 307		
gccactaatc gtaccagtt		20
SEQ ID NO: 308	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	note = source = /note="target seq for Cardiovascular phenotype"
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 308		
gatattccac tggtggtcga		20
SEQ ID NO: 309	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	note = source = /note="target seq for Pyruvate kinase deficiency of red cells"
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 309		
qqaagactcc tqgcataag		20

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SEQ ID NO: 310	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Autosomal dominant progressiveexternal ophthalmoplegia with mitochondrial DNAdeletions 1"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 310		
gaagtcgttg atggatctgg ccaat		25
SEQ ID NO: 311	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Very long chain acyl-CoAdehydrogenase deficiency"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 311		
gaagatcaca gcttttgtgg		20
SEQ ID NO: 312	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Cystinosis, ocular nonnephropathic"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 312		
gtccatctcc ttctaccctc		20
SEQ ID NO: 313	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Pyruvate kinase deficiency of redcells"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 313		
gctcagccca gcttctgtct		20
SEQ ID NO: 314	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Distal myopathy, Tateyama type"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 314		
gattgacctg gtgaaccaag acccc		25
SEQ ID NO: 315	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Malignant hyperthermia,susceptibility to, 1"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 315		
ggccctgcgg atccgcgccca tcctc		25
SEQ ID NO: 316	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Malignant hyperthermia,susceptibility to, 1"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	

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SEQUENCE: 316
ggatccgcgc catctccgc tcctc                                     25

SEQ ID NO: 317      moltype = DNA  length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="target seq for Myopathy, Central
                    Core"
source              1..25
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 317
ggtctgatca tcgacgcttt tgggtg                                    25

SEQ ID NO: 318      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                    note = source = /note="target seq for Myopathy, Central
                    Core"
source              1..20
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 318
ggtctgatca tcgacgcttt                                     20

SEQ ID NO: 319      moltype = DNA  length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="target seq for Myopathy, Central
                    Core"
source              1..25
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 319
gatcatcgac accttttggtg agctc                                    25

SEQ ID NO: 320      moltype = DNA  length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="target seq for Malignant
                    hyperthermia,susceptibility to, 1"
source              1..25
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 320
ggcggaggca ttggggacga gatcg                                     25

SEQ ID NO: 321      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                    note = source = /note="target seq for Ceroid lipofuscinosis
                    neuronal 2"
source              1..20
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 321
gtgacagtgg ggcggggtgt                                         20

SEQ ID NO: 322      moltype = DNA  length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="target seq for Ceroid lipofuscinosis
                    neuronal 2"
source              1..25
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 322
ggggccgggt attggtctgt ctctg                                    25

SEQ ID NO: 323      moltype = DNA  length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="target seq for Niemann-Pick disease
                    type C1"
source              1..25
                    mol_type = genomic DNA

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                organism = Homo sapiens
SEQUENCE: 323
ggatcgacga ttatttcgac tgggt                25

SEQ ID NO: 324      moltype = DNA  length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="target seq for Glutaric aciduria,
                    type 1"
source              1..25
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 324
gagtatcacg tgatccggca cgcca                25

SEQ ID NO: 325      moltype = DNA  length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="target seq for CAPN3-Related
                    Disorders"
source              1..25
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 325
gccctgatgc agaagaaccg gcgga                25

SEQ ID NO: 326      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                    note = source = /note="target seq for CAPN3-Related
                    Disorders"
source              1..20
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 326
gatgcagaag aaccggcgga                20

SEQ ID NO: 327      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                    note = source = /note="target seq for Glycogen storage
                    disease, type II"
source              1..20
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 327
gtcccagaac agctcccctc                20

SEQ ID NO: 328      moltype = DNA  length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="target seq for Nonsyndromic
                    Oculocutaneous Albinism"
source              1..25
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 328
gtacagggat ctgccaacga tccta                25

SEQ ID NO: 329      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                    note = source = /note="target seq for Familial
                    hypercholesterolemia"
source              1..20
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 329
gcgaagatgg ctcggtgag                20

SEQ ID NO: 330      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                    note = source = /note="target seq for Familial
                    hypercholesterolemia"
source              1..20

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SEQUENCE: 330	mol_type = genomic DNA	
gagtggccgc agcgctgtag	organism = Homo sapiens	20
SEQ ID NO: 331	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Deafness, autosomal recessive 7"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 331		
ggtctcaagc tctctcttcg		20
SEQ ID NO: 332	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Chronic infantile neurological, cutaneous and articular syndrome"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 332		
gacggcttcg atgagctgca		20
SEQ ID NO: 333	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Eichsfeld type congenital muscular dystrophy"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 333		
gactctccgg gagactgtcc		20
SEQ ID NO: 334	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Inborn genetic diseases"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 334		
gctcaaaagt ggatgagcgc		20
SEQ ID NO: 335	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Hyperkalemic Periodic ParalysisType 1"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 335		
gcagctgcgg gtcttcaagc		20
SEQ ID NO: 336	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Inclusion body myopathy 2"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 336		
ggacgtggat gtggtggttt		20
SEQ ID NO: 337	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Severe APG05083.1, APG07433.1, APG07513.1, APG08290.1 immunodeficiency due to	

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source	ADADeficiency"	
	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 337		
gaagagcggc attcaccgta ctgtc		25
SEQ ID NO: 338	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Severe APG05083.1, APG07433.1, APG07513.1, APG08290.1 immunodeficiency due to ADADeficiency"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 338		
gcggtacagt ccgcacctgc tggcc		25
SEQ ID NO: 339	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Cardiovascular phenotype"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 339		
gtgtccgagc agcgccactg		20
SEQ ID NO: 340	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Cystic fibrosis"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 340		
ggacacttcg tgccttcgga		20
SEQ ID NO: 341	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Cystic fibrosis"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 341		
ggacacttcg tgccttcgga		20
SEQ ID NO: 342	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Adrenocortical carcinoma, pediatric"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 342		
gggcgtgagc acttcagat gttcc		25
SEQ ID NO: 343	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Fumarase deficiency"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 343		
gattggacgt actcatactc		20
SEQ ID NO: 344	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Adenocarcinoma of prostate"	

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source          1..20
                 mol_type = genomic DNA
                 organism = Homo sapiens

SEQUENCE: 344
gaacactgtc cattggcatg                               20

SEQ ID NO: 345    moltype = DNA length = 20
FEATURE          Location/Qualifiers
misc_feature      1..20
                  note = source = /note="target seq for Familial hypertrophic
                  cardiomyopathy 1"

source          1..20
                 mol_type = genomic DNA
                 organism = Homo sapiens

SEQUENCE: 345
gcatacctcta cggggacttc                               20

SEQ ID NO: 346    moltype = DNA length = 25
FEATURE          Location/Qualifiers
misc_feature      1..25
                  note = source = /note="target seq for Adult
                  hypophosphatasia"

source          1..25
                 mol_type = genomic DNA
                 organism = Homo sapiens

SEQUENCE: 346
gacaacgaga tgccccctga ggcct                          25

SEQ ID NO: 347    moltype = DNA length = 25
FEATURE          Location/Qualifiers
misc_feature      1..25
                  note = source = /note="target seq for Adult
                  hypophosphatasia"

source          1..25
                 mol_type = genomic DNA
                 organism = Homo sapiens

SEQUENCE: 347
ggtactcaga caacaagatg ccccc                          25

SEQ ID NO: 348    moltype = DNA length = 25
FEATURE          Location/Qualifiers
misc_feature      1..25
                  note = source = /note="target seq for Adult
                  hypophosphatasia"

source          1..25
                 mol_type = genomic DNA
                 organism = Homo sapiens

SEQUENCE: 348
gacaacaaga tgccccctga ggcct                          25

SEQ ID NO: 349    moltype = DNA length = 25
FEATURE          Location/Qualifiers
misc_feature      1..25
                  note = source = /note="target seq for Inborn genetic
                  diseases"

source          1..25
                 mol_type = genomic DNA
                 organism = Homo sapiens

SEQUENCE: 349
gccttcttgg acaacgtcac cacca                          25

SEQ ID NO: 350    moltype = DNA length = 25
FEATURE          Location/Qualifiers
misc_feature      1..25
                  note = source = /note="target seq for Inborn genetic
                  diseases"

source          1..25
                 mol_type = genomic DNA
                 organism = Homo sapiens

SEQUENCE: 350
ggtttgccat ccgaggggta gtgct                          25

SEQ ID NO: 351    moltype = DNA length = 20
FEATURE          Location/Qualifiers
misc_feature      1..20
                  note = source = /note="target seq for Crouzon syndrome"

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source          1..20
                mol_type = genomic DNA
                organism = Homo sapiens

SEQUENCE: 351
gcaaatgcct ccacagtggg                                20

SEQ ID NO: 352    moltype = DNA length = 20
FEATURE          Location/Qualifiers
misc_feature      1..20
                  note = source = /note="target seq for Propionyl-CoA
                  carboxylase deficiency"

source          1..20
                mol_type = genomic DNA
                organism = Homo sapiens

SEQUENCE: 352
gacatcatag gcacctccat                                20

SEQ ID NO: 353    moltype = DNA length = 25
FEATURE          Location/Qualifiers
misc_feature      1..25
                  note = source = /note="target seq for Cardiovascular
                  phenotype"

source          1..25
                mol_type = genomic DNA
                organism = Homo sapiens

SEQUENCE: 353
ggtggaaggg atcggctacg acttc                            25

SEQ ID NO: 354    moltype = DNA length = 25
FEATURE          Location/Qualifiers
misc_feature      1..25
                  note = source = /note="target seq for Dysostosis multiplex"

source          1..25
                mol_type = genomic DNA
                organism = Homo sapiens

SEQUENCE: 354
ggagcagctc tgggccgaag tgctg                            25

SEQ ID NO: 355    moltype = DNA length = 20
FEATURE          Location/Qualifiers
misc_feature      1..20
                  note = source = /note="target seq for Multiple sulfatase
                  deficiency"

source          1..20
                mol_type = genomic DNA
                organism = Homo sapiens

SEQUENCE: 355
ggcgactcct ttgtctttga                                20

SEQ ID NO: 356    moltype = DNA length = 20
FEATURE          Location/Qualifiers
misc_feature      1..20
                  note = source = /note="target seq for Bifunctional
                  peroxisomal enzymedeficiency"

source          1..20
                mol_type = genomic DNA
                organism = Homo sapiens

SEQUENCE: 356
ggtggtactg gtcaccggcg                                20

SEQ ID NO: 357    moltype = DNA length = 25
FEATURE          Location/Qualifiers
misc_feature      1..25
                  note = source = /note="target seq for Bifunctional
                  peroxisomal enzymedeficiency"

source          1..25
                mol_type = genomic DNA
                organism = Homo sapiens

SEQUENCE: 357
gtactgggtca ccagcgcggg ggcag                            25

SEQ ID NO: 358    moltype = DNA length = 20
FEATURE          Location/Qualifiers
misc_feature      1..20
                  note = source = /note="target seq for Hereditary
                  cancer-predisposingsyndrome"

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source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 358		
ggaaacttgt aagggttcg		20
SEQ ID NO: 359	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Cardiovascular phenotype"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 359		
gtgcagacat tgacgaatgc cgcac		25
SEQ ID NO: 360	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Cardiovascular phenotype"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 360		
gacattgacg aatgccgcat atctc		25
SEQ ID NO: 361	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Limb-girdle muscular dystrophy,type 2L"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 361		
gcatagtagt aaactagacg		20
SEQ ID NO: 362	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Familial hypercholesterolemia"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 362		
gcacgatgt caacgagggc aaccg		25
SEQ ID NO: 363	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Spastic Paraplegia, Recessive"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 363		
gcactgctgc tcggccccc		20
SEQ ID NO: 364	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Isovaleryl-CoA dehydrogenasedeficiency"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 364		
ggagcagtgg ggctcagtta		20
SEQ ID NO: 365	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Familial	

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source	hypercholesterolemia"	
	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 365		
gaggagctgc ctcacaggtg		20
SEQ ID NO: 366	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Biotinidase deficiency"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 366		
gtccagcgc ctgagttgta		20
SEQ ID NO: 367	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Biotinidase deficiency"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 367		
gcctgagttg tatggccatc		20
SEQ ID NO: 368	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Leber congenital amaurosis"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 368		
gcataatttg tacagctcta agagt		25
SEQ ID NO: 369	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Familial hyperinsulinism"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 369		
ggtggctgag cccagcccg cccccc		25
SEQ ID NO: 370	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Familial cancer of breast"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 370		
ggatgcttgg cagtgggaaa aactt		25
SEQ ID NO: 371	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Cohen syndrome"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 371		
ggaggtctac tacaggtctg		20
SEQ ID NO: 372	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Cardiovascular	

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source	phenotype"	
	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 372		
gctccctctg tactgtgcag gagtc		25
SEQ ID NO: 373	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Wilson disease"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 373		
ggcaatgaac acaaagagca		20
SEQ ID NO: 374	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Wilson disease"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 374		
gccctggatg ggctcagcgg ccatg		25
SEQ ID NO: 375	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Wilson disease"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 375		
gcagccctgg atgggctcag		20
SEQ ID NO: 376	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Familial hypercholesterolemia"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 376		
gtggcccagc gaagatgcga		20
SEQ ID NO: 377	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Familial hypercholesterolemia"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 377		
gaagatgcga aggtgatttc		20
SEQ ID NO: 378	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Floating-Harbor syndrome"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 378		
gctgaagctg gagtgggtcg		20
SEQ ID NO: 379	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Congenital long QT syndrome"	
source	1..25	

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	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 379		
gatgctacac gtcgaccgcc aggga		25
SEQ ID NO: 380	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Congenital long QT syndrome"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 380		
gatgctacac gtcgaccgcc		20
SEQ ID NO: 381	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Andersen Tawil syndrome"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 381		
gaggaagaga tcaaggctct ttcca		25
SEQ ID NO: 382	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Cardiovascular phenotype"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 382		
gagctgataa ccaccctgta catcg		25
SEQ ID NO: 383	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Familial hypercholesterolemia"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 383		
ggaaaactgc ggtatgggcg		20
SEQ ID NO: 384	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Cardiovascular phenotype"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 384		
gtatttgaag gtctctccg gggtc		25
SEQ ID NO: 385	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Gastrointestinal stroma tumor"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 385		
gagggctaaa gtgggctcgg		20
SEQ ID NO: 386	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Dyskeratosis congenita"	

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source          1..20
                 mol_type = genomic DNA
                 organism = Homo sapiens

SEQUENCE: 386
gctgccaagc ctgctggcaa                                20

SEQ ID NO: 387    moltype = DNA length = 25
FEATURE          Location/Qualifiers
misc_feature      1..25
                 note = source = /note="target seq for Dyskeratosis
                 congenita"

source          1..25
                 mol_type = genomic DNA
                 organism = Homo sapiens

SEQUENCE: 387
gccttgctg tgacttccag cactg                            25

SEQ ID NO: 388    moltype = DNA length = 25
FEATURE          Location/Qualifiers
misc_feature      1..25
                 note = source = /note="target seq for Glycogen storage
                 disease IIIa"

source          1..25
                 mol_type = genomic DNA
                 organism = Homo sapiens

SEQUENCE: 388
gatgagtaga acagaaaaat acaag                            25

SEQ ID NO: 389    moltype = DNA length = 25
FEATURE          Location/Qualifiers
misc_feature      1..25
                 note = source = /note="target seq for Dilated
                 cardiomyopathy 1DD"

source          1..25
                 mol_type = genomic DNA
                 organism = Homo sapiens

SEQUENCE: 389
gtctcgtagt ccggtgagcc ggtca                            25

SEQ ID NO: 390    moltype = DNA length = 20
FEATURE          Location/Qualifiers
misc_feature      1..20
                 note = source = /note="target seq for Renal carnitine
                 transport defect"

source          1..20
                 mol_type = genomic DNA
                 organism = Homo sapiens

SEQUENCE: 390
gtggccatgg aatagcgccg                                20

SEQ ID NO: 391    moltype = DNA length = 25
FEATURE          Location/Qualifiers
misc_feature      1..25
                 note = source = /note="target seq for Baraitser-Winter
                 syndrome 1"

source          1..25
                 mol_type = genomic DNA
                 organism = Homo sapiens

SEQUENCE: 391
gatcctcacc gagcgcggtt acagc                            25

SEQ ID NO: 392    moltype = DNA length = 25
FEATURE          Location/Qualifiers
misc_feature      1..25
                 note = source = /note="target seq for Very long chain
                 acyl-CoA dehydrogenase deficiency"

source          1..25
                 mol_type = genomic DNA
                 organism = Homo sapiens

SEQUENCE: 392
gggcatgcat gaccttggcg tgggc                            25

SEQ ID NO: 393    moltype = DNA length = 20
FEATURE          Location/Qualifiers
misc_feature      1..20
                 note = source = /note="target seq for Familial

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source	hypercholesterolemia"	
	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 393		
gtcagatttg tccttgccagt		20
SEQ ID NO: 394	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Limb-girdle muscular dystrophy,type 2A"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 394		
gtacgggggtt gctctgccgg		20
SEQ ID NO: 395	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Familial hypercholesterolemia"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 395		
gacagtagcc cctgctcggc cttcg		25
SEQ ID NO: 396	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Aortic aneurysm, familial thoracic 6"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 396		
gccccatgcc atcatgcgtc		20
SEQ ID NO: 397	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Acromicric dysplasia"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 397		
gctttgtcat cgacatttat accgg		25
SEQ ID NO: 398	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Charcot-Marie-Tooth disease type 2C"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 398		
ggcggacatg cggcaccagg actcg		25
SEQ ID NO: 399	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Breast-ovarian cancer, familial 2"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 399		
gtgtaactgt atacgtatgg cgttt		25
SEQ ID NO: 400	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Breast-ovarian	

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source	cancer, familial 1"	
	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 400		
gatgcctgga cagaggacaa		20
SEQ ID NO: 401	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Charcot-Marie-Tooth disease type 2C"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 401		
gcctaactga tgaggagttt cgagg		25
SEQ ID NO: 402	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Charcot-Marie-Tooth disease type 2C"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 402		
gcctaactga tgaggagttt caagg		25
SEQ ID NO: 403	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Hereditary cancer-predisposingsyndrome"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 403		
gagcgctgct cagatagcga		20
SEQ ID NO: 404	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Early infantile epileptencephalopathy"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 404		
gatgatccgc atggaccggc gggga		25
SEQ ID NO: 405	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Early infantile epileptencephalopathy"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 405		
ggatgatccg catggaccgg		20
SEQ ID NO: 406	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Early infantile epileptencephalopathy"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 406		
gcggatgatc cgcattggacc agcgg		25
SEQ ID NO: 407	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	

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source	note = source = /note="target seq for Acromicric dysplasia"	
	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 407		
gtacgtgatc catcctaggt		20
SEQ ID NO: 408	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Hypertrophic cardiomyopathy"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 408		
gacaccaagg tctccctcaa gtggc		25
SEQ ID NO: 409	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Cardiovascular phenotype"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 409		
gctgtccggg gtagcccaaa gatag		25
SEQ ID NO: 410	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Cardiovascular phenotype"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 410		
gtccacgggtg agggggccct ggtgt		25
SEQ ID NO: 411	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Cardiovascular phenotype"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 411		
gcctagactg caggacacag ggact		25
SEQ ID NO: 412	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Familial hypertrophiccardiomyopathy 1"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 412		
gcatctgcag acatagagac ctgtg		25
SEQ ID NO: 413	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Benign scapuloperoneal muscular dystrophy with cardiomyopathy"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 413		
gccctccaag agcttgccgt		20
SEQ ID NO: 414	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	

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	note = source = /note="target seq for Glycogen storage disease, type II"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 414		
ggccctcacc ctgcgctacg cactc		25
SEQ ID NO: 415	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Diffuse mesangial sclerosis"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 415		
gttttaggcac ttgttttacc tgtat		25
SEQ ID NO: 416	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Colobomatous microphthalmia"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 416		
gggtcgaggc ggacgccata		20
SEQ ID NO: 417	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Ataxia-telangiectasia syndrome"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 417		
gtaaatacat atttactact		20
SEQ ID NO: 418	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Familial cancer of breast"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 418		
ggctacagca cacagctcgt		20
SEQ ID NO: 419	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Limb-girdle muscular dystrophy, type 2A"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 419		
gctacgagat gcaaaatgca gtcaa		25
SEQ ID NO: 420	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Hereditary cancer-predisposingsyndrome"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 420		
gccgcttacc aaaaggagta		20
SEQ ID NO: 421	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	

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misc_feature	1..25	
	note = source = /note="target seq for Asymmetric septal hypertrophy"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 421		
gtgctgcagg ttgttggtgc atggc		25
SEQ ID NO: 422	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="target seq for Hereditary cancer-predisposingsyndrome"	
source	1..20	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 422		
gtaattctat aactccttag		20
SEQ ID NO: 423	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Familial hypertrophiccardiomyopathy 2"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 423		
gctaaagtca ccgggcgctg gaaat		25
SEQ ID NO: 424	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Familial hypertrophiccardiomyopathy 2"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 424		
ggctaaagtc accgggcgct agaaa		25
SEQ ID NO: 425	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Familial hypertrophiccardiomyopathy 2"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 425		
gctaaagtca ccgggcgcta gaaat		25
SEQ ID NO: 426	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Erythrocytosis, familial, 2"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 426		
gaaagagcaa tgctccagg ttgtc		25
SEQ ID NO: 427	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="target seq for Death in infancy"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 427		
gtattggaat ccggagcatc cacgt		25
SEQ ID NO: 428	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	

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misc_feature      1..20
                  note = source = /note="target seq for Muscular Diseases"
source            1..20
                  mol_type = genomic DNA
                  organism = Homo sapiens

SEQUENCE: 428
gcaccttccg gatctttgag                                20

SEQ ID NO: 429      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                  note = source = /note="target seq for Familial
                  hypercholesterolemia"
source             1..25
                  mol_type = genomic DNA
                  organism = Homo sapiens

SEQUENCE: 429
gctgcgagca tggggccctg aggct                            25

SEQ ID NO: 430      moltype = DNA length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                  note = source = /note="target seq for Familial cancer of
                  breast"
source             1..20
                  mol_type = genomic DNA
                  organism = Homo sapiens

SEQUENCE: 430
ggcagcggtta cccagtcgga                                20

SEQ ID NO: 431      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                  note = source = /note="target seq for Hereditary
                  cancer-predisposingsyndrome"
source             1..25
                  mol_type = genomic DNA
                  organism = Homo sapiens

SEQUENCE: 431
gtgtaggcaa cttctatagc cacag                            25

SEQ ID NO: 432      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                  note = source = /note="target seq for Familial
                  hypercholesterolemia"
source             1..25
                  mol_type = genomic DNA
                  organism = Homo sapiens

SEQUENCE: 432
ggaactcccg ccaagatcaa gaaag                            25

SEQ ID NO: 433      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                  note = source = /note="target seq for Familial
                  hypercholesterolemia"
source             1..25
                  mol_type = genomic DNA
                  organism = Homo sapiens

SEQUENCE: 433
gcggctgccg gtatctgtgc ctccc                            25

SEQ ID NO: 434      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                  note = source = /note="target seq for Familial
                  hypercholesterolemia"
source             1..25
                  mol_type = genomic DNA
                  organism = Homo sapiens

SEQUENCE: 434
gaatgatctg caggtgagcg tcgcc                            25

SEQ ID NO: 435      moltype = DNA length = 25
FEATURE            Location/Qualifiers

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misc_feature      1..25
                  note = source = /note="target seq for Inclusion body
                  myopathy 2"
source            1..25
                  mol_type = genomic DNA
                  organism = Homo sapiens

SEQUENCE: 435
gtgctgtagt gcaccaatgt aatct                                25

SEQ ID NO: 436      moltype = DNA length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                  note = source = /note="target seq for Ataxia-telangiectasia
                  syndrome"
source              1..20
                  mol_type = genomic DNA
                  organism = Homo sapiens

SEQUENCE: 436
gctttgatgt taacaatcgc                                      20

SEQ ID NO: 437      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                  note = source = /note="target seq for Benign
                  familialneonatal-infantile seizures"
source              1..25
                  mol_type = genomic DNA
                  organism = Homo sapiens

SEQUENCE: 437
gtcagttctc cgatcattcc agctg                                25

SEQ ID NO: 438      moltype = DNA length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                  note = source = /note="target seq for Marfan
                  Syndrome/Loeys-DietzSyndrome/Familial Thoracic Aortic
                  Aneurysms andDissections"
source              1..20
                  mol_type = genomic DNA
                  organism = Homo sapiens

SEQUENCE: 438
gtatctccat tgtctcctcg                                      20

SEQ ID NO: 439      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                  note = source = /note="target seq for Dilated
                  cardiomyopathy 1G"
source              1..25
                  mol_type = genomic DNA
                  organism = Homo sapiens

SEQUENCE: 439
gtttcagatt cccctggccc acgtg                                25

SEQ ID NO: 440      moltype = DNA length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                  note = source = /note="target seq for Familial
                  hypercholesterolemia"
source              1..20
                  mol_type = genomic DNA
                  organism = Homo sapiens

SEQUENCE: 440
gcggtgagtc tcggtgcagg                                      20

SEQ ID NO: 441      moltype = DNA length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                  note = source = /note="target seq for Breast-ovarian
                  cancer, familial 2"
source              1..20
                  mol_type = genomic DNA
                  organism = Homo sapiens

SEQUENCE: 441
gatacagtat taattgactg                                      20

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SEQ ID NO: 442      moltype = DNA length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                    note = source = /note="target seq for Familial
                    hypercholesterolemia"
source              1..20
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 442
gcgagtgcac ccaactccagc                                20

SEQ ID NO: 443      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="target seq for Familial
                    hypercholesterolemia"
source              1..25
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 443
ggctacaagt gccagtgatg ggaag                            25

SEQ ID NO: 444      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="target seq for Familial
                    hypercholesterolemia"
source              1..25
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 444
gggtgggtac aagtgccagc gtgag                            25

SEQ ID NO: 445      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="target seq for Familial
                    hypercholesterolemia"
source              1..25
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 445
gtacaccagc ctcacccccca acctg                          25

SEQ ID NO: 446      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="target seq for Familial cancer of
                    breast"
source              1..25
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 446
gcctgcccga tagacgagcc tcccg                            25

SEQ ID NO: 447      moltype = DNA length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                    note = source = /note="target seq for Hereditary
                    hemorrhagictelangiectasia type 2"
source              1..20
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 447
gatccgcagc gcggtgagtc                                20

SEQ ID NO: 448      moltype = length =
SEQUENCE: 448
000

SEQ ID NO: 449      moltype = length =
SEQUENCE: 449
000

SEQ ID NO: 450      moltype = AA length = 415
FEATURE            Location/Qualifiers

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REGION 1..415
 note = source = /note="Description of Artificial Sequence:
 Syntheticpolypeptide"

REGION 1..415
 note = source = /note="ADAT"

source 1..415
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 450

MAKRPAATKK	AGQAKKKKSE	VEFSHEYWMR	HALTLAKRAW	DEREVPVGAV	LVHNNRVIGE	60
GWNRPIGRHD	PTAHAEIMAL	RQGGGLVMQNY	RLIDATLYVT	LEPCVMCAGA	MIHSRIGRVV	120
FGARDAKTGA	AGSLMDVLHH	PGMNNHRVEIT	EGILADECAA	LLSDFFRMR	QEIKAKKKAQ	180
SSTDGSGSSG	GSSGSETPGT	SESATPESSG	GSSGGSSEVE	FSHEYWMRHA	LTLAKRARDE	240
REVPVGAVLV	LNNRVIGEGW	NRAIGLHDPT	AHAEIMALRQ	GGLVMQNYRL	IDATLYVTFE	300
PCVMCAGAMI	HSRIGRVVFG	VRNAKTGAAG	SLMDVLHYPG	MNNHRVEITEG	ILADECAALL	360
CYFFRMPRVQ	FNAQKKAQSS	TDSGSSSGGS	SGSETPGTSE	SATPESSGGS	SGGSS	415

SEQ ID NO: 451 moltype = AA length = 1494
 FEATURE Location/Qualifiers
 REGION 1..1494
 note = source = /note="Description of Artificial Sequence:
 Syntheticpolypeptide"

REGION 1..1494
 note = source =
 /note="Nuc-ADAT-Linker-dAPG08290.1-Linker-SV40"

source 1..1494
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 451

MAKRPAATKK	AGQAKKKKSE	VEFSHEYWMR	HALTLAKRAW	DEREVPVGAV	LVHNNRVIGE	60
GWNRPIGRHD	PTAHAEIMAL	RQGGGLVMQNY	RLIDATLYVT	LEPCVMCAGA	MIHSRIGRVV	120
FGARDAKTGA	AGSLMDVLHH	PGMNNHRVEIT	EGILADECAA	LLSDFFRMR	QEIKAKKKAQ	180
SSTDGSGSSG	GSSGSETPGT	SESATPESSG	GSSGGSSEVE	FSHEYWMRHA	LTLAKRARDE	240
REVPVGAVLV	LNNRVIGEGW	NRAIGLHDPT	AHAEIMALRQ	GGLVMQNYRL	IDATLYVTFE	300
PCVMCAGAMI	HSRIGRVVFG	VRNAKTGAAG	SLMDVLHYPG	MNNHRVEITEG	ILADECAALL	360
CYFFRMPRVQ	FNAQKKAQSS	TDSGSSSGGS	SGSETPGTSE	SATPESSGGS	SGGSSMSELD	420
YRIGLAIGTN	SIGWGVIELF	WNKDREYER	VRIVDKGVRM	FDKAEIPNKG	ASLAEPERRIA	480
RSSRRRLNRK	SQRKKEIRNL	LVQHGMITQE	ELDLLYPLSK	KSIDIWDIRL	DGLDRLLNHL	540
EWARLLIHLA	QRRGFKSNRK	SELKDAETGK	VLSSIQVNEK	RLFLYRTVGE	MWIKDAEFSK	600
YDRRRNSPNE	YVFSVSRADL	SELKIVTLFEA	QRKFQSSYAS	KNLQETYLQI	WAHQLPFASG	660
NAILNKVGYC	SLLKGKERRI	PKATYTFQYF	SALDQVNRTR	LGPDPFQPTQ	EQKEIILDKM	720
PQRTDYYKKK	TIPEVSYIDI	RKWLLEDETI	QPKGLNYDPN	EELKKIEKKP	FINLKAFYEI	780
KKVVANYAER	TNEAFSTLDY	DAIAYALTVY	KTDKDIRSYL	KKSNNLSKRC	YDDQLIEELF	840
TLSYTKFGHL	SFKAINHVLP	IMQEGRTYQE	AIHQLGYYDT	NLKKENRSMF	LPLIPDEITN	900
PIVKRAITQA	RKVVAIIR	YGSFNSVHIA	LARELSKSHD	ERKKIMTAHD	ENYKKNKGAI	960
SILIEGILN	PTGDIVIRYK	LWKEQGERCA	YSLKEIPDPT	FFNELKKERN	GSPILEVDHI	1020
LPYSQSFDIS	YHNKVLVYSD	ENRNKGNRIP	YTYFLETNKD	WEAFERYVRS	NKLFSKKKRE	1080
YLLKKTYLPR	ESELIKERHL	NDTRYASTFL	KNFIEQNLQF	KEVEVNLRRK	RVQTVNGVIT	1140
AHLRKRWGLE	KRQETYLH	AMDAIIVACT	DHMHVTRITE	YYQIKESNKS	VKKPYFMPMPW	1200
EGFRDELLSH	LASQPIAKKI	SEELKAGYQS	SDYIFVSRMP	KRSVTGAAMD	QTIRRKGGID	1260
KKGKTIIIRK	VRDKIKFDE	NGDFKVMGKE	QDLATYEAIK	QRYLEHRKNS	KKAFETPLYK	1320
PSKKGTEGNI	KRVKLEGGTK	AFVREVNGGV	AQNSDLVRVD	LFEKDDKYVM	VPIYVPDTCV	1380
SELPKKVVKK	GKGYEQWLTL	DNSFTFKSSL	YPYDLVRLVK	GNEDRFLYFG	TLDIDSDRLN	1440
PKDVNPKSKQ	NEYRYSLKI	ENLEKYEYGV	LGDLRLVKQE	TRRIFNRPKK	KRKV	1494

SEQ ID NO: 452 moltype = AA length = 1494
 FEATURE Location/Qualifiers
 REGION 1..1494
 note = source = /note="Description of Artificial Sequence:
 Syntheticpolypeptide"

REGION 1..1494
 note = source =
 /note="Nuc-ADAT-Linker-nAPG08290.1-Linker-SV40"

source 1..1494
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 452

MAKRPAATKK	AGQAKKKKSE	VEFSHEYWMR	HALTLAKRAW	DEREVPVGAV	LVHNNRVIGE	60
GWNRPIGRHD	PTAHAEIMAL	RQGGGLVMQNY	RLIDATLYVT	LEPCVMCAGA	MIHSRIGRVV	120
FGARDAKTGA	AGSLMDVLHH	PGMNNHRVEIT	EGILADECAA	LLSDFFRMR	QEIKAKKKAQ	180
SSTDGSGSSG	GSSGSETPGT	SESATPESSG	GSSGGSSEVE	FSHEYWMRHA	LTLAKRARDE	240
REVPVGAVLV	LNNRVIGEGW	NRAIGLHDPT	AHAEIMALRQ	GGLVMQNYRL	IDATLYVTFE	300
PCVMCAGAMI	HSRIGRVVFG	VRNAKTGAAG	SLMDVLHYPG	MNNHRVEITEG	ILADECAALL	360
CYFFRMPRVQ	FNAQKKAQSS	TDSGSSSGGS	SGSETPGTSE	SATPESSGGS	SGGSSMSELD	420
YRIGLAIGTN	SIGWGVIELF	WNKDREYER	VRIVDKGVRM	FDKAEIPNKG	ASLAEPERRIA	480
RSSRRRLNRK	SQRKKEIRNL	LVQHGMITQE	ELDLLYPLSK	KSIDIWDIRL	DGLDRLLNHL	540

SEQ ID NO: 453	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
source	note = source = /note="Hurler target 1"	
	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 453		
ggagcagctc taggccgaag tgctcg		25
SEQ ID NO: 454	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
source	note = source = /note="Hurler target 2"	
	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 454		
taggccgaag tgctcgaggc cggga		25
SEQ ID NO: 455	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
source	note = source = /note="Hurler target 3"	
	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 455		
gctctaggcc gaagtgtcgc aggcc		25
SEQ ID NO: 456	moltype = DNA length = 135	
FEATURE	Location/Qualifiers	
misc_feature	1..135	
	note = source = /note="Description of Artificial Syntheticpolynucleotide"	
misc_feature	1..135	
source	note = source = /note="sgRNA Hurler target 1"	
	1..135	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 456		
ggagcagctc taggccgaag tgctcggtcat agttccatga aagccaaaag tggctttgat	60	
gtttctatga taagggtttc ggcccggtgc gtcggggatc gcttgcccat tccgatgggc	120	
ttctcccat ttatt	135	
SEQ ID NO: 457	moltype = DNA length = 135	
FEATURE	Location/Qualifiers	
misc_feature	1..135	
	note = source = /note="Description of Artificial Syntheticpolynucleotide"	
misc_feature	1..135	
source	note = source = /note="sgRNA Hurler target 2"	
	1..135	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 457		
taggccgaag tgctcgaggc cgggagtcac agttccatga aagccaaaag tggctttgat	60	
gtttctatga taagggtttc ggcccggtgc gtcggggatc gcttgcccat tccgatgggc	120	
ttctcccat ttatt	135	

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SEQ ID NO: 458      moltype = DNA  length = 135
FEATURE            Location/Qualifiers
misc_feature        1..135
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticpolynucleotide"
misc_feature        1..135
                    note = source = /note="sgRNA Hurler target 3"
source              1..135
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 458
ggcctgcgac acttcggcct agagcgtcat agttccatga aagccaaaag tggctttgat   60
gtttctatga taagggtttc ggcccgtggc gtcggggatc gcctgcccac tccgatgggc   120
ttctcccatc ttatt                                     135

SEQ ID NO: 459      moltype = DNA  length = 15
FEATURE            Location/Qualifiers
misc_feature        1..15
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..15
                    note = source = /note="Hurler Forward Amplification Primer"
source              1..15
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 459
gactccttca ccaag                                     15

SEQ ID NO: 460      moltype = DNA  length = 14
FEATURE            Location/Qualifiers
misc_feature        1..14
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..14
                    note = source = /note="Hurler Reverse Amplification Primer"
source              1..14
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 460
gtagatcagc accg                                     14

SEQ ID NO: 461      moltype = DNA  length = 14
FEATURE            Location/Qualifiers
misc_feature        1..14
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..14
                    note = source = /note="Hurler Wild Type Probe"
source              1..14
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 461
ctctgggccc aagt                                     14

SEQ ID NO: 462      moltype = DNA  length = 14
FEATURE            Location/Qualifiers
misc_feature        1..14
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..14
                    note = source = /note="Hurler W402X Probe-beta"
source              1..14
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 462
ctctaggccc aagt                                     14

SEQ ID NO: 463      moltype = DNA  length = 13
FEATURE            Location/Qualifiers
misc_feature        1..13
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..13
                    note = source = /note="Hurler NGS forward"
source              1..13

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SEQUENCE: 463	mol_type = other DNA	
acttcctcca gcc	organism = synthetic construct	13
SEQ ID NO: 464	moltype = DNA length = 13	
FEATURE	Location/Qualifiers	
misc_feature	1..13	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..13	
	note = source = /note="Hurler NGS Reverse"	
source	1..13	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 464		
gaaccccgcc tta		13
SEQ ID NO: 465	moltype = DNA length = 31	
FEATURE	Location/Qualifiers	
misc_feature	1..31	
	note = source = /note="Human W402X"	
source	1..31	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 465		
ggagcagctc taggccgaag tgtcgcaggc c		31
SEQ ID NO: 466	moltype = DNA length = 31	
FEATURE	Location/Qualifiers	
misc_feature	1..31	
	note = source = /note="Mouse W392X"	
source	1..31	
	mol_type = genomic DNA	
	organism = Mus musculus	
SEQUENCE: 466		
agaacaactc taggcagagg tctcaaaggc t		31
SEQ ID NO: 467	moltype = DNA length = 31	
FEATURE	Location/Qualifiers	
misc_feature	1..31	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..31	
	note = source = /note="humanized mouse"	
source	1..31	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 467		
ggagcagctc taggccgaag tgtcgcaggc c		31
SEQ ID NO: 468	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="FRDA target 1"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 468		
atcacctgag gtccggagtt caaga		25
SEQ ID NO: 469	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="FRDA target 2"	
source	1..25	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 469		
gtcttgaact ccggacctca ggtga		25
SEQ ID NO: 470	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = source = /note="FRDA target 3"	
source	1..25	

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mol_type = genomic DNA
organism = Homo sapiens

SEQUENCE: 470
tgaactccgg acctcaggtg atcca 25

SEQ ID NO: 471      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = source = /note="FRDA target 4"
source              1..25
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 471
gaaaagtttag ccgggcgtgg tgtcg 25

SEQ ID NO: 472      moltype = DNA length = 500
FEATURE            Location/Qualifiers
misc_feature        1..500
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticpolynucleotide"
misc_feature        1..500
                    note = source = /note="BCL11A enhancer region"
source              1..500
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 472
aaaaatggac aattatgagg aggggagagt gcagacaggg gaagcttcac ctcttttaca 60
attttgggag tccacacggc atggcataca aattatttca ttccattga gaaataaaat 120
ccaattctcc atcaccaaga gagccttccg aaagaggccc cctggggcaa acggccaccg 180
atggagaggt ctgccagttc tcttctaccc caccacgccc cccaccctaa tcagaggcca 240
aacccttcct ggagcctgtg ataaaagcaa ctgttagctt gcactagact agcttcaaag 300
ttgtattgac cctgggtgtg tatgtctaag agtagatgcc atatctcttt tctggcctat 360
gttattacct gtagggactt tgcactggaa tcagctatct gctcttactt atgcacacct 420
ggggcataga gccagccctg tatcgctttt cagccatctc actacagata actcccaagt 480
cctgtctagc tgccttcctt

SEQ ID NO: 473      moltype = DNA length = 26
FEATURE            Location/Qualifiers
misc_feature        1..26
                    note = source = /note="SCD target seq 1"
source              1..26
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 473
gcactagact agcttcaaag ttgtag 26

SEQ ID NO: 474      moltype = DNA length = 26
FEATURE            Location/Qualifiers
misc_feature        1..26
                    note = source = /note="SCD target seq 2"
source              1..26
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 474
cctaatacaga ggccaaaccc ttctctg 26

SEQ ID NO: 475      moltype = DNA length = 26
FEATURE            Location/Qualifiers
misc_feature        1..26
                    note = source = /note="SCD target seq 3"
source              1..26
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 475
caagctaaca gttgctttta tcacag 26

SEQ ID NO: 476      moltype = DNA length = 26
FEATURE            Location/Qualifiers
misc_feature        1..26
                    note = source = /note="SCD target seq 4"
source              1..26
                    mol_type = genomic DNA
                    organism = Homo sapiens

SEQUENCE: 476
gcactagact agcttcaaag ttgtag 26

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SEQ ID NO: 477	moltype = DNA length = 26	
FEATURE	Location/Qualifiers	
misc_feature	1..26	
	note = source = /note="SCD target seq 5"	
source	1..26	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 477		
cctaatacaga ggccaaaccc ttctcg		26
SEQ ID NO: 478	moltype = DNA length = 26	
FEATURE	Location/Qualifiers	
misc_feature	1..26	
	note = source = /note="SCD target seq 6"	
source	1..26	
	mol_type = genomic DNA	
	organism = Homo sapiens	
SEQUENCE: 478		
caagctaaca gttgctttta tcacag		26
SEQ ID NO: 479	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
misc_feature	1..22	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..22	
	note = source = /note="RelA FWD"	
source	1..22	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 479		
cttagtttca ccgcaggttc ta		22
SEQ ID NO: 480	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
misc_feature	1..22	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..22	
	note = source = /note="RelA REV"	
source	1..22	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 480		
ctgtgcactc aacactgatc ta		22
SEQ ID NO: 481	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..21	
	note = source = /note="AurkB FWD"	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 481		
cccagcccta ggttgtttat t		21
SEQ ID NO: 482	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
misc_feature	1..23	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..23	
	note = source = /note="AurkB REV"	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 482		
ctggctacat cttccttgac tac		23
SEQ ID NO: 483	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="Description of Artificial Sequence:	

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misc_feature      Syntheticoligonucleotide"
1..20
note = source = /note="HPRT1 FWD"
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 483
gtggcagaag cagtgagtaa                                20

SEQ ID NO: 484      moltype = DNA length = 22
FEATURE            Location/Qualifiers
misc_feature        1..22
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..22
                    note = source = /note="HPRT1 REV"
source              1..22
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 484
tcccatctag gcactaggtta aa                            22

SEQ ID NO: 485      moltype = DNA length = 54
FEATURE            Location/Qualifiers
misc_feature        1..54
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..54
                    note = source = /note="FWD_Guide 134, Guide 135, Guide 136,
                    Guide 137"
source              1..54
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 485
gtctcgtggg ctcgagatg tgtataagag acagacaatt ctctgcctc agcc 54

SEQ ID NO: 486      moltype = DNA length = 53
FEATURE            Location/Qualifiers
misc_feature        1..53
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..53
                    note = source = /note="REV_Guide 134, Guide 135, Guide 136,
                    Guide 137"
source              1..53
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 486
tcgtcggcag cgtcagatgt gtataagaga cagactgccca tgggaagaag gtg 53

SEQ ID NO: 487      moltype = DNA length = 54
FEATURE            Location/Qualifiers
misc_feature        1..54
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..54
                    note = source = /note="FWD_Guide 138, Guide 139, Guide 140,
                    Guide 141"
source              1..54
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 487
gtctcgtggg ctcgagatg tgtataagag acagacaatt ctctgcctc agcc 54

SEQ ID NO: 488      moltype = DNA length = 53
FEATURE            Location/Qualifiers
misc_feature        1..53
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
misc_feature        1..53
                    note = source = /note="REV_Guide 138, Guide 139, Guide 140,
                    Guide 141"
source              1..53
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 488

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tcgtcggcag	cgtcagatgt	gtataagaga	cagactgcc	tggaagaag	gtg	53
SEQ ID NO: 489 moltype = DNA length = 53						
FEATURE Location/Qualifiers						
misc_feature	1..53					
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"					
misc_feature	1..53					
	note = source = /note="REV_Guide 142, Guide 143, Guide 144, Guide 145"					
source	1..53					
	mol_type = other DNA					
	organism = synthetic construct					
SEQUENCE: 489						
tcgtcggcag	cgtcagatgt	gtataagaga	caggagctgc	acatttgacg	agc	53
SEQ ID NO: 490 moltype = DNA length = 54						
FEATURE Location/Qualifiers						
misc_feature	1..54					
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"					
misc_feature	1..54					
	note = source = /note="FWD_Guide 142, Guide 143, Guide 144, Guide 145"					
source	1..54					
	mol_type = other DNA					
	organism = synthetic construct					
SEQUENCE: 490						
gtctcgtggg	ctcggagatg	tgtataagag	acagattaca	ggtgtgagcc	acgg	54
SEQ ID NO: 491 moltype = DNA length = 53						
FEATURE Location/Qualifiers						
misc_feature	1..53					
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"					
misc_feature	1..53					
	note = source = /note="REV_Guide 164, Guide 165, Guide 166, Guide 167"					
source	1..53					
	mol_type = other DNA					
	organism = synthetic construct					
SEQUENCE: 491						
tcgtcggcag	cgtcagatgt	gtataagaga	cagctgacct	caggtgatac	gcc	53
SEQ ID NO: 492 moltype = DNA length = 54						
FEATURE Location/Qualifiers						
misc_feature	1..54					
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"					
misc_feature	1..54					
	note = source = /note="FWD_Guide 164, Guide 165, Guide 166, Guide 167"					
source	1..54					
	mol_type = other DNA					
	organism = synthetic construct					
SEQUENCE: 492						
gtctcgtggg	ctcggagatg	tgtataagag	acagctttgg	gaggctgaga	cagg	54
SEQ ID NO: 493 moltype = DNA length = 53						
FEATURE Location/Qualifiers						
misc_feature	1..53					
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"					
misc_feature	1..53					
	note = source = /note="REV_Guide 168, Guide 169, Guide 170, Guide 171"					
source	1..53					
	mol_type = other DNA					
	organism = synthetic construct					
SEQUENCE: 493						
tcgtcggcag	cgtcagatgt	gtataagaga	cagtgtctta	ttgtccaggc	tgg	53
SEQ ID NO: 494 moltype = DNA length = 54						
FEATURE Location/Qualifiers						
misc_feature	1..54					
	note = source = /note="Description of Artificial Sequence:					

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misc_feature      Syntheticoligonucleotide"
                  1..54
                  note = source = /note="FWD_Guide 168, Guide 169, Guide 170,
                  Guide 171"
source            1..54
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 494
gtctcgtggg ctcggagatg tgtataagag acagtccagc aggtcagcaa agaa      54

SEQ ID NO: 495      moltype = DNA length = 53
FEATURE            Location/Qualifiers
misc_feature        1..53
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..53
                  note = source = /note="REV_Guide 172, Guide 173, Guide 174,
                  Guide 175"
source            1..53
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 495
tcgtcggcag cgtcagatgt gtataagaga caggcagtat aactggccag cct      53

SEQ ID NO: 496      moltype = DNA length = 56
FEATURE            Location/Qualifiers
misc_feature        1..56
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..56
                  note = source = /note="FWD_Guide 172, Guide 173, Guide 174,
                  Guide 175"
source            1..56
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 496
gtctcgtggg ctcggagatg tgtataagag acagtcagtt gaggagtcca gcttaa      56

SEQ ID NO: 497      moltype = DNA length = 53
FEATURE            Location/Qualifiers
misc_feature        1..53
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..53
                  note = source = /note="REV_Guide 185, Guide 186, Guide 187,
                  Guide 188"
source            1..53
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 497
tcgtcggcag cgtcagatgt gtataagaga cagtggcccc tatgtggaga tca      53

SEQ ID NO: 498      moltype = DNA length = 54
FEATURE            Location/Qualifiers
misc_feature        1..54
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..54
                  note = source = /note="FWD_Guide 185, Guide 186, Guide 187,
                  Guide 188"
source            1..54
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 498
gtctcgtggg ctcggagatg tgtataagag acagggcaga gctcagcctc atag      54

SEQ ID NO: 499      moltype = DNA length = 53
FEATURE            Location/Qualifiers
misc_feature        1..53
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..53
                  note = source = /note="REV_Guide 189, Guide 190, Guide 191,
                  Guide 192"
source            1..53
                  mol_type = other DNA

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                                organism = synthetic construct
SEQUENCE: 499
tcgtcggcag cgtcagatgt gtataagaga cagatatccc cacttcccct gct          53

SEQ ID NO: 500                moltype = DNA length = 54
FEATURE                      Location/Qualifiers
misc_feature                  1..54
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
misc_feature                  1..54
                                note = source = /note="FWD_Guide 189, Guide 190, Guide 191,
                                Guide 192"
source                        1..54
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 500
gtctcgtggg ctcgagatg tgtataagag acagcacctc aaggacagct ctgg          54

SEQ ID NO: 501                moltype = DNA length = 53
FEATURE                      Location/Qualifiers
misc_feature                  1..53
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
misc_feature                  1..53
                                note = source = /note="REV_Guide 193, Guide 194, Guide 195,
                                Guide 196"
source                        1..53
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 501
tcgtcggcag cgtcagatgt gtataagaga cagatatccc cacttcccct gct          53

SEQ ID NO: 502                moltype = DNA length = 54
FEATURE                      Location/Qualifiers
misc_feature                  1..54
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
misc_feature                  1..54
                                note = source = /note="FWD_Guide 193, Guide 194, Guide 195,
                                Guide 196"
source                        1..54
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 502
gtctcgtggg ctcgagatg tgtataagag acagcacctc aaggacagct ctgg          54

SEQ ID NO: 503                moltype = DNA length = 54
FEATURE                      Location/Qualifiers
misc_feature                  1..54
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
misc_feature                  1..54
                                note = source = /note="FWD_Guide 146"
source                        1..54
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 503
gtctcgtggg ctcgagatg tgtataagag acagacaatt ctctgcctc agcc          54

SEQ ID NO: 504                moltype = DNA length = 53
FEATURE                      Location/Qualifiers
misc_feature                  1..53
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
misc_feature                  1..53
                                note = source = /note="REV_Guide 146"
source                        1..53
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 504
tcgtcggcag cgtcagatgt gtataagaga cagactgccca tgggaagaag gtg          53

SEQ ID NO: 505                moltype = DNA length = 54
FEATURE                      Location/Qualifiers
misc_feature                  1..54
                                note = source = /note="Description of Artificial Sequence:

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misc_feature      Syntheticoligonucleotide"
1..54
note = source = /note="FWD_Guide 147"
source            1..54
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 505
gtctcgtggg ctcggagatg tgtataagag acagacaatt ctctgcctc agcc      54

SEQ ID NO: 506      moltype = DNA length = 53
FEATURE            Location/Qualifiers
misc_feature        1..53
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..53
                  note = source = /note="REV_Guide 147"
source              1..53
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 506
tcgtcggcag cgtcagatgt gtataagaga cagactgccca tgggaagaag gtc      53

SEQ ID NO: 507      moltype = DNA length = 53
FEATURE            Location/Qualifiers
misc_feature        1..53
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..53
                  note = source = /note="REV_Guide 148"
source              1..53
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 507
tcgtcggcag cgtcagatgt gtataagaga cagactgccca tgggaagaag gtc      53

SEQ ID NO: 508      moltype = DNA length = 54
FEATURE            Location/Qualifiers
misc_feature        1..54
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..54
                  note = source = /note="FWD_Guide 148"
source              1..54
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 508
gtctcgtggg ctcggagatg tgtataagag acagacaatt ctctgcctc agcc      54

SEQ ID NO: 509      moltype = DNA length = 53
FEATURE            Location/Qualifiers
misc_feature        1..53
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..53
                  note = source = /note="REV_Guide 176"
source              1..53
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 509
tcgtcggcag cgtcagatgt gtataagaga caggtagctc actgcagcct caa      53

SEQ ID NO: 510      moltype = DNA length = 54
FEATURE            Location/Qualifiers
misc_feature        1..54
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..54
                  note = source = /note="FWD_Guide 176"
source              1..54
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 510
gtctcgtggg ctcggagatg tgtataagag acagtgggtg atgaacatac gggt      54

SEQ ID NO: 511      moltype = DNA length = 57
FEATURE            Location/Qualifiers

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misc_feature      1..57
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..57
                  note = source = /note="REV_Guide 177"
source            1..57
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 511
tcgtcggcag cgtcagatgt gtataagaga cagtcagact gaagagctat tgtgtga      57

SEQ ID NO: 512      moltype = DNA length = 54
FEATURE            Location/Qualifiers
misc_feature      1..54
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..54
                  note = source = /note="FWD_Guide 177"
source            1..54
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 512
gtctcgtggg ctcgagatg tgtataagag acagccccac aaaccgatgt agct          54

SEQ ID NO: 513      moltype = DNA length = 53
FEATURE            Location/Qualifiers
misc_feature      1..53
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..53
                  note = source = /note="REV_Guide 209"
source            1..53
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 513
tcgtcggcag cgtcagatgt gtataagaga cagtgaggatg cagtgggtgtg atc        53

SEQ ID NO: 514      moltype = DNA length = 54
FEATURE            Location/Qualifiers
misc_feature      1..54
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..54
                  note = source = /note="FWD_Guide 209"
source            1..54
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 514
gtctcgtggg ctcgagatg tgtataagag acagttgagg ctgcagttag ctac          54

SEQ ID NO: 515      moltype = DNA length = 53
FEATURE            Location/Qualifiers
misc_feature      1..53
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..53
                  note = source = /note="REV_Guide 197"
source            1..53
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 515
tcgtcggcag cgtcagatgt gtataagaga cagtggcccc tatgtggaga tca          53

SEQ ID NO: 516      moltype = DNA length = 54
FEATURE            Location/Qualifiers
misc_feature      1..54
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature      1..54
                  note = source = /note="FWD_Guide 197"
source            1..54
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 516
gtctcgtggg ctcgagatg tgtataagag acagggcaga gctcagcctc atag          54

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SEQ ID NO: 517 moltype = DNA length = 53
 FEATURE Location/Qualifiers
 misc_feature 1..53
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 misc_feature 1..53
 note = source = /note="REV_Guide 198"
 source 1..53
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 517
 tcgtcggcag cgtcagatgt gtataagaga cagccaagcc gaccaaaca gtc 53

SEQ ID NO: 518 moltype = DNA length = 54
 FEATURE Location/Qualifiers
 misc_feature 1..54
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 misc_feature 1..54
 note = source = /note="FWD_Guide 198"
 source 1..54
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 518
 gtctcgtggg ctcggagatg tgtataagag acaggatgcg ctgactgata gcct 54

SEQ ID NO: 519 moltype = DNA length = 53
 FEATURE Location/Qualifiers
 misc_feature 1..53
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 misc_feature 1..53
 note = source = /note="REV_Guide 199"
 source 1..53
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 519
 tcgtcggcag cgtcagatgt gtataagaga cagtggcccc tatgtggaga tca 53

SEQ ID NO: 520 moltype = DNA length = 54
 FEATURE Location/Qualifiers
 misc_feature 1..54
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 misc_feature 1..54
 note = source = /note="FWD_Guide 199"
 source 1..54
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 520
 gtctcgtggg ctcggagatg tgtataagag acagggcaga gctcagcctc atag 54

SEQ ID NO: 521 moltype = DNA length = 54
 FEATURE Location/Qualifiers
 misc_feature 1..54
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 misc_feature 1..54
 note = source = /note="FWD_Guide 149"
 source 1..54
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 521
 gtctcgtggg ctcggagatg tgtataagag acagggtaag gggaaactgg aggc 54

SEQ ID NO: 522 moltype = DNA length = 53
 FEATURE Location/Qualifiers
 misc_feature 1..53
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 misc_feature 1..53
 note = source = /note="REV_Guide 149"
 source 1..53
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 522

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tcgtcggcag	cgtcagatgt	gtataagaga	cagtcactt	tctaagcagg	cg	53
SEQ ID NO: 523 moltype = DNA length = 54						
FEATURE Location/Qualifiers						
misc_feature 1..54						
note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"						
misc_feature 1..54						
note = source = /note="FWD_Guide 150"						
source 1..54						
mol_type = other DNA						
organism = synthetic construct						
SEQUENCE: 523						
gtctcgtggg	ctcggagatg	tgtataagag	acaggtgggc	ctggatttcg	atct	54
SEQ ID NO: 524 moltype = DNA length = 53						
FEATURE Location/Qualifiers						
misc_feature 1..53						
note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"						
misc_feature 1..53						
note = source = /note="REV_Guide 150"						
source 1..53						
mol_type = other DNA						
organism = synthetic construct						
SEQUENCE: 524						
tcgtcggcag	cgtcagatgt	gtataagaga	cagatgaaca	ggcaggaagt	cg	53
SEQ ID NO: 525 moltype = DNA length = 53						
FEATURE Location/Qualifiers						
misc_feature 1..53						
note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"						
misc_feature 1..53						
note = source = /note="REV_Guide 151"						
source 1..53						
mol_type = other DNA						
organism = synthetic construct						
SEQUENCE: 525						
tcgtcggcag	cgtcagatgt	gtataagaga	cagcggttct	gagttgcctt	cct	53
SEQ ID NO: 526 moltype = DNA length = 54						
FEATURE Location/Qualifiers						
misc_feature 1..54						
note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"						
misc_feature 1..54						
note = source = /note="FWD_Guide 151"						
source 1..54						
mol_type = other DNA						
organism = synthetic construct						
SEQUENCE: 526						
gtctcgtggg	ctcggagatg	tgtataagag	acagggtaag	gggaaactgg	aggc	54
SEQ ID NO: 527 moltype = DNA length = 53						
FEATURE Location/Qualifiers						
misc_feature 1..53						
note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"						
misc_feature 1..53						
note = source = /note="REV_Guide 179"						
source 1..53						
mol_type = other DNA						
organism = synthetic construct						
SEQUENCE: 527						
tcgtcggcag	cgtcagatgt	gtataagaga	cagtcagtgg	tccacatgca	agt	53
SEQ ID NO: 528 moltype = DNA length = 54						
FEATURE Location/Qualifiers						
misc_feature 1..54						
note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"						
misc_feature 1..54						
note = source = /note="FWD_Guide 179"						
source 1..54						
mol_type = other DNA						

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                                organism = synthetic construct
SEQUENCE: 528
gtctcgtggg ctcggagatg tgtataagag acagaaggca agttgatcgc tcga          54

SEQ ID NO: 529                moltype = DNA  length = 53
FEATURE                      Location/Qualifiers
misc_feature                  1..53
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
misc_feature                  1..53
                                note = source = /note="REV_Guide 180"
source                        1..53
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 529
tcgtcggcag cgtcagatgt gtataagaga cagagtgggt ctgtgcaca cgt          53

SEQ ID NO: 530                moltype = DNA  length = 54
FEATURE                      Location/Qualifiers
misc_feature                  1..54
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
misc_feature                  1..54
                                note = source = /note="FWD_Guide 180"
source                        1..54
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 530
gtctcgtggg ctcggagatg tgtataagag acagtcgctt aagcccagga gttc          54

SEQ ID NO: 531                moltype = DNA  length = 55
FEATURE                      Location/Qualifiers
misc_feature                  1..55
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
misc_feature                  1..55
                                note = source = /note="REV_Guide 181"
source                        1..55
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 531
tcgtcggcag cgtcagatgt gtataagaga cagtctgggt gatccgtact aatcc          55

SEQ ID NO: 532                moltype = DNA  length = 54
FEATURE                      Location/Qualifiers
misc_feature                  1..54
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
misc_feature                  1..54
                                note = source = /note="FWD_Guide 181"
source                        1..54
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 532
gtctcgtggg ctcggagatg tgtataagag acagtaggct caagggatgc tcct          54

SEQ ID NO: 533                moltype = DNA  length = 53
FEATURE                      Location/Qualifiers
misc_feature                  1..53
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
misc_feature                  1..53
                                note = source = /note="REV_Guide 200"
source                        1..53
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 533
tcgtcggcag cgtcagatgt gtataagaga cagccagagc tgtccttgag gtg          53

SEQ ID NO: 534                moltype = DNA  length = 54
FEATURE                      Location/Qualifiers
misc_feature                  1..54
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
misc_feature                  1..54
                                note = source = /note="FWD_Guide 200"

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source 1..54
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 534
 gtctcgtggg ctcggagatg tgtataagag acaggtccat acccaccttg gcaa 54

SEQ ID NO: 535 moltype = DNA length = 53
 FEATURE Location/Qualifiers
 misc_feature 1..53
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 misc_feature 1..53
 note = source = /note="REV_Guide 201"
 source 1..53
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 535
 tcgtcggcag cgtcagatgt gtataagaga cagccaagcc gaccaaaca gtg 53

SEQ ID NO: 536 moltype = DNA length = 54
 FEATURE Location/Qualifiers
 misc_feature 1..54
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 misc_feature 1..54
 note = source = /note="FWD_Guide 201"
 source 1..54
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 536
 gtctcgtggg ctcggagatg tgtataagag acaggatgcg ctgactgata gcct 54

SEQ ID NO: 537 moltype = DNA length = 53
 FEATURE Location/Qualifiers
 misc_feature 1..53
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 misc_feature 1..53
 note = source = /note="REV_Guide 202"
 source 1..53
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 537
 tcgtcggcag cgtcagatgt gtataagaga cagaggctat cagtcagcgc atc 53

SEQ ID NO: 538 moltype = DNA length = 54
 FEATURE Location/Qualifiers
 misc_feature 1..54
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 misc_feature 1..54
 note = source = /note="FWD_Guide 202"
 source 1..54
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 538
 gtctcgtggg ctcggagatg tgtataagag acagagcagg ggaagtgggg atat 54

SEQ ID NO: 539 moltype = DNA length = 54
 FEATURE Location/Qualifiers
 misc_feature 1..54
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 misc_feature 1..54
 note = source = /note="FWD_Guide 152"
 source 1..54
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 539
 gtctcgtggg ctcggagatg tgtataagag acagattaca ggtgtgagcc acgg 54

SEQ ID NO: 540 moltype = DNA length = 53
 FEATURE Location/Qualifiers
 misc_feature 1..53
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"

-continued

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misc_feature      1..53
                  note = source = /note="REV_Guide 152"
source            1..53
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 540
tcgtcggcag cgtcagatgt gtataagaga caggagctgc acatttgacg agc          53

SEQ ID NO: 541      moltype = DNA length = 54
FEATURE            Location/Qualifiers
misc_feature        1..54
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..54
                  note = source = /note="FWD_Guide 153"
source             1..54
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 541
gtctcgtggg ctcggagatg tgtataagag acagacaatt ctctgcctc agcc          54

SEQ ID NO: 542      moltype = DNA length = 53
FEATURE            Location/Qualifiers
misc_feature        1..53
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..53
                  note = source = /note="REV_Guide 153"
source             1..53
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 542
tcgtcggcag cgtcagatgt gtataagaga cagactgccg tgggaagaag gtc          53

SEQ ID NO: 543      moltype = DNA length = 54
FEATURE            Location/Qualifiers
misc_feature        1..54
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..54
                  note = source = /note="FWD_Guide 154"
source             1..54
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 543
gtctcgtggg ctcggagatg tgtataagag acagattaca ggtgtgagcc acgg          54

SEQ ID NO: 544      moltype = DNA length = 53
FEATURE            Location/Qualifiers
misc_feature        1..53
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..53
                  note = source = /note="REV_Guide 154"
source             1..53
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 544
tcgtcggcag cgtcagatgt gtataagaga caggagctgc acatttgacg agc          53

SEQ ID NO: 545      moltype = DNA length = 54
FEATURE            Location/Qualifiers
misc_feature        1..54
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..54
                  note = source = /note="FWD_Guide 182"
source             1..54
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 545
gtctcgtggg ctcggagatg tgtataagag acagtgggtg atgaacatac gggt          54

SEQ ID NO: 546      moltype = DNA length = 53
FEATURE            Location/Qualifiers
misc_feature        1..53

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note = source = /note="Description of Artificial Sequence:
  Syntheticoligonucleotide"
misc_feature      1..53
source            1..53
                  note = source = /note="REV_Guide 182"
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 546
tcgtcggcag cgtcagatgt gtataagaga caggtagctc actgcagcct caa      53

SEQ ID NO: 547      moltype = DNA length = 54
FEATURE            Location/Qualifiers
misc_feature        1..54
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..54
                  note = source = /note="FWD_Guide 183"
source              1..54
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 547
gtctcgtggg ctcggagatg tgtataagag acagtgcctg tagtcccagc tact      54

SEQ ID NO: 548      moltype = DNA length = 53
FEATURE            Location/Qualifiers
misc_feature        1..53
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..53
                  note = source = /note="REV_Guide 183"
source              1..53
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 548
tcgtcggcag cgtcagatgt gtataagaga cagtcagctc agtgcaacct ctg      53

SEQ ID NO: 549      moltype = DNA length = 54
FEATURE            Location/Qualifiers
misc_feature        1..54
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..54
                  note = source = /note="FWD_Guide 184"
source              1..54
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 549
gtctcgtggg ctcggagatg tgtataagag acagctttgg gaggctgaga cagg      54

SEQ ID NO: 550      moltype = DNA length = 53
FEATURE            Location/Qualifiers
misc_feature        1..53
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..53
                  note = source = /note="REV_Guide 184"
source              1..53
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 550
tcgtcggcag cgtcagatgt gtataagaga cagctgacct caggtgatac gcc      53

SEQ ID NO: 551      moltype = DNA length = 54
FEATURE            Location/Qualifiers
misc_feature        1..54
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
misc_feature        1..54
                  note = source = /note="FWD_Guide 203"
source              1..54
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 551
gtctcgtggg ctcggagatg tgtataagag acaggatgcy ctgactgata gcct      54

SEQ ID NO: 552      moltype = DNA length = 53

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FEATURE	Location/Qualifiers	
misc_feature	1..53	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..53	
	note = source = /note="REV_Guide 203"	
source	1..53	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 552		
tcgtcggcag cgtcagatgt	gtataagaga cagccaagcc gaccaaaca gtg	53
SEQ ID NO: 553	moltype = DNA length = 54	
FEATURE	Location/Qualifiers	
misc_feature	1..54	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..54	
	note = source = /note="FWD_Guide 204"	
source	1..54	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 553		
gtctcgtggg ctcggagatg	tgtataagag acagcacctc aaggacagct ctgg	54
SEQ ID NO: 554	moltype = DNA length = 53	
FEATURE	Location/Qualifiers	
misc_feature	1..53	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..53	
	note = source = /note="REV_Guide 204"	
source	1..53	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 554		
tcgtcggcag cgtcagatgt	gtataagaga cagatatccc cacttcccct gct	53
SEQ ID NO: 555	moltype = DNA length = 54	
FEATURE	Location/Qualifiers	
misc_feature	1..54	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..54	
	note = source = /note="FWD_Guide 205"	
source	1..54	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 555		
gtctcgtggg ctcggagatg	tgtataagag acagggcaga gctcagcctc atag	54
SEQ ID NO: 556	moltype = DNA length = 53	
FEATURE	Location/Qualifiers	
misc_feature	1..53	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
misc_feature	1..53	
	note = source = /note="REV_Guide 205"	
source	1..53	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 556		
tcgtcggcag cgtcagatgt	gtataagaga cagtgggccc tatgtggaga tca	53
SEQ ID NO: 557	moltype = length =	
SEQUENCE: 557		
000		
SEQ ID NO: 558	moltype = length =	
SEQUENCE: 558		
000		
SEQ ID NO: 559	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30	
	note = source = /note="Description of Artificial Sequence:	

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misc_feature      Syntheticoligonucleotide"
                  1..30
                  note = source = /note="sequence 2 for cleavage site assay"
source            1..30
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 559
ccatgatata gacgttgtagg ctgttgtagt                               30

SEQ ID NO: 560      moltype = AA length = 6
FEATURE            Location/Qualifiers
REGION            1..6
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic6xHis tag"
source            1..6
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 560
HHHHHHH                                                    6

SEQ ID NO: 561      moltype = AA length = 10
FEATURE            Location/Qualifiers
REGION            1..10
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic10xHis tag"
source            1..10
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 561
HHHHHHHHHHH                                              10

SEQ ID NO: 562      moltype = DNA length = 34
FEATURE            Location/Qualifiers
misc_feature      1..34
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
source            1..34
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 562
cctgggtgtg aggctgggcc attaaaacct ctcc                               34

SEQ ID NO: 563      moltype = DNA length = 15
FEATURE            Location/Qualifiers
misc_feature      1..15
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
source            1..15
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 563
cctgaaaacc tctcc                                              15

SEQ ID NO: 564      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature      1..33
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 564
cctgggtgtg agctgggccca ttaaaacctc tcc                               33

SEQ ID NO: 565      moltype = DNA length = 22
FEATURE            Location/Qualifiers
misc_feature      1..22
                  note = source = /note="Description of Artificial Sequence:
                  Syntheticoligonucleotide"
source            1..22
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 565
cctgggccat taaaacctct cc                                              22

SEQ ID NO: 566      moltype = DNA length = 11

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FEATURE	Location/Qualifiers	
misc_feature	1..11	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
source	1..11	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 566		
cctggctctc c		11
SEQ ID NO: 567	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 567		
cgggccatta aaacctctcc		20
SEQ ID NO: 568	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
misc_feature	1..40	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 568		
cctgggtgtg aggccagacc tgggccatta aaacctctcc		40
SEQ ID NO: 569	moltype = DNA length = 79	
FEATURE	Location/Qualifiers	
misc_feature	1..79	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
source	1..79	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 569		
cctgggtgtg aggggaagct gacgtccttt ccatggctgc tcgcctgtgt tgccaccgct		60
gggccattaa aacctctcc		79
SEQ ID NO: 570	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 570		
cctgggtgtg acctggggcca ttaaaacctc tcc		33
SEQ ID NO: 571	moltype = DNA length = 35	
FEATURE	Location/Qualifiers	
misc_feature	1..35	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
source	1..35	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 571		
cctgggtgtg aggactgggc cattaaaacc tctcc		35
SEQ ID NO: 572	moltype = DNA length = 34	
FEATURE	Location/Qualifiers	
misc_feature	1..34	
	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
source	1..34	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 572		
agttggcaga tgctctaagt tactgccatg ggaa		34

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SEQ ID NO: 573 moltype = DNA length = 31
FEATURE Location/Qualifiers
misc_feature 1..31
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
source 1..31
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 573
agttggcaga tgcaatgtac tgccatggga a 31

SEQ ID NO: 574 moltype = DNA length = 30
FEATURE Location/Qualifiers
misc_feature 1..30
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
source 1..30
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 574
agttggcaga tgcattgtact gccatgggaa 30

SEQ ID NO: 575 moltype = DNA length = 31
FEATURE Location/Qualifiers
misc_feature 1..31
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
source 1..31
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 575
agttggcaga tgctatgtac tgccatggga a 31

SEQ ID NO: 576 moltype = DNA length = 32
FEATURE Location/Qualifiers
misc_feature 1..32
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
source 1..32
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 576
agttggcaga tgctaagtga ctgccatggg aa 32

SEQ ID NO: 577 moltype = DNA length = 31
FEATURE Location/Qualifiers
misc_feature 1..31
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
source 1..31
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 577
agttggcaga tgctatgtaa tgccatggga a 31

SEQ ID NO: 578 moltype = DNA length = 33
FEATURE Location/Qualifiers
misc_feature 1..33
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
source 1..33
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 578
agttggcaga tgctctatgt actgccatgg gaa 33

SEQ ID NO: 579 moltype = DNA length = 39
FEATURE Location/Qualifiers
misc_feature 1..39
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
source 1..39
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 579

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cgacctgaat gctgtgcggc gctctggett cattcaatc 39

SEQ ID NO: 580 moltype = DNA length = 35
 FEATURE Location/Qualifiers
 misc_feature 1..35
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 source 1..35
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 580
 cgacctgaat gctgtgcggc tctgcttcca ggtga 35

SEQ ID NO: 581 moltype = DNA length = 36
 FEATURE Location/Qualifiers
 misc_feature 1..36
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 source 1..36
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 581
 cgacctgaat gctgtgcggc atctgcttcc aggtga 36

SEQ ID NO: 582 moltype = DNA length = 25
 FEATURE Location/Qualifiers
 misc_feature 1..25
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 source 1..25
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 582
 cgacctgaat gctgcttcca ggtga 25

SEQ ID NO: 583 moltype = DNA length = 28
 FEATURE Location/Qualifiers
 misc_feature 1..28
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 source 1..28
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 583
 cgacctgaat gcttctgctt ccagggtga 28

SEQ ID NO: 584 moltype = DNA length = 22
 FEATURE Location/Qualifiers
 misc_feature 1..22
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 source 1..22
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 584
 cgacctgaat gcttcaggt ga 22

SEQ ID NO: 585 moltype = DNA length = 35
 FEATURE Location/Qualifiers
 misc_feature 1..35
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 source 1..35
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 585
 gcgtggggac tacgacctga atgctgtgcg gctct 35

SEQ ID NO: 586 moltype = DNA length = 24
 FEATURE Location/Qualifiers
 misc_feature 1..24
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
 source 1..24
 mol_type = other DNA
 organism = synthetic construct

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SEQUENCE: 586
gcgacctgaa tgctgtgcgg ctct 24

SEQ ID NO: 587 moltype = DNA length = 28
FEATURE Location/Qualifiers
misc_feature 1..28
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
source 1..28
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 587
gcgtggggac tacgagctgt gcggctct 28

SEQ ID NO: 588 moltype = DNA length = 28
FEATURE Location/Qualifiers
misc_feature 1..28
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
source 1..28
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 588
gcgtggggac tgaatgctgt gcggctct 28

SEQ ID NO: 589 moltype = DNA length = 31
FEATURE Location/Qualifiers
misc_feature 1..31
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
source 1..31
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 589
gcgtggggac tcctgaatgc tgtgcggctc t 31

SEQ ID NO: 590 moltype = DNA length = 36
FEATURE Location/Qualifiers
misc_feature 1..36
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
source 1..36
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 590
gcgtggggac tacgaacctg aatgctgtgc ggctct 36

SEQ ID NO: 591 moltype = DNA length = 30
FEATURE Location/Qualifiers
misc_feature 1..30
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
source 1..30
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 591
gcgtggggac tacgaatgct gtgcggctct 30

SEQ ID NO: 592 moltype = DNA length = 32
FEATURE Location/Qualifiers
misc_feature 1..32
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
source 1..32
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 592
gcgtggggac tacctgaatg ctgtgcggct ct 32

SEQ ID NO: 593 moltype = DNA length = 21
FEATURE Location/Qualifiers
misc_feature 1..21
 note = source = /note="Description of Artificial Sequence:
 Syntheticoligonucleotide"
source 1..21
 mol_type = other DNA

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organism = synthetic construct
SEQUENCE: 593
gcgtgaatgc tgtgcggctc t 21

SEQ ID NO: 594      moltype = DNA length = 24
FEATURE            Location/Qualifiers
misc_feature        1..24
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
source              1..24
                    mol_type = other DNA
                    organism = synthetic construct
SEQUENCE: 594
gcgacctgac tgctgtgcgg ctct 24

SEQ ID NO: 595      moltype = DNA length = 34
FEATURE            Location/Qualifiers
misc_feature        1..34
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
source              1..34
                    mol_type = other DNA
                    organism = synthetic construct
SEQUENCE: 595
gcgtggggac tacgcctgaa tgctgtgcgg ctct 34

SEQ ID NO: 596      moltype = DNA length = 42
FEATURE            Location/Qualifiers
misc_feature        1..42
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
source              1..42
                    mol_type = other DNA
                    organism = synthetic construct
SEQUENCE: 596
atggaggagt tggcagatgc tctaattgac tgccatggga ag 42

SEQ ID NO: 597      moltype = DNA length = 40
FEATURE            Location/Qualifiers
misc_feature        1..40
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
source              1..40
                    mol_type = other DNA
                    organism = synthetic construct
SEQUENCE: 597
atggaggagt tggcagatgc taatgtactg ccatgggaag 40

SEQ ID NO: 598      moltype = DNA length = 34
FEATURE            Location/Qualifiers
misc_feature        1..34
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
source              1..34
                    mol_type = other DNA
                    organism = synthetic construct
SEQUENCE: 598
atggaggagt tggcagatgt actgccatgg gaag 34

SEQ ID NO: 599      moltype = DNA length = 18
FEATURE            Location/Qualifiers
misc_feature        1..18
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
source              1..18
                    mol_type = other DNA
                    organism = synthetic construct
SEQUENCE: 599
atgtactgcc atgggaag 18

SEQ ID NO: 600      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
source              1..30

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mol_type = other DNA
organism = synthetic construct
SEQUENCE: 600
atggaggagt tgggtgtactg ccatgggaag 30
SEQ ID NO: 601
FEATURE
misc_feature
note = source = /note="Description of Artificial Sequence:
Syntheticoligonucleotide"
source
mol_type = other DNA
organism = synthetic construct
SEQUENCE: 601
atggaggagt tggcagatgc tctaaatgta ctgccatggg aag 43
SEQ ID NO: 602
FEATURE
misc_feature
note = source = /note="Description of Artificial Sequence:
Syntheticoligonucleotide"
source
mol_type = other DNA
organism = synthetic construct
SEQUENCE: 602
atggaggagt tggcagatgc tcttctaatag tactgccatg ggaag 45
SEQ ID NO: 603
FEATURE
misc_feature
note = source = /note="Description of Artificial Sequence:
Syntheticoligonucleotide"
source
mol_type = other DNA
organism = synthetic construct
SEQUENCE: 603
atggaggagt tggcagatgc cctctaatagt actgccatgg gaag 44
SEQ ID NO: 604
FEATURE
misc_feature
note = source = /note="Description of Artificial Sequence:
Syntheticoligonucleotide"
source
mol_type = other DNA
organism = synthetic construct
SEQUENCE: 604
atggaggagt tggcagataa tgtactgccca tgggaagaag 40
SEQ ID NO: 605
FEATURE
misc_feature
note = source = /note="Description of Artificial Sequence:
Syntheticoligonucleotide"
source
mol_type = other DNA
organism = synthetic construct
SEQUENCE: 605
atggcgtact gccatgggaa gaag 24
SEQ ID NO: 606
FEATURE
misc_feature
note = source = /note="Description of Artificial Sequence:
Syntheticoligonucleotide"
source
mol_type = other DNA
organism = synthetic construct
SEQUENCE: 606
atggaggagt tggcagatgc ttctaatagta ctgccatggg aagaag 46
SEQ ID NO: 607
FEATURE
misc_feature
note = source = /note="Description of Artificial Sequence:
Syntheticoligonucleotide"

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source                1..32
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 607
atggaggagt tggcatctgc catgggaaga ag                               32

SEQ ID NO: 608        moltype = DNA length = 42
FEATURE              Location/Qualifiers
misc_feature          1..42
                      note = source = /note="Description of Artificial Sequence:
                      Syntheticoligonucleotide"

source                1..42
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 608
atggaggagt tggcagatgc aatgtactgc catgggaaga ag                               42

SEQ ID NO: 609        moltype = DNA length = 124
FEATURE              Location/Qualifiers
misc_feature          1..124
                      note = source = /note="Description of Artificial Sequence:
                      Syntheticpolynucleotide"

source                1..124
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 609
atggaggagt tggcagatgc caaactgaaa aacaaatcaa agcactctta ttgagtgtctg   60
cgcatccccg acgccacggg ccgaaaccct tatcatagaa actctaattgt actgccatgg   120
gaag                                           124

SEQ ID NO: 610        moltype = DNA length = 84
FEATURE              Location/Qualifiers
misc_feature          1..84
                      note = source = /note="Description of Artificial Sequence:
                      Syntheticoligonucleotide"

source                1..84
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 610
atggaggagt tggcagatgc tgcttatata gacctccac cgtacacgcc taccgcccat   60
tttctaattgt actgccatgg gaag                                           84

SEQ ID NO: 611        moltype = DNA length = 34
FEATURE              Location/Qualifiers
misc_feature          1..34
                      note = source = /note="Description of Artificial Sequence:
                      Syntheticoligonucleotide"

source                1..34
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 611
atggaggagt tgtctaattgt actgccatgg gaag                               34

SEQ ID NO: 612        moltype = DNA length = 16
FEATURE              Location/Qualifiers
misc_feature          1..16
                      note = source = /note="Description of Artificial Sequence:
                      Syntheticoligonucleotide"

source                1..16
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 612
ctgccatggg aagaag                                           16

SEQ ID NO: 613        moltype = DNA length = 90
FEATURE              Location/Qualifiers
misc_feature          1..90
                      note = source = /note="Description of Artificial Sequence:
                      Syntheticoligonucleotide"

source                1..90
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 613
atggaggagt tggcagatgc gcggtgttc ctgtacagaa ccgtgggcga gatgtggatc   60
aaggatgctc taatgtactg ccatgggaag                                           90

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SEQ ID NO: 614      moltype = DNA  length = 41
FEATURE            Location/Qualifiers
misc_feature        1..41
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source              1..41
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 614
atggaggagt tggcagatgc ctaatgtact gccatgggaa g                41

SEQ ID NO: 615      moltype = DNA  length = 97
FEATURE            Location/Qualifiers
misc_feature        1..97
                     note = source = /note="Description of Artificial Sequence:
                     Syntheticoligonucleotide"
source              1..97
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 615
atggaggagt tggcagatgc tgtcatgac ttttccgct cgtcgtggga cttgctcagt 60
tctctggcca gctcgtctaa tgtactgcc tgggaag                        97

SEQ ID NO: 616      moltype = DNA  length = 41
FEATURE            Location/Qualifiers
misc_feature        1..41
                     note = source = /note="Description of Artificial Sequence:
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source              1..41
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 616
atggaggagt tggcagatgc tctatgtact gccatgggaa g                41

SEQ ID NO: 617      moltype = DNA  length = 32
FEATURE            Location/Qualifiers
misc_feature        1..32
                     note = source = /note="Description of Artificial Sequence:
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source              1..32
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 617
cagggacagt gcgcattctcc ctggtcacca ag                        32

SEQ ID NO: 618      moltype = DNA  length = 17
FEATURE            Location/Qualifiers
misc_feature        1..17
                     note = source = /note="Description of Artificial Sequence:
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source              1..17
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 618
cagggacagt caccaag                17

SEQ ID NO: 619      moltype = DNA  length = 31
FEATURE            Location/Qualifiers
misc_feature        1..31
                     note = source = /note="Description of Artificial Sequence:
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source              1..31
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 619
cagggacagt gcgcattctcc tggtcaccaa g                31

SEQ ID NO: 620      moltype = DNA  length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                     note = source = /note="Description of Artificial Sequence:
                     Syntheticoligonucleotide"
source              1..30
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 620

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cagggacagt gcgcattctt ggtcaccaag                               30

SEQ ID NO: 621      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
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                    Syntheticoligonucleotide"
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 621
cagggacagt gcgcattccc tctggtcacc aag                             33

SEQ ID NO: 622      moltype = DNA  length = 29
FEATURE            Location/Qualifiers
misc_feature        1..29
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
source              1..29
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 622
cagggacagt gcgcattctg gtcaccaag                                   29

SEQ ID NO: 623      moltype = DNA  length = 17
FEATURE            Location/Qualifiers
misc_feature        1..17
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
source              1..17
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 623
cagggacggt caccaag                                              17

SEQ ID NO: 624      moltype = DNA  length = 31
FEATURE            Location/Qualifiers
misc_feature        1..31
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
source              1..31
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 624
cggggacagg gcgcattccc tggtcaccaa g                               31

SEQ ID NO: 625      moltype = DNA  length = 32
FEATURE            Location/Qualifiers
misc_feature        1..32
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
source              1..32
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 625
cgacctgaat gctgtgcggc tctgcttcca gg                             32

SEQ ID NO: 626      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 626
cgacctgaat gctgtgcggc atctgcttcc agg                             33

SEQ ID NO: 627      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = source = /note="Description of Artificial Sequence:
                    Syntheticoligonucleotide"
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 627
cgacctgaat gctgtgcggc ttctgcttcc agg 33

SEQ ID NO: 628 moltype = DNA length = 28
FEATURE Location/Qualifiers
misc_feature 1..28
note = source = /note="Description of Artificial Sequence:
Syntheticoligonucleotide"
source 1..28
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 628
cgacctgaat gctgtgtctg cttccagg 28

SEQ ID NO: 629 moltype = DNA length = 33
FEATURE Location/Qualifiers
misc_feature 1..33
note = source = /note="Description of Artificial Sequence:
Syntheticoligonucleotide"
source 1..33
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 629
cgacctgcat gctgtgcggc atctgcttcc agg 33

SEQ ID NO: 630 moltype = DNA length = 33
FEATURE Location/Qualifiers
misc_feature 1..33
note = source = /note="Description of Artificial Sequence:
Syntheticoligonucleotide"
source 1..33
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 630
cgacctgcat gctgtgcggc ttctgcttcc agg 33

SEQ ID NO: 631 moltype = DNA length = 28
FEATURE Location/Qualifiers
misc_feature 1..28
note = source = /note="Description of Artificial Sequence:
Syntheticoligonucleotide"
source 1..28
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 631
cgacctgcat gctgtgtctg cttccagg 28

SEQ ID NO: 632 moltype = DNA length = 33
FEATURE Location/Qualifiers
misc_feature 1..33
note = source = /note="Description of Artificial Sequence:
Syntheticoligonucleotide"
source 1..33
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 632
cgacctgaat gctgtgcgac atctgcttcc agg 33

SEQ ID NO: 633 moltype = DNA length = 32
FEATURE Location/Qualifiers
misc_feature 1..32
note = source = /note="Description of Artificial Sequence:
Syntheticoligonucleotide"
source 1..32
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 633
tggggactac gacctgaatg ctgtgcggt ct 32

SEQ ID NO: 634 moltype = DNA length = 27
FEATURE Location/Qualifiers
misc_feature 1..27
note = source = /note="Description of Artificial Sequence:
Syntheticoligonucleotide"
source 1..27
mol_type = other DNA

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                                organism = synthetic construct
SEQUENCE: 634
tggggactac gaatgctgtg cggtctct                                27

SEQ ID NO: 635                moltype = DNA length = 66
FEATURE                        Location/Qualifiers
misc_feature                  1..66
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
source                        1..66
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 635
tggggactac gagcaggcag aagtatgcaa agcatgcatc tcaattcctg aatgctgtgc 60
ggctct                                                                66

SEQ ID NO: 636                moltype = DNA length = 72
FEATURE                        Location/Qualifiers
misc_feature                  1..72
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
source                        1..72
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 636
tggggactac gaagaaggcg atagaaggcc atgcgctgcg aatcgggagc ggctgaatg 60
ctgtgcggct ct                                                        72

SEQ ID NO: 637                moltype = DNA length = 99
FEATURE                        Location/Qualifiers
misc_feature                  1..99
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
source                        1..99
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 637
tggggactac gatgactcgc tgcgctcggc cgttcggctg cggcgagcgg tatcagctca 60
ctcaaaggcg gtaatacggc ctgaatgctg tgcgctct                        99

SEQ ID NO: 638                moltype = DNA length = 14
FEATURE                        Location/Qualifiers
misc_feature                  1..14
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
source                        1..14
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 638
tgggggtgcgg ctct                                                  14

SEQ ID NO: 639                moltype = DNA length = 27
FEATURE                        Location/Qualifiers
misc_feature                  1..27
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
source                        1..27
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 639
tggggactac gactgctgtg cggtctct                                27

SEQ ID NO: 640                moltype = DNA length = 39
FEATURE                        Location/Qualifiers
misc_feature                  1..39
                                note = source = /note="Description of Artificial Sequence:
                                Syntheticoligonucleotide"
source                        1..39
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 640
cgacctgaat gctgtgcggc gctctggctt cattcaatc                    39

SEQ ID NO: 641                moltype = DNA length = 35
FEATURE                        Location/Qualifiers
misc_feature                  1..35

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	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
source	1..35 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 641		
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SEQ ID NO: 642	moltype = DNA length = 36 Location/Qualifiers	
FEATURE	1..36	
misc_feature	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
source	1..36 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 642		
cgacctgaat gctgtgcggc atctgcttcc aggtga		36
SEQ ID NO: 643	moltype = DNA length = 25 Location/Qualifiers	
FEATURE	1..25	
misc_feature	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
source	1..25 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 643		
cgacctgaat gctgcttcca ggtga		25
SEQ ID NO: 644	moltype = DNA length = 28 Location/Qualifiers	
FEATURE	1..28	
misc_feature	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
source	1..28 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 644		
cgacctgaat gcttctgctt ccaggtga		28
SEQ ID NO: 645	moltype = DNA length = 22 Location/Qualifiers	
FEATURE	1..22	
misc_feature	note = source = /note="Description of Artificial Sequence: Syntheticoligonucleotide"	
source	1..22 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 645		
cgacctgaat gcttccaggt ga		22
SEQ ID NO: 646	moltype = AA length = 4 Location/Qualifiers	
FEATURE	1..4	
source	mol_type = protein organism = Homo sapiens	
SEQUENCE: 646		
EEQL		4
SEQ ID NO: 647	moltype = AA length = 6 Location/Qualifiers	
FEATURE	1..6	
source	mol_type = protein organism = Homo sapiens	
SEQUENCE: 647		
AEVSQA		6
SEQ ID NO: 648	moltype = AA length = 4 Location/Qualifiers	
FEATURE	1..4	
source	mol_type = protein organism = Mus musculus	
SEQUENCE: 648		
GEQL		4

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SEQ ID NO: 649	moltype = AA length = 6
FEATURE	Location/Qualifiers
source	1..6
	mol_type = protein
	organism = Mus musculus
SEQUENCE: 649	
AEVSKA	6
SEQ ID NO: 650	moltype = AA length = 4
FEATURE	Location/Qualifiers
REGION	1..4
	note = source = /note="Description of Artificial Sequence: Syntheticpeptide"
source	1..4
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 650	
GEQL	4
SEQ ID NO: 651	moltype = AA length = 6
FEATURE	Location/Qualifiers
REGION	1..6
	note = source = /note="Description of Artificial Sequence: Syntheticpeptide"
source	1..6
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 651	
AEVSQA	6

That which is claimed:

1. A nucleic acid molecule comprising a polynucleotide encoding an RNA-guided nuclease (RGN) polypeptide, wherein said RGN polypeptide comprises an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 11, 45, or 54;

wherein said polynucleotide encoding an RGN polypeptide is operably linked to a promoter heterologous to said polynucleotide.

2. The nucleic acid molecule of claim 1, wherein said RGN polypeptide comprises the amino acid sequence set forth as SEQ ID NO: 11, 45, or 54.

3. The nucleic acid molecule of claim 1, wherein said RGN polypeptide is nuclease dead or functions as a nickase.

4. The nucleic acid molecule of claim 1, wherein the RGN polypeptide is operably fused to a base-editing polypeptide.

5. The nucleic acid molecule of claim 4, wherein the base-editing polypeptide is a deaminase.

6. The nucleic acid molecule of claim 1, wherein the RGN polypeptide is operably fused to an effector domain.

7. The nucleic acid molecule of claim 6, wherein the effector domain is a cleavage domain, a deaminase domain, or an expression modulator domain.

8. The nucleic acid molecule of claim 7, wherein the expression modulator domain is an epigenetic modification domain, a transcriptional repressor domain, or a transcriptional activation domain.

9. The nucleic acid molecule of claim 6, wherein the effector domain is operably fused at the N-terminus or at the C-terminus of said RGN polypeptide.

10. The nucleic acid molecule of claim 6, wherein the effector domain is operably fused at an internal location of said RGN polypeptide.

11. The nucleic acid molecule of claim 1, wherein the RGN polypeptide is capable of binding a guide RNA capable of hybridizing to a target DNA sequence.

12. The nucleic acid molecule of claim 11, wherein said target DNA sequence is located adjacent to a protospacer adjacent motif (PAM) site of:

- (a) nnnncc (SEQ ID NO: 6), wherein the RGN polypeptide comprises an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 11;
- (b) nnraaty (SEQ ID NO: 50), wherein the RGN polypeptide comprises an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 45; or
- (c) nnnnnc (SEQ ID NO: 59), wherein the RGN polypeptide comprises an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 54.

13. A vector comprising the nucleic acid molecule of claim 1.

14. The vector of claim 13, wherein said vector further comprises at least one nucleotide sequence encoding a guide RNA, and wherein the guide RNA comprises:

- (a) a CRISPR RNA comprising a CRISPR repeat sequence having at least 95% sequence identity to SEQ ID NO: 12, wherein the RGN polypeptide comprises an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 11;
- (b) a CRISPR RNA comprising a CRISPR repeat sequence having at least 95% sequence identity to SEQ ID NO: 46, wherein the RGN polypeptide comprises an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 45; or
- (c) a CRISPR RNA comprising a CRISPR repeat sequence having at least 95% sequence identity to SEQ ID NO: 55, wherein the RGN polypeptide comprises an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 54.

15. The vector of claim 14, wherein the guide RNA comprises:

- (a) a tracrRNA comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 13,

- wherein the RGN polypeptide comprises an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 11;
- (b) a tracrRNA comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 47, wherein the RGN polypeptide comprises an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 45; or
- (c) a tracrRNA comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 56, wherein the RGN polypeptide comprises an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 54.
16. A cell comprising the nucleic acid molecule of claim 1.
17. A system, said system comprising:
- a) one or more guide RNAs (gRNAs), or one or more nucleotide sequences encoding the one or more gRNAs; and
- b) an RNA-guided nuclease (RGN) polypeptide comprising an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 11, 45, or 54, or a nucleotide sequence encoding the RGN polypeptide; wherein at least one of said nucleotide sequence encoding the one or more gRNAs or encoding the RGN polypeptide is operably linked to a promoter heterologous to said nucleotide sequence.
18. The system of claim 17, wherein said RGN polypeptide comprises the amino acid sequence set forth as SEQ ID NO: 11, 45, or 54.
19. The system of claim 17, wherein the one or more gRNAs hybridize to a target DNA sequence within a eukaryotic cell.
20. The system of claim 17, wherein said RGN polypeptide is nuclease dead or functions as a nickase.
21. The system of claim 17, wherein the RGN polypeptide is operably linked to a base-editing polypeptide.
22. The system of claim 21, wherein the base-editing polypeptide is a deaminase.
23. The system of claim 17, wherein the RGN polypeptide is operably fused to an effector domain.
24. The system of claim 23, wherein the effector domain is a cleavage domain, a deaminase domain, or an expression modulator domain.
25. The system of claim 24, wherein the expression modulator domain is an epigenetic modification domain, a transcriptional repressor domain, or a transcriptional activation domain.
26. The system of claim 23, wherein the effector domain is operably fused at the N-terminus or at the C-terminus of said RGN polypeptide.
27. The system of claim 23, wherein the effector domain is operably fused at an internal location of said RGN polypeptide.
28. The system of claim 17, wherein said system further comprises one or more donor polynucleotides.
29. A method for binding a target DNA sequence comprising delivering a system of claim 17, to said target DNA sequence or a cell comprising the target DNA sequence.
30. The method of claim 29, wherein said system binds to and cleaves and/or modifies said target DNA sequence.
31. A method for cleaving and/or modifying a target DNA sequence, comprising contacting the target DNA sequence with:
- a) an RNA-guided nuclease (RGN) polypeptide, wherein said RGN polypeptide comprises an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 11, 45, or 54; and
- b) one or more guide RNAs capable of hybridizing to the target DNA sequence, thereby cleaving and/or modifying said target DNA sequence to obtain a modified target DNA sequence.
32. The method of claim 31, wherein said modified target DNA sequence comprises insertion of heterologous DNA into the target DNA sequence, deletion of at least one nucleotide from the target DNA sequence, or mutation of at least one nucleotide in the target DNA sequence.
33. The method of claim 31, wherein the target DNA sequence is within a cell.
34. The method of claim 33, wherein the cell is a eukaryotic cell.
35. The method of claim 33, further comprising culturing the cell; and selecting cells comprising said modified DNA sequence.
36. An RNA polynucleotide comprising at least one synthetic ribonucleotide analogue, wherein said RNA polynucleotide comprises a nucleotide sequence encoding an RNA-guided nuclease (RGN) polypeptide comprising an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 11, 45, or 54.
37. The RNA polynucleotide of claim 36, wherein said RGN polypeptide comprises the amino acid sequence set forth as SEQ ID NO: 11, 45, or 54.
38. The RNA polynucleotide of claim 36, wherein the RNA polynucleotide further comprises a nucleotide sequence encoding a base-editing polypeptide operably linked to the nucleotide sequence encoding the RGN polypeptide.
39. The RNA polynucleotide of claim 36, wherein the RNA polynucleotide is an mRNA.
40. A eukaryotic cell comprising a nucleic acid molecule comprising a polynucleotide encoding an RNA-guided nuclease (RGN) polypeptide, wherein said RGN polypeptide comprises an amino acid sequence having at least 90% to SEQ ID NO: 11, 45, or 54.
41. The eukaryotic cell of claim 40, wherein said RGN polypeptide comprises the amino acid sequence set forth as SEQ ID NO: 11, 45, or 54.
42. The eukaryotic cell of claim 40, wherein said polynucleotide is an mRNA.
43. The eukaryotic cell of claim 40, wherein the RGN polypeptide is operably fused to a base-editing polypeptide.
44. The eukaryotic cell of claim 43, wherein the base-editing polypeptide is a deaminase.
45. The eukaryotic cell of claim 40, wherein the RGN polypeptide is operably fused to an effector domain.
46. The eukaryotic cell of claim 45, wherein the effector domain is a cleavage domain, a deaminase domain, or an expression modulator domain.
47. The eukaryotic cell of claim 46, wherein the expression modulator domain is an epigenetic modification domain, a transcriptional repressor domain, or a transcriptional activation domain.
48. The eukaryotic cell of claim 45, wherein the effector domain is operably fused at the N-terminus or at the C-terminus of said RGN polypeptide.

49. The eukaryotic cell of claim 45, wherein the effector domain is operably fused at an internal location of said RGN polypeptide.

50. A eukaryotic cell comprising an RNA-guided nuclease (RGN) polypeptide, wherein said RGN polypeptide comprises an amino acid sequence having at least 90% to SEQ ID NO: 11, 45, or 54.

51. The eukaryotic cell of claim 50, wherein said RGN polypeptide comprises an amino acid sequence having at least 95% to SEQ ID NO: 11, 45, or 54.

52. The eukaryotic cell of claim 50, wherein said RGN polypeptide comprises the amino acid sequence set forth as SEQ ID NO: 11, 45, or 54.

53. The eukaryotic cell of claim 50, wherein said RGN polypeptide is capable of cleaving a target DNA sequence of a DNA molecule upon binding said target DNA molecule.

54. The eukaryotic cell of claim 53, wherein said target DNA sequence is located adjacent to a protospacer adjacent motif (PAM).

55. The eukaryotic cell of claim 53, wherein cleavage by said RGN polypeptide generates a double-stranded break.

56. The eukaryotic cell of claim 53, wherein cleavage by said RGN polypeptide generates a single-stranded break.

57. The eukaryotic cell of claim 50, wherein said RGN polypeptide is nuclease inactive or a nickase.

58. The eukaryotic cell of claim 50, wherein the RGN polypeptide is operably fused to a base-editing polypeptide.

59. The eukaryotic cell of claim 58, wherein the base-editing polypeptide is a deaminase.

60. The eukaryotic cell of claim 50, wherein the RGN polypeptide is operably fused to an effector domain.

61. The eukaryotic cell of claim 60, wherein the effector domain is a cleavage domain, a deaminase domain, or an expression modulator domain.

62. The eukaryotic cell of claim 61, wherein the expression modulator domain is an epigenetic modification domain, a transcriptional repressor domain, or a transcriptional activation domain.

63. The eukaryotic cell of claim 60, wherein the effector domain is operably fused at the N-terminus or at the C-terminus of said RGN polypeptide.

64. The eukaryotic cell of claim 60, wherein the effector domain is operably fused at an internal location of said RGN polypeptide.

65. The eukaryotic cell of claim 50, wherein the RGN polypeptide comprises one or more nuclear localization signals.

66. A delivery vector comprising: (A) a polynucleotide encoding an RNA-guided nuclease (RGN) polypeptide; or (B) an RGN polypeptide,

wherein said RGN polypeptide comprises an amino acid sequence:

- (i) having at least 95% sequence identity to SEQ ID NO: 11, 45, or 54;
- (ii) having at least 98% sequence identity to SEQ ID NO: 1 or 19; or
- (iii) set forth as SEQ ID NO: 36;

67. The delivery vector of claim 66, wherein the delivery vector comprises said polynucleotide of (A), and wherein the delivery vector further comprises a polynucleotide encoding a guide RNA capable of binding to said RGN polypeptide.

68. The delivery vector of claim 67, wherein at least one of said polynucleotide encoding an RGN polypeptide and

said polynucleotide encoding a guide RNA is operably linked to a promoter heterologous to said polynucleotide.

69. The delivery vector of claim 67, wherein the delivery vector further comprises one or more donor polynucleotides.

70. The delivery vector of claim 66, wherein the delivery vector comprises said polynucleotide of (A), and wherein the delivery vector is an adeno-associated virus (AAV) particle.

71. The delivery vector of claim 66, wherein the delivery vector comprises a lipid nanoparticle.

72. The delivery vector of claim 71, wherein the delivery vector comprises said polynucleotide of (A), wherein said polynucleotide of (A) is an mRNA, wherein the delivery vector further comprises a polynucleotide encoding a guide RNA capable of binding to said RGN polypeptide, and wherein said polynucleotide encoding a guide RNA is an mRNA.

73. The delivery vector of claim 71, wherein the delivery vector comprises said RGN polypeptide of (B), wherein the delivery vector further comprises a guide RNA, and wherein said guide RNA is capable of binding to said RGN polypeptide to form a ribonucleoprotein complex.

74. The delivery vector of claim 66, wherein said RGN polypeptide comprises the amino acid sequence set forth as SEQ ID NO: 11, 45, 54, 1, or 19.

75. A nucleic acid molecule comprising a polynucleotide encoding a CRISPR RNA (crRNA), wherein the crRNA comprises a spacer and a CRISPR repeat, wherein the CRISPR repeat comprises a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 2, 28, 37, 46, or 55; and wherein said polynucleotide encoding a crRNA is operably linked to a promoter heterologous to said polynucleotide.

76. The nucleic acid molecule of claim 75, wherein the crRNA comprises:

- (a) a CRISPR repeat comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 2, wherein the RGN polypeptide comprises an amino acid sequence having at least 98% sequence identity to SEQ ID NO: 1;
- (b) a CRISPR repeat comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 28, wherein the RGN polypeptide comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 27;
- (c) a CRISPR repeat comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 37, wherein the RGN polypeptide comprises an amino acid sequence set forth as SEQ ID NO: 36;
- (d) a CRISPR repeat comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 46, wherein the RGN polypeptide comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 45; or
- (e) a CRISPR repeat comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 55, wherein the RGN polypeptide comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 54.

77. A vector comprising the nucleic acid molecule of claim 75.

78. The vector of claim 77, wherein said vector further comprises a polynucleotide encoding a trans-activating CRISPR RNA (tracrRNA).

79. The vector of claim **78**, wherein said tracrRNA comprises:

- (a) a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 3, wherein the crRNA comprises a CRISPR repeat comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 2;
- (b) a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 29, wherein the crRNA comprises a CRISPR repeat comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 28;
- (c) a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 38, wherein the crRNA comprises a CRISPR repeat comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 37;
- (d) a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 47, wherein the crRNA comprises a CRISPR repeat comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 46; or
- (e) a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 56, wherein the crRNA comprises a CRISPR repeat comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 55.

80. A nucleic acid molecule comprising a polynucleotide encoding a trans-activating CRISPR RNA (tracrRNA) comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NOs: 13, 21, 29, 38, 47, or 56; and wherein said polynucleotide encoding a tracrRNA is operably linked to a promoter heterologous to said polynucleotide.

81. The nucleic acid molecule of claim **80**, wherein the tracrRNA comprises:

- (a) a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 13, wherein the RGN polypeptide comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 11;
- (b) a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 21, wherein the RGN polypeptide comprises an amino acid sequence having at least 98% sequence identity to SEQ ID NO: 19;
- (c) a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 29, wherein the RGN polypeptide comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 27;
- (d) a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 38, wherein the RGN polypeptide comprises an amino acid sequence set forth as SEQ ID NO: 36;
- (e) a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 47, wherein the RGN polypeptide comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 45; or
- (f) a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 56, wherein the RGN polypeptide comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 54.

82. A vector comprising the nucleic acid molecule of claim **80**.

83. The vector of claim **82**, wherein the vector further comprises a polynucleotide encoding a CRISPR RNA (crRNA) comprising a spacer and a CRISPR repeat.

84. The vector of claim **83**, wherein the CRISPR repeat of the crRNA comprises:

- (a) a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 12, wherein the tracrRNA comprises a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 13;
- (b) a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 20, wherein the tracrRNA comprises a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 21;
- (c) a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 28, wherein the tracrRNA comprises a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 29;
- (d) a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 37, wherein the tracrRNA a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 38;
- (e) a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 46, wherein the tracrRNA comprises a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 47; or
- (f) a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 55, wherein the tracrRNA comprises a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 56.

85. A method for producing a genetically modified mammalian hematopoietic progenitor cell having decreased BCL11A mRNA and protein expression, the method comprising introducing into an isolated human hematopoietic progenitor cell:

- a) an RNA-guided nuclease (RGN) polypeptide, wherein the RGN polypeptide comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 1, 11, 19, 27, 36, 45, or 54, or a polynucleotide encoding said RGN polypeptide, wherein said polynucleotide encoding the RGN polypeptide is operably linked to a promoter to enable expression of the RGN polypeptide in the cell; and
- b) a guide RNA (gRNA), wherein the gRNA comprises a CRISPR repeat comprising a nucleotide sequence having at least 95% sequence identity to SEQ ID NO: 2, 12, 20, 28, 37, 46, or 55, or a polynucleotide encoding said gRNA, wherein said polynucleotide encoding the gRNA is operably linked to a promoter to enable expression of the gRNA in the cell,

wherein the RGN polypeptide and gRNA are expressed in the cell and cleave at the BCL11A enhancer region, resulting in genetic modification of the human hematopoietic progenitor cell and reducing the mRNA and/or protein expression of BCL11A.

86. The method of claim **85**, wherein the gRNA further comprises a spacer sequence that targets SEQ ID NO: 473, 474, 475, 476, 477, or 478.

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